

Protein Engineering

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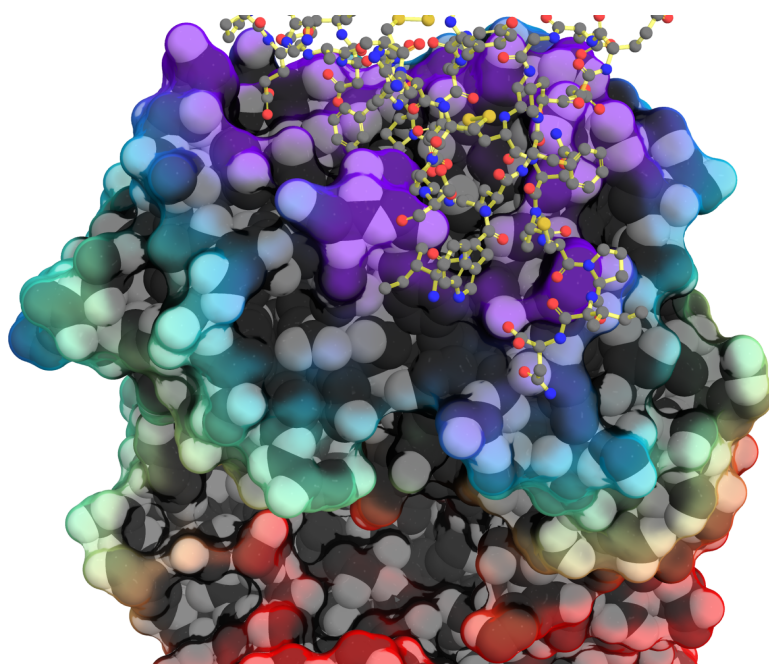
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Preface



Welcome to protein engineering, the art and science of improving proteins. This text will teach you how to better frame your protein engineering problem and to choose approaches to reach your goal. Both online and pdf versions of this text are free to view and download here (<https://betterenzyme.com>).

The cover image shows part of the antibody drug trastuzumab (Herceptin) in a space-filling representation bound to its target

Preface

receptor HER2, shown in a ball-and-stick representation. Protein engineering transformed a mouse antibody into a humanized therapeutic while preserving its ability to recognize HER2, making trastuzumab one of the first successful antibody drugs.

1 Setting Protein Engineering Goals

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Summary. Protein engineering is the modification of proteins to improve their properties, usually by replacing amino acid residues with other amino acids. Proteins evolved in nature for their biological role, but for other applications, they usually require improvements. The four molecular goals are improvements in stability, binding, reactivity and selectivity. Biotechnology uses engineered proteins in three main application areas: medicine, agriculture, and industry. Protein engineering in medicine has tuned the bioavailability of insulin and increased the potency and plasma half-life of monoclonal antibodies. Protein engineering in agriculture has increased the insect resistance of crops and stabilized enzymes for use as feed additives. Protein engineering for industrial applications include stabilizing detergent enzymes and engineering enzymes for pharmaceutical synthesis. Replacing chemical methods for pharmaceutical manufacture with biocatalysis usually yields a greener, more environmentally friendly process.

Learning goals

- While evolution in nature improves proteins for their natural function, application of proteins to non-natural functions requires protein engineering.

1 *Setting Protein Engineering Goals*

- The four molecular protein engineering goals are improvements in stability, binding, reactivity and selectivity. Identifying the goal is a critical first step of any protein engineering project.
- Achieving these molecular goals leads to a broad range of user benefits including lower costs, safer drugs, greener chemistry, improved diagnostics, or accessible healthcare.
- Engineered proteins have applications in medicine, agriculture and industry. Your home likely contains engineered proteins.

1.1 Why proteins need to be engineered

Over millennia humans have domesticated crops and animals by breeding to select for desired traits. Current wheat produces more grain than wild wheat; dogs are better companions than wolves. The molecular understanding of biology in recent decades created more powerful approaches to improve biology for human applications. These applications of biology to make useful products in medicine, agriculture and industry are collectively called biotechnology. The worldwide sales of biotechnology were ~US\$160 billion in 2017.^[1] The most significant application area is medicine, sometimes called red biotechnology, **Figure 1.1**. The next largest is agriculture, called green biotechnology and the smallest is industry, called white biotechnology.

The two primary technologies within biotechnology are genetic engineering and protein engineering. Genetic engineering changes the genetic code of an organism. The most common genetic engineering is adding a gene from one organism to another organism so that it makes a protein that it did not make previously. Such recombinant organisms gain new abilities. For example, adding the gene

1.1 Why proteins need to be engineered

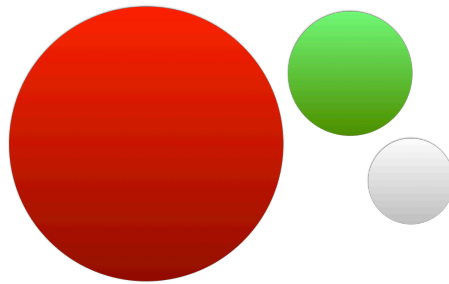


Figure 1.1. Biotechnology includes applications in medicine (red), agriculture (green) and industry (white). The areas of the circles approximate the relative worldwide sales in each area.

encoding human insulin to *E. coli* bacteria allowed manufacture of this essential therapeutic protein. Adding genes for proteins toxic to insects into crop plants made them resistant to insects. Combining genes to degrade different hydrocarbons into one bacteria enabled it to degrade oil spills more effectively.

Protein engineering is the modification of proteins to improve them for specific applications. Altering the gene that encodes the natural protein creates a new gene that encodes an improved variant protein. Changing the protein can improve its stability, binding, catalytic activity or even give it new functions.^[2] The modifications are typically substitution of one amino acid residue with another one, but sometimes include adding or removing amino acid residues, even entire domains of a protein. For example, engineering of subtilisin, a detergent protease, replaced an easily oxidized amino acid with one less easily oxidized, thereby increasing its stability to bleach during washing,^[3] **Figure 1.2.**

Applications that differ from the natural function usually require protein engineering. Proteins have evolved in nature for specific natural functions. When proteins are used for another function or

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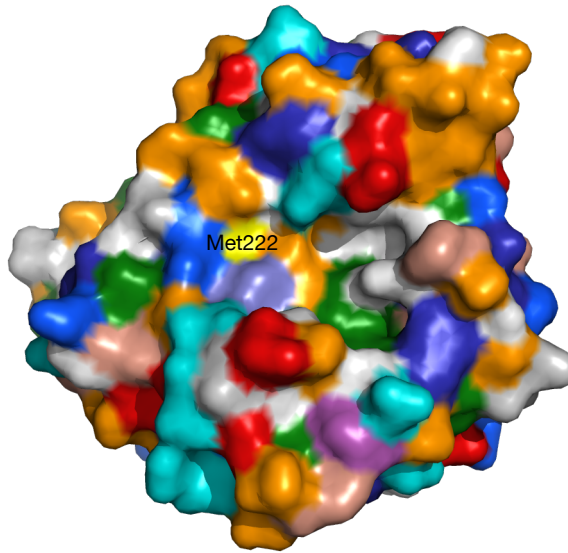


Figure 1.2. Surface representation of the protease subtilisin where the twenty amino acids are colored in one of twelve different colors according to traditional amino acid properties. Protein engineering increased the stability of this protease to bleach by replacing methionine 222 (labelled yellow patch near the center) with alanine.

1.1 Why proteins need to be engineered

in a different way, they will likely be not good enough. Natural selection chooses proteins that improve fitness in nature, not for various applications that humans identify. Natural proteins may be accidentally suitable for these applications, but won't be optimized for them. For example, nature has not evolved proteins tolerant to organic solvents since no organisms live in organic solvents, but some proteins evolved to tolerate high temperatures often also tolerate organic solvents. In some cases, the different requirements are obvious. For example, an enzyme-catalyzed manufacture of a pharmaceutical intermediate may involve an unnatural substrate, organic cosolvents, elevated temperatures and high concentrations of substrates and products. In these cases, protein engineering can adjust the active site to fit the new substrate to increase reaction rate and stabilize the protein to organic solvents, elevated temperatures and high substrate and product concentrations.

In other cases, the differences between the natural protein function and application may be subtle. For example, human insulin regulates the glucose levels in the blood. Patients with type I diabetes cannot make insulin and require injections of human insulin. Although the goal is the same - to maintain glucose levels, the dosing method differs. Patients inject insulin several times per day in contrast to the pancreas continuously secreting varying amounts of insulin. This difference in dosing required engineering fast-acting variants of human insulin for injection before meals and slow-acting variants to maintain glucose levels overnight.

Evolution does not maximize the properties of a protein, but only maintains the minimum necessary. Natural selection cannot distinguish between a protein that is improved just enough so that another protein limits function and a protein that is much better. Selection favors both equally since the much better property does not contribute to fitness. A hammer made from extra-hard steel does not drive nails any better than a hammer with steel just hard enough. Since most random mutations are detrimental, a much-

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better protein will eventually lose its extra abilities and return to just good enough.

An example of ‘just enough’ is that most proteins are just stable enough for their natural function.^[4,5] Proteins from mesophiles (organisms that grow at moderate temperatures) are not stable at high temperatures. In contrast, homologous proteins from thermophiles (organisms that grow at high temperatures) are stable at high temperatures. This stability shows that proteins from mesophiles could be more stable if needed.^[6] The reason for the marginal stability is that evolution cannot select for more than the minimum needed stability.^[7] An extra-stable protein in a mesophile does not give it any selective advantage, so this extra stability is slowly lost through genetic drift. One can also generalize this ‘just enough’ quality to other protein properties. Enzymes are just fast enough for their function, they bind their substrates just tightly enough and discriminate between substrates just as much as needed.

1.2 Four molecular-level protein engineering goals

To the person who does not know where he wants to go
there is no favorable wind. — Seneca

The application improvements enabled by engineered proteins vary widely because they depend on the function of the protein being engineered. As described below, some improvements make vaccines more effective, others improve the synthesis of pharmaceutical, still others enable plants to grow faster under heat stress. Despite this wide range of applications, the improvements themselves stem from only four types of protein property changes: stability, binding, reactivity or selectivity, **Table 1.1**. Later tables in this chapter list a wide range of applications, but also classify

1.2 Four molecular-level protein engineering goals

each application according to one of these four protein property changes.

Table 1.1. Four molecular-level goals in protein engineering.

Goal	Examples of changes
stability	tolerate heat, organic solvents, extremes of pH or other harsh conditions; tolerate storage or use for a long time at normal conditions; tolerate destabilizing substitutions; longer plasma lifetime; resistant to digestive proteases
binding	bind target ligand more tightly or less tightly; avoid binding to antibodies to avoid allergic reaction
catalytic activity	faster catalysis of existing substrates; expand catalysis to new substrates; inactivate undesired catalytic activity; enable catalysis of a new chemical reaction
selectivity	favor binding one ligand over another in the same solution; favor reaction of one of several competing substrates; favor formation of one of several possible products

Stability refers to the ability of a protein to maintain its function, usually binding to a target or catalyzing a reaction. Increasing protein stability allows it to tolerate harsh conditions such as high temperatures or the presence of bleach as in the detergent protease example above. It may make the protein last longer under normal application or storage conditions. For biopharmaceuticals, stability may also refer to extending the plasma lifetime of the protein. Most proteins are rapidly removed from the blood thus limiting

1 Setting Protein Engineering Goals

their therapeutic benefit. Increasing the plasma lifetime enhances the therapeutic benefit.

Binding refers to the affinity of the protein for a target molecule. Increases in the affinity of a protein for its target can lower the effective dose of a biopharmaceutical or lower the detection limit of a diagnostic test. For biopharmaceuticals, *decreases* in binding can also be important. Biopharmaceuticals should not bind to human antibodies to avoid causing an allergic reaction. Humanization of proteins engineers them so that they do not bind to human antibodies.

Reactivity refers to the ability of an enzyme to catalyze chemical reactions. Increases in reaction rates lowers the amount of enzyme needed for the chemical transformation. Increases in reactivity may improve reactions for existing substrates or may create new catalytic abilities by expanding reactivity to new substrates. Improvements in reactivity may enable an enzyme to catalyze new reaction types of chemical reaction, even those that enzymes in nature do not catalyze.

Selectivity refers to the relative binding or relative reactivity of two or more competing molecules in the solution. Selectivity may refer to the ability of a protein to bind one of several ligands or to catalyze the reaction of one of several substrates. It can also refer to the selective formation of one of several possible products from a single substrate. A common type of selectivity exhibited by enzymes is enantioselectivity where one enantiomer reacts faster than the other.

Identifying the protein engineering goal is a critical first step in any protein engineering project because it defines the approaches that one can use. Typically the target protein already has some desirable properties, but needs improvement on one of them. One seeks to improve this one property (stability, binding, reactivity or selectivity) while not degrading the already desirable properties. To

1.2 Four molecular-level protein engineering goals

fully define that goal one should also state the amount of change needed (e.g., a two-fold increase in stability) because it reveals how difficult the engineering will be.

Sometimes protein engineering changes several protein properties so that it can be hard to identify which property is the true protein engineering goal. The true goal is the one property that *must* change to improve the protein for the application. The other changes are side effects of engineering to achieve this goal.

For example, researchers wanted to engineer a protease that could degrade the protein gluten in the human digestive track so they could add it to the diet of gluten-intolerant patients.^[8] The researchers started with the protease KumaWT, which favored hydrolysis after a ProArg or ProLys sequence and wanted to change it to favor hydrolysis after a ProGln sequence since gluten contains many ProGln amino acid pairs in its sequence. The engineered protein showed changes in three properties: KumaWT was more reactive toward ProGln-containing peptides, more selective for ProGln versus ProArg sequences and bound the ProGln sequence more tightly to the protease binding site.

Which of these properties was the true goal? Imagine if the catalytic activity toward the ProGln sequence increased, but there was no change in the selectivity (the ProLys sequence also reacted) and no change in the binding (the increase in reactivity was not associated with better binding). The engineered protein would still be improved because it would degrade the ProGln containing peptides. Increasing reactivity is the true goal. In contrast, imagine if selectivity improved without changing the catalytic activity. (A decrease in the activity toward the ProLys sequence could increase the selectivity for the ProGln sequence without increasing the activity toward the ProGln sequence.) No more ProGln would be cleaved than with the starting enzyme. Thus, selectivity is not the true goal and the increase in selectivity is a side effect of increased

1 Setting Protein Engineering Goals

reactivity. Finally, consider the case where the binding improved, but not the selectivity or reactivity. Again, without an increase in reactivity, no more ProGln sequences would be cleaved. The authors hypothesized that the reason for the poor reactivity was that the ProGln sequence binds poorly. Thus, increasing binding was an approach to increase reactivity. If they achieved increased binding without a concomitant increase in reactivity, they would have failed and tried another approach to increase reactivity. Increased binding alone would not have yielded a better enzyme. Thus, protein engineering can simultaneously change several protein properties. The true protein engineering goal is the property that *must* change in order to see an improvement in the proposed application.

1.3 Application areas of engineered proteins

Examples from different application areas show how changes in protein stability, binding, reactivity and selectivity can improve proteins in widely different applications.

1.3.1 Engineering proteins for medicine

Biopharmaceuticals are biomolecules used as drugs. They can be proteins such as antibodies or enzymes; they can also be nucleic acids such as micro RNAs and even complete microbes such as attenuated live vaccines. This text considers only protein biopharmaceuticals.

The first protein biopharmaceuticals were recombinant equivalents of the natural protein where the biotechnology advantage was the ability to produce large amounts of protein. For example, blood factors that were isolated from donated blood could now

1.3 Application areas of engineered proteins

be produced in cell cultures due to genetic engineering. The subsequent generations of blood factors were improved engineered variants. For example, engineering increased the potency of the blood coagulation protein, Factor VII used to treat hemophilia, so that each dose required less protein,^[9] **Table 1.2.** The action of this blood coagulation protein depends on its binding to a membrane surface containing acidic phospholipids, which is characteristic of damaged vascular cells or of platelets that adhere to the damage as the clot forms. Mutagenesis of the membrane-binding domain of Factor VII yielded variants that bound more tightly to the membrane and therefore were more potent. The best variant contained five substitutions and was 149-296 fold better than wild-type Factor VII. Other researchers increased its potency with three amino acid substitutions that increased its catalytic activity 30-fold.^[10] Proteolytic activity of Factor VII activates the next factor in the coagulation cascade.

Table 1.2. Examples of protein engineering to improve bio-pharmaceuticals.

Category	Example of improvement	Property change
blood factors	more potent coagulation factor VII	increase binding to damaged cells; increased reactivity (proteolysis)
thrombolytics & anticoagulants	more potent and longer plasma half-life of tissue plasminogen activator	increase binding to fibrin; reduce binding to receptor that removes proteins from blood
hormones	fast-acting or long-lasting insulins	alter binding with other monomers

1 Setting Protein Engineering Goals

Category	Example of improvement	Property change
monoclonal antibodies	reduced immunogenicity of mouse-derived antibody	reduce binding to immune response proteins
growth factors	longer serum half-life of granulocyte colony-stimulating factor	increase binding to receptors that recycle proteins to blood stream
vaccines	less toxic pertussis (whooping cough) vaccine	inactivate reactivity that causes toxicity
other	less immunogenic asparaginase	reduce binding to immune response proteins

Insulin variants engineered for altered bioavailability are even better than human insulin at controlling blood sugar levels. The engineering relied on the fact that monomeric insulin is the active form, while multimeric forms are inactive. Insulin lispro is a fast-acting insulin analog for injections before meals because it favors the monomeric form.^[11] The amino acid sequence of lispro differs only in the reversal of the penultimate lysine and proline residues on the C-terminal end of the B-chain. (Insulin consists of 51 amino acids arranged as a dimer of an A-chain and a B-chain linked by disulfide bonds.) This residue change blocks the formation of insulin dimers and hexamers and increases the amount of monomeric insulin (the active form), thereby creating a faster-acting variant. In contrast, substitutions to create a slow-acting insulin analog, glargine, promote the association of the monomers into an insoluble precipitate, which then slowly

1.3 Application areas of engineered proteins

releases monomeric insulin to maintain basal levels of insulin for an extended time.

An essential class of therapeutic proteins is monoclonal antibodies and monoclonal antibody drug conjugates. Common goals of antibody engineering are increasing binding to the target and minimizing immunogenicity.^[12] One example of a therapeutic monoclonal antibody is trastuzumab (Herceptin®) used to treat breast cancer. This antibody interferes with the HER2 receptor (human ErbB2 receptor tyrosine kinase), which signals cell proliferation. Cancer cells, usually breast cancer, overexpress the HER2 receptor causing the cancer cells to grow uncontrollably. Interfering with the receptor stops uncontrolled growth.

Trastuzumab is a humanized version of a mouse antibody, **Figure 1.3**. Humanization prevents an allergic reaction to a mouse antibody. This humanization involved first, the transplantation of the complementarity-determining regions from the mouse antibody into a human IgG antibody and second, an optimization of the surrounding regions, called framework regions.^[13] The transplantation of the complementarity-determining regions changed 24 amino acids in the light chain and 32 in the heavy chain. This substitution did not yield a useful antibody. The engineered version bound less tightly than the mouse antibody and did not block cancer cell proliferation upon binding. Next, an additional seven substitutions in the surrounding (framework) regions increased the binding to the target 100-fold making it three-fold tighter than the mouse antibody. This tighter binding restored the ability of the antibody to block cell proliferation. Trastuzumab was approved in the USA in 1998 to treat metastatic breast cancer.

Growth factors and cytokines, such as vascular endothelial growth factor, epidermal growth factor and granulocyte colony-stimulating factor, stimulate cell recruitment, proliferation, morphogenesis, and differentiation. Inhibitors of these growth

1 Setting Protein Engineering Goals

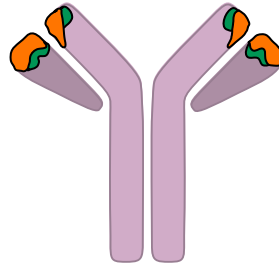


Figure 1.3. To engineer trastuzumab, researchers combined the complementarity-determining regions from a mouse antibody (orange; 56 amino acids) with a human IgG antibody (lavender). Full activity required seven additional substitutions (green) outside the binding region.

factors act as anticancer drugs, while enhanced growth factors promote wound repair. Natural growth factors lack stability and specificity; they are prone to degradation in serum and have multiple activities. In one example of protein engineering of a cytokine, Sarkar and coworkers increased the half-life and potency of granulocyte colony-stimulating factor by modifications that increased its recycling back to the bloodstream.^[14]

Cantor and coworkers engineered asparaginase, an enzyme for leukemia treatment, to be less immunogenic.^[15] Treatment of childhood acute lymphoblastic leukemia involves injections of asparaginase, which cleaves asparagine to aspartate, thereby depriving the cancerous cells of asparagine. Healthy cells can make asparagine, but these cancer cells cannot. Humans do not have an asparaginase enzyme, so the treatment uses enzyme from the bacteria *Escherichia coli*. One limitation is that some patients' immune systems recognize this enzyme as a non-human protein and inhibit it by binding it with an antibody or even inducing a severe allergic reaction. Cantor and coworkers identified amino

1.3 Application areas of engineered proteins

acid sequences on the surface of *E. coli* asparaginase that are likely to be recognized by human T-cells and replaced them with non-recognized amino acid sequences. The replacements maintained the catalytic activity of the asparaginase.

Vaccines against pertussis (whooping cough) contain engineered proteins. The bacteria that causes pertussis, *Bordetella pertussis*, secrete pertussis toxin protein, which consists of five subunits. Immunization with the pertussis toxin or just its S1 subunit protects against disease. However, the S1 subunit is an enzyme that catalyzes the ADP-ribosylation of GTP-binding proteins. This catalytic activity is the origin of the toxicity of the pertussis toxin. Two amino acid substitutions in the S1 subunit - Glu129Gly, Arg9Lys - deactivate its catalytic activity. This inactivated version of the S1 subunit is the vaccine against pertussis.^[16]

The mRNA vaccines against the SARS-CoV-2 virus encode the spike protein, which sits on the virus surface and binds to cell-surface proteins during infection. Upon binding to the target cell, the spike protein changes conformation. Researchers reasoned that the spike protein in the pre-binding conformation would make a better vaccine than the spike protein in the post-binding conformation. Binding antibodies to the spike protein before it binds to the cell could prevent infection. To stabilize the pre-binding conformation, researchers replaced six residues in the spike protein with proline.^[17] Proline limits the flexibility at those sites due to its ring structure and keeps the spike protein in the pre-binding conformation. This stabilization of the protein in the pre-binding conformation improved the vaccine.

1.3.2 Engineering proteins for agriculture

Agricultural biotechnology aims mainly to improve productivity of crops and animals. Some products, like bovine somatotrophin

1 Setting Protein Engineering Goals

(a growth hormone), are copies of the natural protein produced in recombinant *E. coli* bacteria. The pituitary gland produces small amounts of somatotrophin, but production in bacteria increases the availability and lowers the cost. Treating cows with somatotrophin increases milk production. In other cases, natural proteins are improved by engineering, **Table 1.3**.

Table 1.3. Examples of protein engineering to improve agriculture.

Category	Example of improvement	Property change
herbicide-resistant crops	glyphosate-resistant soybeans	add new reactivity (glyphosate oxidase)
insect-resistant crops	cotton resistant to bollworm	increase binding of <i>Bt</i> toxins to target proteins in insect gut
enhanced photosynthesis	faster growth of <i>Arabidopsis</i> at higher temperatures	increase stability of rubisco activase
modified products	algae producing an increased fraction of medium-chain fatty acids	increased reactivity toward medium-chain fatty acid precursors
feed additives	animal feed with higher phosphorus availability	increase stability of phytase

Glyphosate is a broad-spectrum, systemic herbicide that acts by inhibiting the enzyme EPSP synthase in aromatic amino acid biosynthesis. Farmers can control weeds by spraying glyphosate

1.3 Application areas of engineered proteins

on fields, but only if the crops are insensitive to glyphosate. Adding a glyphosate-insensitive variant of EPSP synthase from *Agrobacterium* sp. created the first glyphosate-resistant soybean plants.^[18] The next generation of glyphosate-resistant crops contain enzymes such as oxidases or acetyl transferases, which inactivate glyphosate. Protein engineering to increase the catalytic activity of these enzymes is an important goal.^[19]

Genetic modification of crops such as cotton, corn, and rice to express insecticidal crystal proteins produced by *Bacillus thuringiensis* protects the crops against insect pests. The toxicity of these proteins varies toward different insects, and some insects develop resistance to these toxins. Enhancing the toxicity of these proteins by increasing their binding to target proteins in the insect gut is an essential goal in agricultural biotechnology.^[20]

Heat stress hinders plant growth and lowers crop yields mainly due to reduced photosynthesis. This reduction is mainly due to an inactivation of a thermolabile ATPase, Rubisco activase, whose role is to maintain ribulose-1,5-bisphosphate carboxylase/oxygenase (rubisco) in its active state. Kurek and coworkers engineered a thermostable rubisco activase, which improved the growth of *Arabidopsis* plants at higher temperatures.^[21] The leaf area increased approximately 20% during moderate heat stress as compared to wild-type plants.

Engineering the selectivity of fatty acid synthesis enzymes changes the composition of the lipids that accumulate in plants. Whittle and Shanklin altered the chain length selectivity of a plant fatty acid desaturase to accept a 16-carbon instead of an 18-carbon substrate.^[22] Lin and Lee engineered algae to produce more medium chain length fatty acids relative to long-chain-length fatty acids for use in biodiesel production.^[23]

Engineering a more stable phytase, a feed additive enzyme, increased the nutritional value of feed and reduced pollution.

1 Setting Protein Engineering Goals

Phosphorus in grains and oilseeds often occurs as phytic acid (*myo*-inositol hexakisphosphate). Non-ruminants such as pigs and chicken digest phytic acid incompletely leading to high phosphorus in wastewater. The enzyme phytase in animal feed aids digestion of phytic acid by catalyzing its hydrolysis. Forming animal feed pellets requires pressing at high temperatures, so the phytases have been engineered for higher thermal stability to withstand this step.^[24]

1.3.3 Engineering proteins for industry

Industry uses enzymes for manufacturing and similar non-natural applications, **Table 1.4**. The three main application areas are detergents, food and beverage, and other, which includes enzymes for the synthesis of biofuels, fine chemicals and pharmaceuticals. The advantages of using enzymes over chemical reagents or catalysts are that they are faster, greener and more selective. The primary engineering goals are stability and faster catalysis, both of which lower the cost of the enzyme.

Table 1.4. Examples of protein engineering to improve industrial enzymes.

Category	Example of improvement	Property change
detergents	bleach-tolerant protease	stabilize subtilisin to oxidation
food and beverage	simplify starch hydrolysis to maltodextrins and glucose	stabilize α -amylase to low pH

1.3 Application areas of engineered proteins

Category	Example of improvement	Property change
other (e.g., biofuels, pharmaceuticals)	engineer transaminase for manufacture of diabetes drug	increase reactivity toward target substrate; increase stability to reaction conditions

Detergent enzymes such as proteases speed removal of food, blood and other stains on clothing. Subtilisin was the first industrial enzyme to be engineered.^[3] Subtilisin tolerated hot water and surfactants, but bleach rapidly inactivated it, which limited its usefulness. Replacement of an oxidation-sensitive methionine 222 in the active site (**Figure 1.2**) by alanine stabilized subtilisin while maintaining high activity. The cost of these industrial proteins are typically \$100/kg. In contrast, proteins for medical applications can cost 10⁵-fold more, \$10,000/g, because their manufacture and regulatory approval are more complex.

The largest volume application in food and beverages is the conversion of cornstarch to glucose catalyzed by α -amylase and glucoamylase. Glucose isomerase can isomerize the resulting glucose to high fructose corn syrup, which tastes sweeter than glucose. This process uses high temperatures to prevent microbial growth and high concentrations to minimize reactor size. Both α -amylase^[25] and glucose isomerase^[26] have been engineered to tolerate high temperatures and high glucose concentration. Glucose isomerase has a half-life = 50-100 d at the operating temperature of 55 °C. More than 500 tons of glucose isomerase produce millions of tons of high fructose corn syrup each year, corresponding to productivities of >10,000 kg corn syrup per kg enzyme.

The use of enzymes for chemical synthesis is known as biocatalysis.^[27] In pharmaceutical manufacture, the most significant application

1 Setting Protein Engineering Goals

is the penicillin G amidase-catalyzed manufacture of penicillin and cephalosporin antibiotics. This process yields more than 10,000 tons of antibiotics annually with a productivity of >600 kgs product/kg enzyme.^[28] These antibiotics are natural products, so they are the natural substrates for this enzymes. However, many other pharmaceuticals are not natural products, so the ability of enzymes to act on them is accidental. Engineering of enzymes to improve their ability to act on unnatural substrates and tolerate the harsh conditions of a chemical reactor is a common goal.

Engineered enzymes can also create new biochemical pathways within cells, and the whole cells can be used for synthesis. For example, Ran and Frost expanded the substrate range of an aldolase to create a new metabolic pathway to make shikimic acid for an influenza drug synthesis.^[29] Keasling and coworkers combined enzymes from different organisms and biochemical pathways to create a new biochemical pathway for the synthesis of artemisinin, an anti-malarial.^[30]

Green chemistry

One reason to use enzymes for manufacture of pharmaceuticals is to decrease the environmental impact of the process. The risk associated with a chemical depends both on how dangerous it is (hazard) and on one's contact with it (exposure), Equation 1.1.

$$risk = hazard \cdot exposure \quad (1.1)$$

In the past, governments and industry focused on reducing risk by minimizing exposure. Lab coats, safety glasses, and other chemical handling rules limit the exposure of workers to hazardous chemicals. Green chemistry instead focuses on the hazard and tries to minimize it instead of exposure.^[31] Preventing problems

1.3 Application areas of engineered proteins

is inevitably easier and less expensive than contending with difficulties after they occur. Green chemistry is the design of chemical products and processes that reduce or eliminate hazardous substances. Replacing chemical manufacturing steps with enzyme-catalyzed steps is an essential tool in green chemistry. Enzyme-catalyzed steps often replace hazardous solvents with water and hazardous catalysts with biodegradable enzymes. One example of greener pharmaceutical manufacture is the improved synthesis of sitagliptin, the active ingredient in an oral type 2 diabetes drug. The difficult step is the addition of the amino group with the correct orientation, **Figure 1.4**.

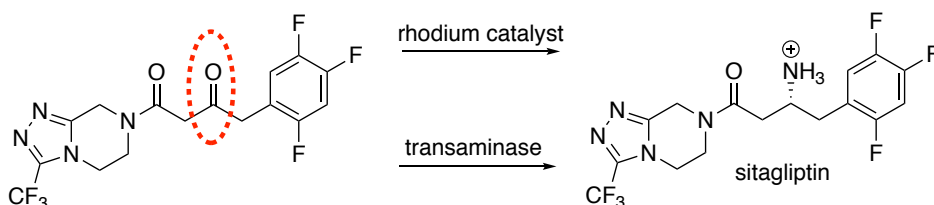


Figure 1.4. Two improved routes to sitagliptin showing the difficult step of inserting the amino group with the correct configuration. Improvements in the chemocatalytic process (rhodium catalyst) won a Green Chemistry award in 2006. Further improvements by switching to a biocatalytic process (transaminase) won a second Green Chemistry award in 2010.

Researchers at Merck & Co. had already dramatically improved the original synthesis of sitagliptin by using a catalytic asymmetric hydrogenation with a rhodium catalyst. This improved synthesis won a Green Chemistry award in 2006. Biocatalysis further improved this synthesis. Researchers at Codexis and Merck & Co. used a transaminase-catalyzed formation of an amine, which eliminated four steps including one that required a high pressure reactor.^[32] The biocatalysis approach resulted in a 10-13% higher

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overall yield and 19% less total waste. This improved synthesis won a second Green Chemistry award in 2010. Enabling this synthesis required extensive protein engineering of the transaminase. The starting transaminase did not react at all with the required substrate and was unstable under the reaction conditions. The engineering replaced 27 amino acids creating a highly active, highly selective and stable enzyme.

1.4 A look ahead: two main strategies of protein engineering

Protein engineering typically begins with proteins that already possess some, but not all, of the properties required for a particular application. These starting points are often natural proteins, but they may also be proteins proposed through computational design, although computational methods do not yet reliably generate proteins with all the properties needed for most applications.

The next chapter introduces the experimental methods used in protein engineering. Genetic modification of microorganisms enables the production of desired proteins in large quantities. Molecular biology techniques can then be used to alter the DNA encoding these proteins and create variants with improved properties.

Chapter 3 introduces the logic of protein engineering: protein states, defined as particular structural and functional conditions of proteins, determine protein properties, and changes in the relative Gibbs energies of these states alter those properties. Chapter 4 introduces protein structure and the molecular modeling of protein states.

The remaining chapters focus on the two principal strategies of protein engineering: rational design and directed evolution. Rational

design seeks beneficial changes on the basis of protein structure, molecular mechanism, and computational modeling. Chapters 5–8 examine the rational design of protein stability, binding, reactivity, and selectivity, explaining both how these properties are measured and how they can be improved. Chapters 9–11 cover directed evolution, in which protein variants are generated, screened for improved properties, and subjected to repeated cycles of optimization.

Glossary

Binding is the ability of a protein to associate with a target molecule.

Protein engineering goal is the protein property that *must* change in order to reach the desired application improvement. Most protein engineering goals are one of these four: stability, binding, reactivity or selectivity. The specific application improvement depends on the function of the protein being improved.

Reactivity is the ability of an enzyme to catalyze a chemical transformation.

Selectivity is the ability of a protein to discriminate between competing molecules in solution. This discrimination can involve selectivity in binding the competing molecules or selectivity in reacting with the competing molecules. Selectivity can also refer to the favored formation of one of several competing products from a single substrate.

Stability is the ability of a protein to remain folded and functional. The function can be binding to a target or catalyzing a reaction. Stability can also refer to the ability of a biopharmaceutical to persist in plasma.

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Problems

1. *The application benefits depend on the function of the target protein.* Three examples of improvements achieved by engineering a more stable protein mentioned in this chapter were rubisco activase, Covid-19 vaccine and subtilisin. For each of these three target proteins: a) identify the function of this protein in the application, b) describe the type of stability that was improved by the engineering, and c) how this increased stability improved the function of the target protein in the application.

2. *Proteins in nature are just good enough for the natural function.* Proteins acquire point mutations through mistakes in DNA replication. Natural selection selects against deleterious mutations - those that hinder the function of a protein. Although mutations can also yield improvements in protein properties, these improvements are lost if they do not contribute to the fitness of the organism. For example, imagine a metabolic enzyme that acquires a substitution that makes it ten-fold faster. The organism does not grow ten-fold faster, but only two-fold faster because a second enzyme in the pathway now limits the rate. What will happen as these two protein accumulate additional substitutions? Keep in mind that beneficial substitutions are rare, while detrimental substitutions are common.

3. *Distinguish between binding as a goal and as a hypothesis for how to increase reactivity.* Tumors derived from the nervous system such as glioblastomas are more sensitive than normal tissues to L-methionine starvation. Injections of bacterial methionine- γ -lyase can deplete methionine and inhibit tumor growth. Researchers sought to avoid using a bacterial enzyme by engineering human cystathionine- γ -lyase (hCGL) to accept a new substrate, L-methionine,^[33] **Figure 1.5.** hCGL showed no detectable activity with L-methionine. The researchers hypothesized that L-methionine does not react because it does not bind to the active

1 Setting Protein Engineering Goals

site of hCGL. Three substitutions in the active site of hCGL (E59N, R119L, E339V) created variant hCGL-NLV with a more hydrophobic binding site. This variant protein improved both binding and catalysis; it could both bind L-methionine and catalyze its cleavage. Although the protein engineering changed two properties only one of these was the true goal. Identify the true goal by considering the possibility that the researchers changed only one of these properties: improved binding with no change in catalysis or improved catalysis with no change in binding. Which of these hypothetical improvements would yield an enzyme that could be useful as a cancer treatment?

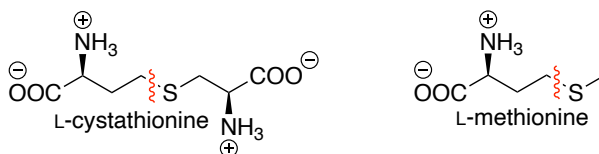


Figure 1.5. Cystathionine- γ -lyase catalyzes the cleavage of cystathionine at the bond marked with a wavy red line. L-Methionine is smaller and more hydrophobic than cystathionine. Methionine- γ -lyase catalyzes the cleavage of L-methionine at the bond marked with a wavy red line.

4. *Industrial enzymes in your kitchen* Automatic dishwasher detergents contain mixtures of enzymes. Suggest three enzymes that catalyze different reactions that may be useful to include in dishwasher detergents and explain their potential role. Suggest improvements in two different types of stability that may be required for each enzyme to be useful in dishwasher detergents.

Answers

1. Rubisco activase: a) Function: An enzyme that remodels Rubisco active sites to maintain CO₂ fixation in photosynthesis. b) Stability improved: Thermal stability — resistance to inactivation at elevated leaf temperatures. c) Benefit: Plants engineered with heat-stable rubisco activase maintain photosynthetic efficiency under heat stress, improving crop productivity in warmer climates. COVID-19 vaccine antigen (spike protein) a) Function: Viral surface glycoprotein used as the immunogen to elicit protective antibodies. b) Stability improved: Conformational stability — spike engineered into a prefusion-stabilized form. c) Benefit: The stabilized antigen better mimics the native viral surface structure, increasing neutralizing antibody responses and vaccine effectiveness. Subtilisin a) Function: A protease used in industrial applications, especially as an active ingredient in laundry detergents. b) Stability improved: Chemical stability — resistance to autolysis, oxidation, and inactivation by surfactants or high pH. The requirements are broader than just oxidative stability. c) Benefit: The stabilized enzyme retains activity longer in harsh detergent conditions, improving cleaning performance and extending product shelf life.
2. As mutations accumulate, most substitutions will be neutral or deleterious rather than beneficial. The initially improved enzyme will gradually lose its advantage because destabilizing or activity-reducing mutations are much more frequent than further rare beneficial ones. Since the organism's growth is constrained by other steps in the pathway, selection pressure to maintain the extra efficiency of the faster enzyme is weak. Thus, the first enzyme will tend to drift back toward “good enough” performance, while the newly rate-limiting enzyme may now experience stronger selection for stabilizing or activity-enhancing mutations. Overall, proteins converge on the minimum performance sufficient for organismal fitness, rather than maximizing their possible catalytic or stability potential.
3. The true goal was improved catalysis, not just binding. Bind-

1 Setting Protein Engineering Goals

ing alone, without catalytic turnover, would not deplete methionine and thus would be useless as a cancer treatment. In contrast, if catalysis improved while binding affinity remained unchanged, the enzyme could still process L-methionine effectively at physiological concentrations. Therefore, binding was a hypothesis for how to enable catalysis, but the actual engineering goal was to create an enzyme that catalyzes methionine cleavage.

4. Protease (e.g., subtilisin): Cleaves peptide bonds in food residues such as egg or meat; Amylase (e.g., α -amylase): Hydrolyzes starch from foods like potatoes or pasta into soluble sugars; Lipase: Breaks down triglycerides and fat residues into glycerol and fatty acids; Cellulase: Hydrolyzes cellulose and microfibrils from plant-based food soils (e.g., vegetable fibers). Also helps prevent redeposition of small particles on dishes; Mannanase (β -mannanase): Degrades galactomannans found in thickeners like guar gum, common in sauces and ice cream residues. All enzymes require alkaline stability (stable and active at detergent pH of 9–11), oxidative stability (some contain bleach or other oxidants to remove stains), thermal stability (hot wash cycles), surfactant stability (resistance to denaturation by detergent surfactants), shelf-life stability (resistance to long storage in formulated detergent powders/liquids).

2 Making Protein Variants

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Summary. Protein expression converts DNA into protein. It involves two steps: the transcription of the coding region of the DNA into mRNA by RNA polymerase followed by the translation of the mRNA into protein by the ribosome. Proteins are manufactured by modification of the natural protein synthesis pathways in microbes like *Escherichia coli*. The modifications specify the protein to make and increase the amount of protein made. The DNA encoding the target proteins is carried on plasmids which are circular, independently replicating genetic elements. Protein synthesis also requires DNA regions outside the coding sequence to start and stop transcription of the DNA to mRNA and translation of the mRNA to protein. Experimental techniques for construction of plasmids include template-independent DNA synthesis, cutting of DNA with restriction enzymes and assembly of DNA fragments using the polymerase chain reaction.

Key learning goals

- Protein overexpression plasmids encode the target proteins, the control elements needed to make large amounts of protein, and other regions required for plasmids to replicate in microbes.
- The main control of protein expression is at transcription of DNA to mRNA step. The *lac* operator (a region of DNA) and

2 Making Protein Variants

lac repressor (a protein that binds to the *lac* operator) serve as on/off switches for transcription of DNA to mRNA. Addition of a non-hydrolyzable lactose analog, IPTG, turns on transcription. Other control points are the plasmid copy number and the strength of the ribosome binding site.

- The pET21 plasmid contains the gene encoding the target protein, T7 RNA polymerase promoter & terminator, *lac* operator, *lacI* gene and an ampicillin resistance gene. The pET plasmids use T7 RNA polymerase, which is 5-10-fold more efficient than endogenous *E. coli* RNA polymerase.
- Site-directed mutagenesis uses the polymerase chain reaction with chemically or enzymatically synthesized DNA fragments (primers) to alter the DNA sequence to encode protein variants.

2.1 Protein expression (DNA → mRNA → protein)

Protein expression is the conversion of a DNA sequence into the corresponding protein. It consists of two steps. First, the DNA encoding the target protein is transcribed into messenger RNA (mRNA) by RNA polymerase. Second, this mRNA is translated into protein by the ribosome.

In biotechnology, proteins are manufactured by microbes, typically bacteria or yeast, that have been modified to make large amounts of the target protein. In the laboratory, researchers grow these microbes in 2-liter shake flasks containing approximately 400 mL of culture media (a mixture of amino acids, sugars, and other nutrients) in a warm incubator. Industrial manufacture uses large fermenters similar to those in a brewery. The details below refer to protein expression in prokaryotes like *Escherichia coli*.

2.1 Protein expression ($DNA \rightarrow mRNA \rightarrow protein$)

Protein expression requires control elements outside the region that encodes the target protein. One set of control elements bind and release the RNA polymerase that transcribes the DNA, while another set of control elements bind and release the ribosome that translates the genetic code into protein. Thus, the DNA surrounding the gene that encodes the target protein also contains four additional regions (two start signals, two stop signals) outside the DNA sequence that encodes the target protein.

Transcription starts by binding the RNA polymerase to the double-stranded DNA upstream of the coding sequence, **Figure 2.1**. The recognition site, called the promoter, includes of two regions each 6 bases long. One occurs approximately 10 base pairs upstream of the start codon and the other occurs 35 base pairs upstream of the start codon. The RNA polymerase is larger than this region so that it covers a larger region when it binds - about 40 base pairs upstream and 20 base pairs downstream of the start codon.

Upon binding, the RNA polymerase unwinds approximately 16 bases of DNA around the start codon to expose single-stranded DNA as a template for transcription. The RNA polymerase starts the synthesis of an RNA strand that is complementary to the DNA. As the synthesis proceeds the polymerase unwinds the template DNA ahead of it and rewinds the DNA behind it, maintaining an unwound region of about 16 base pairs in the region of transcription.

RNA synthesis continues until the polymerase encounters a termination signal. Here the transcription stops, the RNA is released from the polymerase, and the polymerase dissociates from its DNA template. The termination signal in *E. coli* consists of a symmetrical inverted repeat of a GC-rich sequence followed by four or more A residues. Transcription of the GC-rich inverted repeat results in the formation of a segment of RNA that can form a stable stem-loop structure by complementary base pairing. This structure in

2 Making Protein Variants

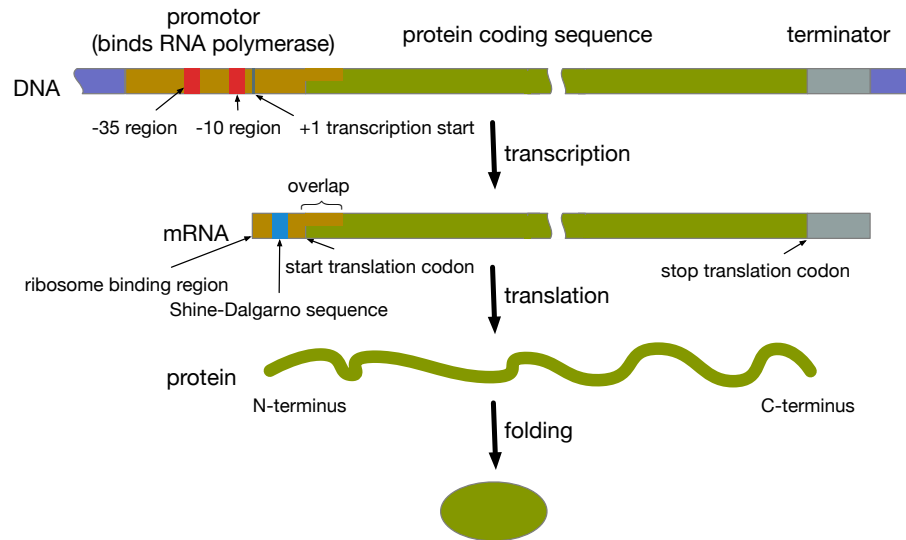


Figure 2.1. Protein expression requires control elements outside the protein coding region. The transcription of DNA into RNA requires an upstream region to bind the RNA polymerase (promoter region) and a downstream region to release the RNA polymerase (terminator region). The promoter region includes two six-base regions at 10 bp upstream and 35 bp upstream that recognize the RNA polymerase. The translation of RNA to protein requires a ribosome binding site, a start codon and a stop codon. Both the RNA polymerase and the ribosome are large molecular machines. Although they recognize and bind to the regions marked, they also cover a much larger region upon binding including part of the protein coding region marked overlap.

2.2 Protein overexpression

the RNA disrupts its base pairing with the DNA template, which causes the termination. Termination occurs *after* the terminator region because it is the RNA, not the DNA that forms the stem-loop structure. The DNA must first be transcribed into RNA to form the stem-loop structure that causes termination.

Translation starts when the ribosome binds to the mRNA. The ribosome binding site (Shine-Dalgarno sequence) is approximately 8 bases upstream of the start codon (methionine codon), but the ribosome covers a much larger region including the start codon and the beginning of the coding region. Protein synthesis occurs as the ribosome moves along the RNA sequence in groups of three nucleotides and attaches the next amino acid to the growing protein chain. Termination occurs when the ribosome encounters a stop codon and detaches from the RNA and releases the completed protein. Thus, translation requires a ribosome binding site before the coding sequence and a stop codon after the coding region. The start codon, also required, is within the coding region.

The DNA sequence encoding the control elements for expression of a β -lactamase protein, **Figure 2.2**, shows a specific example of these elements.

2.2 Protein overexpression

Cells typically make small amounts of each protein, but to manufacture proteins for biotechnology applications one wants to make as much protein as possible. This overproduction of the target protein is called protein overexpression. The two common strategies to increase protein expression are to use a more efficient RNA polymerase and to include multiple copies of the DNA encoding the target protein. These approaches are very effective; up to 50% of the protein in an *E. coli* culture can be the target protein.

2 Making Protein Variants

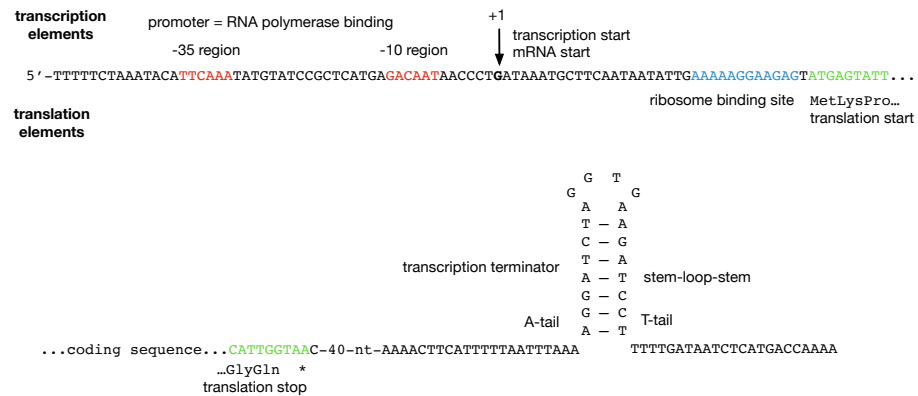


Figure 2.2. DNA sequence for the expression of a β -lactamase protein showing the transcription and translation control elements. The -35 and -10 regions (red text) in the promoter bind the endogenous RNA polymerase, but the polymerase extends beyond these regions. The ribosome binds to the ribosome binding site (blue text), but also extends beyond this region. During transcription, the terminator region of the RNA forms the stem-loop-stem structure shown, which releases the mRNA from the polymerase and stops transcription. The DNA does not form the stem-loop-stem structure.

2.2 Protein overexpression

2.2.1 A better RNA polymerase

E. coli bacteria contain an endogenous RNA polymerase, but RNA polymerases from bacteriophage are often faster. Bacteriophage RNA polymerases hijack the cellular machinery of the host bacteria to make phage proteins instead of bacterial proteins. The most common replacement polymerase for increased protein expression is the T7 RNA polymerase, which originates from bacteriophage T7. The T7 RNA polymerase catalyzes RNA synthesis 5-10-fold faster than the *E. coli* RNA polymerase, so more than the normal amount of mRNA will be produced, which will lead to more protein.

Overexpressing the target protein with T7 RNA polymerase requires a gene that encodes and expresses the T7 RNA polymerase protein. Some strains of *E. coli* (such as BL21(DE3) and BL21(DE3)pLysS) have been engineered to contain the T7 RNA polymerase gene in their genomic DNA. These strains should be used for protein overexpression that requires the T7 RNA polymerase.

Overexpressing the target protein with T7 RNA polymerase also requires a different promoter upstream of the target gene so that it binds the T7, not the *E. coli*, RNA polymerase. The DNA sequence of the T7 promoter differs from the endogenous promoters in *E. coli*.

2.2.2 Multiple copies of the target protein gene

A second way to increase the amount of protein is to have many copies of the DNA that encodes the target protein and its control elements. Protein overexpression uses plasmids, which are small (5-10 kb), independently replicating, circular DNA molecules. The pET plasmid, which will be discussed later in this chapter, makes 15-20 copies of itself in *E. coli*. These multiple copies of the target protein DNA increase the amount of protein that is produced. pET

2 Making Protein Variants

plasmids are classified as low copy number plasmids; some high copy number plasmids make 500-700 copies of themselves in each cell.

2.2.3 Ribosome binding site

The slow step of translating mRNA into protein is the initiation of protein synthesis, which depends on binding of the ribosome to the ribosome binding site (Shine-Dalgarno sequence). Optimization of the initiation step can include optimizing the distance between the Shine-Dalgarno sequence and the start codon, eliminating any secondary structure in the mRNA that could hinder binding of the ribosome, and adding additional binding elements upstream of the Shine-Dalgarno sequence.^[1]

2.3 Turning protein overexpression on or off

Strong overexpression of the target protein hinders the growth of microbes. Diverting most of the cell's energy and resources to synthesis of the target protein can even prevent growth entirely. The solution is to initially turn off the overexpression of the target protein to allow the bacteria to grow, then turn it on to produce the protein. During overexpression phase the bacteria no longer multiply, but instead make the target protein. The most commonly used on/off switch is the *lac* operator.

The *lac* operator originates from the lactose operon or *lac* operon. An operon is a set of genes whose expression is regulated together. The *lac* operon consists of three structural genes and a *lac* operator, **Figure 2.3**. The *lac* operator is a section of DNA that controls the expression of the three genes by reversibly binding the *lac* repressor protein, which is encoded by the *lacI* gene elsewhere in the

2.3 Turning protein overexpression on or off

genome. The *lac* operator is in the promoter region where the RNA polymerase must bind, so when the *lac* repressor protein binds to the *lac* operator, it blocks the RNA polymerase from binding. By default, the expression of the three genes in the lactose operon is off. Binding of allolactose to the *lac* repressor protein changes its shape so that it dissociates from the *lac* operator. Expression is now turned on because the RNA polymerase can bind to the promoter.

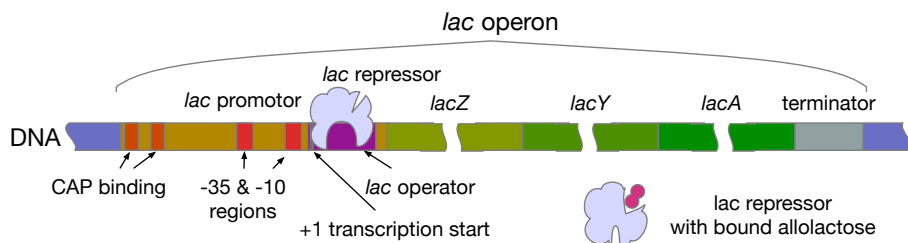


Figure 2.3. The *lac* operon consists of three structural genes and control elements that turn on expression of these genes only when lactose is present in the cell. By default the *lac* repressor protein binds to the *lac* operator and prevents the RNA polymerase from binding in this region. When allolactose is present, it binds to the *lac* repressor protein and changes its shape so that it no longer binds to the *lac* operator. Upon release of the *lac* repressor protein, the RNA polymerase can bind and start transcription of the three structural genes.

The off state of the lactose operator is not perfect. The *lac* repressor protein sometimes dissociates from the *lac* operator even with no lactose present leading to leakiness, that is, constant low-level expression. Given the strong expression from the T7 promoter, the imperfect off state can lead to difficulties in cell growth.

To improve the off state, researchers use a low copy number plasmid and two copies of the *lac* operator. The low copy number of the plasmid reduces the probability that the *lac* repressor protein

2 Making Protein Variants

spontaneously dissociates from the *lac* operator to turn on protein synthesis.

The two *lac* operator switches are used to turn off expression of both the target gene and the T7 RNA polymerase gene. Overexpression of the target protein first requires expression of the T7 RNA polymerase. Placing the *lac* operator in the promoter region blocks this first step. (Expression of the T7 RNA polymerase uses the endogenous *E. coli* promoter.) The second location for the *lac* operator is in the T7 promoter region upstream of the target gene. In the no lactose state, the *lac* repressor protein should prevent expression of the T7 RNA polymerase and of the target protein thus decreasing the leakiness of the off state.

To further ensure that protein overexpression is turned off, some researcher add a third switch - an inhibitor of the T7 RNA polymerase. This inhibitor is a protein, T7 lysozyme. It requires an additional plasmid, pLysS, that encodes synthesis of the T7 lysozyme protein.

In the presence of lactose or an analog, the switches are on. Lactose binds to the lactose repressor protein, which changes its shape in a way that releases it from the two *lac* operators. Upon release of the lactose repressor protein, the endogenous *E. coli* polymerase binds and starts to transcribe the gene encoding the T7 RNA polymerase. After it is expressed, the T7 RNA polymerase can bind to the promoter upstream of the target gene to initiate protein expression.

Lactose induces the *lac* promoter indirectly. Lactose can be isomerized into allolactose, which induces the *lac* promoter.^[2] The natural inducer of the *lac* promoter is likely galactosyl glycerol found in plant galactolipids and common in the diet of ruminants and herbivores,^[3] **Figure 2.4.**

For biotechnology applications, researchers use a non-hydrolyzable analog of galactosyl glycerol to induce the *lac* promoter. Isopropyl

2.3 Turning protein overexpression on or off

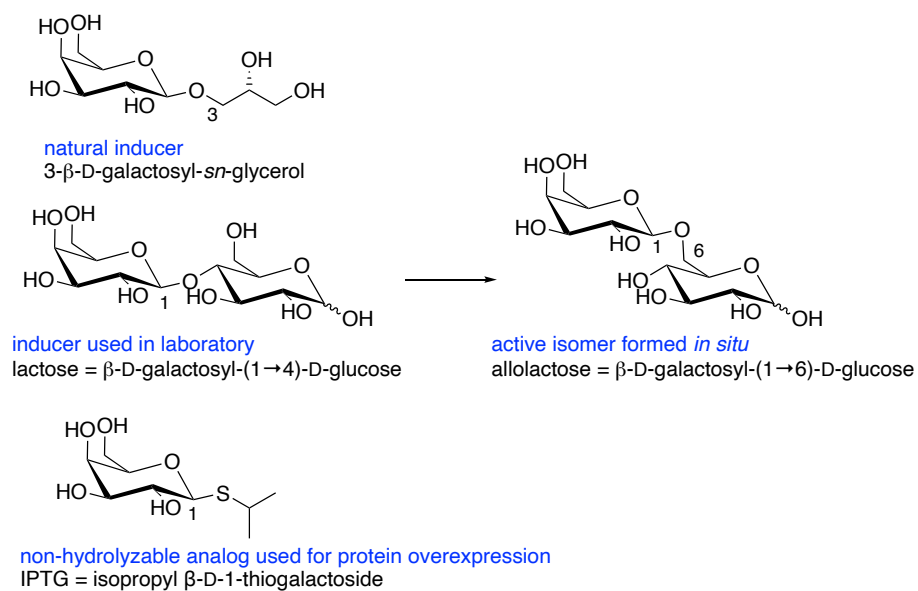


Figure 2.4. Inducers of the *lac* promoter. The natural inducer is likely galactosyl glycerol found in plant galactolipids. Lactose itself is not an inducer. IPTG is a non-hydrolyzable analog of galactosyl glycerol and is used to induce the *lac* promoter in biotechnology.

2 Making Protein Variants

β -D-thiogalactopyranoside, IPTG, contains a sulfur atom at the glycosidic link, which prevent hydrolysis. Induction starts upon addition of IPTG and continues indefinitely since the compound does not hydrolyze.

A less expensive alternative to the *lac* promoter often used in industry is the λ promoter, which is turned on by heating the culture to 42 °C for 30 min instead of adding IPTG.^[4] Initially the culture is grown at 30 °C and a thermolabile variant of the λ cI857 repressor protein binds to the operator site of the λ promoter and blocks transcription. Upon heating the repressor irreversibly denatures and no longer binds to the operator allowing protein expression to proceed.

In summary, there are two or three off switches for protein expression using pET plasmids. Two of the off switches are *lac* operators in the promoter region: one in the T7lac promoter upstream of the coding region on the plasmid and one in the endogenous promoter upstream of the gene for the T7 RNA polymerase on chromosome. Both of these switches control the synthesis of mRNA. The addition of IPTG turns these switches on. The final switch is the low level expression of T7 lysozyme by the pLysS plasmid to inhibit the T7 RNA polymerase. The BL21(DE3)pLysS strain of *Escherichia coli* includes this plasmid.

Figure 2.5 shows an example gene and its associated control elements for expression in *Escherichia coli*. This coding region encodes a poly(ethyleneterephthalate)-degrading enzyme, a PETase.^[5] The gene originates from the bacterium *Ideonella sakaiensis*, but the DNA sequence has been optimized for expression in *Escherichia coli*. This optimization replaces codons that are rarely used in *E. coli* with a synonymous, more commonly used codon. The start and stop elements for the RNA polymerase are the T7 promoter and the T7 terminator. The T7 promoter contains the *lac* operator, which serves

2.4 pET plasmids

as an on/off switch. The start elements for translation are the ribosome binding site (T7_trans_en_RBS) and the start codon (ATG) and the stop element for translation is the stop codon (TGA).

The PETase gene contains eight additional amino acid residues at the C-terminus: a linker (not labelled) and a His-tag. The linker (CTCGAG) encodes the dipeptide Leu-Glu, while the His-tag (CAC-CACCATCACCACCAC) encodes six histidine residues. The addition of six histidine residues to the protein simplifies purification. After protein expression, the mixture of proteins is added to a column containing bound nickel or cobalt ions, which have a high affinity for the imidazole ring of histidine. Most proteins bind poorly to the column and elute quickly, but the His-tagged protein binds tightly. After washing away the unwanted proteins, eluting with a buffer containing imidazole releases the His-tagged protein.

2.4 pET plasmids

The pET family of plasmids are often used for protein overexpression. The pET family of plasmids all use the T7 RNA polymerase (pET stands for plasmid for Expression by T7 RNA polymerase), but differ in the some details. The expression plasmid used to make the PETase protein mentioned above is derived from the pET21 plasmid, **Figure 2.6**. The pET21 plasmid contains an origin of replication, which determines its copy number, and genes to express the lactose repressor protein (*lacI*), and an ampicillin resistance protein (AmpR). The lactose repressor protein binds to the *lac* operator to turn off protein expression. The expression of *lacI* and AmpR are constitutive (always on) at a normal level (the promoters bind the *E. coli* RNA polymerase).

The ampicillin resistance protein or β -lactamase is a selectable marker that ensures that only bacteria containing the pET21 plas-

2 Making Protein Variants

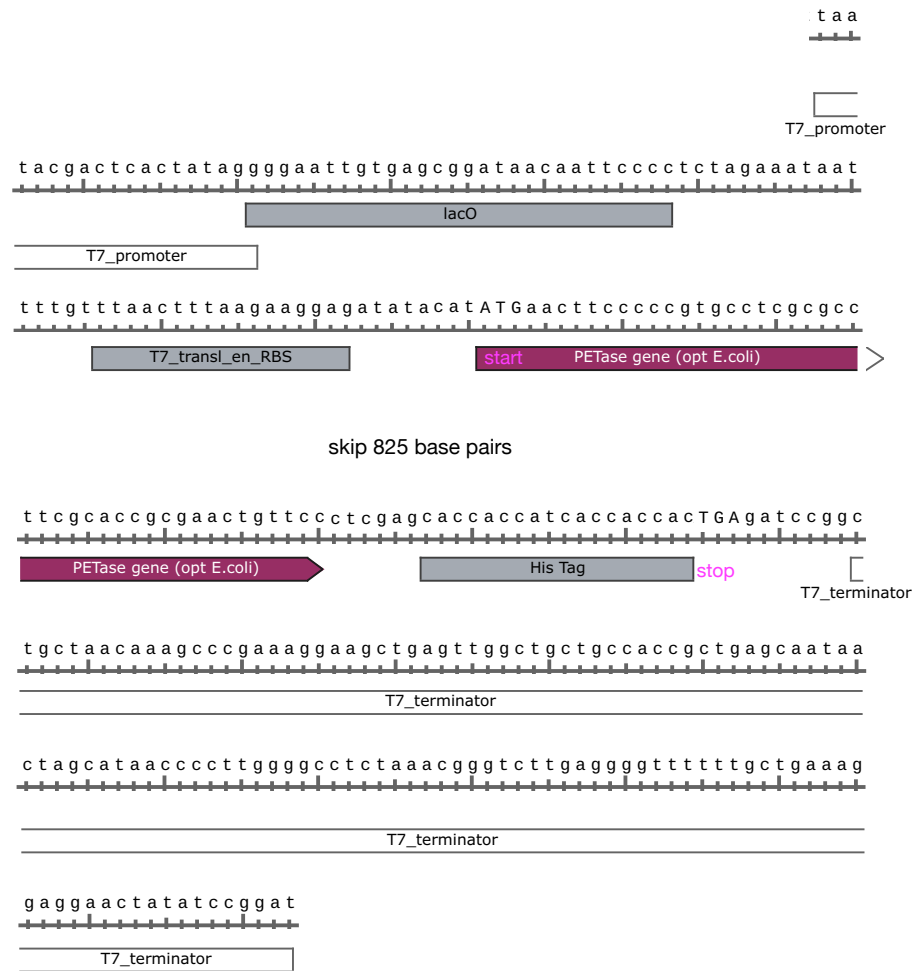


Figure 2.5. Control elements before and after the DNA encoding a PETase gene (maroon fill). The T7 promoter region binds the T7 RNA polymerase. The *lac* operator (*lacO*) is an on/off switch to control expression. The ribosome binding site is labelled T7_trans_en_RBS. The PETase gene is 870 bp long not including the added His-tag; the central 825 bp are not shown to save space. The His-tag encodes a six-histidine C-terminal tag added to the PETase. The T7 terminator forms secondary structures in the RNA that stop the RNA polymerase.

2.4 pET plasmids

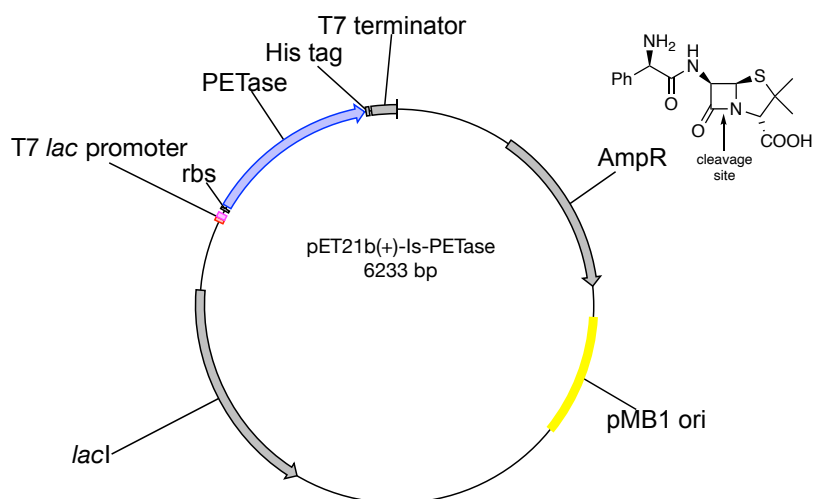


Figure 2.6. A plasmid derived from pET21 to overexpress the PETase protein in *Escherichia coli*. This plasmid contains an origin of replication (yellow section) that controls the copy number of the plasmid. The plasmid also expresses the lactose repressor protein (*lacI*) and the β -lactamase (Amp^R) at normal levels. The structure of the antibiotic ampicillin is at the upper right.

2 Making Protein Variants

mid grow in the culture. Researchers add ampicillin to the growth media, which prevents bacterial growth. Bacteria containing the pET21 plasmid express the β -lactamase, so they grow in the presence of ampicillin. This expression of the AmpR gene is constitutive and at a normal level. **Figure 2.2** above shows the sequences of control elements for the AmpR gene on a pET21 plasmid.

The different pET plasmids (about fifty) differ in mainly in the purification tag. The His-tag may be at the N- or C-terminus; it may be removable with a protease if the linker contains a protease-sensitive site. Some plasmids encode a glutathione-S-transferase (GST) purification tag. GST is a 26 kDa protein. It has a high affinity for glutathione and yields higher purity protein than the His-tag. Another family of plasmids used for protein expression are the pBAD plasmids, which use an arabinose promoter, **Table 2.1**. Protein expression from these plasmids is more tightly controlled than for pET plasmids and can be turned off more completely or tuned to different levels. Protein synthesis is turned on by the addition of different amounts of arabinose.

Table 2.1. Plasmid families used for protein overexpression.

name	antibiotic resistance	origin of replication	promoter, inducer
pET	ampicillin or kanamycin	pMB1 ori, 15-20 copies per cell	strong, T7 or T7lac, IPTG
pBAD	ampicillin or kanamycin	p15A ori, 10-12 copies per cell	tunable, arabinose

2.5 Site-directed mutagenesis

The most common method of protein engineering is site-directed mutagenesis to replace an existing amino acid residue in a protein with a different amino acid that is expected to improve the protein. Site-directed mutagenesis changes the DNA sequence encoding the target protein to encode a different amino acid at the selected location. Although amino acid replacements are the most common change made to improve proteins, one can also insert additional amino acids or delete amino acids using the same methods.

As an example of site-directed mutagenesis, consider the replacement of methionine 222 with alanine to stabilize subtilisin for use in laundry detergent,^[6] Fig **Figure 2.7**. This replacement required replacing the ATG codon for methionine with GCG to encode alanine. In contrast to methionine, alanine lacks a sulfur atom and is unaffected by bleach. Alanine is also smaller than methionine so it does not hinder access of the substrate to the active site.

```
... 218 219 220 221 222 223 224 225 ... amino acid numbering
... AAC GGT ACG TCA ATG GCA TCT CCG ... original DNA sequence (forward strand)
... Asn Gly Thr Ser Met Ala Ser Pro ... original protein encoded

... AAC GGT ACG TCA gcG GCA TCT CCG ... modified DNA sequence
... Asn Gly Thr Ser Ala Ala Ser Pro ... variant protein encoded
```

Figure 2.7. Site-directed mutagenesis to replace methionine at position 222 with alanine involves replacement of two nucleotides in the codon. The original DNA sequence encoding subtilisin (top) specifies methionine at position 222 using the ATG codon. Replacement of this codon with GCG encodes a variant of subtilisin where position 222 is an alanine (bottom). The two substituted nucleotides are shown in lower case.

Smith and coworkers first reported site-directed mutagenesis in 1978,^[7] but advances in molecular biology methods have yielded

2 Making Protein Variants

improved procedures. The section below describes using an inverse polymerase chain reaction (inverse PCR) to replace methionine 222 in subtilisin with alanine. In the original work the researchers at Genentech and Genencor used another method since their experiments were completed before the invention of the polymerase chain reaction.^[6]

This site-directed mutagenesis method uses an inverse polymerase chain reaction (inverse PCR) to copy the whole plasmid using back-to-back primers,^[8] **Figure 2.8**. The primers are designed to bind back-to-back with the DNA synthesis proceeding outward. Since the plasmid is circular, the result is a linear double stranded copy of the plasmid. Copying the whole plasmid avoids the cut and paste steps that complicated older site-directed mutagenesis methods.

The DNA primers are single-stranded 20-30 nucleotide long oligomers. The location of the primers should be such that desired mutation (typically 1-3 base changes) lies approximately in the center of one of the primers, which is the mutagenic primer. The mutagenic primer encodes the new DNA sequence; thus several nucleotides are not complementary to the plasmid DNA. One primer is complementary to one DNA strand; the other primer is complementary to the other DNA strand. The primers should have similar melting temperatures and not form dimers with themselves or each other. **Figure 2.9** shows an example of primers designed to replace methionine 222 in subtilisin with alanine.

An additional incubation with three enzymes circularizes this linear DNA fragment for insertion and replication in bacteria, **Figure 2.8**. Simultaneous treatment with a kinase to add a phosphoryl group to the 5'-ends of the PCR products and a ligase to join the two ends yield a circular double stranded DNA. Also in the same reaction, methylation-specific nuclease fragments the original template, which does not contain the mutation. The template DNA is the circular wild-type plasmid isolated from *E. coli*. When

2.5 Site-directed mutagenesis

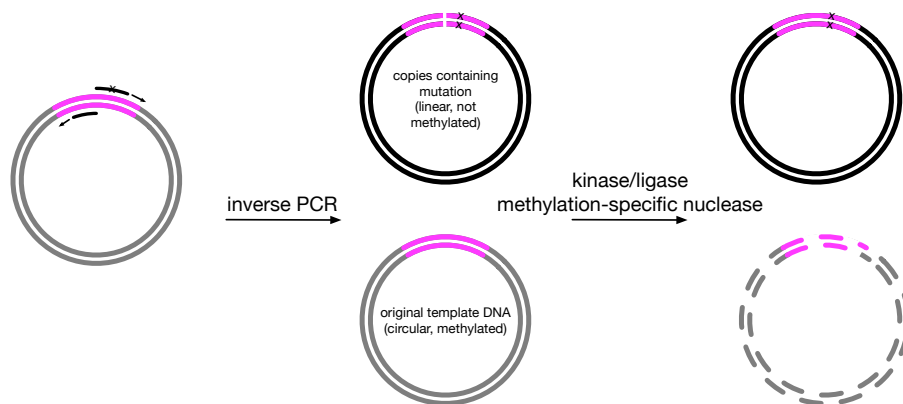


Figure 2.8. Site-directed mutagenesis involves copying the entire plasmid using mutagenic primers followed by a clean-up step. The magenta section represents the target gene for mutagenesis. Extension of two back-to-back primers (black) using the inverse polymerase chain reaction in the direction shown by the arrows yields a linear copy of the plasmid. If one of the primers contains a mismatch (x), then the copies contain this altered DNA sequence. Clean up involves circularization of the linear copies with kinase and ligase enzymes and fragmentation of the original wild type plasmid with a nuclease that only cleaves DNA containing methylation. The wild-type plasmid, prepared by growth in *E. coli*, contains methylation, but the mutated plasmid, created by the polymerase chain reaction, does not contain methylation, so the nucleases does not fragment it.

2 Making Protein Variants

```

DNA synthesis <- 3' - G TTT ATG CCC CGC ATG TT          Met
5' - ... T GGA AAC AAA TAC GGG GCG TAC AAC GGT ACG TCA ATG GCA TCT CCG CAC GTT ...
3' - ... A CCT TTG TTT ATG CCC CGC ATG TTG CCA TGC AGT TAC CGT AGA GGC GTG CAA ...
                    5' - C GGT ACG TCA gCG GCA TCT CCG -> DNA synthesis
                                   Ala
reverse primer, original aa
forward strand
reverse strand
forward primer with mutation
new amino acid

```

Figure 2.9. Mutagenic back-to-back primers to replace methionine 222 in subtilisin with alanine and a section of the circular plasmid DNA sequence. Top: The reverse primer (18 nt, 50% GC, $T_m = 63^\circ\text{C}$) binds to the forward strand and contains no mutations. DNA polymerase extends this primer to the left generating the complement to the forward strand. In subsequent PCR cycles, this primer extension will also create complements to the mutated forward strand. Bottom: The forward primer (22 nt, 68% GC, $T_m = 61^\circ\text{C}$) binds to the reverse strand and contains a two-nucleotide mismatch. DNA polymerase extends this primer to the right generating a complement to the reverse strand. Primers were predicted by NEBaseChanger (<https://nebasechanger.neb.com>).

2.6 Template-independent synthesis of DNA

plasmids replicate in *E. coli*, methyl groups are added to the DNA. The nuclease recognizes and cleaves DNA containing these methyl groups. In contrast, the primers and the DNA synthesized using the polymerase chain reaction do not contain methyl groups and are ignored by this nuclease. The result is a circular copy of the plasmid only containing the altered DNA sequence. Transfer of this plasmid into *E. coli* allows researchers to grow cells, which then make the variant protein encoded.

2.6 Template-independent synthesis of DNA

The site-directed mutagenesis experiment above relies on the ability to synthesize DNA fragments (primers) with a defined sequence. DNA polymerase can synthesize DNA, but only by copying an existing template. To make primers and even entire genes, one needs a template-independent synthesis of DNA.

The first approach to template-independent DNA synthesis was chemical synthesis.^[9] This synthesis is relatively complex and involves reactive intermediates and toxic solvents. Researchers normally order the desired sequence from a commercial service. The synthesis starts with a solid silica support. Reagents such as protected nucleoside phosphoramidites are washed over the silica to carry out the multiple steps needed to add each base. The harsh reaction conditions slowly damage the DNA, which caps the length of the DNA to about 200 bases. More recently, researchers have developed enzymatic approaches for the template-independent synthesis of DNA, **Figure 2.10**. This enzymatic approach relies on an engineered terminal deoxynucleotidyl transferase (Tdt) for the coupling step. The enzymatic approach generates less waste, can make longer oligonucleotides (>1000 bases) and can make repetitive sequences that the chemical methods cannot.^{[10], [11]}

2 Making Protein Variants

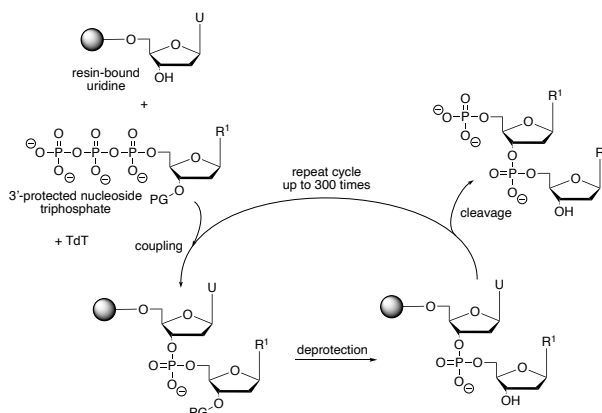


Figure 2.10. Enzymatic synthesis of DNA using 3'-protected nucleoside 5'-triphosphates. Synthesis starts at a uridine attached to a resin bead, which provides a place for terminal deoxynucleotidyl transferase (TdT) to bind and a cleavage site once synthesis is complete. TdT ligates the 3'-protected nucleoside 5'-triphosphate to the 3'-terminus of the growing oligonucleotide chain. Washing removes excess reagents and the pyrophosphate by-product of ligation. Deprotection of the 3'-PG exposed the 3'-hydroxyl for the next synthesis cycle. On completion, the oligonucleotide is cleaved from the resin bead by uracil DNA glycosylase.

Entire genes and longer fragments can be synthesized by chemical or enzymatic synthesis of 300 bp DNA fragments, followed by joining them to create longer oligonucleotides.^[12] The experimental procedures for the methods described in this chapter be found online or in molecular biology handbooks.^[13]

Glossary

Operator is a section of DNA upstream of the a gene that binds a transcription regulator. For example, the *lac* operator binds the *lac* repressor protein, which inhibits transcription of the downstream gene.

Promoter a region of DNA upstream of a region that encodes one or more proteins. The promoter region binds the RNA polymerase that will transcribe the DNA into mRNA.

Plasmid is a circular, independently replicating, DNA within a microbe separate from the chromosomal DNA. For example, pET plasmids are inserted into *Escherichia coli* to overexpress proteins under the control of a T7 promoter.

Primer is a short (10-50 nucleotides), single-stranded DNA fragment. When this fragment bind to a complementary region on a longer DNA fragment, then DNA polymerase can extend the primer making a complementary copy of the longer DNA fragment. If the primer contains a mutation, then the longer copy will also contain that mutation.

T7 promoter is a promoter that binds the T7 RNA polymerase, which is more efficient than the endogenous RNA polymerase in *Escherichia coli*.

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Problems

1. Identify whether the items below are proteins, oligonucleotides or small organic molecules. Explain their role in protein overexpression.

lacI gene product, T7 RNA polymerase, *lac* operator, ampicillin, IPTG, ribosome binding site, stop codon, T7 terminator, His-tag, mutagenic PCR primer, transcription start site

2 Making Protein Variants

2. a) Why does DNA transcription stop after the terminator, but mRNA translation stop at the stop codon? b) What problems could arise when using a T7-lac promoter on a high-copy number plasmid? c) Why do researchers put the gene encoding T7 RNA polymerase on the genome instead of on the expression plasmid?

3. *Plain text files.* In computing, plain text files are files that contain only text characters including white spaces and punctuation. For example, a plain text file could contain the following FASTA-formatted amino acid sequence for subtilisin:

```
>tr|M5B284|M5B284_BACIU Subtilisin OS=Bacillus subtilis OX=1423 GN=aprE
MRNKKLWISLLFALTIFTMAFSNMSAQAAGKSSTEKKYIVGFKQTMSAMSSAKKKDVIS
EKGKQVQKQFKYVNAATATLDEKAVKELKQDPSVAYVEEDHIAHEYAQSVPYGISQIKAP
ALHSQGYTGSNVKVAVIDSGIDSSHPDLNVRGGASFVPSETNPYQDGSSHGTHVAGTIAA
LNSIGVLGVAPSASLYAVKVLDTGSGQYSWIINGIEWAISNNMDVINMSLGGPSGSTA
LKTVDKAVSSGIVVAAAAGNEGSSGSSSTVGYPKYPSTIavgavNssnQRASFSSAGS
ELDVMAPGVSIQSTLPGGTYGAYNGTSMATPHVAGAAALILSKHPTWTNAQVRDRLESTA
TYLGSSFYYGKGLINVEAAAQ
```

In contrast, formatted text files include additional characters within the text to indicate font, bold, line spacing, and other information. For example, `<b>` and `</b>` indicate bold in the example below. The program that you use to open this file must know that these formatting characters are not part of the text, but indicate formatting commands. If not, it may misinterpret these characters as data. The same subtilisin sequence with HTML bold tags added to the first line:

```
>tr|M5B284|M5B284_BACIU Subtilisin OS=Bacillus subtilis OX=1423 GN=aprE
<b>MRNKKLWISLLFALTIFTMAFSNMSAQAAGKSSTEKKYIVGFKQTMSAMSSAKKKDVIS</b>
EKGKQVQKQFKYVNAATATLDEKAVKELKQDPSVAYVEEDHIAHEYAQSVPYGISQIKAP
ALHSQGYTGSNVKVAVIDSGIDSSHPDLNVRGGASFVPSETNPYQDGSSHGTHVAGTIAA
```



```

LNNSIGVLGVAPSASLYAVKVLDTGSGQYSWIINGIEWAISNNMDVINMSLGGPSGSTA
LKTVDKAVSSGIVVAAAAGNEGSSGSSSTVGYPAYKYPSTIAVGAVNSSNQRAFSSAGS
ELDVMAPGVSIQSTLPGGTYGAYNGTSMATPHVAGAAALILSKHPTWTNAQVRDRLESTA
TYLGSSFFYYGKGLINVEAAAQ

```

Bioinformatics and protein engineering often use plain text files to store information and commands. You can open plain text files using word processing programs like Word or Pages, but to keep them as plain text files, that is, not add formatting text to them, be sure to use a plain text format when you save them (usually ‘txt’ extension) and not a word processing format (e.g., ‘pages’ or ‘docx’ extension). More commonly researchers use text editor programs, which are programs intended to manipulate plain text files: Mac: TextEdit,* BBEdit; Windows: Notepad, Textpad; Linux: pico, vim, nano). The next question requires you to work with a plain text file similar to the example above, which is a FASTA format file. FASTA format files are plain text files used to store DNA or amino acid sequences.

Practice: The and indicate bold in web pages, which use ‘html’ format files. Copy the second amino acid sequence text above into a text editor, save it with the extension ‘html,’ then open that file with a web browser. The and characters should be hidden and the characters in between should be displayed as bold. Next, change the extension of this file from ‘html’ to ‘txt’ and open with the web browser again. The browser now displays all the characters in the file, but does not show anything in bold. A web browser can read both plain text files and web pages, which have the html format. The file extension tells the browser whether to interpret the characters as text or as formatting information. Most bioinformatics and computational chemistry programs expect

*Unfortunately, the default settings use rich text formatting which includes formatting characters. Change the default by selecting Format → Make Plain Text option in the menu bar. This change will also set the extension of files saved in the future to ‘txt’ from the default ‘rtf’.

2 Making Protein Variants

text files as input. If the file contains formatting or other codes, they will change the meaning of the file and the program will not accept them.

4. PCR exponentially amplifies DNA in a sample through repeated cycles of denaturation, primer annealing, and extension. In site-directed mutagenesis using PCR with a mutagenic primer, almost all newly synthesized DNA strands contain the mutation because they either incorporate the mutagenic primer sequence or copy a DNA strand that already contains the mutation (**Figure 2.11**). In the first cycle, extension from the mutagenic primer yields a daughter strand containing the mutation, while extension from the complementary primer produces a strand without the mutation. In subsequent cycles, the mutagenic primer can anneal to and extend from both wild-type and already-mutated template strands. Non-mutated strands increase only by copying the original wild-type template using the non-mutagenic primer. Each cycle increases the proportion of mutated strands. After approximately 25–35 cycles, nearly every amplified DNA molecule incorporates the mutagenic primer sequence and therefore contains the mutation. The original wild-type template remains as a small fraction of the total (it is not amplified, only diluted by the exponentially increasing mutated products).

As an exercise, show that the 32 strands after the next cycle (including both template and synthesized strands) would consist of 26 strands with the mutation and 6 strands without the mutation. Further show that 11 of the resulting 16 double-stranded DNA molecules would contain mutations on both strands.

5. *Site-directed mutagenesis*. Gluten, a protein found in wheat, rye, and barley, contains a high proportion of proline and glutamine. Digestive proteases cleave gluten to oligopeptides enriched in the dipeptide sequence proline-glutamine because peptides containing this sequence are resistant to digestive proteases. In celiac

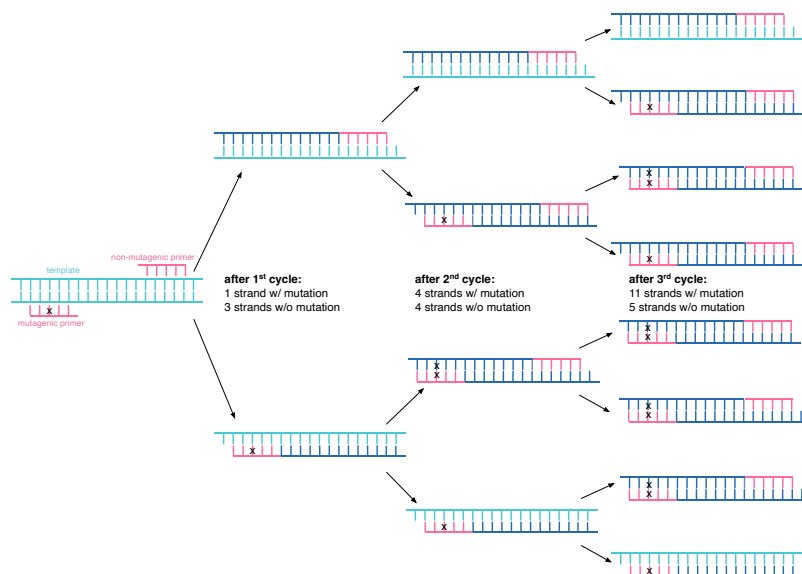


Figure 2.11. Site-directed mutagenesis with the polymerase chain reaction using a mutagenic primer. The mutation is marked by an 'x.' Among the PCR-synthesized DNA strands (dark blue), mutation-containing strands increase exponentially (from 1 to 4 to 11), while non-mutated strands increase linearly (from 1 to 2 to 3). The exponential increase in mutated strands results from extending the mutagenic primer and copying existing mutated strands by extensions with the non-mutagenic primer. Non-mutated strands increase linearly because they arise only from copying one of the template strands. The exponential accumulation of mutation-containing strands dilutes the non-mutated strands to insignificance after the typical 30 cycles of PCR.

2 Making Protein Variants

disease patients, these oligopeptides are believed to trigger an autoimmune response, which causes diarrhea and other gastrointestinal problems. One potential solution is adding a protease that can degrade Pro-Gln containing peptide to the meals of celiac patients. Gordon and coworkers describe protein engineering to create such a protease, which they called KumaMAX.[†] You do not need to read that paper to answer this question. Some of the authors of that paper formed the company PVP Biologics to further improve KumaMAX. Recently this improved protease passed Phase I clinical trials and in 2020 Takeda Pharmaceutical Company acquired PVP Biologics. In the future (after 2025), it may be a treatment for celiac disease.

KumaMAX contains seven amino acid substitutions as compared to KumaWT. This question focuses on the Gly319Ser substitution, which contributes to the switch in selectivity from Arg/Lys to Gln by decreasing the amount of space available and by adding a hydrogen bond partner for Gln.

- a) *Find the DNA sequence for KumaWT in GenBank.* GenBank is found at <https://www.ncbi.nlm.nih.gov/nucleotide/>. Search for kumamolisin; the sequence should be the first hit. The enzyme is synthesized as a 552 amino acid residue precursor, which corresponds to $552 \cdot 3 = 1656$ nucleotides. The DNA sequence contains 5841 nucleotides, so we must find the section that codes for the protein. In the search results, click on the text 'GenBank' below the name of the gene. This page contains the DNA sequence and information about the DNA sequence. In the 'FEATURES' section, it mentions that the coding region (CDS) corresponds to nucleotides 2940 to 4598. How many nucleotides does that correspond to? (Your answer

[†]Non-linear least squares fitting of eq. Equation 6.5 to the data starts with an initial guess for the dissociation constant followed by iteration to find the best value. The supporting information includes a Python function for this fit.

should be a number.) Why does that number not match $552 \cdot 3$?
Hint: The length of an amino acid or nucleotide sequence is $\text{last_number} - \text{first_number} + 1$. The '+ 1' is needed to include the starting amino acid or nucleotide. For example, a sequence of five nucleotides numbered 1 through 5 has a length of $5 - 1 + 1 = 5$.

- b) *Save the coding region as a FASTA-formatted file on your computer.* Click on the text 'CDS' in the features section to show the DNA sequence with the coding region highlighted. On this page, click on 'FASTA' at the bottom right (near the pop-up with the translation to amino acids) to view only the coding region in FASTA format. FASTA-formatted files are plain text files, often named using .txt or .fasta as the extension, but this extension is not required. The FASTA format requires at least two lines. The first line starts with '>' (greater than symbol) and contains a description of the sequence. This line is followed by one or more lines containing single-letter codes for the amino acid or nucleotide sequence. Copy and paste the FASTA file into an empty text file on your computer. Save the file with the name 'KumaWT.fasta'. (Do not copy the entire page; only the FASTA part.) Your file should contain about 1660 nucleotides, not about 5800. It must be a text file.
- c) Design primers for site directed mutagenesis of Gly319 to Ser in KumaWT using the web tool NEBaseChanger (<https://nebasechanger.neb.com>). Upload the KumaWT.fasta file in the 'enter your sequence' section and G319S in the 'select input mode and set mutations' section. Keep the other entries at default values, then click 'build primers'. What is the DNA sequence that codes for Gly318Gly319Pro320? What will be the new sequence that encodes Gly318Ser319Pro320?
- d) Sketch a diagram of how the primers will bind to a circular plasmid containing this gene; include the sequence of the

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primers in your diagram oriented correctly and indicate the direction of DNA synthesis.

Answers

1. oligonucleotides: ribosome binding site (RNA), stop codon (RNA), *lac* operator (DNA), T7 terminator (DNA), transcription start site (DNA)

proteins: *lacI* gene product, T7 RNA polymerase, His-tag (part of a protein)

small molecules: ampicillin, IPTG

2. a) In bacteria, transcription termination requires that the RNA polymerase transcribe through the terminator sequence to produce the RNA hairpin and poly-U tract. This RNA secondary structure destabilizes the RNA–DNA hybrid and causes polymerase dissociation. Because the RNA structure only forms after the terminator has been transcribed, transcription stops after the terminator. By contrast, in translation, a stop codon is not decoded into an amino acid but instead recruits release factors that trigger termination. Thus translation ends at the stop codon. b) The *lac* operator is leaky, so some transcription occurs even without IPTG induction. On a high-copy plasmid, many T7 promoters combined with leaky expression can produce enough toxic or burdensome protein to overwhelm host metabolism and inhibit growth. c) Placing the T7 RNA polymerase gene in the chromosome limits its copy number to one (two during replication). If it were carried on a plasmid, the multiple copies would amplify leaky expression from the T7–*lac* promoter, leading to uncontrolled polymerase production, runaway target gene expression, and severe metabolic stress.

Answers

5. a) $4598 - 2940 + 1 = 1659$. The extra three nucleotides in the DNA sequence encode the stop codon.

b) text file containing the text below

```
>AB070740.2:2940-4598 Bacillus sp. MN-32 kscp gene for kuma ...
ATGAGCGACATGGAGAAGCCTTGGAAGGAAGAGGAGAAGCGCGAGGTCCTCGCGGGACACGCTCGCAGGC
AGGCGCCGCAGGCTGTGCGATAAGGGACCGGTGACCGGGGACCAGCGGATTTCCGTTACGGTCGTGCTGCG
CCGCCAACGAGGCGATGAACTCGAGGCGCATGTGGAACGCCAGGCCGCGCTCGCACCTCACGCGCGAGTG
CATCTGGAGCGAGAAGCGTTTGCCGCTTCGCACGGCGCTTCGCTCGACGACTTTGCGGAGATTGAAAAGT
TCGCGGAAGCGCACGGGCTCACGCTCGATCGCGCCACGTGGCTGCGGGCACCGCGGTGCTGAGCGGCCC
GGTGGACGCCGTTAACCAAGCGTTTGGGGTCGAGTTGCGCCATTTTCGATCATCCAGACGGATCCTATCGA
AGCTACGTCGCGCGACGTGCGTGTGCCGGCTTCCATCGCGCCTCTGATTGAAGCGGTGTTTGGCCTGGACA
CGCGCCCGGTGGCGCGGCCCCACTTTCGGCTGCGGAGGCGCGCCGAGGGCGAGTTTGAGGCGAGATCGCA
GTCCGCGGCGCCGACTGCGTACACGCCGCTCGACGTGCGCAGGCGTACCAATTTCCCGAGGGGCTCGAC
```

plus 15 more lines

c)

```
GGTGGCCCT and GGTagcCCT
GlyGlyPro      GlySerPro
```

d) answer should also show a circular diagram.

```
<-- cggcaggtagcacaggta
5'- GCCGTCCATCGTGTCCATCAGCTGGGGTGGCCCTGAGGACA
3'- CGGCAGGTAGCACAGGTAGTCGACCCACCGGACTCCTGT
      cagctggggtagccctgaggaca -->
```


3 The Logic of Protein Engineering

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Summary. Gibbs energy diagrams are the conceptual tool that connects the goals of protein engineering to the changes in amino acid sequence. Gibbs energy diagrams show the states available to a protein, the relative energies of these states and barriers between them. Protein states (macrostates) are different protein forms, for example, folded and unfolded protein states. Protein states differ in their flexibility and in their non-covalent interactions within the protein and between protein, solvent and any ligands. These differences in molecular interactions in different protein states create Gibbs energy differences between them. The Gibbs energy differences between protein states determine the properties of proteins, including those usually targeted for protein engineering: stability, binding, reactivity and selectivity. Protein engineering works by selecting amino acid replacements that alter the relative Gibbs energy of protein states.

Key learning goals

- Gibbs energy diagrams show the options available to a protein including the available states, their relative energies and the barriers between the states.
- Protein states (macrostates) are protein forms that have macroscopic or bulk properties that one can measure. For

3 The Logic of Protein Engineering

example, the folded and unfolded protein states differ in their fluorescence properties.

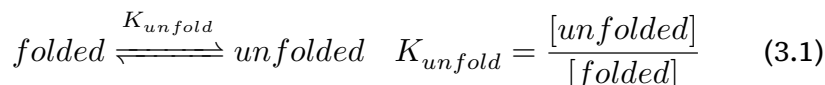
- Protein states consist of countless numbers of microstates, which are individual conformations.
- Protein states differ in Gibbs energy due to differences in non-covalent interactions (electrostatic and van der Waals) and in entropy. Amino acids substitutions alter these interactions.
- The Gibbs energy difference between various states determine a protein's properties. The Gibbs energy difference between folded and unfolded protein, ΔG_{unfold} , determines protein stability. The Gibbs energy difference between bound protein and target and free protein and target, ΔG_{diss} , determine binding strength. The Gibbs energy difference between the substrate state and the transition state for the reaction, $\Delta G^\ddagger$, determine the reaction rate.
- Replacing amino acid residues in a protein changes the relative energies of the states and therefore the properties of the protein. This change in Gibbs energy of the protein states is the basis of protein engineering.

3.1 Gibbs energy diagrams

Protein states. Proteins exist in different forms called macrostates or more commonly states. Protein states differ in their non-covalent interactions and flexibility, but not in their covalent structure. For example, the folded state of an enzyme is the native, catalytically active form and the unfolded state is the denatured, catalytically inactive form. States refer to protein forms that have macroscopic or bulk properties that one can measure. For example, the folded and unfolded states of a protein have different fluorescence properties and migrate differently in a gel electrophoresis experiment.

3.1 Gibbs energy diagrams

Protein states can interconvert with each other. For example, a solution may contain interconverting folded and unfolded states, Equation 3.1. The relative amounts of each state corresponds to the equilibrium constant for the interconversion. This balance between the folded and unfolded states determines protein stability.



Gibbs energy diagrams. Gibbs energy diagrams show the options available to a protein. The diagram shows the available states, their relative energies and the barriers between the states, **Figure 3.1**. The relative Gibbs energies of the states determine the relative amounts of protein in each state; in other words, the relative Gibbs energies determine the equilibrium constant between the two states, Equation 5.1. The difference in Gibbs energy, ΔG , between two states is proportional to the natural logarithm of the equilibrium constant between the two states. The constant R is the gas constant, which corresponds to 1.987 cal/mol·K when ΔG has the units of calories and to 8.314 J/mol·K when ΔG has the units of joules. The constant R connects the temperature scale to the units of energy used in physics (calories or joules). R is a molar quantity suitable for chemistry calculations, while Boltzmann's constant is used when working with particles. T represents the temperature in degrees Kelvin.

$$\Delta G = -RT \ln K_{eq} \quad (3.2)$$

If the reaction is favorable, then the equilibrium constant is >1 , the natural logarithm of the equilibrium constant is positive, and the Gibbs energy change is negative. For example, in **Figure 3.1**, state 1 could represent the denatured state and state 2 could represent

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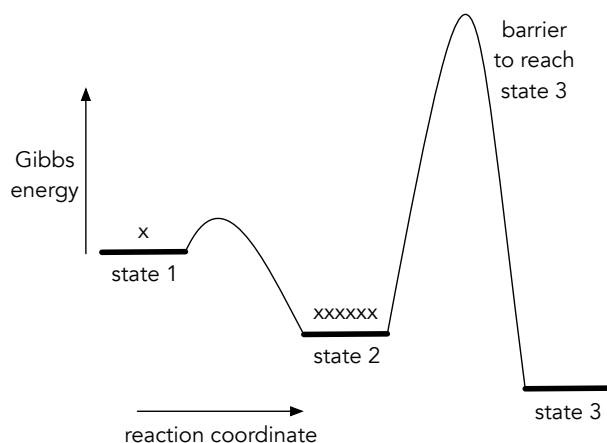


Figure 3.1. Gibbs energy diagrams for protein states show the available states, their relative energies and the barriers between them. The proteins (x) equilibrate between states 1 and 2 with more proteins in state 2 because it is lower in energy. The barrier between states 1 and 2 is low so that the proteins equilibrate between them on the time scale of the experiment. State 3 is even lower in energy than states 1 and 2, but a high barrier prevents proteins from reaching that state. The x-axis on these diagrams, reaction coordinate, refers to changes in protein conformations.

3.1 Gibbs energy diagrams

the native state. The native state is more stable, so most of the protein exists in that form. The equilibrium constant for unfolding is unfavorable (<1) and the Gibbs energy change for unfolding is positive. At room temperature (298 K) an equilibrium constant of 10 corresponds to a 1.36 kcal/mol Gibbs energy difference. An equilibrium constant of 100 corresponds to 2.73 kcal/mol, which is an additional 1.36 kcal/mol. Each additional factor of 10 in the equilibrium constant corresponds to an additional 1.36 kcal/mol of Gibbs energy difference.

The barriers between protein states indicate how fast the proteins interconvert between the states. The barrier between states 1 and 2 is small, so those two states interconvert readily on the time scale of the experiment. The barrier to reach state 3 is high so that state, although lower in energy, is not populated because there has not been enough time for the proteins to equilibrate. Since protein states differ in their conformation and many conformational changes are fast, many barriers between states are low. When the conformational change requires collective movements of many atoms it can be slow. For example, large scale movement of a loop to open an active site can be slow because it requires two hinge movements and release of existing interactions between amino acids. State 3 could represent a very slow-to-form protein conformation such as one with several knots.

Chemical reactions, such as the conversion of a substrate to product, are also described by Gibbs energy diagrams. The relative Gibbs energies of the substrate and product correspond to the equilibrium constant between them. Enzymes cannot change this equilibrium; it depends only on the properties of substrate and product. The height of the barrier corresponds to rate at which they reach this equilibrium. Enzymes can change this rate by stabilizing the transition state for the reaction. This text will use Gibbs energy diagrams for both protein states, where only conformational changes occur and for chemical reactions where

3 The Logic of Protein Engineering

covalent bonds are broken and formed. For most chemical reactions, this text will use the standard Gibbs energy, indicated by the superscript $^\circ$. This standard free energy refers to the standard state of 1 M concentration for both [S] and [P] and is used for simplicity. In reality, the Gibbs energies differ as the concentrations of [S] and [P] change. At the beginning of a reaction (high [S], low [P]) the Gibbs energy of the product is lower than the Gibbs energy of the substrate. As the reaction proceeds, the concentrations of substrate and product change, the Gibbs energy levels change such that at equilibrium (low [S], high [P]) the Gibbs energy levels are equal.

3.2 The relative Gibbs energies of states determine protein properties

Proteins properties depend on the relative energies of different protein or chemical states, **Figure 3.2**. Protein stability depends on the difference between the native state and denatured state, Equation 3.1 above. The denatured state is less stable (ΔG_{unfold} is positive) so most of the protein exists in the native state.

The binding strength of an antibody, Ab, for an antigen, Ag, depends on the energy difference between that bound and unbound states, Equation 3.3. A positive ΔG_{diss} corresponds to unfavorable dissociation, thus favorable binding, which means the Gibbs energy of free Ag + Ab is higher than the Gibbs energy of Ag·Ab.



Enzyme catalysis involve a chemical reaction, Equation 3.4, which has a certain ΔG_{rxn} that reflects the energy difference between

3.2 The relative Gibbs energies of states determine protein properties

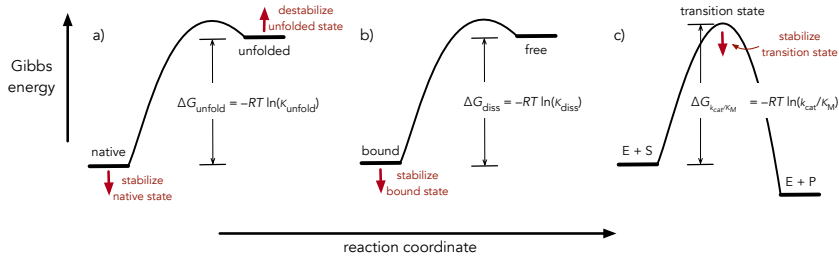


Figure 3.2. Gibbs energy differences determine protein properties. a) The Gibbs energy difference between the native and unfolded states, ΔG_{unfold} , sets the equilibrium constant between the native and unfolded states, K_{unfold} , to determine the stability of a protein. Increasing ΔG_{unfold} by either stabilizing the native state or destabilizing the unfolded state yields a more stable protein because unfolding becomes less favorable. b) The Gibbs energy difference, ΔG_{diss} , sets the equilibrium constant between the bound and free states to determine the binding affinity of a protein. Increasing ΔG_{diss} by stabilizing the bound state yields tighter binding because dissociation becomes less favorable. c) The Gibbs energy difference, $\Delta G_{k_{\text{cat}}/K_M}$, between enzyme and starting material, E + S, and the transition state sets the rate constant, k_{cat}/K_M , that determines how fast the reaction proceeds. Decreasing $\Delta G_{k_{\text{cat}}/K_M}$ by stabilizing the transition state speeds up the reaction because it lowers the barrier separating enzyme and starting material from enzyme and product, E + P.

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starting material and product. Enzymes cannot change this ΔG_{rxn} ; it is determined by the starting material and product. Enzymes only speed up the approach to equilibrium.

$$\text{starting material} \xrightleftharpoons{K_{eq,rxn}} \text{product} \quad \Delta G_{rxn} = \frac{[\text{product}]}{[\text{starting material}]} \quad (3.4)$$

Transition state theory proposes that the rate of a chemical reaction depends on the energy difference between starting materials and the transition state, $\Delta G^\ddagger$, Equation 3.5. The transition state lies at higher energy than the starting material, so $\Delta G^\ddagger$ is positive. This equilibrium is a pseudo-equilibrium because some of the transition state species are continuing on to product.

$$\text{starting material} \xrightleftharpoons{K_{eq,ts}} \text{transition state} \quad \Delta G_{uncatalyzed}^\ddagger = \frac{[\text{transition state}]}{[\text{starting material}]} \quad (3.5)$$

Enzymes catalyze reactions by stabilizing the transition state. Enzymes bind the transition state and stabilize the conformation and charge distribution needed for the bond-breaking and bond-making, Equation 3.6. This stabilization lowers the barrier between starting material and product so the reaction proceeds faster. The two states that determine enzyme catalysis are the *starting material + enzyme state* and the *transition state · enzyme state*.

3.3 Protein engineering manipulates the Gibbs energies of states

$$\begin{aligned} \text{starting material} + \text{enzyme} &\xrightleftharpoons{K_{eq,ts,cat}} \text{transition state} \cdot \text{enzyme} \\ \Delta G_{catalyzed}^{\ddagger} &= \frac{[\text{transition state} \cdot \text{enzyme}]}{[\text{starting material}][\text{enzyme}]} \end{aligned} \quad (3.6)$$

3.3 Protein engineering manipulates the Gibbs energies of states

Improving protein properties requires changing the relative Gibbs energies of different protein states, **Table 3.1**. Increasing the value of ΔG_{unfold} stabilizes proteins, increasing the value of ΔG_{diss} strengthens binding, and decreasing $\Delta G_{k_{cat}/K_M}$ speeds up catalysis. Changing the relative Gibbs energies is called differential stabilization. The structural changes introduced by protein engineering must stabilize some states more than others so that the difference between them changes, which results in a change in the protein properties.

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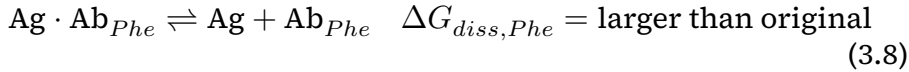
Table 3.1. Changes in Gibbs energy needed to improve protein function.

Protein Func- tion	Gibbs		Engineering Approaches for Improvement
	En- ergy Change	Comparison States	
stability	ΔG_{unfold}	folded protein vs. unfolded ensemble	increase ΔG_{unfold} by stabilizing folded protein or destabilizing unfolded ensemble
binding	ΔG_{diss}	protein bound to ligand vs. solvated protein & ligand	increase ΔG_{diss} by stabilizing the bound complex
catalysis	$\Delta G_{k_{cat}/K_M}$	enzyme and starting materials vs. transition state	decrease $\Delta G_{k_{cat}/K_M}$ by stabilizing the transition state

For example, altering the relative Gibbs energies of the two states involved in binding, the unbound state (free Ag + Ab) and the bound state (Ag·Ab), alters binding strength, Equation 3.3 above. To create a differential stabilization, the substitution must alter the Gibbs energies of the two states by different amounts. The substitution of an alanine residue with phenylalanine in the antigen-binding region of the antibody might strengthen binding if the antigen interacts with the added phenyl group. Thus, the bound state would be stabilized. In the free antibody, exposing an additional hydrophobic phenyl group to water is unfavorable, thus the free state would be destabilized. The net result is a larger Gibbs energy of disso-

3.3 Protein engineering manipulates the Gibbs energies of states

ciation, which corresponds to stronger binding, Equation 3.7 and Equation 3.8.



In contrast, the substitution of an alanine residue with phenylalanine in the core of the antibody, remote from the antigen-binding site, would alter both states similarly so the difference in Gibbs energy for Equation 3.3 would remain the same. The binding strength would not improve.

The improvement factor of a protein engineering experiment is the factor by which the stability, binding, reaction rate or selectivity has improved. This improvement corresponds to a ratio of equilibrium constants or rate constants for the property being improved. The value for the variant protein is divided by the value for the original protein, Equation 3.9.

$$\text{improvement factor} = \frac{\text{variant } K_{eq} \text{ or rate constant}}{\text{original } K_{eq} \text{ or rate constant}} \quad (3.9)$$

The improvement factor should be >1 if the protein has improved. If the improvement factor is less than one when it seems that it should be >1 , consider whether the equilibrium or rate constants being compared should increase or decrease to improve the protein. For example, protein stability depends on the unfolding equilibrium constant. An increase in the unfolding equilibrium constant corresponds to more unfolding and decreased stability. An increase in the unfolding equilibrium constant is not an

3 The Logic of Protein Engineering

improvement, but a degradation of stability. The equilibrium constant that should be compared to identify improvement is the inverse, or $1/K_{unfold}$. The ratio of $1/K_{unfold}$ for the variant divided by that for the original corresponds to the improvement factor, Equation 3.10. If the variant unfolds less readily, then $1/K_{unfold}$ for the variant will be larger and the improvement will be >1 .

$$\text{stability improvement factor} = \frac{1/K_{unfold, variant}}{1/K_{unfold, original}} \quad (3.10)$$

One can also assign a Gibbs energy to the improvement. Protein properties in **Figure 3.2** correspond to Gibbs energy difference between two states, a ΔG . The Gibbs energy for an improvement contains two Δ 's because it is the difference between ΔG for the variant and the ΔG for the original, Equation 3.11.

$$\Delta\Delta G_{improvement} = \Delta G_{variant} - \Delta G_{original} \quad (3.11)$$

An improvement can be a positive or a negative $\Delta\Delta G$ depending on how the property is defined. An increase in protein stability corresponds to a positive value for $\Delta\Delta G_{unfold}$, Equation 3.12, because making unfolding less favorable corresponds to a more stable protein.

$$\Delta\Delta G_{unfold} = \text{larger than original} - \text{original} = \text{positive value} \quad (3.12)$$

An increase in antibody binding affinity corresponds to a positive value for $\Delta\Delta G_{diss}$, Equation 3.13, because making dissociation less favorable corresponds to tighter binding.

3.3 Protein engineering manipulates the Gibbs energies of states

$$\Delta\Delta G_{diss} = \text{larger than original} - \text{original} = \text{positive value} \quad (3.13)$$

In contrast, an increase in reaction rate corresponds to a negative value for $\Delta\Delta G^\ddagger$, Equation 3.14, because lowering the energy of the transition state speeds up the reaction.

$$\Delta\Delta G^\ddagger = \text{smaller than original} - \text{original} = \text{negative value} \quad (3.14)$$

Selectivity is another protein property that is often a target for protein engineering. Selectivity refers to selective binding or selective catalysis. Binding selectivity compares the binding of a protein to two targets, so it is the difference in Gibbs energies for binding of the two targets, Equation 3.15.

$$\begin{aligned} \text{binding selectivity} &= \Delta G_{diss, target\ 1} - \Delta G_{diss, target\ 2} \\ &= \Delta\Delta G_{diss, target\ 1\ vs.\ target\ 2} \end{aligned} \quad (3.15)$$

There are two Δ 's in the Gibbs energy for binding selectivity. Similarly, selectivity for reaction compares the reactivity of two substrates or the formation of two different products from an enzyme. Since the selectivity of binding or reaction compare two reactions, it already contains two Δ 's: $\Delta\Delta G$. The Gibbs energy change for the improvement in selectivity will have three Δ 's: $\Delta\Delta\Delta G$, Equation 3.16.

improvement in binding selectivity

$$\begin{aligned} &= \Delta\Delta G_{diss, variant, target\ 1\ vs.\ target\ 2} - \Delta\Delta G_{diss, original, target\ 1\ vs.\ target\ 2} \\ &= \Delta\Delta\Delta G_{diss, original \rightarrow variant, target\ 1\ vs.\ target\ 2} \end{aligned} \quad (3.16)$$

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One can write a similar equation for reaction selectivity.

In Chapter 1 we learned to define protein engineering goal in terms of protein properties. Now we learned that relative Gibbs energies of specific protein states must change to improve these protein properties. The next sections describes the non-covalent interactions and entropy contributions that we can modify to engineer changes in these Gibbs energies.

3.4 Non-covalent interactions & entropy

To engineer changes in the Gibbs energies, one needs to alter the molecular features that contribute to differences in the Gibbs energies of different protein states. The two contributors to differences in Gibbs energy are enthalpy, ΔH , and entropy, ΔS , Equation 3.17.

$$\Delta G = \Delta H - T\Delta S \quad (3.17)$$

Protein states differ not in their covalent bonds, but in their conformations and associations with solvent and other molecules. These non-covalent interactions differ between states so manipulating these interactions can change the properties of a protein. For protein states, the enthalpic contributions come from favorable or unfavorable non-covalent interactions within a protein, between protein and solvent and between protein and target molecules. Stabilizing non-covalent interactions lowers the Gibbs energies of the corresponding states, while destabilizing interactions raise these energies. The entropic contributions come from differences in molecular flexibility and solvent interactions between the states. Entropy is the degree of disorder in a system so states with higher disorder are more favorable and have lower Gibbs energy.

3.4 Non-covalent interactions & entropy

Non-covalent interactions. The two fundamental non-covalent interactions are electrostatic interactions and van der Waals interactions. Charged or partially charged atoms attract or repel each other. Interactions between oppositely charged atoms are attractive (negative energies), while interactions between like-charged atoms are repulsive (positive energies). These energies vary inversely with the distance between them, r , according to Coulomb's law, Equation 3.18, where q_1 and q_2 are the charges on the atoms and ϵ_0 is the dielectric constant of the medium separating them.

$$\text{electrostatic energy contribution} = \frac{q_1 \cdot q_2}{4\pi \cdot \epsilon_0 \cdot r} \quad (3.18)$$

Since the energy varies inversely with r , they persist over long distances, **Figure 3.3**. Hydrogen bonds are a special type of electrostatic interaction that includes contributions from covalent bonding.

All atoms interact via van der Waals interactions. The interaction is zero at long distances, attractive at intermediate distances and repulsive at short distances. The Lennard-Jones potential, Equation 3.19, describes the energetics of van der Waals interactions where r is the distance between the atoms and r_o is the distance where their interaction energy is lowest (optimal distance or van der Waals distance). Here the constant, ϵ , is *not* the dielectric constant, but the maximum interaction energy (depth of the attractive well). When $r_o = r$, the van der Waals energy is $-\epsilon$. Different values of ϵ and r_o account for different interactions between different pairs of atoms. The 12th- and 6th-power inverse dependence on r indicates that both favorable and unfavorable interactions act only at short distances. Molecular modeling programs typically include van der Waals interactions between atoms separated by less than 10 Å. This approach saves computing interactions that are zero or very close to zero.

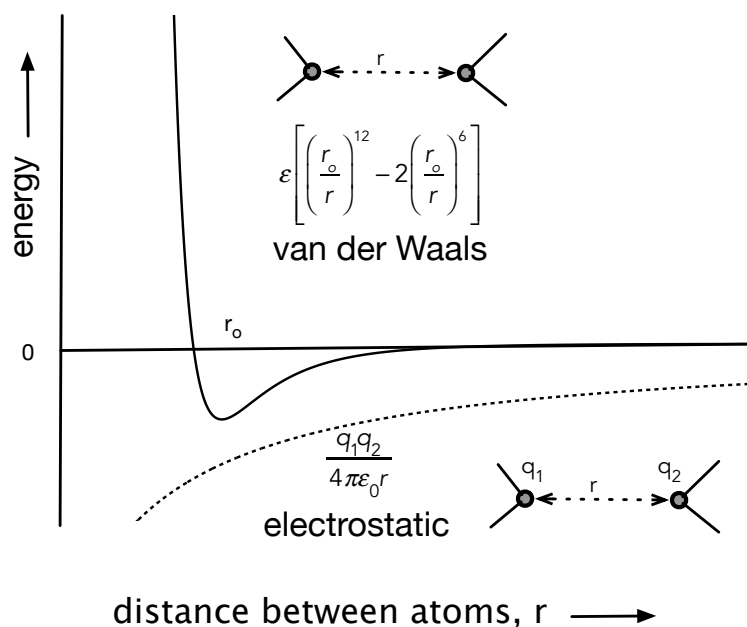


Figure 3.3. The two fundamental non-covalent interactions are electrostatic interactions (dashed line) and van der Waals interactions (solid line). The attractive electrostatic interactions between oppositely charged atoms become increasingly favorable (lower energy) as the two atoms approach each other. The energy decreases according to $1/r$, which allows electrostatic interactions to persist over long distances. A similar unfavorable interaction occurs between two like charges (not shown). The van der Waals interactions are unfavorable at short distances due to steric bumping, slightly favorable at intermediate distances due to the interaction of permanent and induced dipoles. The favorable interaction decreases according to $1/r^6$, so it dissipates to zero more quickly than electrostatic interactions.

3.4 Non-covalent interactions & entropy

$$\text{van der Waals energy contribution} = \epsilon \left[\left(\frac{r_o}{r} \right)^{12} - 2 \left(\frac{r_o}{r} \right)^6 \right] \quad (3.19)$$

The physical basis for the repulsion of neutral atoms at short distances is the Pauli exclusion principle. This principle states that electrons cannot have the same quantum numbers. This principle prevents atoms from collapsing since outer electrons of an atom cannot move closer to the nucleus into shells already occupied by other electrons. One can imagine this interaction as atoms bumping into one another.

The physical basis of the attractive interaction can be orientation, dispersion or induction depending on whether the molecules are non-polar or polar, **Figure 3.4**. If the atoms are in a polar bond, then they create a permanent dipole. Orientations between the bonds that cancel the dipoles create an attractive interaction, while orientations that reinforce the dipoles create repulsive interactions. If the atoms are non-polar (lack a permanent dipole), then fluctuation of their electron clouds creates instantaneous, temporary dipoles. Favorable alignment of these dipoles creates a weak net attraction, called London dispersion. For large, non-polar atoms, this attraction between instantaneous, temporary dipoles is the main origin of van der Waals attraction. The final possibility for van der Waals forces is between a non-polar atom and a polar bond. The permanent dipole in the polar bond induces a dipole in the non-polar atom, which creates an attractive interaction, sometimes called a Debye force.

Atoms approach each other until attractive interactions between them (electrostatic and van der Waals attractions) balance the repulsive interactions between them (electrostatic repulsion and bumping).

3 The Logic of Protein Engineering

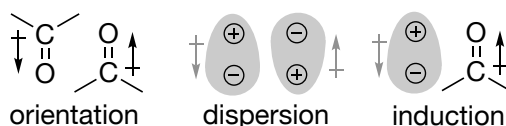


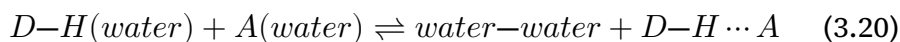
Figure 3.4. Three contributions to the attractive van der Waals interaction. *Orientation*: Polar molecules contain dipoles (black arrows), which create favorable electrostatic interactions when they orient to cancel each other. *Dispersion*: Fluctuating polarization of electron clouds (gray) in non-polar atoms creates temporary dipoles (gray arrows) that attract each other. *Induction*: A polar bond induces a temporary dipole in the electron cloud of a nearby non-polar atom to create a attractive interaction.

Non-covalent interactions in water. Moving from vacuum to water changes the interaction between atoms in two ways. First, the net strength of the interactions between atoms decreases because in water the atoms always interact with something - water or each other - instead of nothing or each other in vacuum. Second, the hydrophobic effect strengthens the association between non-polar atoms to minimize the unfavorable association of water with non-polar atoms.

The equations above describe interactions between atoms in vacuum where the atoms have a choice between interacting with nothing (vacuum) or another atom. In water, the choice changes. Atoms interact with either water or other atoms; interacting with nothing is no longer an option. For polar and charged atoms electrostatic interactions, including hydrogen bonds, are weaker in water than they would be in vacuum, **Table 3.2**, because the atoms also make favorable electrostatic interactions and hydrogen bonds to water when they are not interacting with each other. For example, hydrogen bonds between solutes in water are only 0-2 kcal/mol, not

3.4 Non-covalent interactions & entropy

the 3-5 kcal/mol in vacuum. Hydrogen bonds in water are simply traded to different partners, Equation 3.20, so the advantage is any net gain in the new set of hydrogen bonds. Two solutes start with hydrogen bonds to water. If the solutes trade those hydrogen bonds for hydrogen bond to each other, they break the solute-water hydrogen bond and create water-water hydrogen bonds from the released water and solute-solute hydrogen bonds. The net gain depends on the relative strength of all these changing hydrogen bonds.



Here A is the hydrogen bond acceptor atom and $D-H$ is the hydrogen bond donor. In a similar manner van der Waals interactions are weaker in water than in vacuum because van der Waals interactions between water and solutes offsets the advantage of van der Waals interactions between solutes.

Table 3.2. Noncovalent interactions between atoms in vacuum and in water.^a

Interaction	Type of Atom	Origin	Strength in Vacuum	Strength in Water
electrostatic interactions	charged & partially charged atoms	attraction between opposite charges, repulsion between like charges	3-7 kcal/mol	1-5 kcal/mol
hydrogen bonds	H + O, N, halogen	sharing a hydrogen atom between two electronegative atoms, an electrostatic interaction with a covalent contribution	3-5 kcal/mol	0-2 kcal/mol
van der Waals repulsion (bumping)	all	unfavorable bumping between atoms	0-large	0-large
van der Waals attraction	non-polar, polar bonds	orientation, dispersion, induction	≈1 kcal/mol for Me-Me	≤1 kcal/mol for Me-Me

Interaction	Type of Atom	Origin	Strength in Vacuum	Strength in Water
hydrophobic effect	non-polar, only occurs in water	burying of hydrophobic surfaces minimizes unfavorable water orientations & creates van der Waals attraction	0	≈2 kcal/mol for Me-Me

^a Estimates from Lodish, H. F., Berk, A., Zipursky, S. L., Matsudaira, P., Baltimore, D., & Darnell, J. (2000). Molecular cell biology (4th ed). W.H. Freeman. Section 2.2

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The unique structure of water creates the hydrophobic effect, which is the tendency of non-polar solutes to cluster in water. The hydrophobic effect is not a bond or interaction between non-polar atoms; instead, it is a consequence of the behavior of water in the presence of a hydrophobic solute. The quote below describes the dynamic structure of water.

Water isn't unique in forming hydrogen bonds, but it is the only common substance that can be joined by these gentle, frangible molecular handclasps into a three-dimensional network. Most liquids are little more than a disorderly scrum of jostling molecules. But water is delicately poised between order and disorder, constantly adopting a defective version of the framework structure that, in ice, immobilizes the water molecules into crystalline regularity. - Philip Ball^[1]

Adding a hydrophobic solute disrupts this dynamic structure of water. The balance between order and disorder tips toward an ordered structure surrounding the hydrophobic solute. Water molecules at the non-polar surface cannot form hydrogen bonds with the non-polar surface, so they make fewer total hydrogen bonds, **Figure 3.5**. These hydrogen bonds are stronger than those in bulk water, immobilizing the water molecules at the surface of a non-polar solute into an ice-like cage structure. The restricted mobility of the water molecules in this cage decreases their entropy, making this arrangement less stable than bulk water. Clustering of non-polar solutes reduces the contact area between water and non-polar surface, releases water molecules from the cage-like structure and lowers their energy. The clustering of hydrophobic solutes occurs not due to an attractions between the non-polar atoms, but because of the change in water structure when they cluster. Weak van der Waals interactions between the non-polar atoms contribute a small amount to the hydrophobic effect, but

3.4 Non-covalent interactions & entropy

the main origin is the behavior of the water molecules. The spontaneous separation of oil and water in a mixture is a macroscopic demonstration of the hydrophobic effect.

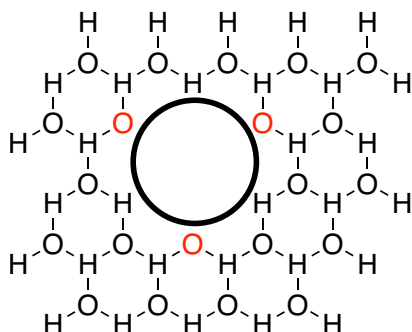


Figure 3.5. Water surrounds hydrophobic solutes with an ice-like structure. In this 2-D diagram of a hydrophobic circle in water, three oxygens near the circle (red color) have only two hydrogen partners, while the rest of the oxygen atoms have three hydrogen partners. Fewer partners strengthen the remaining bonds leading to a rigid, ice-like structure at the hydrophobic interface. This 2-D diagram simplifies the true 3-D structure, where each oxygen atom has four hydrogen partners in the bulk water and two or three hydrogen partners in the water at the hydrophobic interface.

Entropy. Besides the non-covalent interactions between atoms, entropy also contributes to the Gibbs energy of the different protein states. Entropy is the degree of disorder in a system. For molecules, entropy is related to the number of microstates that the molecule can adopt. Assuming equal probability for each microstates, the entropy of a molecular state is proportional to the natural logarithm of the number of microstates within that state, Equation 3.21.

3 The Logic of Protein Engineering

$$S = R \ln(\# \text{ of microstates}) \quad (3.21)$$

A typical example of microstates is different conformations along a single bond, such as the three different conformations of the side chain of serine. Another typical case is two enantiomeric possibilities, such as the gauche+ and gauche- conformations of butane.

The microstate of a protein refers to the a specific conformation or arrangement of atoms and bonds at a given moment. Microstates can differ in the position of each amino acid main chain, the orientation of individual side chains and the hydrogen bonds that form between them. Both the amino acid sequence of the protein and environmental factors such as temperature, pH, and ionic strength determine which microstates are accessible to a protein.

Proteins exist in countless numbers of possible microstates; fortunately, we do not need to count them because we are concerned only with *differences* in entropy and therefore only the relative numbers of microstates. For example to compare the entropy of two states, such as folded and unfolded protein, one needs the relative number of available microstates in each state, Equation 3.22.

$$\Delta S = R \ln \left(\frac{\# \text{ of microstates}_{\text{unfolded protein}}}{\# \text{ of microstates}_{\text{folded protein}}} \right) \quad (3.22)$$

A configuration that has more microstates will have a lower Gibbs energy than one with fewer microstates, assuming all other things are equal. For example, if the product state allows free rotation of the side-chain atoms, while the starting material state does not, then this difference lowers the Gibbs energy of the product state. The contribution of entropy to the Gibbs energy is $-T\Delta S$, Equation 3.23.

3.4 Non-covalent interactions & entropy

$$\begin{aligned} \text{entropy contribution to } \Delta G &= -T\Delta S \\ &= -TR \ln \left(\frac{\# \text{ of microstates}_{\text{product}}}{\# \text{ of microstates}_{\text{starting material}}} \right) \end{aligned} \quad (3.23)$$

If the flexible side chain in the product has three rotatable bonds, each of which can adopt three stable conformations, then there are nine possible microstates in the product state. The fixed side chain in the starting material can adopt only one microstate. At 25 °C,

$$\text{entropy contribution to } \Delta G = -298 \text{ K} \cdot 1.987 \text{ cal/mol} \cdot \text{K} \cdot \ln(9/1) = -1.3 \text{ kcal/mol}$$

Thus, amino acid substitutions that increase the disorder or flexibility of a state stabilize that state. While protein structures show stabilizing non-covalent interactions like hydrogen bonds, static structures do not show the flexibility of a state. One must remember to consider differences in flexibility in addition to structural differences when estimating Gibbs energy differences between states.

Protein folding. Folded structures are the low-energy state for most proteins. This low energy comes mainly from the hydrophobic effect. Burying a $-\text{CH}_2$ group contributes 1.1 ± 0.5 kcal/mol to protein stability. The hydrophobic effect provides ~60% of the driving force to collapse the amino acid chain into a compact structure.^[2] The remaining ~40% comes from attractive interactions between the amino acids. The chains orient to maximize the hydrophobic effect and the attractive electrostatic and van der Waals interactions including hydrogen bonds. This favorable contact between non-polar atoms tightly packs amino acids within the hydrophobic core of proteins. The only empty space is small cavities where the packing is imperfect.

The size, hydrophobicity, charge, and hydrogen bonding abilities of the twenty proteinogenic amino acids differ from one another.

Polar amino acids favor the outside of the structure to interact with the surrounding polar water molecules, while non-polar amino acids favor the interior of the structure due to the hydrophobic effect. The shape, charge, and hydrogen bonding determine the specificity of the protein folding; that is, which sections fold into helices and strands and subsequently how these secondary structures associate into the tertiary structures that form protein domains.

3.5 Logic of protein engineering

Protein engineering will involve substituting one, or more likely, many amino acids in a protein. Amino acid substitutions change non-covalent interactions and entropy, which in turn change the Gibbs energy of states, which changes protein properties. Gibbs energy differences connect protein properties to these amino acid substitutions, **Figure 3.6**. Protein properties like stability, binding, and catalysis depend on Gibbs free energies differences between different states. Amino acid substitutions change the molecular interactions (flexibility and non-covalent interactions) that determine these Gibbs energies. Thus, replacing amino acids changes interactions, which changes Gibbs energies, which changes protein properties.

Protein engineering involves changing the structure of the protein to change interactions within each state such that protein properties improve. In most cases, the goal is to improve some protein properties while keeping others unchanged. Amino acid replacements must both improve the targeted properties and not ruin the existing favorable properties. The next chapter describes working with protein structures and computer modeling approaches to predict how amino acids substitutions affect the Gibbs energies of various states.

3.5 Logic of protein engineering

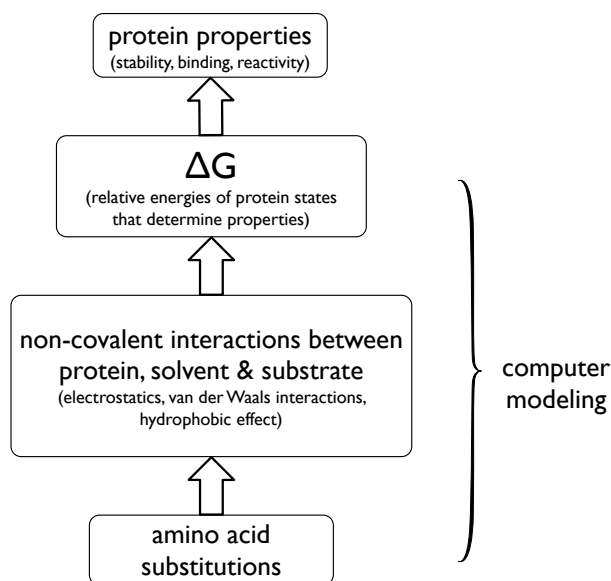


Figure 3.6. The logic of protein engineering. Differences in Gibbs energies of various states determine protein properties like stability, binding, and catalysis. For example, the relative Gibbs energies of the folded and unfolded protein states determine a protein's stability. The states differ in energy because the unfolded state of a protein is flexible and the amino acid residues interact mainly with solvent, while the folded state is less flexible and amino acid residues interact mostly with each other. Protein engineering involves changing the amino acid sequence of the protein to change the flexibility and non-covalent interactions to change the relative Gibbs energies of the states that determine protein properties. Computer modeling keeps track of the complex interactions between amino acids to predict the Gibbs energy of the different states.

3.6 Amino acid structures

Most protein engineering will involve substituting one amino acid for another with the expectation that the replacement will improve the desired property. The basis for this expectation is the differences in the side chains, R, of the different α -amino acids. Substitutions change the size and shape of the side chain and may remove and/or introduce new functional groups.

The structure below shows the general structure of an L- α -amino acid, **Figure 3.7**. Biochemists use the Fischer nomenclature (D or L) instead of the Cahn-Ingold-Prelog nomenclature (R or S) for amino acids and sugars. In the Fischer nomenclature all proteinogenic amino acids have the L-configuration, while the Cahn-Ingold-Prelog nomenclature assigns R to cysteine and S to the others making it less convenient. The number in red indicates the pK_a values for the protonated amino and carboxylic acid groups. At a pH above this value, these groups are deprotonated. At pH 7, α -amino acids exist as zwitterions: a positive charge at the α -amino group and a negative charge at the carboxylate. The carboxylic acid deprotonates to the carboxylate at pH 7 because this pH is above the pK_a of the carboxylic acid group, but the α -amino group does not deprotonate because pH 7 is below the pK_a of the protonated amino group.

Planning amino acid replacements requires knowing the structures and properties of the amino acid side chains. **Figure 3.8** shows the side chains of the twenty standard amino acids grouped according to their classification as hydrophobic, polar, or charged. The figures also include mnemonics for the one-letter codes for the amino acids.

The nine amino acids with hydrophobic side chains are typically buried inside the protein core when the protein folds. The six

3.6 Amino acid structures

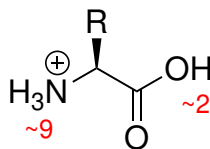


Figure 3.7. The general structure of an L- α -amino acid in its fully protonated form. The amino group is attached at the α -position of the carboxylic acid. All proteinogenic amino acids have the absolute configuration shown, which is called L in the Fischer nomenclature. The fully protonated form shown exists below pH 2; at pH 7, the carboxylic acid is deprotonated to the carboxylate and the net charge on the amino acid is zero.

amino acids with polar side chains form hydrogen bonds as proton donors or acceptors. The number indicates the pK_a values for these residues in typical unfolded peptides. At high pH or in unusual protein environments, cysteine and tyrosine may deprotonate and become negatively charged.

The five amino acids with charged side chains often form salt bridges. **Figure 3.8** shows these five in their protonated forms. At pH 7, aspartate and glutamate would be negatively charged, histidine would be mostly neutral, and lysine and arginine would be positively charged. Histidine often occurs in enzyme active sites because the pK_a of the side chain near-neutral pH allows it to both accept and donate protons readily.

Figure 3.8 also shows the atom naming for the side chains of several amino acids. The first character of the atom name is the chemical symbol for the atom type. For example, all names for carbon atoms begin with C. The next character indicates the remoteness from the carboxylic acid according to the Greek alphabet: α = alpha, β = beta, γ = gamma, δ = delta, ϵ = epsilon, ζ = zeta, η = eta. For exam-

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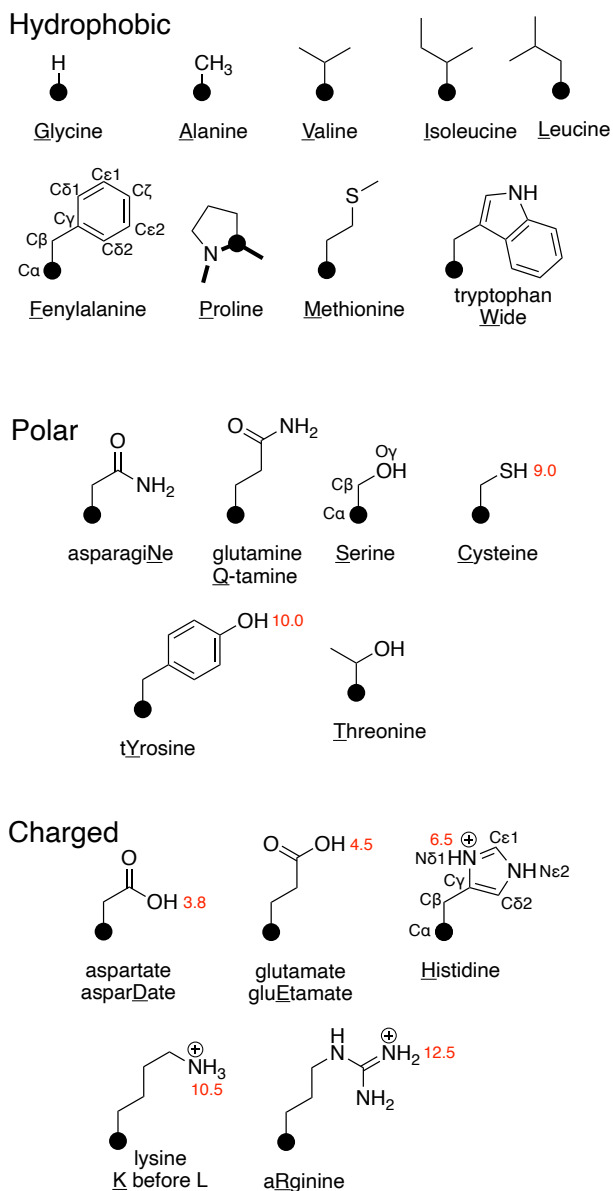


Figure 3.8. The twenty standard amino acids. The large black dot marks the alpha carbon. The drawing of proline includes the backbone nitrogen and backbone bonds shown in bold. The amino acids names (or corresponding mnemonics) contain underlined letters signifying their single-letter abbreviations. The drawings of phenylalanine, serine and histidine include the atom names of the carbons. The Greek letters in the atom names indicate the remoteness from the carboxylic acid carbon atom: α = alpha, β = beta, γ = gamma, δ = delta, ϵ = epsilon, ζ = zeta. The number in the atom name indicates the branch. For example, C δ 1 and C δ 2 indicate the equivalent carbon atoms along two different branches. The red numbers indicate the pK_a of this group in a typical unfolded peptide.

ple, the side chain atoms in serine are C β and O γ . The last character of the atom name is a number to indicate the branch if required. For example, O δ 1 and O δ 2 indicate the carboxyl oxygens in aspartate, which are equivalent due to resonance but correspond to different branches. Greek letters are inconvenient in protein structure files, so they are transliterated with Roman letters according to $\alpha = A$, $\beta = B$, $\gamma = G$, $\delta = D$, $\epsilon = E$, $\zeta = Z$, $\eta = H$. For example, the side chain atoms in serine are named CB and OG in protein structure files. The main chain atoms in the protein structure files are named N, CA, C, and O corresponding to the amino nitrogen, α carbon, carbonyl carbon, and carbonyl oxygen, respectively.

Acid/base behavior, resonance structures and tautomers of imidazole. The versatile nature of the imidazole ring of histidine make it important for catalysis, but the versatility also creates complexity. Below is a review of the acid/base behavior, resonance structures and tautomers for the imidazole ring.

The neutral, uncharged imidazole can act both as an acid and as a base, but only the basic behavior is relevant to most biochemistry, **Figure 3.9**. Imidazole often acts as a base because it accepts a proton at pH 6.5, which is a common pH in biochemistry. The protonated form is also called an imidazolium cation; its pK_a is 6.5. Because histidine contains the imidazole ring in its side chain, it can act as a base or, in the imidazolium form, as an acid near neutral pH. This ability allows it to catalyze proton transfers near neutral pH so histidine often occurs in the active site of enzymes. Note that neutral imidazole rarely acts as an acid because it loses its proton only at pH 15, which is an extreme pH for biochemistry. The pK_a of imidazole is 15.

The imidazolium cation is a single structure that is best represented by two resonance structures, **Figure 3.10**. These structures show that two nitrogen atoms are equivalent because the positive charge

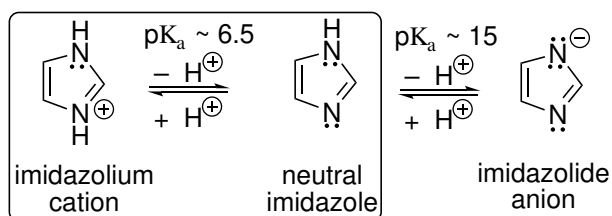


Figure 3.9. Protons transfer readily between neutral imidazole and the imidazolium cation near neutral pH because the pK_a of the imidazolium cation is 6.5. Imidazole is the base (proton acceptor) while the imidazolium cation is the acid (proton donor). Imidazole can also donate a proton, thereby acting as an acid, but this behavior is rare in biochemistry. The pK_a of imidazole is 15, so harsh base is required for imidazole to act as an acid. Expect to see only the neutral imidazole and the imidazolium cation (boxed) in biochemical mechanisms.

of the imidazolium cation is equally distributed between both nitrogen atoms. One Lewis structure is inadequate to accurately represent the imidazolium cation. The resonance structures are two Lewis structures that contribute to the true structure. An alternative to resonance structures is to draw dotted lines to represent delocalized electrons and partial charge indicators. Note that resonance is *not* a rapid equilibrium between the two structures; instead, the true structure is a weighted average of the two resonance structures.

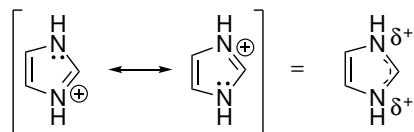


Figure 3.10. Resonance structures of the imidazolium cation show that the positive charge is delocalized on both nitrogen atoms and both C-N bonds have partial double bond character. The two nitrogen atoms are equivalent. Resonance structures are represented by double-headed arrows and represent a single molecular species, not equilibrium between separate species. Atom locations are identical between resonance structures, but the electron locations differ. (Other resonance structures of the imidazolium cation, not shown, make minor contributions to the structure because they are much higher in energy.)

In contrast to the imidazolium cation, the two nitrogens in neutral imidazole differ from one another - one has an added proton, while the other does not. Neutral imidazole exists as two equivalent isomers where the proton is on different nitrogen atoms. The two isomers exchange rapidly by transferring a proton. These two equivalent isomers of imidazole are tautomers, which are rapidly equi-

brating isomers, usually by a proton transfer. In histidine, the imidazole ring contains a substituent making the two two tautomers different, **Figure 3.11**. The tautomers of histidine differ by which nitrogen is protonated. Catalysis may require one of the two tautomers of histidine in the active site.

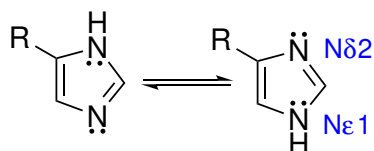


Figure 3.11. Histidine with a neutral imidazole ring exists as a pair of tautomers, which differ by which nitrogen is bonded to a proton. In one tautomer, the proton sits on N δ 2; in the other it sits on N ϵ 1. The tautomers equilibrate by proton transfer via an imidazolium cation (not shown).

Glossary

Conformation see microstate

Differential stabilization of states alters the Gibbs energy difference between states and therefore the equilibrium constant between the states. Protein engineering seeks to differentially stabilize the states that determine protein stability, binding, reactivity, or selectivity in order to improve protein properties.

Entropy is the degree of disorder quantified by the number of accessible microstates. Entropy is stabilizing contribution to the Gibbs energy.

Gas constant (R) is the conversion factor used in chemistry to connects energy to the temperature of the system for a mole of particles. The units are calories (or joules)/°K·mole. Physicists

use the Boltzmann constant, which makes the same connection, but per particle. Its units are calories (or joules)/°K.

Hydrophobic effect is the tendency of non-polar solutes to cluster in water. The origin of this clustering is the unfavorable ordering of water molecules at a non-polar interface. Minimizing non-polar interface by clustering releases the ordered water molecules.

Microstates are specific conformations or arrangements of atoms and bonds at a given moment. Protein microstates can differ in the position of each amino acid main, the orientation of individual side chains and the hydrogen bonds that form between them. The relative number of microstates between two protein states determines the entropy difference between them.

States are different forms of a protein that differ in their conformation and/or association with other molecules, but not in covalent structure. States, sometimes called macrostates, consist of countless microstates. States refer to protein forms that have macroscopic or bulk properties that one can measure. The folded and unfolded states of a protein have different fluorescence properties.

Tautomers are rapidly equilibrating isomers, most often by transfer of a proton to a different atom. The keto and enol forms of carbonyl compounds are tautomers. The imidazole ring of histidine exists in two tautomeric forms.

References

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2. Pace, C. N., Scholtz, J. M., & Grimsley, G. R. (2014). Forces stabilizing proteins. *FEBS Lett.*, 588(14), 2177–2184. <https://doi.org/10.1016/j.febslet.2014.05.006>

Problems

1. To increase the binding of an antibody to its target, do you need to increase or decrease the value of ΔG_{diss} ? How large must the change be in kcal/mol for the binding to increase by a factor of one hundred? Explain whether the change is a ΔG_{diss} , a $\Delta\Delta G_{diss}$ or a $\Delta\Delta\Delta G_{diss}$.
2. Two stable conformations of butane are the anti and gauche conformation. The gauche conformation exists as a pair of enantiomers. Explain how entropy favors the gauche conformation over the anti conformation. How large, in kcal/mol, is this entropy contribution?
3. Draw resonance structures for the deprotonated form of imidazole (the imidazolate anion). Draw an arrow-pushing reaction mechanism for the tautomerization of histidine.
4. Draw the structure of amino acid His showing the correct absolute configuration and the correct protonation state at pH 7 for any acidic or basic functional groups. The pK_a of a carboxylic acids like acetic acid is ~5, but the pK_a of the carboxylic acid in free amino acids is ~2. Rationalize this difference in pK_a .
5. *Optional.* Use the Python script below to calculate the change in Gibbs energy associated with an increase in reaction rate for an enzyme variant as compared to the wild-type enzyme. To use the script, copy the text below and save it as a text file named GibbsEnergyChange.py on your computer Desktop, then follow the instructions below.

Problems

```
#!/usr/bin/env python3
'''
This script calculates the Gibbs energy change in kcal/mol associated
with the improvement of the reaction rate of an enzyme.

To use the script:
1. place the script on your Desktop
2. open a terminal window
3. change the working directory to the Desktop by typing at the prompt:
    cd ~/Desktop
4. in the terminal type: python3 GibbsEnergyChange.py
5. enter the rates for the wild type and variant protein when prompted.

To test the script enter 1 for the rate of wt enzyme and 10 for the
rate of the variant enzyme; the output should be -1.4 kcal/mol.
The negative value indicates that the transition state is lower in
the variant, so the reaction is faster.

'''
#load math function needed to calculate logarithms
import math

# define R in units of cal/mol*K
# to change the units of R to J/mol*K, replace this value
# with 8.314 and also change the units in the print
# statement below to kJ/mol
R = 1.987

# temperature of the comparison in units of degrees Kelvin
T = 298

# asks for user input
rate_wt = float(input("What is the rate of the wt enzyme?"))
```

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```
rate_variant = float(input("What is the rate of the mutant enzyme?"))

# calculates Gibbs energy change
def GibbsEnergyChange(rate_wt,rate_variant):
    return (- R*T*math.log(rate_variant / rate_wt))/1000.0

# print result
print('The mutation changed the free energy of the transition '
      'state by %.1f kcal/mol.' % (GibbsEnergyChange(rate_wt,
      rate_variant)))
```

Answers

1. For a 100-fold increase in binding, the K_d of the new complex should be 100-fold lower (less favorable) than the starting complex. The value of ΔG_{diss} should be higher. The relationship between free energy change and equilibrium constant for bound/unbound is $\Delta G_{diss} = -RT \ln(K_{diss})$

To compare two complexes: $\Delta\Delta G_{diss}(newvsold) = -RT \ln(K_{diss}(new)/K_{diss}(old))$

For a 100-fold improvement in binding:

$$\Delta\Delta G_{diss}(newvsold) = -RT \ln((x/100)/x) \text{ where } x \text{ is } K_{diss}(old)$$

$$\Delta\Delta G_{diss}(newvsold) = -RT \ln(1/100) = +RT \ln(100)$$

$$\text{At 298K: } \Delta\Delta G_{diss} = +(1.987 \times 10^{-3} \text{ kcal/mol/K})(298 \text{ K})(4.605) = +2.73 \text{ kcal/mol}$$

2. Entropy favors the gauche conformation because it has more accessible microstates (two enantiomeric conformations (gauche+ and gauche-)) than the anti conformation (one conformation). The contribution of microstates to entropy is given by $S = R \ln(\Omega)$

where Ω is the number of accessible microstates. For the reaction $anti \rightleftharpoons gauche$, the difference in entropy is,

$$\Delta S_{(g \text{ vs. } a)} = R \ln(\Omega_g/\Omega_a) = R \ln(2/1) = R \ln(2)$$

The free energy contribution from this entropy term is: $\Delta G_{(g \text{ vs. } a)} = -T\Delta S = -RT \ln(2)$

At room temperature (298K): $\Delta G_{(g \text{ vs. } a)} = -(1.987 \times 10^{-3} \text{ kcal/mol/K})(298\text{K})(0.693) = -0.41 \text{ kcal/mol}$

So entropy favors the gauche conformation by approximately 0.41 kcal/mol due to the availability of two equivalent conformations versus one for the anti form. This entropic advantage is why the gauche form, despite potential steric interactions, is significantly populated at room temperature.

3.

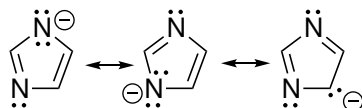


Figure 3.12. Resonance structures of the imidazolid anion.

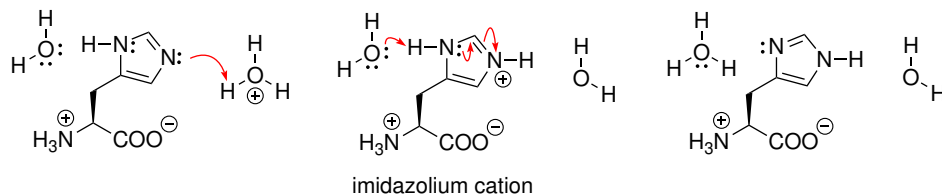


Figure 3.13. The tautomerization of histidine via the imidazolium cation is the most likely mechanism near neutral pH. The alternative mechanism via the imidazolid anion is unfavorable because the anion is unlikely to form near neutral pH.

3 *The Logic of Protein Engineering*

4. The 3-unit difference in pK_a ($5 \rightarrow 2$) is a 1000-fold increase in acidity of the histidine carboxylate as compared to that in acetic acid. This increase is due to the neighboring ammonium cation, which stabilizes the carboxylate form by electrostatics and to a lesser extent by inductive effects. At pH 7, the predominant form of histidine is COO^- (deprotonated), NH_3^+ (protonated), imidazole ring: neutral (its pK_a is 6.5); net charge = 0, it has the (S) absolute configuration at the α -carbon.

4 Protein Structure and Molecular Modeling

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Summary. Protein structures show the fold of the protein, which amino acid residues interact with each other and the shape of cavities and tunnels. Protein structure files specify the locations (x, y, z coordinates) of the atoms in a protein as well as any bound molecules and solvent. Protein visualization programs such as PyMOL create images from these files. Different representations highlight different aspects of these intricate structures including shape, electrostatic charge and flexibility. Computer modeling programs such as AMBER or Rosetta predict protein structures using simplified mechanical models of molecules. Accurately modeling protein properties remains difficult because it is difficult to model protein states. Protein states are collections of interconverting conformations that are accessible to a particular protein form. Molecular dynamics simulations reveal how proteins move on short time scales, but finding a representative sample of all the conformations that contribute to a state is difficult. In practice, researchers use approximations to model protein states, which lowers the accuracy of the prediction of protein properties.

Key learning goals

4 Protein Structure and Molecular Modeling

- Protein structure files are text files containing the x,y,z coordinates of the protein atoms as well as other structural information.
- Protein visualization software displays the data in protein structure files. This software simplifies the complexity of protein structures by using different representations for different parts of the protein. This software can also measure distances between atoms within a protein and predict structures of variants with amino acid substitutions.
- Computer modeling of proteins uses simplified mechanical models of molecules called force fields to calculate the energy of a structure. Most predictions of protein structures from their amino acid sequence are extrapolations from known structures. These extrapolations include machine learning models and correctly predict most protein folds and are increasingly accurate in predicting side chain orientations.
- Protein move constantly and all of the accessible protein conformations contribute to the protein states. Gibbs energy differences between protein states determine protein properties.
- Protein states are difficult to model because important conformations may be difficult to find. For example, an open conformation important for catalysis may occur rarely or not at all in molecular dynamics simulations because the conformation requires simultaneous orientation of multiple bonds.
- Most predictions rely on approximations to model protein states. Their accuracy varies according to the validity of these approximations.

4.1 Protein structure

4.1.1 Determining protein structures

X-ray crystallography determines the structure of proteins by measuring the electron density distribution within protein crystals, **Figure 4.1**. Fitting a model of the known amino acid sequence of the protein to this electron density data yields the structure.

One can also determine the structures of proteins by NMR (nuclear magnetic resonance). NMR measures the distances between various nuclei in the protein. Fitting a model of the known amino acid sequence to these distances yields a set of matching structures with differing conformations. While x-ray structures typically yield a single structure, NMR structures yield a collection of similar structures that represent conformations explored by the protein.

Homology modeling is a computational approach to predict protein structures. Extrapolation from a known protein structure (the template) yields predicted structures for homologous proteins. If the two proteins share >80% identical amino acids, then the model is typically within 1-2 Å root mean square deviation of the correct structure. For more distant homologs, the reliability depends on the degree of sequence identity, the choice of the best template, and the alignment of the template and target protein sequences. Web tools such as SwissModel^[1] and the machine learning model AlphaFold^[2] automate this extrapolation.

4.1.2 Structure data files

One way to describe the structure of a molecule is to specify the location of each atom using Cartesian coordinates (x,y,z). The example structure file for aspirin lists these coordinates as well as

4 Protein Structure and Molecular Modeling

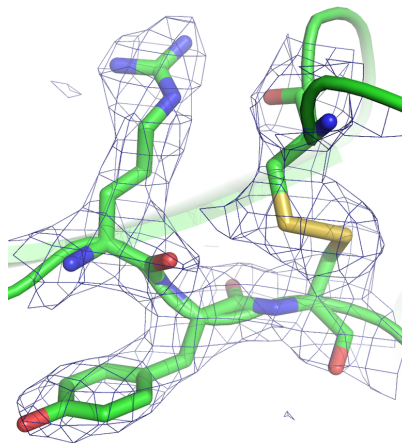


Figure 4.1. Measuring electron density reveals a protein's structure. The mesh diagram represents the observed electron density in an x-ray crystallography experiment for four amino acids in a protein. The mesh is contoured at 1σ ; that is, it encloses the regions that contain more than one standard deviation above the mean electron density. The line diagram is the best fit of the atoms in the known amino acid sequence (arginine, tyrosine, and two cysteines linked by a disulfide bond) to the observed electron density. Image from: https://en.wikipedia.org/wiki/File:example_of_electron_density_map.png; CC BY-SA 3.0 license.

4.1 Protein structure

some additional information, **Figure 4.2**. The first line states the total number of atoms. Lines 2-22 list the information for each atom: the element, serial number, (x,y,z) coordinates, atom type number (see molecular mechanics in Section 4.2.1). The final columns list the other atoms bonded to the atom. For example, line 2 describes carbon atom numbered 1 located at $x = -1.573$, $y = 0.146$ and $z = -0.7046$. The units are Ångstroms. It is a carbon atom type 2 (sp^2 -carbon, alkene) and it makes bonds to the atoms numbered 2, 6, and 12. Upon rotation of the molecule, these (x, y, z) coordinates change.

[illegible]

Figure 4.2. Cartesian coordinate file for aspirin and, at the right, a line diagram of the structure showing the atom numbering. The first line gives the total number of atoms. The subsequent lines specify the element, a serial number for each atom (1-21), the three x, y, and z coordinates of its location, the atom type number (1, 2, 3, 5, 6, 7 or 24; these numbers are used by the molecular modeling program) and the last four columns list the serial numbers of the atoms bonded to it.

An alternative file format is internal coordinates or a Z-matrix (not shown). The position of the first atom is arbitrary. Subsequent

4 Protein Structure and Molecular Modeling

atom positions are indicated relative to this first atom by specifying measurements (distance, bond angle, dihedral angle) relative to previous atoms. The advantage of this format is that the coordinates do not change when the molecule rotates.

Protein structure files contain thousands of atoms, use the protein data bank (pdb) file format, and are freely available at <https://www.rcsb.org>. These pdb format files specify the Cartesian coordinates (x, y, z) of each atom and include additional information. Protein structure files are text files, which can be opened by text editors or word processing programs, **Figure 4.3**. The files are large (5,524 atom locations, 115 pages for the example in **Figure 4.3**). The information in these files can be grouped into introduction, protein atom (and nucleic acid atom, if present) coordinates, and non-protein atom coordinates.

The file partly shown in **Figure 4.3** contains 6,391 lines; 5,524 lines contain x, y, z coordinates of atoms and the remaining lines contain additional information. The introductory information describes the structure and the experimental methods. It includes the title of the structure, journal reference, remarks, amino acid sequence, list of heteroatoms in the structure, list of disulfide bonds and crystallographic information.

The protein coordinates section lists the atoms numbered sequentially. The atom naming follows the convention described in the previous chapter and includes the amino acid name, amino acid number, and the protein chain indicated by a letter. For example, atom 1 in **Figure 4.3** is the amino nitrogen of serine, which is the first amino acid in chain A. The next numbers are the Cartesian (x, y, z) coordinates for this atom. Most x-ray crystal structure files do not include hydrogen atoms because most protein crystals do not diffract well enough to allow hydrogen atoms to be resolved. Molecular modeling programs will add hydrogen atoms by inferring their positions from the positions of other atoms. In contrast,

4.1 Protein structure

```

introduction {
HEADER      OXIDOREDUCTASE                      30-JAN-01   1I0Z
TITLE       HUMAN HEART L-LACTATE DEHYDROGENASE H CHAIN, TERNARY
TITLE       2 COMPLEX WITH NADH AND OXAMATE
.
JRNL        AUTH   J.A.READ,V.J.WINTER,C.M.ESZES,R.B.SSESSIONS,
JRNL        AUTH 2  R.L.BRADY
JRNL        TITL   STRUCTURAL BASIS FOR ALTERED ACTIVITY OF M- AND
JRNL        TITL 2  H-ISOZYME FORMS OF HUMAN LACTATE DEHYDROGENASE.
JRNL        REF    PROTEINS                      V. 43   175 2001
.
protein coordinates {
ATOM        1  N   ALA  A   1      10.019  82.794  38.679  1.00 55.58  N
ATOM        2  CA  ALA  A   1      8.866  82.324  37.874  1.00 54.98  C
ATOM        3  C   ALA  A   1      8.975  80.832  37.653  1.00 54.13  C
ATOM        4  O   ALA  A   1      8.581  80.342  36.603  1.00 54.75  O
ATOM        5  CB  ALA  A   1      8.811  83.066  36.530  1.00 55.52  C
ATOM        6  N   THR  A   2      9.510  80.122  38.643  1.00 52.88  N
ATOM        7  CA  THR  A   2      9.707  78.670  38.572  1.00 51.63  C
.
.
.
occupancy
TER         5106      ASP  B 332
.
non-protein coordinates {
HETATM      5107  PA  NAI  A 401      15.594  42.646  59.976  1.00 13.36  P
HETATM      5108  O1A NAI  A 401      15.067  42.558  61.345  1.00 14.33  O
HETATM      5109  O2A NAI  A 401      16.899  43.424  59.848  1.00 14.62  O
HETATM      5110  O5B NAI  A 401      14.531  43.286  58.976  1.00 12.18  O
.
.
.
CONNECT     5205 5204
CONNECT     5206 5204
.
END

```

Figure 4.3. Three sections from the x-ray structure file for human heart lactate dehydrogenase from the protein data bank (pdb id = 1i0z). The first section, introduction, contains the name of the molecule, a reference to the peer-reviewed article describing this structure and experimental details about the structure determination. The second section lists the coordinates of protein atoms. The third section lists coordinates for non-protein atoms (solvent and other co-crystallized molecules). It also contains 'CONNECT' lines to specify the connectivity of non-protein molecules such as NADH (named NAI) in this structure. The END record marks the end of the file.

4 Protein Structure and Molecular Modeling

protein structure files from NMR experiments do include hydrogen atom positions because the NMR experiments measure locations of hydrogens.

The (x, y, z) coordinates are followed by the atom occupancy, which is 1.00 in most cases, indicating that the atom is located at one position. For flexible regions, the atoms may lie in multiple positions. In these cases, the structure file lists coordinates for each position and assign each one an occupancy. If an atom occupies two positions equally, then the pbb file contains separate lines for each position and each position has an atom occupancy of 0.50.

The numbers in the column after occupancy are the temperature factors, or B-factors, for each atom. The B-factor has units of $\text{\AA}^2$ and describes the displacement of the atomic positions from a mean value. Typical values are 15-30 $\text{\AA}^2$ in the protein core, but more flexible or disordered regions (such as the N-terminal serine in the structure above) have higher values (57 $\text{\AA}^2$).

The last section lists the coordinates and connectivity for non-protein atoms within the structure, which are called heteroatoms (HETATM). These include water, other solvent molecules (e.g., glycerol) and any inhibitors or substrates co-crystallized with the protein. The solvent water molecules are those that remain in fixed positions within the crystal so that they diffract some of the x-rays. In reality, the protein is completely surrounded by water molecules. The records starting with CONECT specify the connectivity for the heteroatoms. In contrast, the protein coordinates section assumes standard atom connectivity for amino acid residues and nucleotides, so it does not contain CONECT records. The end of the pdb structure file contains a record for bookkeeping (MASTER) and an END record.

4.1.3 Visualizing proteins

Opening a structure file with molecular visualization software instead of a text editor reveals a three-dimensional image that can be rotated and zoomed on the screen. PyMOL is popular molecular visualization software for proteins, which is available as an open-source version, a free educational version, and also paid versions. Tutorials are available online and a publication gives an overview of the different types of visualizations that are possible.^[3] PyMOL includes a Python interpreter so that Python commands can extend and automate PyMOL. Later chapters include PyMOL scripts that automate highlighting features like surface residues, holes in the protein core or ion pairs.

Molecules look like the blobs of electron density shown in **Figure 4.1**, but molecular visualization software represents molecules in ways that simplify and emphasize different aspects of their structure. For example, the ball and stick representation of aspirin resembles mechanical molecular models used for conformational analysis, **Figure 4.4**. The wireframe view focuses on the connectivity of the atoms, while the space-filling representation consists of overlapping van der Waals spheres to show overall size and shape of the molecule. The model may be colored by element, as shown in **Figure 4.4**, but colors can also indicate the electrostatic charge, hydrophobicity, or other atomic property.

Since proteins contain so many more atoms than small organic molecules, researchers use additional simplifying representations that hide atoms. The default settings in PyMOL show the protein as a ribbon that traces the main chain of a protein without showing any individual protein atoms, **Figure 4.5**. This representation, called a cartoon, uses flat arrows to indicate β -sheet secondary structures, helices to indicate α -helices, and thin tubes to indicate loops. This representation emphasizes the protein fold while

4 Protein Structure and Molecular Modeling

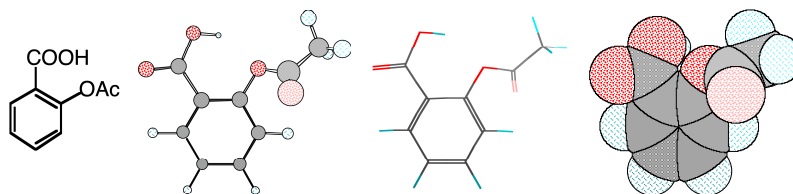


Figure 4.4. Line drawing of aspirin (left) and three representations of its three-dimensional structure: ball and stick, wireframe, and space-filling.

ignoring the atomic details of the structure. The center structure in **Figure 4.5** adds gray and white colors to the ribbon to indicate different domains as well as stick representations of the bound cofactor NADH and inhibitor. The right-most representation in **Figure 4.5** shows a protein property of the same protein. The tubes are colored and scaled to indicate the degree of flexibility.

4.1.4 Intuitive protein engineering

In some cases, protein engineers use only inspection of the protein structure combined with chemical reasoning to choose substitutions. Looking at the three-dimensional structures of proteins using a visualization program such as PyMOL reveals the overall fold of the protein, the location of the active site, which residues line the active site, and many other details. Molecular visualization software can measure distances and angles in a structure and even create models of where amino acids have been replaced. For example, inspection of the structure identifies substitutions that could expand the active site to accommodate a larger substrate.

This intuitive approach is a good first step, but one should not be disappointed if it fails. First, this approach typically examines only

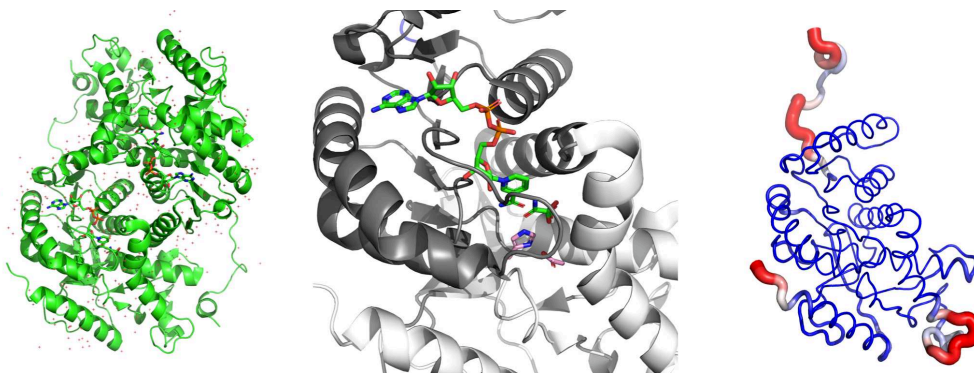


Figure 4.5. Three images of lactate dehydrogenase (pdb id = 1i0z) as displayed in PyMOL. Left: Default settings show the dimer observed in the crystal. The green ribbon represents the protein atoms. Helical flat ribbons indicate α -helices, flat arrows indicate β -sheets, and thin lines indicate turns. The red crosses represent solvent water molecules, and the bound NADH and inhibitor oxamate are shown in stick representation with the carbon atoms colored green. Center: A view of the active site region with the catalytic domain in white ribbons and the Rossmann fold domain in gray ribbons. The side chains of the catalytic residue (His193 and Asp166) are shown as sticks with pink carbons. Right: A ribbon representation where the colors and thickness of the tubes indicate the temperature factors of the atoms. The central part of the protein (blue, thin tubes) has well-defined locations, but the N- and C-termini and residues 213-228 (red and white, thick tubes) have poorly defined positions suggesting flexibility. The text version of this pdb file was shown in **Figure 4.3**.

one structure while protein properties depend on differences between two protein states. The intuitive approach ignores one state completely and also assumes that the single structure examined is a good representation of the folded state. In the example above, substitutions to expand the active site may cause the surrounding residues to readjust such that the active site contracts instead. Finally, some protein features such as relative energies, solvation, protein movements, and pK_a of catalytic groups are not evident from simple inspection of the structure and require calculations.

4.2 Computer modeling of proteins

All models are wrong, but some are useful.

— George Box, statistician

The goal of computer modeling in protein engineering^[4] is to predict which substitutions will improve the target properties of the protein. This prediction requires models of the protein states that contribute to the target properties. The two components of modeling a protein state are first to estimate the Gibbs energy of individual protein structures in each protein state and second to collect a representative sample of conformations that contribute to that state. Estimating the Gibbs energy of protein structures is relatively reliable, but collecting a representative sample of all contributing conformation remains difficult. Some important conformations occur infrequently and are difficult to find with computational methods.

4.2.1 **Calculating the energy of a structure**

The theoretically correct approach to model molecules is quantum mechanics where the electron distributions in molecules are calculated using approximations to solve the Schrödinger equation. Quantum mechanics is a first-principles method without experimentally derived parameters, but is far too complicated and slow to model entire proteins. Quantum mechanics is required to model the bond-making and bond-breaking steps of a reaction. These steps involve electron redistribution that cannot be modeled with simpler approaches. In these cases, quantum mechanics is used to model the active-site region (<100 atoms) while simpler methods are used for the rest of the protein.

Most computer modeling of proteins uses molecular mechanics, which simplifies molecules by treating them like macroscopic mechanical objects. Molecular mechanics uses classical physics to model molecules where spherical balls represent atoms and distance-dependent attractions between them represent bonding interactions. This theoretical model of molecules is incorrect because bonds between atoms arise from delocalization of electrons and interaction of electrons with nuclei. These electron interactions do not behave according to classical physics. Nevertheless, molecular mechanics yields accurate structures of molecules. Empirical adjustment of the equations and parameters of molecular mechanics to experimental structures created predictions that match the known structures of molecules.

Different molecular mechanics software uses different force fields to calculate the energy of a molecular structure. The force field is the mechanical model used to represent molecules in molecular mechanics calculations. It consists of 1) an equation with multiple terms that yields the energy of a structure, 2) atom types to describe elements with different bonding arrangements, and 3) parameters

(constants) adjusted so that the equations yield the correct energies and structures. Force fields differ in the complexity of the terms in the energy equation, the number and definition of atom types, and the number and value of the parameters. Each force field is best suited for specific types of molecules. Force fields like MM3 (Molecular Mechanics 3) are best to predict the structures of small organic molecules, while force fields like AMBER (Assisted Model Building with Energy Refinement) or Rosetta are better suited for biomolecules.

Energy equation for physics-based force fields. The energy equation of a force field consists of multiple terms, some of which describe interactions between bonded atoms and others describe the interactions between non-bonded atoms, Equation 4.1. Bonded interaction terms describe how energy varies with bond length, bond angle, and torsion angle along single bonds. Non-bonded interaction terms describe repulsive and attractive van der Waals and electrostatic interactions. The total energy, v , varies with the structure of the molecule as defined by the distances between the atoms, r^N .

$$v(r^N) = \sum_{\text{bonded atoms}} \text{bonded interactions} + \sum_{\text{all atom pairs}} \text{non-bonded interactions} \quad (4.1)$$

The force field energy equation includes at least three terms to describe bonded interactions: bond stretching, angle bending and torsion along single bonds, **Figure 4.6**. Each term defines an ideal value and the energy cost for deviations from this ideal. The bond stretch term defines an optimum bond length, l_0 between two bonded atoms. When $l = l_0$, this term is zero indicating no energy penalty. Bond lengths shorter or longer than l_0 increase the energy of the molecule by the square of the deviation multiplied

4.2 Computer modeling of proteins

by k_b , which represents the stiffness of the bond. The angle bend term applies to a group of three bonded atoms and similarly has an optimum angle, θ_0 , and angle stiffness, k_θ . The torsion angle term defines the orientation along a single bond for a group of four bonded atoms. There are usually several torsion angle minima. For example, the H-C-C-H torsion angle in ethane has three minima, which correspond to the three staggered conformations. The constant V_n defines the amplitude of the curve, the n defines its periodicity (number of minima), γ shifts the entire curve along the torsion angle (ω) axis. Multiple substituents at the ends of a single bond require several torsional angle terms to include the different interactions created as the torsion angle varies.

The non-bonded interactions depend on the distance between atoms, r , as shown in the bottom two terms in **Figure 4.6**. These terms were explained in the previous chapter. Pairs of atoms connected by chemical bonds are normally excluded from computation of non-bonded interactions because bonded energy terms replace non-bonded interactions. In biomolecular force fields all pairs of connected atoms separated by up to 2 bonds (1-2 and 1-3 pairs) are excluded from non-bonded interactions.

The force field calculates the sum of all the possible interactions to estimate the energy of a particular conformation of a molecule, Equation 4.2. The sum includes terms for interaction between bonded atoms (all bonds, all angles, all torsions) and terms for interactions of non-bonded atoms (all possible pairwise interactions between atoms).

4 Protein Structure and Molecular Modeling

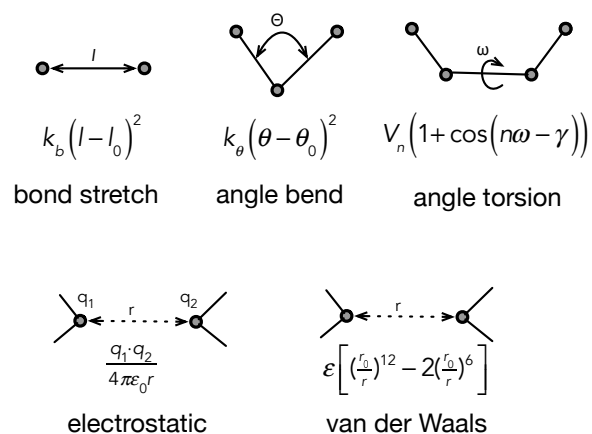


Figure 4.6. Simple force field terms that represent bonded (top three) and non-bonded interactions (bottom two). Bonded interactions describe distances, angle, and torsions between two, three, and four atoms connected by bonds. Non-bonded interactions describe the pairwise interactions between atoms that are nearby but not bonded covalently. All force fields model these five interactions but may modify these terms or add additional terms to describe them more accurately.

4.2 Computer modeling of proteins

$$\begin{aligned}
 E_{total} = & \sum_{\text{bonds}} k_{b,i}(l_i - l_{i,0})^2 + \sum_{\text{angles}} k_{\theta,i}(\theta_i - \theta_{i,0})^2 + \sum_{\text{torsions}} V_n(1 - \cos(n\omega - \gamma)) \\
 & + \sum_{i=1}^N \sum_{j=i+1}^N \left(\epsilon_{ij} \left[\left(\frac{r_{i,j,0}}{r_{i,j}} \right)^{12} - 2 \left(\frac{r_{i,j,0}}{r_{i,j}} \right)^6 \right] + \frac{q_i \cdot q_j}{4\pi \cdot \epsilon_0 \cdot r_{i,j}} \right)
 \end{aligned}
 \tag{4.2}$$

For example, calculating the energy of *n*-butane would contain bond stretch terms for each C-C and C-H bond in the molecule, angle bending terms for each H-C-H, H-C-C, and C-C-C bond, and torsion angle terms for each H-C-C-H, H-C-C-C, and C-C-C-C bond. Non-optimal bond lengths or angles would raise the energy of the structure. The van der Waals interaction term would calculate bumping or weak attractive interactions between all atom pairs. The calculation of the non-bonded interactions may ignore electrostatic interactions for this non-polar molecule.

The number of bonded interactions increases linearly with the number of atoms in the molecule, N . For example, if each atom is bonded to an average of three other atoms, then the number of bonded interactions is $3N$. In contrast, the number of non-bonded interaction increase with the square of number of atoms. Each atom interacts with every other atom so the number of non-bonded interactions is $N(N - 1)/2$. Division by two eliminates double counting of the interaction between atoms 1 and 2 and the interaction between atom 2 and 1 as separate interactions.* This different scaling with the number of atoms means that the number of non-bonded interactions is much larger than the number of

*The precise number of non-bonded interaction is $2N$ less than this value because connected atoms separated by up to two bonds are excluded from the calculation, but this small correction does not change the conclusion that there are many more non-bonded than bonded interactions.

bonded interactions for large molecules like proteins. For example, the dehydrogenase in **Figure 4.5** above contains 374 amino acids per monomer. Each monomer contains approximately 2800 atoms and the software will add 2900 hydrogen atoms for a total of 5700 atoms. There are $5700 \cdot 3 = 17,100$ bonded interactions to calculate and $5700 \cdot 5699/2 = 16$ million non-bonded interactions. The number of non-bonded interaction is ~1000-fold more than the number of bonded interactions. The number of non-bonded interactions will further increase if the calculation includes water molecules. To save computation time, programs usually ignore non-bonded interactions between atoms that are more than 6 Å from one another.

Atom types. Atom types are the molecular subunits in a molecular mechanics calculation and include both element type and its bonding arrangement. While elements are the authentic subunits of molecules, molecular mechanics uses atom types to distinguish different bonding arrangements of the same element. This approach compensates for the fact that molecular mechanics does not explicitly include electrons in the calculations. For example, in the aspirin example in **Figure 4.2** above, C1-C6 are all atom-type 2, which in the MM2 force field is an sp^2 -hybridized carbon in an alkene (a non polar bond). C12 and C16 are atom-type 3, which is an sp^2 -hybridized carbon in a carbonyl or imine (a polar bond), while C17 is atom type 1, which is an sp^3 -hybridized carbon. Even though these atoms are all carbon atoms, molecular mechanics requires different parameters to describe their different bonding arrangements. In contrast, a quantum mechanics approach would assign all these atoms as carbon atoms and the calculated differences in the electron distribution density of each carbon would lead to different bonding arrangements.

Using atom types instead of elements as the subunits of molecules leads to some unusual atom types. For example, the MM2 force field models lone pairs of electrons as a special atom types. These

4.2 Computer modeling of proteins

pseudo atoms account for the non-spherical nature of atoms containing lone pairs, such as oxygen and nitrogen. These lone-pair pseudo atoms improve the model of van der Waals interactions, torsional potentials and hydrogen bonding for these atoms. Other force fields (e.g., MM3, MM4) do not use lone-pair pseudo atoms and instead model lone pairs with a more complex electrostatic treatment and adjusted torsional potentials. These unusual atoms types are a reminder that molecular mechanics oversimplifies chemical bonding.

Different force fields use different definitions for atoms types. For example, AMBER, a force field optimized for biopolymers, includes united atoms types which include hydrogen atoms implicitly. For example, the AMBER-assigned atom types for tyrosine include CD (sp^2 aromatic carbon in 6-membered ring with 1 hydrogen), C2 (sp^3 carbon with 2 hydrogens), and CH (sp^3 carbon with 1 hydrogen), **Figure 4.7**. The rationale for this approximation is that large structures adjust to avoid small distortions in C-H subunits, so they are all similar in biopolymers. Force fields for small organic molecules never use united atom types for a methyl group. Strained organic molecules may distort methyl groups, and these distortions contribute to the overall energy of a molecule.

A force field for protein modeling such as AMBER^[5] differs from a force field for modeling small organic molecules such as MM3,^[6] **Table 4.1**. MM3 includes 153 atom types to model a wide range of elements and bonding arrangement in organic molecules. In contrast, AMBER includes only 41 atom types, which is enough to describe proteins, nucleic acids and carbohydrates. MM3 includes elaborate functions for the bond stretch, bond angle and cross terms to accurately model the structures of a wide range of organic molecules, including strained structures. In contrast, AMBER uses a simplified treatment of bond stretch, bond angle and van der Waals interactions to be fast enough to calculate large structures. Distortions of bonds, angles and torsions are uncom-

4 Protein Structure and Molecular Modeling

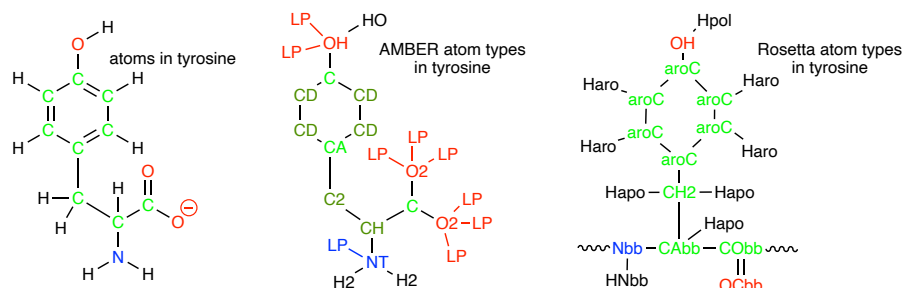


Figure 4.7. The AMBER and Rosetta force fields assign the atoms in tyrosine differently (middle and right structure) from the actual elements of the atoms in in tyrosine (left structure). The different atom types for the same element to indicate different bonding or polarity. The AMBER hydrogen atoms include HO (hydrogen on oxygen) and H2 (amino hydrogen in NH_2); the oxygen atoms include OH (alcohol oxygen) and O2 (carboxyl or phosphate nonbonded oxygen); the carbon atoms include C (sp^2 carbonyl carbon and aromatic carbon with hydroxyl substituent in tyrosine) and CA (sp^2 aromatic carbon in 6-membered ring with 1 substituent). Tyrosine contains only one nitrogen, which is assigned AMBER atom type NT (sp^3 nitrogen with 3 substituents). In other cases, several atoms are combined into a single unit called a united atom type: CD (sp^2 aromatic carbon in 6-membered ring with 1 hydrogen), C2 (sp^3 carbon with 2 hydrogens), and CH (sp^3 carbon with 1 hydrogen). AMBER assigns lone pairs, which are not atoms, to the atom type LP. Rosetta hydrogen atoms include Hapo (apolar hydrogen), Haro (hydrogen on an aromatic ring), Hpol (polar hydrogen), and HNbb (backbone amino hydrogen). Rosetta carbon atoms include aroC (carbons in an aromatic ring), CH2 (methylene carbons), CAbb (backbone C alpha), and CObb (backbone carbonyl carbon). Rosetta oxygen atoms are OH (hydroxyl oxygen) and OCbb (backbone carbonyl oxygen). The backbone amide nitrogen is assigned atom type Nbb.

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mon in proteins since the proteins can relieve strain by moving the backbone. MM3 uses a simplified treatment of electrostatic interactions using only formal charges and bond dipoles, since most organic molecules are non-polar. MM3 adds a point charge term for charged molecules, but errors can occur in highly polar molecules. AMBER uses partial point charges for all atoms to more accurately model electrostatic interactions in biomolecules, which are highly polar.

Table 4.1. Comparison of a force field for small organic molecules (MM3) with one for proteins and nucleic acids (AMBER).

Force field	Number of atom types	Application	Stretch	Bend	vdW	Electrostatic
MM3	153	general, or- ganic molecules	r^2, r^3, r^4	6 terms	e^r, r^2	dipole or charge
AMBER	41	biopolymers	r^2	2 terms	$r^{12}, r^6, r^{12}, r^{10}$	charge
Rosetta	59	biopolymers	statistical poten- tial	statistical poten- tial	r^{12}, r^6 with modifi- cations	charge with modifi- cations

Rosetta, a statistics-based force field. The MM3 and AMBER force fields rely on rules that describe physical interactions and are known as physics-based force fields. Another class of force fields, called statistics- or knowledge-based, replaces some of

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the rules with statistical data from databases. Statistics-based force fields replace rules of how to build structures with examples of structures. The best known statistics-based force field is Rosetta developed by the Baker laboratory at the University of Washington. Rosetta replaces the physics-based equations for bonded interactions with statistically derived energies from structures in the protein data bank.^[7] For example, instead of defining optimum bond distances and bond angles, Rosetta assigns backbone conformation according to how frequently they occur for similar amino acids. Frequently-occurring structures are scored as more stable than rarely-occurring structures. Similarly, Rosetta includes the propensity of amino acids to be buried in protein structures, which approximates solvation effects, and the propensity of amino acids to occur next to each other to weight the overall energy score of a protein structure. This approach is much faster than a physics-based force field. Rosetta uses similar physics based energy terms to calculate non-bonded interactions. One disadvantage of knowledge-based force fields as compared to physics-based force fields is the loss of insight as to why some interactions are favored over others.

Modeling bond stretching. The best description of how energy varies with bond length is the Morse potential, Equation 4.3, where l is the bond length and l_0 is the ideal bond length.

$$v(l)_{\text{bond}} = D \left(1 - e^{-a(l-l_0)} \right)^2 - D \quad (4.3)$$

D is a constant that describe the depth of the energy well (bond strength), while a is a constant that describes the width of the well (bond stiffness). At large bond lengths, the expression inside the parentheses approaches one and the energy approaches zero, **Figure 4.8**. This behavior indicates no interaction at large distances. When $l = l_0$, the expression inside the parentheses is zero and the

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energy is $-D$. This behavior indicates that as the two atoms approach each other the energy decreases due to the formation of the bond reaching a minimum at l_0 , which is the optimal bond length for this atom pair. If the distance between the atoms decreases further, the energy increases sharply. The computer evaluation of the exponential term in the Morse potential equation requires expansion to a series and evaluation of multiple terms, which is computationally costly.

To speed up calculation, force fields replace the Morse potential with polynomials, which are faster to calculate. AMBER uses a single quadratic term, eq, Equation 4.4. This parabola equation describes a harmonic oscillator or spring where the energy increases when the distance deviates from the optimal distance. This simple harmonic matches the Morse potential closely near the optimal distance, l_0 , but deviates when the bond is distorted. The simple harmonic increases in energy symmetrically when the bond is shortened or lengthened, so it underestimates the energy when the distance is significantly shortened and overestimates the energy when the distance is significantly lengthened. AMBER tolerates these deviation because it is intended to model proteins where the bond distances are always near their optimal distances.

$$v(l)_{\text{bond}} = k_b(l - l_0)^2 \quad (4.4)$$

In contrast, the bonding and interatomic interactions within small organic molecules often forces bond lengths to adopt shorter or longer distances than the optimum. Accurate modeling of organic structures requires accurate modeling of a wider range of bond lengths. MM3 adds cubic and quartic polynomial terms to match the Morse potential over a wider range of interatomic distances, Equation 4.5. The cubic polynomial term increases the energy at distances shorter than l_0 and decreases the energy at distances longer than l_0 to extend the range of bond lengths that match

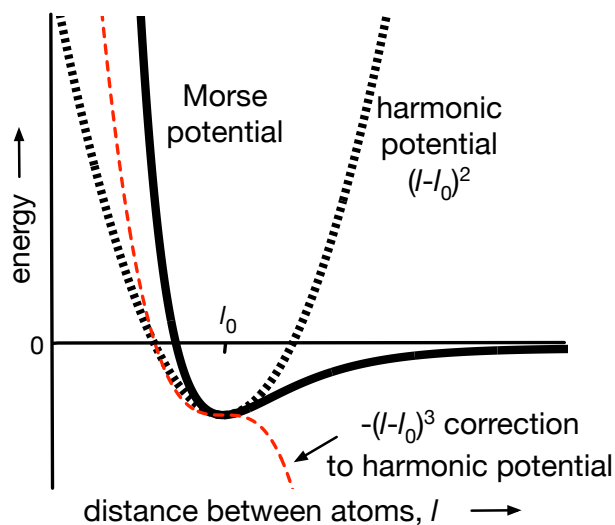


Figure 4.8. Equations to model the bond stretch in molecular mechanics. The most accurate equation is the Morse potential (solid line), which predicts no interaction at long distances, stabilization at an intermediate distances near l_0 due to formation of the bond, and a sharply increasing repulsive interaction at short distances. A harmonic potential (parabola, short dashed line) matches the Morse potential near l_0 , but deviates at shorter and longer distances. A cubic correction to the harmonic potential (red dash line) decreases the energy at long distances and increases the energy at short distance to match the Morse potential over a wider range of bond lengths.

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the Morse potential. The positive quartic term offsets the large negative values of the cubic term at long bond distances. The constants CS (cubic stretch) and QS (quartic stretch) adjust the magnitude of these additional terms. Including these corrections allows MM3 to yield accurate bond distances for even strained organic molecules.

$$v(l)_{\text{bond}} = k_b(l - l_0)^2 - CS(l - l_0)^3 + QS(l - l_0)^4 \quad (4.5)$$

MM3 also includes cross terms that account for interactions between the terms. For example, bond angles can affect the energy of the bond stretch. As a bond angle (e.g, H-O-H) narrows, the two H atoms start to bump each other. Opening the angle relieves the strain, but lengthening the O-H bond also relieves the strain. The stretch-bend cross term accounts for this interaction between angle bending and bond stretching for more accurate energy calculations. AMBER does not use cross terms.

Rosetta does not model bond stretching at all, but uses statistical potentials to define the conformation of backbone and the side chain orientations.

Modeling electrostatic interactions. In contrast to the bond stretching term where MM3 adds additional terms, AMBER uses more complex calculations for electrostatic interactions. Proteins and nucleic acids are charged molecules so electrostatic interactions are an important contributions to their structure. Most small organic molecules are not charged, so electrostatic interactions are less important. AMBER assigns point charges to all atoms and calculates the interactions between all atom pairs using Coulomb's law plus an additional term for hydrogen bond interactions, Equation 4.6. Hydrogen bonds are partially covalent and have preferred bonds angles. If hydrogen bonds were modeled using a purely electrostatic interaction, then this partial covalent character would be

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lost. AMBER calculate pairwise electrostatic interactions for all atoms within 30-40 Å of each other. The diameter of a spherical 30 kDa protein is ~40 Å, so charges on the protein surface weakly interact with charges on the opposite surface.

$$v(r)_{\text{electrostatics}} = \frac{q_1 q_2}{4\pi\epsilon_0 r} + \left(\frac{A}{r^{12}} - \frac{C}{r^{10}} \right) \quad (4.6)$$

Most organic molecules are uncharged, so MM3 ignores electrostatic interactions between the partial atomic charges. Instead, MM3 assigns bond dipoles to polar bonds and calculates the interactions between them, Equation 4.7. This interaction decreases with the third power of the distance between them so pairwise interactions beyond 18 Å are ignored. In cases where an organic molecule is charged, MM3 adds a Coulombic interaction term for the charged atoms like the first term in Equation 4.6 above.

$$v(r)_{\text{electrostatics}} = \frac{\mu_1 \mu_2}{\epsilon_0 r^3} (\cos \chi - \cos \alpha_1 \cdot \cos \alpha_2) + \text{if molecule is charged } \frac{q_1 q_2}{4\pi\epsilon_0 r} \quad (4.7)$$

Rosetta also uses Coulomb's law with partial charges to model pairwise electrostatic interactions, but adds modifications to improve accuracy.^[7] Rosetta modifies the dielectric constant to vary between the protein core (a low dielectric of 6) and the solvent-exposed surface (a high dielectric of 80). Rosetta also modifies the electrostatic interactions at short and long distances and when the atoms are within four bonds of each other. These empirical modifications improve its ability to match known protein structures.

4.2.2 Optimizing the geometry of a structure

The calculated energy of a structure is a score, and the geometry optimization is the strategy to get the best score, which correspond to a minimum energy structure. Predicting molecular structures using molecular mechanics involves cycles of energy calculations followed by adjustments of the structure (bond lengths, angles, etc.) and energy calculation of the new structure. This calculation finds structures that lie at the bottom of an energy well and is called a geometry optimization or an energy minimization. The convergence criterium is an energy difference between successive calculation, which determines when the cycles of calculation and structure adjustment should stop. When the energy difference drops below the convergence criterium, the calculation stops. A lower convergence criteria finds a structure closer to the center of the energy minimum and requires a longer calculation.

For example, a geometry optimization of *n*-butane yielded the structure in **Figure 4.9**. This calculated structure matches the experimental structure of the gauche conformation of butane. The constants in the molecular mechanics energy equation have been adjusted so that the energy minima matches the experimental structures.

Note that gauche conformation of butane is not the most stable conformation of *n*-butane. The geometry optimization yielded the gauche conformation because the initial structure (not shown) was closest to the gauche conformation. Geometry optimization yields a minimum energy structure that lies closest to the starting structure. The algorithm only moves down the energy slope to find the closest energy minimum. This energy minimum may not be the global minimum, which is the lowest energy conformation among all possible conformations. The gauche conformation is not the global minimum because it has a C-C-C-C torsion angle of

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$\sim -60^\circ$, which retains some bumping interaction between the two methyl groups.

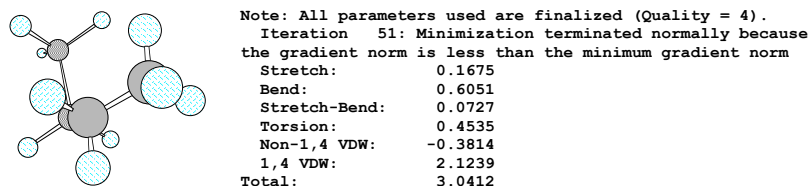


Figure 4.9. Geometry optimization of *n*-butane using the MM2 molecular mechanics force field in the software Chem3D. The calculation, called an geometry optimization or energy minimization, involved calculating the energy of the starting *n*-butane structure, adjusting the geometry slightly to lower energy and repeating until the decrease in energy was insignificant (51 iterations). The list shows the energy components of the final structure shown as a ball and stick model. This software refers to the convergence criterion as the ‘minimum gradient norm.’ The units for energy are kcal/mol.

The global minimum energy conformation of *n*-butane is the anti conformation, where the two methyl groups lie across from each other with a torsion angle of 180° , **Figure 4.10**. A second geometry optimization of *n*-butane starting from a structure similar to the anti conformation yielded the anti conformation, **Figure 4.11**. It has a lower energy showing that the first optimized structure was not the global minimum, but a local minimum. In both cases the geometry optimization yielded the *closest* energy minimum, not necessarily the lowest energy minimum.

The energy from a molecular mechanics calculation, called a steric energy, is the energy of the molecule relative to a hypothetical strain-less molecule where all bond lengths, angles, torsions, and

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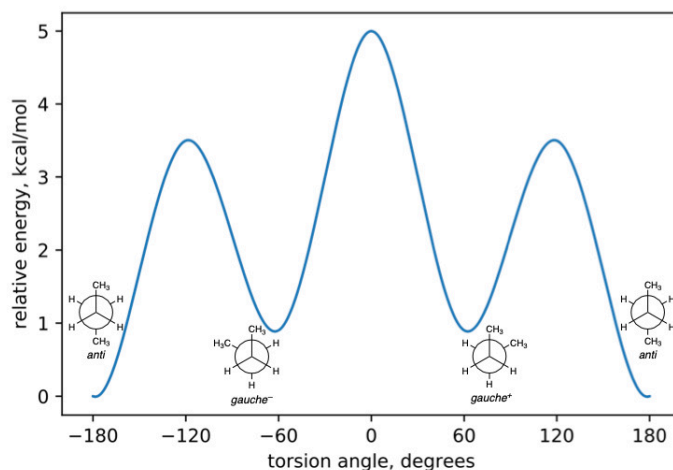


Figure 4.10. The relative energy of *n*-butane varies with the C-C-C torsion angle. The maxima correspond to eclipsed conformations along the central carbon-carbon bond, while the minima correspond to staggered conformations. The lowest energy minima is the anti conformation where the torsion angle between the two methyl groups is 180°. The two minima with energies of +0.9 kcal/mol relative to the anti conformation are enantiomeric gauche conformations where the torsion angle is $\pm 60^\circ$. The two gauche conformations differ by an either clockwise or counter clockwise orientation of the methyl groups.

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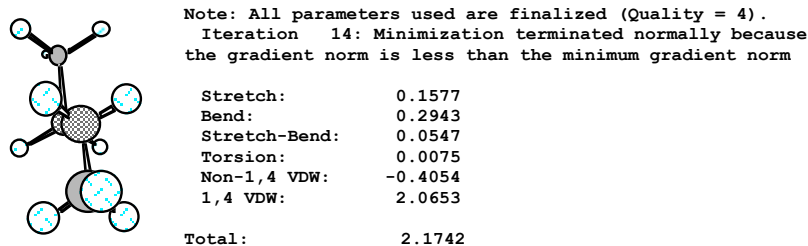


Figure 4.11. The geometry-optimized structure of the anti conformation of *n*-butane. The energy of the anti conformation is 0.87 kcal/mol lower than for the gauche conformation in **Figure 4.9** above.

non-bonded interactions are at their optimum values, **Figure 4.12**. The absolute values of steric energy have no physical significance, but differences in steric energy correspond to enthalpy differences. Thus, the steric energy value of 3.04 kcal/mol calculated for gauche butane above is not significant by itself, but the difference in steric energies between gauche and anti butane ($3.04 - 2.17 = 0.87$ kcal/mol) predicts the enthalpy difference between these two conformations. Differences in steric energy are only valid for different conformations or configurations of the same system. The steric energies of butane and pentane cannot be compared since they differ in molecular formula. One should not over interpret the individual energy contributions to the steric energy shown in **Figure 4.9** and **Figure 4.11**. The parameters are adjusted so the total energies and the structures match experimental values, but there are no experimental values for the individual energy contributions, so they should not be used for analysis.

Molecular properties depend on Gibbs energy differences, but the molecular mechanics calculations in **Figure 4.10** yield enthalpies, but not entropies, for anti and gauche butane. The torsion diagram also provides the information needed to estimate the entropy difference between the anti and gauche conformations of butane.

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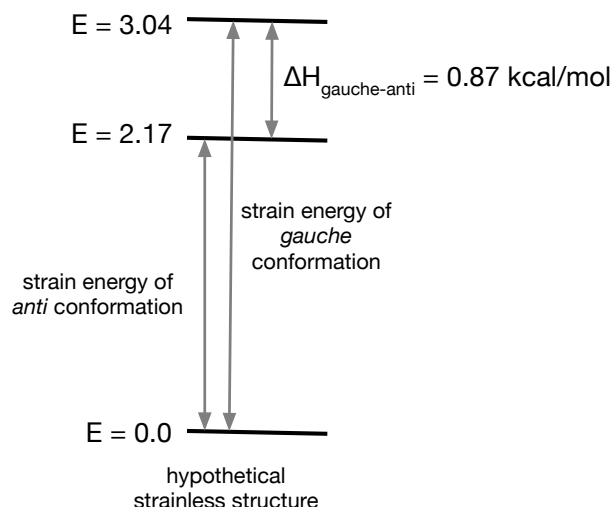


Figure 4.12. Using molecular mechanics to calculate enthalpy differences between conformations. Geometry optimization of the gauche conformation of butane yielded a strain energy of 3.04 kcal/mol. This strain energy compares the energy of this gauche conformation to a hypothetical strainless structure of butane. A similar geometry optimization yielded a lower strain energy for the anti-conformation of butane, 2.17 kcal/mol. While the absolute values of these strain energies have no physical significance, the difference between them, 0.87 kcal/mol, corresponds to the enthalpy difference between the two conformations. Rosetta uses a different approach where energy differences between conformations correspond to Gibbs energy differences.

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Butane can adopt one anti conformation, but two enantiomeric gauche conformations. Entropy favors the gauche conformation because there are two possibilities. Problem 1 at the end of the chapter asks the reader to estimate the Gibbs energy difference between anti and gauche butane and compare to the experimental equilibrium constant. The Gibbs energy estimate comes from the steric energy, which corresponds to enthalpy, and from an estimate of the entropy difference.

Rosetta differs from MM3 and AMBER in yielding Gibbs energies instead of enthalpies as the result of an energy calculation. The assumption of statistics-based potentials like Rosetta is that the data in the protein structure database follows a Boltzmann distribution. The Boltzmann distribution is a probability distribution developed for gases that varies exponentially with Gibbs energy of the state. States with lower energy are more likely to be occupied. If the database contains a random distribution of structures and the interactions within them act independently, then one expects that the more stable interactions will occur more frequently than less favorable interactions. The statistical frequency of an interaction in a protein structure database reveals its Gibbs energy relative to other, similar interactions. Interactions that occur frequently in the database are assigned a low energy, while interactions that occur rarely are assigned a high energy. The probability of finding a specific conformation, p_i , within an equilibrium mixture of conformations decreases exponentially as the Gibbs energy of the conformation, G_i , increases. k is Boltzmann's constant and T is the temperature, Equation 4.8.

$$p_i \propto e^{-G_i/kT} \quad (4.8)$$

The RosettaDesign tool is useful for protein engineering and a web interface to this software is available at <http://rosettadesign.med.unc.edu>.

4.2.3 Searching for conformations

The most important reason to search for conformations (microstates) is to create a model of the protein state, which is needed to predict protein properties. Finding all conformation is impossible; the goal is to find a representative sample of conformations for that state. If one seeks to estimate protein stability, then finding all the low-energy conformations is important. If one seeks to estimate reactivity, then one needs to find the conformations that contribute to reactivity, even if they are high energy conformations. Other reasons to search for conformation is to find the global energy minimum calculation or to estimate entropy when using AMBER for modeling.^[8] Modeling with Rosetta does not require a separate estimate for entropy.

Three approaches to conformational search are systematic search, random search, and molecular dynamics. The systematic search tests all possible orientations along rotatable bonds. A systematic search identifies all of the minimum conformations, including the global minimum. For butane, a systematic adjustment of the C-C-C-C torsion angle by 30° followed by geometry optimization of each of the six structures to the closest local minimum identified the three minima including the global minimum, **Figure 4.10**.

The number of possible conformations in a systematic search increases exponentially as the number of rotatable bonds increases. In general, the number of possibilities is given by the number of choices at each location raised to the power of the number of locations, Equation 4.9.

$$\text{possibilities} = \text{choices at each location}^{\text{locations}} \quad (4.9)$$

For the number of possible conformations:

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$$\text{possible conformations} = \text{positions along each bond}^{(\text{bonds})} \quad (4.10)$$

Even with a small molecule like a five-amino-acid peptide, a systematic search for the global minimum is difficult. If an amino acid has an average of five rotatable bonds (two in the main chain, three in the side chain), then a five-amino-acid peptide has 25 rotatable bonds. If there are six conformational possibilities at each bond, then the number of possible conformations is $(6)^{25}$ or $\sim 10^{19}$. This large number of possible conformations limits the application of systematic searches to small molecules.

The random search approach, called Monte Carlo search to suggest gambling, generates random structures by a torsion angle rotation followed by geometry optimization to yield a local energy minimum conformation. The Monte Carlo search prioritizes low energy structures because these will be most abundant. If after a random torsion angle rotation and geometry optimization, the energy of the structure is lower than the previous structure, then the current structure and the starting point for another random torsion angle rotation. If the current structure is not lower, then the previous structure is used again as a starting point. Occasionally, higher energy structures are accepted to avoid getting stuck in a local minimum. The search stops when structures with lower energy minima can no longer be found. The Monte Carlo search is more efficient than the systematic search because it focuses on the low energy structures instead of all the structures.

For compact structures like proteins, random search methods like Monte Carlo are inefficient. Most random changes of a torsion angle in a protein create unfavorable bumping interactions and are not realistic possibilities. To limit the search to more plausible possibilities, researchers limit the changes to side chain orientations. This approach generates plausible structures, but makes it difficult

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to escape local minima because alternative conformations of proteins often require cooperative adjustments in several amino acid side chains combined with main chain adjustments.

Molecular dynamics is the most common conformational search method for proteins. Molecular dynamics models molecular motions by allowing the atoms to move according to the force field. Newton's equation of motion, $a = F/m$, describes how a force, F , on an object with mass, m , accelerates the object in certain direction and by a certain amount, a . The force field describes the force on each atom due to distortions of the bonding interactions from their ideal values and due to non-bonded interactions with other atoms. The step size for the calculation of atom movements must be shorter than molecular vibrations (femtoseconds) to remain realistic. Calculating one nanosecond of protein movement may take several hours of computer time. Combining these steps creates a movie of molecular motion.

A typical molecular dynamic calculation for protein is a simulation. It seeks to model the folded protein state to show the conformations that a protein explores in solution. Its goal is not find all conformations, but to find a representative sample of conformations that the protein explores. Molecular dynamics simulations do not have an end point where the simulation is finished. Instead the researchers continue the simulation for an arbitrary time. For example, the simulation of a thermostable adenylate kinase below modeled 1 nanosecond of protein movement, but required hours of computation time. This simulation modeled the folded protein state in solution to identify which of the four ion pairs on its surface contributed to its stability.^[9] Close contacts in the x-ray structure suggested that all four ion pairs contribute to stability, **Table 4.2**. The authors reasoned that these distances could be misleading. First, the packing of the protein within the crystal creates additional electrostatic interactions that are not present in solution. Second, ion pairs on the surface are often flexible

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due to side-chain motion leading to uncertain positions. The molecular dynamics simulation of this protein modeled how the protein moves in solution. One ion pair had the shortest average distance because close contact persisted throughout the simulation. This behavior identified it as a stabilizing interaction. Two ions pairs had longer average distances because they maintained close contact through only parts of the simulation. The authors classified these two pairs as slightly stabilizing. The last ion pair had the longest distance because the pair stayed apart throughout the simulation. The authors classified this pair as destabilizing. Transferring these ion pairs into a less stable homolog confirmed these assignments. This molecular dynamics simulation revealed how the folded protein behaved in solution, which was different from that suggested by the single structure shown by x-ray.

Table 4.2. Molecular dynamic simulation of ion pairs on the surface of adenylate kinase identified the Arg116-Glu198 pair as the strongest.

Amino acid residues	Close N-O contacts in x-ray structure	Average N-O distance in MD simulation	Close contact maintained during simulation
Lys19-Glu202	2.8 Å, 3.2 Å	3.2 Å	partly
Arg116-Glu198	3.0 Å	2.7 Å (strongest)	yes
Arg131-Glu156	3.0 Å, 3.2 Å	3.8 Å	partly
Lys180-Asp114	4.0 Å, 4.8 Å	10.9 Å (weakest)	no

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Typical molecular dynamics simulations model 1-10 nanoseconds of protein movement; in unusual cases simulations extend to one microsecond. These simulation times are short as compared to the times required a protein state. A protein state consists of all the conformations (microstates) that contribute to a protein property. For example, the turnover time for an enzyme catalyzed reaction may be a millisecond, so simulations shorter than a millisecond may omit conformations that occur rarely, but nevertheless contribute to catalysis. An example of a puzzle solved by long-timescale molecular dynamics simulations is the effect of mutations on the substrate scope and catalytic activity of monoamine oxidase.^[10] The x-ray structure showed a closed conformation which could not explain the effects of mutations. The molecular dynamics simulations identified partially open and fully open conformations that could account for the changes in catalysis.

Since Gibbs energy consists of enthalpy and entropy, there are two ways that conformations can be rare or unfavorable. The first is high enthalpies or internal energies. Conformations with distorted bond torsions, angles, or lengths or with unfavorable bumping or electrostatic interactions have high enthalpies. The second type of unfavorable conformation are those that require several things to occur simultaneously. For example, moving a buried side chain requires not only rotating along one of the single bonds in the side chain, but simultaneously rotating along single bonds in adjacent side chains to create space for buried side chain to move. The buried side chain only has an opportunity to move when the adjacent side chain has moved and therefore occurs more rarely than conformations that don't require simultaneous movements. This requirement for simultaneous movements is an entropy cost since it imposes a requirement for order - simultaneous movement instead of movement anytime.

Some computational approaches to find rare conformations during

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a molecular dynamics simulation are 1) to extend the simulation for a longer time, 2) to repeat short simulations from several different starting conformations, and 3) to temporarily raise the temperature unrealistically high (500 °C) to speed up slower conformational changes, then return the modeling to normal temperatures. While molecular dynamics is the best way to search for protein conformations, researchers recognize that the search may miss rarely-occurring conformations.

Molecular dynamics simulations typically include hundreds of explicit water molecules to more accurately model the solvation of the protein. The models used for the water molecule differ in their complexity. A typical model is called TIP3P (transferable intermolecular potential with 3 points) and represents water with three fixed sites corresponding to the three atoms. Each site has a charge, and the oxygen site also has van der Waals interactions. Explicit solvation models can find specific interactions between water and the protein, but adding hundreds of molecules slows down the simulation. The alternative to explicit solvation is implicit solvation, which models the solvent and counter ions as a continuous medium with a particular dielectric constant. This approach accounts for the average behavior of the solvent but does not find specific interactions with solvent molecules.

4.2.4 Predicting protein properties

The observable properties of a molecule depend not only on the global minimum conformation, but on the relative abundance of all the conformations (microstates) that contribute to the state. For example, accurate prediction of the density of liquid butane, which depends on its molecular volume, must include both the anti and gauche conformations.^[11] The state of liquid butane is an unequal

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mixture of these two conformations and both contribute to the measured property of density.

The properties of a protein are determined not by a single structure, but by states, which are collections of interconverting conformations. Computer modeling must find these conformations, which is more difficult than calculation the energy of individual structures. There are too many conformations of a protein to find all of them, so modeling uses simulations of protein movements to locate a representative sample of these conformations. If this sample of conformations is not representative of the state, then the prediction will be incorrect. For example, the design of a retroaldolase failed when protein adopted not the calculated structure, but a subtly different conformation that was not considered in the calculations.^[12] Finding all the possibilities to consider is still an unsolved and active area of research.^[13]

Because of the difficulties in accurately modeling protein states, researchers rely on approximations to model protein properties, **Table 4.3**. For example, one often assumes that a single x-ray structure is representative of the conformations available to a folded protein. In other example, the unfolded protein's structure is unknown, so researchers assume it is an unfolded random coil, but when this assumption fails, the predictions are incorrect.^[14] These approximations reduce the accuracy of computational predictions making protein engineering challenging. Most computer modeling predictions of substitutions to improve proteins are incorrect, but the fraction of correct predictions, typically 1-10%, is vastly better than random guessing, so computer modeling is a useful tool. Computer modeling narrows the choice of substitutions from astronomical numbers to smaller numbers that can be tested experimentally.

Table 4.3. Computer modeling approaches to common protein engineering problems.

Goal	Approach	Comments
What is the structure of a variant protein?	homology modeling (extrapolation from known structures) SwissModel, AlphaFold	generally reliable single structure
Which conformations contribute to a protein state?	molecular dynamics simulations	challenging to find rare conformations
What is the structure of the enzyme-substrate complex or the protein-target complex?	docking calculations identify binding sites followed by molecular dynamics	challenging to find adjustments upon binding
What is the structure of the transition state?	quantum mechanical modeling to define charges and shape of an energy maximum with partially broken and formed bonds	challenging for multi-step reactions
Which protein variant is more stable?	compare energies calculated by FoldX or Rosetta	low reliability, unfolded protein state is unknown, use single structure for folded protein

Goal	Approach	Comments
How to increase the binding of my protein to its target?	modeling to predict changes that increase challenging, but several successful designs favorable interactions between protein and using Rosetta have been reported target and remove unfavorable interactions	
How to engineer a faster or more selective enzyme?	identify and compare the currently impossible or protein conformations that difficult to predict contribute to each property	

4.3 Conclusions

Examination of protein structures reveal many details about proteins and are the critical first step in protein engineering. However, protein structures alone are not enough to accurately predict protein properties like stability, binding, reactivity and selectivity. These properties depend on differences between two protein states, whose structure and energy may be unknown. One rarely has information about the unfolded protein state, the binding orientation of a target molecule, or the rate-limiting step of a reaction. In addition, protein states consist of numerous conformations, some of which may be critical for the protein property, but difficult to find. Researchers often assume that most features of protein states will cancel out and focus on a few differences between them. The accuracy of the predictions varies with the validity of these assumptions.

Computer models of proteins only approximate reality and in that sense they are all incorrect. Models differ in complexity because they make different assumptions. A good computer model will capture enough of the essential features so that it predicts the behavior in question. The most common assumption is that a single structure is representative of the protein state. A good approach to computer modeling is to first test the calculation on cases where the answer is known. If the predictions from this control calculation match experiments, then the predictions for cases where the result is unknown have a better chance of matching as well.

Glossary

Atom types are the molecular subunits used in molecular mechanics calculations. An atom type includes both an element and a bonding arrangement such as an sp^2 -hybridized carbon in

a polar double bond. Some atom types, called united atom types, include several atoms such as a CH₂ group.

Deep neural network is a multi-layered machine learning model trained on large datasets of protein sequences, structures, and experimental data to learn complex, hierarchical patterns in protein sequence-structure-function relationships for predicting and designing protein properties. These networks excel at tasks like protein fold prediction due to their hierarchical architecture matching protein organization, but often struggle with more precise protein engineering predictions that require understanding subtle sequence-function relationships.

Force field is a mathematical description of how atoms interact with one another. A force field consists of a set of equations that describe how bonded and non-bonded atoms interact, a set of atom types that represent elements in different bonding arrangements and a set of parameters that scale the magnitude of the interactions to fit experimental values. Physics-based force fields, such as MM3 and AMBER, attempt to model the physical interactions between atoms, while knowledge-based force fields, such as Rosetta, include statistical potential that scores geometries that occur frequently in known protein structures as more stable.

Molecular dynamics is a computational method to simulate the motion of interacting atoms. The computation uses Newton's equations of motion with a force field that describes how atoms interact. Molecular dynamics simulations generate a collection of conformations, which can be played as a movie, or, more commonly, analyzed to extract average geometry relevant to a protein property like catalysis.

Molecular mechanics is a computational modeling approach to predict the structure, energetics, and dynamics of molecular. Molecular mechanics treats atoms as balls that interact with other atoms via classical physics interactions like balls on a

spring. Although this model of molecules is incorrect because it does not include electrons explicitly in the model, molecular mechanics yields accurate values because the equations and parameters have been adjusted to fit experimental values.

Quantum mechanics is a computational modeling approach that include electrons explicitly in the calculations. Quantum mechanics are needed to model bond-making and bond-breaking steps in a reaction. Quantum mechanics calculations are much more complex than molecular mechanics, so quantum mechanics is limited to 10-50 atoms in a calculation.

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Problems

1. *Gibbs energy and entropy estimate.* The goal of this question to compare an experimental equilibrium constant to one calculated using molecular mechanics. First, calculate the experimental equilibrium constant between the anti and gauche conformations for the conversion of gauche to anti. At equilibrium in the gas phase experimental measurements show that *n*-butane consists of a mixture of 70% anti and 30% gauche conformations.^[15] Assume this refers to 300 °K. Next, calculate the equilibrium constant from the molecular mechanics calculations. The enthalpy comes from the molecular mechanics calculation (use data from **Figure 4.12**). To estimate the entropy difference between the anti and gauche conformations recall that there is only one anti conformation, but two enantiomeric

gauche conformations. Combine the enthalpy and entropy values to calculate the Gibbs energy difference and convert this energy into an equilibrium constant. Compare the experimental and calculated values and comment on any differences.

2. *Protein Data Bank*. The PDB is a repository for the 3D structural data of proteins and nucleic acids. The data, typically obtained by X-ray crystallography or NMR spectroscopy and submitted by biologists and biochemists from around the world, are freely accessible on the internet.

- a) *Searching for a structure file*. Go to the PDB website at <http://www.rcsb.org>. Each structure has a unique 4-digit pdb id. You can search for structures by name, by pdb id, by author, and by other criteria. The structure we are looking for is subtilisin from *Bacillus amyloliquefaciens* or subtilisin BPN'. How many structures of subtilisin BPN' are contained in the Protein Data Bank? Searching for 'subtilisin BPN'' in the search box at the top yields >120 structures, but specifying that 'subtilisin BPN'' must be in the UniProt molecule name (in the drop-down menu or the Advanced Search panel) returns ~60 structures. For example, the first search results include 3SSI, but the second does not. Is file 1THM a structure of subtilisin BPN'? Explain your answer.
- b) *Viewing the text of the structure file*. Find the entry with pdb id 1S01. Download the file (in PDB Format) to your computer. The file should be named 1s01.pdb; if it is named 1s01.cif.gz, look for a different download link where a menu pops up to allow you to choose PDB Format. Open the file with a text editor or a word processing program. How many protein atoms does the structure contain? How many water molecules (HOH)? What are the x, y, z coordinates for the C α atom of Ser221? Note that the structure data does not include hydrogens.

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3. *Introduction to PyMOL*. Download and install the most recent version (3.1) of PyMOL to your computer from <http://pymol.org/ep>

- a) *Viewing the structure file with PyMOL*. Open the file 1S01.pdb with PyMOL. (Alternatively, you can directly download it from within PyMOL using the sequence: File → Get PDB... → enter 1S01 and click Download.) Use the trackpad or mouse to rotate and zoom. Try the different combinations of keyboard and mouse. Click on an atom, note that PyMOL tells you which atom it is. The protein is shown as a ribbon representation, but there are also water molecules, two isopropyl alcohol solvent molecules, and one calcium ion. Select Display → Sequence to show the amino acid sequence. Click on the calcium ion (large sphere). A small red dot indicates that it is selected. Selection also created an object called (sele); see gray rectangle in panel to the right of the structure. What is its residue number? Note that it is also highlighted in the sequence.
- b) *Viewing the active site residues and saving your work*. Type the commands below, including spaces, but not the comment lines starting with '#'.

```
# create a new object named 'triad' containing
# residues numbered 32, 64 and 221
> select triad, resi 32 + resi 64 + resi 221
# show object triad in sticks representation
> show sticks, triad
# (or click on the 'S(how)' near triad (make sure
# triad is selected and select sticks)
# zoom in on the triad with:
> zoom triad
# hide water molecules
> hide everything, resn hoh
```


or click on the H(ide) near 1S01 and choose waters

4. *Save your work two different ways.* Saving your session allows you to return to this point with all changes that you have made. Saving an image creates a static picture for use in a presentation or report. It loses all molecule information. Since it is an image file, you can adjust it (e.g., add a text label) with an image editing program. The difference between the commands is the extension chosen for the file name: pse versus png. PyMOL matches the information saved to the extension. The files are saved in your current directory. You may not know what your current directory is, so you may have to hunt for them. You can set your working directory under with File → Working Directory... →. Alternatively, you can add a path to the file name like: ~/Desktop/file_name.pse.

```
> save file_name.pse
# (or File -> Save Session, then type file_name.pse)
> save file_name.png
# (or File -> Export Image As -> PNG... ->
# Save PNG image as ... -> file_name.png)
```

5. *Measuring distances and specifying histidine tautomers in PyMOL.* Open your 1s01.pse file that you saved in the previous question. It should show the protein as a cartoon, hidden water molecules, and the catalytic triad as sticks.

- a) *Tautomers of histidine.* Neutral histidine can exist as one of two tautomers, sometimes called HID (hydrogen on N δ 2) and HIE (hydrogen on N ϵ 1). **Figure 4.13** shows catalytic triads showing these two tautomers. Draw potential hydrogen bonding patterns in both and explain why only one of these tautomers creates a catalytic configuration. (Review catalytic triads of serine proteases, if needed.)

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- b) *Identifying hydrogen bonds.* PyMOL can display hydrogen bonds. Select the triad (select triad) A(ction) → find → polar contacts → within selection. Click on the black space to deselect triad. The geometric criteria for the existence of a hydrogen bond between N and O are 1) an N-O distance or 2.9 Å and 2) a nearly linear N-H-O angle ($\geq 120^\circ$). Measure the distance and, if possible, the angle (you will need to add hydrogens: A(ction) → hydrogens → add polar). To measure the angle select Wizard → Measurement. Change the default measurement from “Distances” to “Angles” by clicking on “Distances” and choosing “Angles” from the pop-up menu. Then click on the three atoms that define the angle.
- c) *Structure shows the incorrect tautomer of histidine.* PyMOL does not identify a hydrogen bond between OD1 or OD2 of Asp32 and ND of His 64. Why not? Are the distances between O and N appropriate for forming a hydrogen bond? Does the 1s01 structure contain the catalytic or the non-catalytic tautomers of histidine? Do you think the structures of the two tautomers differ enough that an x-ray experiment could distinguish the two?
- d) *Specifying the correct tautomer of histidine.* To correct this error, you can specify which tautomer of histidine that PyMOL draws. The choices are: HIS = nothing specified, HID = neutral tautomer with hydrogen on ND, HIE = neutral tautomer with hydrogen on NE, HIP = hydrogens on both nitrogens; positively charged. Open pdb file 1S01.pdb and replace the names of all of the atoms in histidine 64 from ‘HIS’ to ‘HID’ as shown in Figs. **Figure 4.14** and **Figure 4.15**. Save the file and reopen it with PyMOL. Remeasure the hydrogen bond distances between the serine, histidine, and aspartate and show that the structure now shows a catalytically productive hydrogen bond arrangement. However, one detail is still incorrect. The angle for the hydrogen bond between His

NE2 and SerOG is 83° , which suggests a poor hydrogen bond. Suggest a reason why this conclusion is likely incorrect.

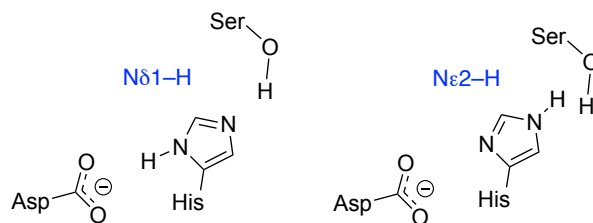


Figure 4.13. Only one tautomer of histidine creates a viable catalytic triad. One tautomer has the hydrogen on N δ 1, while the other has the hydrogen on N ϵ 2.

ATOM	455	N	HIS	A	64	-1.852	38.475	29.575	1.00	6.26	N
ATOM	456	CA	HIS	A	64	-0.431	38.903	29.664	1.00	5.63	C
ATOM	457	C	HIS	A	64	-0.018	39.823	28.532	1.00	4.99	C
ATOM	458	O	HIS	A	64	0.508	40.917	28.774	1.00	4.27	O
ATOM	459	CB	HIS	A	64	0.462	37.666	29.791	1.00	7.00	C
ATOM	460	CG	HIS	A	64	1.882	37.897	30.144	1.00	5.95	C
ATOM	461	ND1	HIS	A	64	2.883	38.035	29.195	1.00	6.43	N
ATOM	462	CD2	HIS	A	64	2.491	37.967	31.341	1.00	6.26	C
ATOM	463	CE1	HIS	A	64	4.038	38.217	29.803	1.00	6.84	C
ATOM	464	NE2	HIS	A	64	3.834	38.154	31.126	1.00	6.08	N

Figure 4.14. Old listing of coordinates for His64 in pdb file 1s01 does not specify which tautomer should be drawn.

6. *Site-directed mutagenesis of subtilisin BPN'*. The Met222Ala substitution in subtilisin BPN' stabilized it to inactivation by bleach.

- Select Met222 and show it as sticks using a different color for the carbon atoms than the one used for the catalytic triad.
- To switch from viewing your protein to editing it, click on '3-button viewing' in the Mouse control legend on the bottom right. This click will toggle the mouse mode to '3-button editing' so that you can modify your protein. Mutate Met222 to alanine (Wizard → Mutagenesis → Protein →) Then follow the

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ATOM	455	N	HID	A	64	-1.852	38.475	29.575	1.00	6.26	N
ATOM	456	CA	HID	A	64	-0.431	38.903	29.664	1.00	5.63	C
ATOM	457	C	HID	A	64	-0.018	39.823	28.532	1.00	4.99	C
ATOM	458	O	HID	A	64	0.508	40.917	28.774	1.00	4.27	O
ATOM	459	CB	HID	A	64	0.462	37.666	29.791	1.00	7.00	C
ATOM	460	CG	HID	A	64	1.882	37.897	30.144	1.00	5.95	C
ATOM	461	ND1	HID	A	64	2.883	38.035	29.195	1.00	6.43	N
ATOM	462	CD2	HID	A	64	2.491	37.967	31.341	1.00	6.26	C
ATOM	463	CE1	HID	A	64	4.038	38.217	29.803	1.00	6.84	C
ATOM	464	NE2	HID	A	64	3.834	38.154	31.126	1.00	6.08	N

Figure 4.15. New listing of coordinates for His64 in pdb file 1s01 specifies the neutral tautomer with hydrogen on ND1.

prompts to select Met222 by clicking on it and choose a new amino acid by clicking on the light purple box labelled 'No Mutation'.

Answers

1. experimental = 70% anti in gas phase; $K_{eq} = 2.3$. A value greater than 1 indicates that anti is favored for the gauche $\rightleftharpoons$ anti equilibrium.

Now we will predict the same equilibrium constant using the molecular mechanics calculation of ΔH and an estimate of ΔS from microstates

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -900 \text{ cal/mol} - 300 \text{ K}(1.987 \text{ cal/mol} \cdot \text{K} \cdot \ln(1/2))$$

$$\Delta G = -900 + 413 = -500 \text{ cal/mol}$$

$$\Delta G = -RT \ln K_{eq}$$

$$-500/1.987 \times 300 = \ln K_{eq}$$

$$-0.83 = \ln K_{eq}; K_{eq} = 2.3!$$

Answers

2. a) 1THM is not a subtilisin, but a subtilisin-type serine proteinase. A search with the keyword 'subtilisin' anywhere in the record will return this protein, but not if the search is limited to the molecule name. b) 1938 protein atoms; note that 1939 is not an atom is has no coordinates; 215 water molecules; ATOM 1529 CA SER A 221 x=7.506 y=40.969 z=32.045

3. a) The calcium ion is residue 301. b) You should have a zoomed-in view of the catalytic triad.

5. a) Catalysis requires the histidine to act as a base and deprotonate the serine O γ in the first step; this requires the N δ 1 tautomer. b) 3.0 Å between H64 NE2 and S221 OG; 134° for the hydrogen attached to H64 NE2, which is a hydrogen bond. The hydrogen on S221 OG does not form a hydrogen bond because the angle is too acute, 83°. This H-O-C-C rotamer is likely not the orientation during catalysis. c) no, PyMOL has not added a hydrogen to N64 ND1, so no hydrogen bond is possible. The distance between H64 ND1 and D32 OD1 is 2.6 Å, which is short enough for a hydrogen bond; similarly the distance between H64 ND1 and D32 OD2 is 3.2 Å, which is also short enough. d) non catalytic; PyMOL has drawn the incorrect tautomer of H64; the pdb file does not specify tautomers because tautomers differ in the location of hydrogens, but the pdb file does not contain hydrogens.

5 Engineering More Stable Proteins

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Summary. Protein function requires the folded protein form, but this form is unstable mainly because it readily unfolds into a flexible, unstructured form. Protein folding is favored by burying of hydrophobic side chains and hydrogen bonding between the amino acids. Protein unfolding is favored by the increase in flexibility of the main chain of amino acids upon unfolding. Protein stability is usually measured by the reversible unfolding of the protein with either heat or chemical additives like urea. Protein stabilization involves making substitutions that shift the folding-unfolding balance toward the folded form. Stabilizing substitutions can either stabilize the folded conformation or destabilize the unfolded ensemble. This tutorial emphasizes web-based tools to identify substitutions that stabilize proteins. Besides unfolding, other sources of protein instability are chemical modifications like oxidations or cleavage by proteases and aggregation of partly unfolded proteins into insoluble particles.

Key learning goals

- Proteins are dynamic, equilibrating collections of structures. Most of the structures are folded, but a small fraction are unfolded. This protein unfolding is the primary source protein instability. Stabilization shifts the folding-unfolding balance toward folding, but does not prevent unfolding.

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- To measure protein stability, one measures the ratio of folded to unfolded protein. Under normal conditions, the amount of unfolded protein is too small to detect. Adding denaturants such as urea or heating the sample increases the amount of unfolded protein to measurable amounts. Extrapolation back to normal conditions reveals the stability of the protein.
- One way to stabilize proteins is to restore amino acid residues that are conserved in homologs. The rationale for this approach is evolution conserves residues that provide some benefit; one benefit is a contribution to stability.
- Another way to stabilize proteins is to destabilize the unfolded form. The main driving force for unfolding is the increased flexibility of the unfolded form. Decreasing the flexibility of unfolded form by adding disulfide crosslinks or introducing proline residues often stabilizes proteins.
- The most direct way to stabilize proteins is to create or strengthen attractive interactions between amino acids in the folded conformation. Although proteins will remain dynamic and still unfold, the stronger interactions either slow down unfolding or speed up refolding.

5.1 Introduction

Protein stability usually refers to resistance to unfolding. Stresses like high temperatures, organic cosolvents, high substrate or product concentrations, extremes of pH or high ionic strength can all cause unfolding. In many cases, stabilizing a protein to one stress also stabilizes it to other stresses. For example, proteins that tolerate high temperatures often also tolerate organic cosolvents.^[1]

One advantage of more stable proteins is the ability to use them as biocatalysts in artificial environments where naturally-occurring

5.1 Introduction

proteins are unstable. For example, using biocatalysis at higher than physiological temperatures yields faster reaction rates, higher substrate solubility, and lower solution viscosities. Reactions at high temperature can also avoid microbial contamination since most contaminating microbes cannot grow at high temperature. Biocatalysts that tolerate organic cosolvents and high concentrations of substrates and products simplify the scale-up of industrial processes by requiring less solvent, smaller equipment, and less complicated product isolation.

The second advantage of more stable proteins is extended lifetime or storage at ordinary temperatures. For example, manufacture of β -lactam antibiotics with penicillin G acylase recycles the immobilized enzyme many times to lower the overall cost. More stable therapeutic proteins last longer during storage and are more resistant to proteases in serum.

A third advantage of more stable proteins is higher yields of soluble protein during manufacture, especially for small, single-domain proteins. Over-expression of recombinant proteins in microbes creates high concentrations of protein. The unfolded forms can aggregate into insoluble particles called inclusion bodies. More stable single-domain proteins aggregate less and yield higher amounts of soluble protein.^[2]

A fourth advantage is that more stable proteins are also more 'evolvable' or able to acquire beneficial traits. Substitutions that change protein function may destabilize proteins. If the destabilized protein fails to fold, then the potential improvement is lost. More stable proteins can tolerate greater numbers of destabilizing mutations and are thus more evolvable than their marginally stable variants.^[3] One source of stable proteins is thermophiles and other extremophiles.^[4] For example, the Pfu DNA polymerase used for the polymerase chain reaction (PCR) comes from the hyperthermophile *Pyrococcus furiosus*, Figure **Figure 5.1**. This

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polymerase catalyzes DNA synthesis at 72 °C and tolerates the high temperatures (95 °C) needed to dissociate complementary DNA strands.



Figure 5.1. The archaeon *Pyrococcus furiosus* is a hyperthermophile that grows best at 100 °C. The heat-stable Pfu DNA polymerase used in PCR comes from this microbe. This painting simulates a scanning electron micrograph. Image by Fulvio314 from https://en.wikipedia.org/wiki/Pyrococcus_furiosus (CC BY-SA 3.0).

One disadvantage of enzymes from thermophiles is the typically low catalytic activity of these enzymes at room temperatures.^[5] Since thermophiles do not live at room temperature, there is no selective pressure for high activity at room temperature. If the application requires activity at room temperature, enzymes from thermophiles may not be suitable. Pfu DNA polymerase has negligible activity at room temperature. Other limitation of enzymes from thermophiles is the lack of other needed protein characteristics. For example, applications may require unusual substrate specificity unavailable in thermophiles or human proteins to minimize immune response.

Many human disease-causing mutations are associated with amino acid substitutions that decrease protein stability.^[6] For example, mutations that decrease the stability of fructose biphosphate al-

5.2 Native and denatured conformations equilibrate

dolase cause hereditary fructose intolerance and mutations of the tumor suppressor protein p53 cause some cancers.^[7] The ability to predict substitutions that alter protein stability may help identify mutations that cause genetic diseases and treat them with genetic editing technologies. For example, sickle cell disease, caused by the lower stability of the Glu6Val variant of β -globin, has been treated by expressing the unmutated fetal β -globin gene.^[8]

5.2 Native and denatured conformations equilibrate

The native protein state, N, is the folded, functional form. It is compact with the hydrophobic side chains mainly buried and the polar side chains mainly exposed to solvent. It has a specific structure (or similar set of structures), typically with α -helices, β -sheets, and turns folded in a particular arrangement, Figure **Figure 5.2**. While the structure is dynamic and moves, it has an overall stable structure. Most of the amino acids interact with each other; only amino acids on the protein surface interact with solvent water. In contrast, the denatured protein state, D, is not functional and is not a single state, but a collection or ensemble of more or less folded states. The main chain makes large, random motions and the amino acids interact mainly with solvent water, not with each other.

The native form of a typical protein is more stable than the denatured state by ~ 10 kcal/mol, Figure **Figure 5.3**. A Gibbs energy of unfolding, $\Delta G^{un\,fold}$, of +10 kcal/mol corresponds to an equilibrium constant $e^{-(10,000/RT)}$ or 4.6×10^{-8} at room temperature indicating that unfolding is rare, eq. Equation 5.1. Folding and unfolding are fast for single domain proteins. Native forms continuously unfold to denatured states and these continuously refold to native forms.

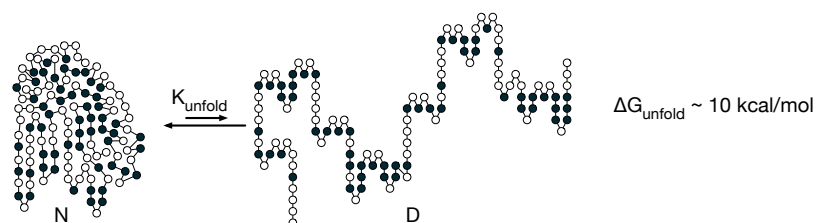


Figure 5.2. Schematic of protein unfolding showing the equilibrium between a compact folded native structure (N) and an ensemble of flexible unfolded states (D). Folding creates a specific structure that mostly buries hydrophobic amino acids (filled circles) in the interior of the folded structure while mostly exposing hydrophilic amino acids (open circles). Unfolding exposes more the amino acids to solvent. For clarity, the figure shows only one example unfolded structure, but in reality it is an ensemble of rapidly interchanging unfolded structures.

5.2 Native and denatured conformations equilibrate

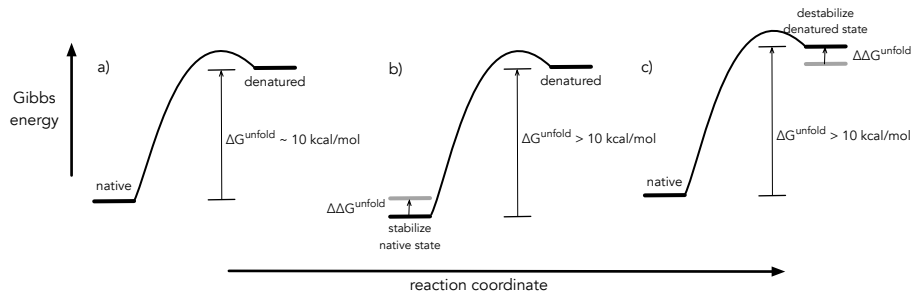


Figure 5.3. The Gibbs energy difference between the folded, native state of a protein and the denatured ensemble of states determines protein stability. The barrier to unfolding for single domain proteins is low. a) The denatured state for the original protein lies $\sim 10 \text{ kcal/mol}$ above the native state. b) Stabilization of the native form stabilizes the protein because it increases the energy difference between the native and denatured forms. c) Destabilization of the denatured form also stabilizes the protein because it increases the energy difference between the native and denatured forms.

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$$\Delta G^{unfold} = -RT \ln(K^{unfold}) \text{ or } K^{unfold} = e^{-\Delta G^{unfold}/RT} \quad (5.1)$$

Protein solutions contain unfolded protein molecules even for stably folded proteins. In the case where the native form is ~ 10 kcal/mol more stable than the unfolded form, one in twenty-two million protein molecules unfolds at any given time in solution at room temperature, eq. Equation 5.2. This tiny fraction is too small to detect by spectroscopic methods, but it is nevertheless a large number of molecules. A solution containing 1 mg of a 30 kDa protein contains 4×10^{16} molecules; of these, nearly a billion (10^9) are in the denatured form at any time.

$$K^{unfold} = \frac{[D]}{[N]} = 4.6 \times 10^{-8} \text{ or } [N] = 22 \times 10^6 \cdot [D] \quad (5.2)$$

The main driving force for protein folding is the hydrophobic effect, which is the tendency of non-polar solutes to cluster in water. Burying a $-\text{CH}_2$ group contributes 1.1 ± 0.5 kcal/mol to protein stability. The hydrophobic effect provides $\sim 60\%$ of the driving force to collapse the amino acid chain into a compact structure.^[9] The folded form of a protein is its lowest energy conformation in dilute solutions. (In concentrated protein solutions, oligomeric or aggregated states may be lower in energy.) The chains orient to maximize the hydrophobic effect and also to make attractive interactions between amino acids: hydrogen bonds and other electrostatic interactions.

Hydrogen bonds contribute the remaining 40% of the driving forces for protein folding. The hydrogen bonds between protein atoms each contribute on average 1.1 ± 0.8 kcal/mol each to protein stability. The unfolded protein makes hydrogen bonds between protein and water, so this energy contribution is the net gain in

5.2 Native and denatured conformations equilibrate

hydrogen bond strength. Hydrogen bonds also define how the protein will fold into helices and sheets.

The dominant driving force for unfolding of the main protein chain is the increase in flexibility. The denatured ensemble is a collection of highly flexible forms. This flexibility creates many conformations or micro-states and the existence of these states is the main driving force for unfolding. The micro-states create a favorable entropy contribution, $-T\Delta S$, to the Gibbs energy. To find the Gibbs energy contribution of entropy, one multiplies the entropy change by temperature and a negative sign since an increase in entropy lowers Gibbs energy, eq. Equation 5.3, where W_{folded} and W_2 are the different numbers of micro-states in the states being compared.

$$\Delta G_{unfolded-folded} = -T\Delta S_{2-1} = -RT\ln\left(\frac{W_2}{W_1}\right) \quad (5.3)$$

To find the effect of a molecule's conformational flexibility on Gibbs energy, one compares the number of conformations in the flexible state to the non-flexible state. For example, one can estimate the difference in Gibbs energy between the folded and unfolded states for one amino acid in a protein at 300 °K only due to differences in backbone flexibility while ignoring any differences in side-chain flexibility. Assume that the backbone has a single conformation in the folded state, N, but can adopt three staggered conformations along each of the two rotatable bonds (ψ , ϕ) in the unfolded state, D, Figure **Figure 5.4**.

The approach is to estimate the difference in the number of conformations available to the folded and unfolded states and then convert this difference to Gibbs energy using eq. Equation 5.3 above. The amino acid in the folded state has one available conformation

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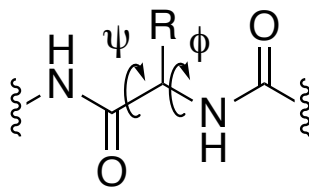


Figure 5.4. Each amino acid residue contains two rotatable bonds (ψ , ϕ) along the main chain. The C-N bond in the amide link does not readily rotate because it has partial double bond character.

while in the unfolded state, an amino acid can adopt $3 \cdot 3 = 9$ possible conformations. Converting this difference in possible conformations into Gibbs energy yields:

$$\Delta G_{N-D} = -RT \ln \left(\frac{W_2}{W_1} \right) = 1.987 \text{ cal/mol} \cdot ^\circ\text{K} \cdot 300^\circ\text{K} \cdot \ln \left(\frac{1}{9} \right) = +1.3 \text{ kcal/mol}$$

Thus, entropy due to differences in the backbone flexibility favors the unfolded state by 1.3 kcal/mol for each amino acid residue. More accurate estimates that account for the unequal likelihood of the nine conformations and differences in side chain flexibility yield a similar number: 1.4 kcal/mol per amino acid residue.^[10] For a typical protein of 300 amino acids, this entropy effect of backbone flexibility contributes 400 kcal/mol, which is large as compared to the balance of ~ 10 kcal/mol in favor of the folded state. Hydrophobic interactions and hydrogen bonds in the native state counterbalance this flexibility advantage of the denatured state with a slightly larger Gibbs energy contribution. Protein stability is a balance between one set of large forces favoring the folded state and another set favoring the unfolded state.

5.3 Measuring the folding-unfolding equilibrium

Measuring the equilibrium constant requires measuring the ratio of folded and unfolded protein at equilibrium. Under normal conditions, the equilibrium amount of unfolded protein is too small to measure. Instead, researchers use destabilizing conditions where easily detectable amounts of both the folded and unfolded protein exist, Figure **Figure 5.5**. Typical destabilizing conditions are additives like urea or guanidinium chloride, changes in the solution pH, or increases in temperature. After measuring the equilibrium constant at these destabilizing conditions, researchers extrapolate back to normal conditions to get the desired equilibrium constant under normal conditions.

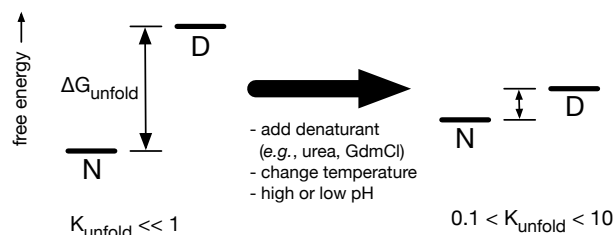


Figure 5.5. Changing the environment shifts the equilibrium between native and denatured states. The equilibrium constant can be measured when the relative amounts of the two forms are in the range of 0.1: 1 to 10:1, that is, when ΔG^{unfold} is within 1.4 kcal/mol of zero. GdmCl = guanidinium chloride.

Most of this book treats protein folding and unfolding as a cooperative, two-state process (Figure **Figure 5.2** above). The entire protein is either folded or denatured, and the process is entirely reversible. There are no partially folded proteins or folding intermediates. The cooperative folding might start with one amino acid adopting an α -helix conformation, quickly followed by neighboring amino acids

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adopting the α -helix structure to create intrachain hydrogen bonds. This assumption of a simple, reversible two-state process is reasonable for single domain monomeric proteins. In these cases, adding stabilizing substitutions anywhere in the protein can contribute to stabilization since the entire protein unfolds simultaneously.

Changing the environment does not gradually change protein structure from folded to unfolded. Proteins remain fully folded or fully unfolded, but their ratio changes with the environment. Unfolding is cooperative because the interactions that stabilize the folded protein are stronger in combination than their individual contributions. The protein remains folded even when some stabilizing interactions break, but breaking too many destabilizes the others leading to complete unfolding of the protein.^[11]

Section 5.5.1 below will briefly consider multiple domain and oligomeric proteins. Multiple domain proteins usually fold and unfold stepwise via intermediates. The same stabilization strategies apply, but the substitutions must be in unfolded regions, not anywhere in the protein.

5.3.1 Denaturation with urea, ΔG_{H_2O}

Urea unfolds proteins because it 1) solvates exposed hydrophobic groups to reduce the hydrophobic effect and 2) forms hydrogen bonds to the backbone to disrupt secondary structures. Typical proteins unfold at 3-5 M urea; the solubility limit of urea is 18 M at room temperature.

The urea-induced denaturation experiment involves preparing solutions of protein in various concentrations of urea, waiting until the folding-unfolding reaches equilibrium, and measuring the amounts of native and denatured protein. The most common method to detect the folded and unfolded proteins is measuring

5.3 Measuring the folding-unfolding equilibrium

the intensity of protein absorbance or fluorescence, Figure **Figure 5.6**, but any method that distinguishes between the native and denatured states is suitable. Some examples are measuring shifts in the wavelength of the fluorescence maximum, changes in the circular dichroism spectra, changes in the NMR spectra, decreases in catalytic activity, or changes in the dye fluorescence upon binding to hydrophobic regions of the unfolded protein.

The reasoning above predicts that the fluorescence emission wavelength increases for the denatured protein, but it is not easy to predict the relative intensity of this emission. The intensity change also depends wavelength chosen to monitor the fluorescence. For the example in Figure **Figure 5.6a**, the fluorescence intensity at 320 nm (near the emission maximum of the native protein) decreases as the protein unfolds, but the fluorescence intensity at 360 nm (near the emission maximum of the denatured protein) increases (not shown).

To extract the constants for unfolding equilibrium at different urea concentrations, $N \rightleftharpoons D$ from the fluorescence changes, one needs to know the fractions of native, F_N , and denatured protein, F_D . The ratio of these fractions is the equilibrium constant, eq. Equation 5.4. One assigns the fluorescence in low or no denaturant, Y_N , to the native protein and the fluorescence at high denaturant concentrations, Y_D , to the denatured protein. The supporting information includes a derivation of this equation. Intermediate values of fluorescence, Y_{obs} , correspond to a mixture of native and denatured states. Only data in the transition region of the denaturation experiment contribute to the analysis because when the protein is nearly completely folded or completely unfolded, the data do not precisely define the ratio of the two forms.

$$K^{unfold} = \frac{F_N}{F_D} = \frac{Y_{obs} - Y_N}{Y_D - Y_{obs}} \quad (5.4)$$

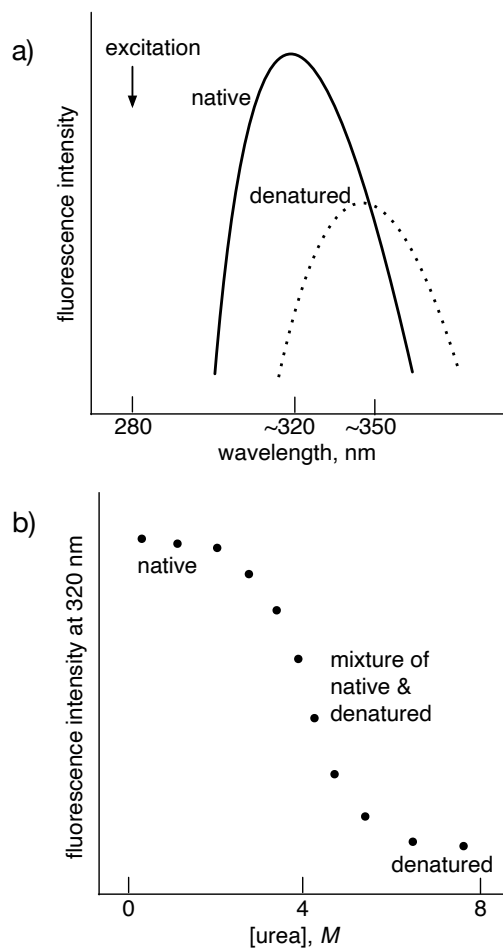


Figure 5.6. Unfolding of a protein in urea as monitored by tryptophan fluorescence. a) The indole ring of tryptophan absorbs light (absorbance maximum 280 nm; not shown), then emits light at lower energy (higher wavelengths). The emission maximum of a folded protein is typically 320 nm, which is similar to the emission maximum of indole in hexane. Upon unfolding the emission maximum shifts to 350 nm, which is similar to the emission maximum of tryptophan in water. b) Hypothetical urea unfolding data showing a decrease in fluorescence intensity at 320 nm as the protein unfolds in increasing concentrations of urea.

5.3 Measuring the folding-unfolding equilibrium

Applying the a linear extrapolation approach^[12] to these measured equilibrium constants reveals the Gibbs energy of unfolding in water. This linear extrapolation approach starts by converting these equilibrium constants to Gibbs energy values according to eq. Equation 5.1 above and plotting them versus urea concentration, Figure **Figure 5.7**. The plot reveals a straight line with a negative slope since adding more urea favors protein unfolding. The line crosses the x-axis at $\Delta G^{unfold} = 0$. The goal of this experiment is to measure ΔG^{unfold} without any urea, that is, $[\text{urea}] = 0$. Extrapolation of the line to y-axis intercept reveals the Gibbs energy of unfolding in pure water, $\Delta G_{H_2O}^{unfold}$.

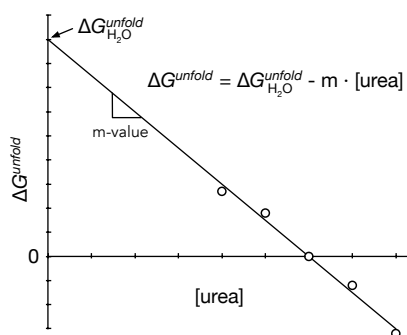


Figure 5.7. Linear extrapolation of the Gibbs energies calculated from the data in Figure **Figure 5.6b** reveals the Gibbs energy of unfolding in water. The data where the protein is nearly completely folded or completely unfolded are not included because the folded to unfolded ratio is not precisely defined by the data.

The equation for this line has a slope of $-m$ and a y-intercept at $\Delta G_{H_2O}^{unfold}$. The m -value is the absolute value of the slope and indicates the sensitivity of the protein to denaturant, which depends on the solvent-accessible surface area exposed upon unfolding.^[12] A protein that unfolds completely has a higher m -value than a similar

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protein that unfolds only partially. Similar proteins with different m -values indicate different degrees of unfolding in their denatured states. Typical m -values for proteins undergoing urea-induced unfolding are ~ 1000 cal/(mol·M). This linear extrapolation approach is a convenient way to measure protein stability and the supporting information includes an example calculation.

$$\Delta G^{unfold} = \Delta G_{H_2O}^{unfold} - m \cdot [\text{urea}] \quad (5.5)$$

Positive Gibbs energies of unfolding in pure water indicate that unfolding of the protein is unfavorable. A higher Gibbs energy of unfolding indicates a more stable folded protein. The molecular basis for the linear relationship between unfolding Gibbs energy and urea concentration may be a binding interaction between urea and protein. The binding of urea destabilizes the protein and the fraction of urea bound increases linearly with concentration.

5.3.2 Heat denaturation, $\Delta G^{unfold}(T)$

Heat also unfolds proteins. Increases in temperature increase the entropy contribution to Gibbs energy of unfolding, eq. Equation 5.6 where ΔH^{unfold} and ΔS^{unfold} are, respectively, the changes in enthalpy and entropy upon unfolding. The entropy contribution due to the increase in flexibility of the unfolded protein increases at higher temperature and eventually leads to unfolding.

$$\Delta G^{unfold} = \Delta H^{unfold} - T\Delta S^{unfold} \quad (5.6)$$

Here we consider reversible heat-induced unfolding. Section 5.6.1 below considers irreversible unfolding caused by aggregation of the unfolded protein. As with urea-induced denaturation, this unfolding can be monitored spectroscopically,

5.3 Measuring the folding-unfolding equilibrium

Figure 5.8. For reversible unfolding, the equilibrium constant yields the Gibbs energy of unfolding at these high temperatures. The melting temperature, T_m , is the temperature where the amounts of folded and unfolded protein are equal; $\Delta G^{unfold} = 0$.

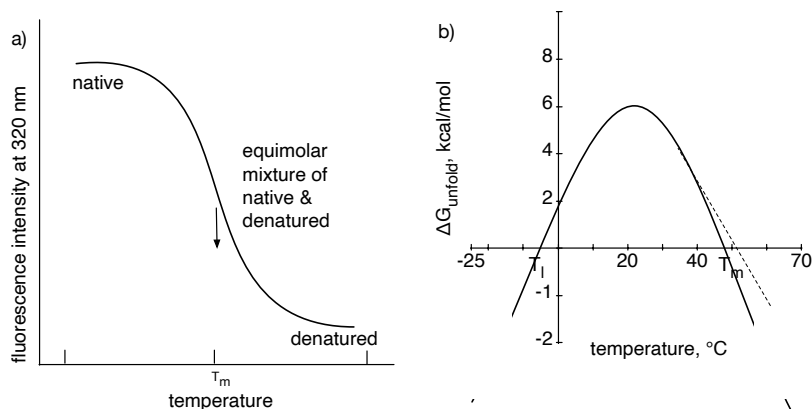


Figure 5.8. Heat induced unfolding of proteins. a) Hypothetical melting curve for a protein measured by the decrease in fluorescence intensity at 320 nm as the protein unfolds. The melting temperature, T_m , is the temperature where the amounts of folded and unfolded protein are equal. b) The Gibbs energy of unfolding of a protein varies non-linearly with temperature according to eq. Equation 5.6. The stability of a protein also decreases at lower temperatures; T_l marks the low temperature unfolding temperature. This plot uses values typical for a 300 aa protein: $T_0 = 300^\circ\text{K}$, $\Delta H_0 = 30 \text{ kcal/mol}$, $\Delta S_0 = 80 \text{ cal/mol-deg}$, $\Delta C_p = 4.2 \text{ kcal/mol-deg}$. The dotted line is an approximation sometimes used near the melting temperature, see eq. Equation 5.8.

These measured Gibbs energies of unfolding at high temperatures can be extrapolated to room temperature, but this extrapolation is

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more complex than extrapolating the Gibbs energy of unfolding to pure water from a urea denaturation curve.^[13] One might expect a plot of ΔG_{unfold} versus temperature to yield a straight line according to eq. 3.7 above, where ΔS_{unfold} is the slope and ΔH_{unfold} is the y-intercept. However, the Gibbs energy of protein unfolding varies with temperature according to a more complex equation, eq. Equation 5.7.^[14] Extrapolating the melting curve data to lower temperatures also requires measuring ΔC_p which is the change in heat capacity upon unfolding, Figure **Figure 5.8**.

$$\Delta G^{\text{unfold}} = \Delta H_0^{\text{unfold}} - T\Delta S_0^{\text{unfold}} + \Delta C_p [(T - T_0) - T \ln(T/T_0)] \quad (5.7)$$

Here T_0 is the reference temperature and $\Delta H_0^{\text{unfold}}$ and $\Delta S_0^{\text{unfold}}$ are the enthalpy and entropy of unfolding at that temperature. For reactions involving small molecules, the heat capacity of starting materials and products are similar, and differences may be ignored. Eq. Equation 5.7 simplifies to eq. Equation 5.6 when $\Delta C_p = 0$.

However, the heat capacities of the folded and unfolded proteins differ significantly. Another way to state the reason for the more complex equation is that for protein unfolding ΔH^{unfold} and ΔS^{unfold} vary with temperature while eq. Equation 5.6 assumes they are constant. The extra terms in eq. Equation 5.7 account for the variation of ΔH^{unfold} and ΔS^{unfold} with temperature.

The unfolded protein has a higher heat capacity than the folded protein because unfolding exposes hydrophobic groups to water. Water surrounds these hydrophobic groups with an ice-like structure, which has slightly higher energy than bulk water. The fluctuation of water between bulk and ice-like structure increases the heat capacity.

The change in heat capacity for protein unfolding, ΔC_p , is always positive with a typical value of 14 cal/mol·deg per residue or 4.2

5.3 Measuring the folding-unfolding equilibrium

kcal/mol·deg for a 300 aa protein.^[15] Differential scanning calorimetry can measure this change in heat capacity. Differential scanning calorimetry measures the heat required to increase the temperature of a sample. The baseline value before unfolding reveals the heat capacity of the folded protein, while the baseline value after unfolding reveals the heat capacity of the unfolded protein. The difference is the change in heat capacity. In the absence of experimental data, the web tool SCooP^[16] predicts the parabolic curve for temperature-dependent unfolding of a protein based on its structure.

The parabolic curve in Figure **Figure 5.8b** predicts that proteins will also denature at low temperatures. For most proteins the predicted cold denaturation temperature, T_l , lies well below the freezing point of water as shown in the sketch, but for some proteins, the cold denaturation temperature lies in the liquid range of water. For example, the yeast protein frataxin denatures at both 7 °C and at 30 °C.^[17] At high temperatures the increase in conformational entropy (flexibility) of the protein atoms drives unfolding, while at low temperatures a decrease of the hydrophobic effect drives unfolding. At lower temperatures, the cost of ordering water molecules around a hydrophobic group decreases. As the tendency of hydrophobic groups to aggregate in water decreases, the folded form loses a stabilizing force.

Increases in melting temperature usually correspond to increases in protein stability. Comparing the unfolding or melting temperatures of protein variants is a common, approximate measure of their stability. One assumes that the unfolding process is similar for protein variants so the parabolic curve of Gibbs energy of unfolding versus temperature (Figure **Figure 5.8b** above) remains similar. One heats a protein sample slowly while monitoring the protein tryptophan fluorescence (Figure **Figure 5.8a**). The temperature corresponding to the midpoint of the change in fluorescence indicates the melting temperature. Usually one is on

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the right side of the parabola describing protein heat stability so an increase in melting temperature corresponds to a increase in protein stability.

A comparison of pairs of homologous proteins from thermophiles and mesophiles showed that the melting temperatures, T_m , were 31.5 °C higher and the Gibbs energies of unfolding 8.7 kcal/mol higher for the thermophiles.^[18] From this comparison, we can derive a rule of thumb that stabilizing a protein by 1 kcal/mol will increase the melting temperature by 3.6 °C, eq. Equation 5.8, see dotted line in Figure **Figure 5.8b**.

$$\Delta\Delta G^{unfold}(\text{kcal/mol}) \sim \Delta T_m(^{\circ}\text{C})/3.6^{\circ}\text{C/kcal/mol} \quad (5.8)$$

In many cases, the Gibbs energy of heat-induced unfolding, $\Delta G^{unfold}(T)$, is comparable to that obtained by urea unfolding. When these values differ, it is likely because the unfolded states differ in the two experiments.

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Stabilizing a dynamic object like a protein differs from stabilizing a rigid object like a chair. To stabilize a chair one can add a supporting brace, which keeps the chair in the desired structure. In contrast, a protein folds and unfolds continuously; it is dynamic. Adding a stabilizing interaction to a protein does not prevent unfolding. Instead, the stabilizing interaction shifts the folding-unfolding equilibrium toward folding so the protein spends more time in the folded form. Since the equilibrium constant is the ratio of the forward and reverse rate constants, eq. Equation 5.9, shifting the equilibrium constant also changes the rates of folding and unfolding. A stabilizing interaction may slow down unfolding,

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speed up folding, or both; but, unlike stabilizing a chair, stabilizing a protein does not *prevent* unfolding.



Stabilizing a protein also differs from stabilizing a rigid object because one can stabilize a protein by destabilizing the unfolded form.^[19] Everyday objects like chairs are not dynamic and cannot be stabilized this way. This book uses the expression ‘stabilize a protein’ or ‘increase protein stability’ to mean either stabilizing the folded form or destabilizing the unfolded form.

One reason that it is hard to predict which substitutions will stabilize a protein is that the structure of the denatured state (an ensemble of rapidly equilibrating structures) is unknown. Stabilizing substitution must affect the folded and unfolded forms differently. A substitution that equally stabilizes (or destabilizes) the folded and unfolded forms does not change stability of the protein. X-ray crystal structures reveal changes to the folded form, but changes to the denatured state are invisible because its structure is unknown. Stabilization can occur without strengthening the folded structure. For example, seven amino acid substitutions dramatically stabilized xylanase (T_m increased by 25 °C), but the structures of the wild-type and stabilized proteins showed similar interactions between amino acids in both structures.^[20] In this case, the stabilizing substitutions likely destabilized the unfolded state and changed its structure, but nature of these changes is unknown because the structure of the unfolded state is unknown.

A second reason that it is hard to predict stabilizing substitutions is that the substitutions must not hinder catalytic or binding ability of the target proteins. Substitutions that increase stability often

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also decrease activity and vice versa.^[21] One reason for this trade-off is that catalysis and binding rely on interactions of the substrate or transition state with unsatisfied hydrogen bonds, exposed hydrophobic groups and unpaired charges in the protein. Satisfying these interactions with amino acid substitutions will stabilize the protein, but also disrupt binding and catalysis. Researchers typically avoid substitutions near the active site when searching for stabilizing substitutions to prevent disrupting binding and catalysis.

Some protein stabilization strategies seek to stabilize the native protein, others destabilize the denatured conformation, and, for a few, the molecular mechanism is unknown, Table **Table 5.1**. The design strategies that seek to stabilize the native protein require a protein structure or at least a homology model to start the design. A homology model is an extrapolated 3D-structure of a protein based on the known structure of a similar protein. Swiss Model and AlphaFold are web tools to generate homology models from protein sequences. Strategies that seek to destabilize the unfolded ensemble also require a protein structure to ensure that the changes do not distort the folded form. Strategies with unknown mechanism do not require a protein structure. For example, restoring residues that are conserved in homologs (consensus sequence approach, see below) requires only a protein sequence. Random mutagenesis followed by screening to find stabilized variants also does not require a protein structure. Chapters 9 and 10 discuss random mutagenesis methods.

5.4 Stabilization to cooperative unfolding

Table 5.1. Design strategies to stabilize proteins and web tools to implement these strategies.

Strategy	Rationale	Example
restore residues conserved in homologs	not specified	Consensus Finder identifies conserved amino acids in homologs that are missing from target protein
add disulfide crosslinks	destabilize unfolded protein	Disulfide by Design identifies suitable locations, additional analysis needed to narrow choices
add Pro residues	destabilize unfolded protein	Analysis of structure to identify locations suitable for proline WHAT-IF
substitutions in or near flexible regions	stabilize folded protein	Random mutagenesis in or near flexible regions
random mutagenesis and screening	not specified	Random mutagenesis anywhere followed by screening for stabilized variants
optimize electrostatic interactions	stabilize folded protein	TKSA-MC identifies destabilizing electrostatic interactions

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Strategy	Rationale	Example
optimize hydrophobic and other interactions	stabilize folded protein	Modeling with Rosetta and FoldX to identify stabilizing substitutions

To engineer a change, one needs to choose both the location of the substitution and the replacement amino acid. The ‘restore residues conserved in homologs’ approach predicts both, so it is easy to implement. The ‘substitutions in or near flexible regions’ approach defines the location, but not the replacements. Additional modeling or experiments are needed to identify the replacements. The ‘add disulfide links’ approach specifies that cysteine is the replacement amino acid, but additional modeling is needed to find suitable locations. Design strategies like ‘improve hydrophobic and other interactions’ are the least specific and require computer modeling to predict both the location and the replacement amino acid. Each of the approaches in **Table 5.1** will be described in more detail below. The effect of each stabilizing mutation is typically small (0.5 kcal/mol), so substantial stabilization of the protein (2-4 kcal/mol) requires multiple substitutions and often more than one design strategy.

5.4.1 Restore conserved amino acids

Evolution conserves amino acids that contribute to protein function. This contribution may be to structure, catalysis, stability or other protein property. The consensus approach hypothesizes that conserved amino acids residues are more likely to contribute to protein stability than a non-conserved amino acid. To use the consensus approach to stabilize a protein, one identifies amino

5.4 Stabilization to cooperative unfolding

acid residues conserved within homologous proteins, but which are missing in the target protein. Restoring these conserved amino acids is more likely to stabilize the protein than random substitutions.

Consensus design works because natural proteins rarely follow the consensus sequence at all structurally important positions. Natural proteins only have to be stable enough to fulfill their biological function; proteins with stabilities above a certain threshold will have no further selection advantage. There are many stabilizing residues, but individual proteins do not need all of them to reach the threshold of being stable enough. Thus, replacing a residue with the corresponding consensus amino acid may improve the stability or folding efficiency of a protein of interest. The same reasoning leads to the conclusion that the stability of a protein designed by consensus can be higher than that of the proteins in the alignment.

The consensus sequences approach does not hypothesize any molecular basis for the stabilization. It relies only on amino acid sequence comparisons from bioinformatics and does not require any structural information.^[22,23] It is easy to implement, and several web-based tool to generate a consensus sequence for a protein are available: Consensus Finder^[24] or FireProt.^[25] Typically researchers restore one or a few conserved amino acids. Usually at least half of these predicted substitutions prove to be stabilizing. Pantoliano and coworkers first suggested that substitutions to the consensus amino acid may stabilize proteins and showed that the consensus-type mutation Met50Phe increased the unfolding temperature of subtilisin BPN' by 1.8 °C^[26] and many others have used this approach. For example, a sequence alignment of 351 homologs of *Staphylococcus* nuclease A identified the His124Glu substitutions as potentially stabilizing, Figure **Figure 5.9**.

One can also restore all the conserved amino acids simultaneously.

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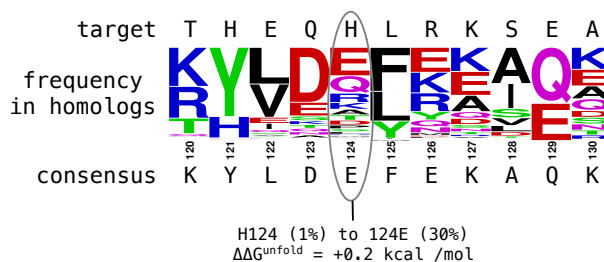


Figure 5.9. The most frequently-occurring amino acid in a multiple sequence alignment defines a consensus sequence. The target sequence (*Staphylococcus* nuclease A) was aligned with 351 homologs. The height of the letters in the logo plot reflects the frequency of the amino acid in the multiple sequence alignment. The sequence of the most common amino acid at each position is the consensus sequence. This consensus sequence predicts that an H124E substitution would stabilize this protein since 30% proteins in the alignment have E, but only 1% have H. Unfolding experiments showed that this substitution slightly stabilized *Staphylococcus* nuclease.^[27] The WebLogo tool generated the logo plot of amino acid frequencies.

5.4 Stabilization to cooperative unfolding

Lehmann and coworkers predicted the consensus sequence of fungal phytases based on the sequence alignment of thirteen sequences, Figure **Figure 5.10**.^[22] The consensus phytase differed by at least 80 amino acids from the parent sequences. A synthetic gene encoded the consensus phytase containing all these changes. The consensus protein was 15-26°C more thermostable than any of its parents.

source	weight	sequence
<i>Aspergillus terreus</i> 9A-1	0.50	54- QVLARHGARS PTHSKEKAYA
<i>Aspergillus terreus</i> cbs	0.50	QVLARHGARS PTDSKTKAYA
<i>Aspergillus niger</i> var. <i>awamori</i>	0.33	QVLSRHGARY PTESKGKKYS
<i>Aspergillus niger</i> T213	0.33	QVLSRHGARY PTESKGKKYS
<i>Aspergillus niger</i> NRRL3135	0.33	QVLSRHGARY PTDSKGKKYS
<i>Aspergillus fumigatus</i> 13073	0.20	QVLSRHGARY PTSSKSKKYK
<i>Aspergillus fumigatus</i> 32722	0.20	QVLSRHGARY PTSSKSKKYK
<i>Aspergillus fumigatus</i> 58128	0.20	QVLSRHGARY PTSSKSKKYK
<i>Aspergillus fumigatus</i> 26906	0.20	QVLSRHGARY PTSSKSKKYK
<i>Aspergillus fumigatus</i> 32239	0.20	QVLSRHGARY PTASKSKKYK
<i>Emericella nidulans</i>	1.0	QVLSRHGARY PTESKSKAYS
<i>Talaromyces thermophilus</i>	1.0	QLLSRHGARY PTSSKTELYS
<i>Myceliophthora thermophila</i>	1.0	QVLSRHGARA PTLKRAASTV
Consensus		QVLSRHGARY PTSSK-KAYS

Figure 5.10. Part of the sequence alignment (amino acid positions 54–73) of fungal phytases and the derived consensus sequence.^[22] Amino acid sequences from the same species are weighted so that each species contributes equally to the consensus sequence. The '-' in the consensus sequence indicates that the consensus amino acid is ambiguous at this position.

At some positions, counting the numbers of occurrences identifies the consensus amino acid. For example, all thirteen sequences start with QVL, and most have S in the next position (boxed), so the consensus sequence begins with QVLS, Figure **Figure 5.10**. At

other positions, identifying the consensus amino acid requires care to ensure unbiased sampling of sequences. Identifying the most common residues requires a uniform sampling of amino acid sequences throughout the evolutionary tree, but the available amino acid sequences may not be evenly distributed. For the phytase example above, researchers reduced the bias of multiple sequences from different strains of the same species by weighting the amino acid sequences so that each *Aspergillus* species had equal weight. For example, the list included five sequences from *Aspergillus fumigatus* strains, Figure **Figure 5.10**. Each of these sequences was weighted by 0.2 in deriving the consensus to avoid bias toward this species. For example, consider the last amino acid in Figure **Figure 5.10** (position 73, boxed). The first two sequences (*A. terreus*) have A at this position, the next three (*A. niger*) and two others (*Emericella* and *Talaromyces*) have S, the five *A. fumigatus* sequences have K, and the last sequence (*Myceliophthora*) has V. Although S and K both occur five times, K occurs only in the five *A. fumigatus* sequences and receives a total weight of 1.0, while S occurs in three different species and gets a total weight of 3.0. The consensus amino acid at this position is S. The consensus sequence is not unique, but depends on the set of comparison sequences. In later work, as more phytase sequences became available, researchers tested a revised consensus sequence, which further increased stability.

In other cases, a database search may yield thousands of similar sequences and it is impractical to assign different weights to the sequences. To minimize phylogenetic bias in these cases, one clusters similar sequences (typically >90% identity) together and uses only one sequence from each cluster for the consensus calculation. The web-based tools to find consensus sequences mentioned above use this clustering to minimize phylogenetic bias.

One disadvantage of the consensus sequence approach is that its conservative nature predicts small numbers of substitutions, often

5.4 *Stabilization to cooperative unfolding*

less than a handful or even none. Although experimental screening reveals that many stabilizing substitutions exist, the consensus sequence approach, by focusing on those conserved among homologs, misses most of them.

Ancestral proteins are extinct proteins that correspond to branch points in a phylogenetic tree. Gene synthesis and expression of these proteins in microbial hosts resurrects these proteins. In many cases, ancestral proteins are more stable than modern proteins. Some ancestral proteins may be more stable because the earth was warmer in prehistoric times. However, ancestral proteins from times when the earth's temperature was only slightly warmer are also more stable, so other effects may contribute.^[28] One contribution is that reconstructed ancestral proteins are often similar to the consensus sequence for that group since many descendants retain the ancestral residues. However, ancestral proteins differ from consensus proteins in two ways. First, ancestral proteins identify amino acids conserved within a related group, while consensus proteins include data from all homologous proteins, including those outside a group. Second, the consensus approach weights all sequences equally, but ancestral sequence reconstruction takes branch lengths into account, so more recent changes count less in an ancestral sequence prediction.

5.4.2 **Destabilize the unfolded ensemble**

The main property favoring the unfolded ensemble is its flexibility. Reducing this flexibility destabilizes the unfolded ensemble, thus shifting the folding-unfolding equilibrium toward folding. Two ways to reduce this flexibility are to add disulfide cross-links and to replace amino acid residues with proline. In both cases the replacement amino acid is defined; the challenge is to find suitable

locations for the replacement. The ideal site in the folded structure would fit the replacement perfectly, so the only effect of the substitution is to reduce the flexibility of the unfolded ensemble.

Add disulfide cross links. Replacing a nearby pair of amino acid residues with cysteines followed by spontaneous oxidation creates a disulfide cross link, Figure **Figure 5.11**. Such disulfide links often stabilize proteins to unfolding. For example, the replacement of Ala43 and Ser80 in *Bacillus* RNase with cysteines followed by spontaneous oxidation yielded a more stable protein that unfolded in 5.77 M urea as compared to 4.58 M urea for the wild type.^[29] This engineering involved both amino acid replacements and the formation of a cross link, both of which could contribute to changes in stability. To measure the effect of the cross link alone, the researchers compared the stabilities of the oxidized disulfide form with the reduced dithiol form of the protein. These two forms differ only in the presence or absence of a cross link. For the example above, the Gibbs energy of unfolding of the oxidized form was 3.2 kcal/mol higher than for the reduced dithiol form demonstrating that the cross link stabilized the protein.

The reason that disulfide cross links stabilize proteins is that cross links reduce the flexibility of the unfolded protein. Upon unfolding, the cross link remains and limits the movement of the unfolded protein. Flexibility creates disorder (entropy), which is stabilizing. Reducing flexibility of the unfolded protein destabilizes it. The cross link constrains the distance between the linked cysteines. In the folded form, the cysteines are already close, so this constraint has little effect. In contrast, this constraint dramatically limits the movement and flexibility of the unfolded protein. Since increased flexibility and the associated increased entropy is the main reason that proteins unfold, reducing the flexibility of the unfolded form shifts the folding-unfolding equilibrium toward folding. This shift stabilizes the protein to unfolding.

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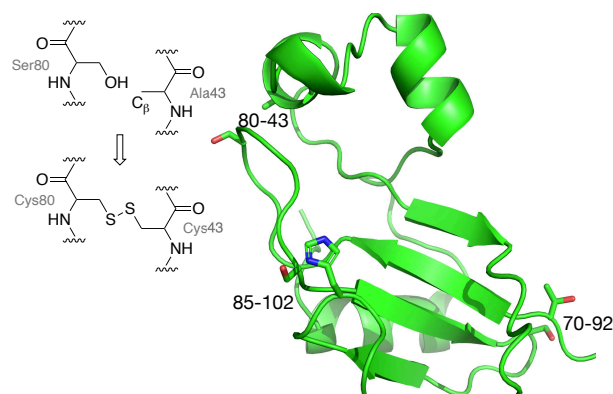


Figure 5.11. Replacement of Ser80 and Ala43 in *Bacillus* RNase with cysteines followed by spontaneous oxidation created a disulfide cross link. This cross link stabilized the protein by 3.2 kcal/mol. Similar cross links introduced at two other locations had variable effects.

A common misconception is that disulfide cross links prevent unfolding; they do not. Proteins remain flexible, dynamic structures even after adding disulfide cross links. They still unfold in urea solutions and upon heating. The RNase protein in the example above still unfolded in urea solutions, although unfolding required higher urea concentrations than for wild-type. A similar misconception is that protein cross links strengthen and stabilize the folded protein state. They do not. The locations chosen for a disulfide cross link are already nearby in the folded protein. The introduction of the cysteines and the cross link may slightly change the folded protein stability, but this change is most often a destabilization, see below.

The chain entropy model predicts the amount of expected stabilization from a disulfide cross link. This model calculates the change in entropy due to reduced flexibility of the unfolded protein state

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caused by the disulfide cross link. This model assumes that the introduction of cysteines and formation of the disulfide cross link has no effect on the stability of the folded protein state. The amount of protein stabilization expected from disulfide cross link depends on the size of the ring created by the cross-link. Larger rings limit the flexibility of more amino acids and are therefore more stabilizing. Equation eq. Equation 5.10 below estimates the chain entropy effect of disulfide cross link on the unfolded state.^[30]

$$\Delta\Delta S^{unfold} = -2.1\text{cal/mol} \cdot ^\circ K - \frac{3}{2}R \ln n \quad (5.10)$$

The $\Delta\Delta S^{unfold}$ refers to change due to the crosslink for the entropy change associated with unfolding; the n is the number of residues in the ring created by the crosslink and R is the gas constant. A link between amino acid residues 43 and 80 creates a ring of 38 residues; one more than the difference between 80 and 43. This equation comes from from polymer theory estimates of the probability that the two ends of the chain will coincide in the same volume element. If linked, the two ends must remain in the same volume element, while if unlinked they may move apart and allow greater numbers of conformations. To estimate the Gibbs energy change of the disulfide cross link using the chain entropy model, one multiplies by $-T$ where T is the temperature (in $^\circ K$) yields the anticipated Gibbs energy contribution, eq. Equation 5.11.

$$\Delta\Delta G^{unfold} (\text{cal/mol}) = -T \cdot \Delta\Delta S^{unfold} = T \left[2.1 + \frac{3}{2} \cdot 1.987 \cdot \ln n \right] \quad (5.11)$$

For example, the chain entropy model predicts that at 25 $^\circ C$ a cross link between amino acids 43 and 80 will stabilize a protein by 3.9 kcal/mol, which is slightly higher than the measured value of 3.2

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kcal/mol in *Bacillus* RNase, Figure **Figure 5.11**. The stabilization predicted by eq. Equation 5.11 increases non-linearly with the number of amino acid residues in the ring, Figure **Figure 5.12**. The estimated stabilization for the smallest possible ring (two amino acids) is 1.2 kcal/mol. The stabilization increases to 3.0 kcal/mol at $n = 15$, which is the average separation of disulfide links in natural proteins.^[31] With larger numbers of amino acid residues in the ring, the increase slows because the cross link restricts a smaller proportion of the motions in larger rings as compared to smaller rings.

The measured values for stabilization often do not match that predicted by the chain entropy model, likely because the chain entropy model ignores any effect on the folded state. The x-ray structure of *Bacillus* RNase 43-80 disulfide link mentioned above shows a small reorganization of backbone structure that may account for the slightly lower than predicted stabilization (3.2 vs. 3.9 kcal/mol).^[29] Most of the examples in Figure **Figure 5.12** seem to fit in this category. Two other disulfide links introduced in *Bacillus* RNase also are unusual cases. The disulfide at positions 85-102 stabilized the protein more than predicted by the chain entropy model, while the one at 70-92 was so much lower that it destabilizes the protein. The x-ray structures of the more-stable-than-expected protein showed no structural changes as compared to wild type, while the destabilized cross-linked protein showed substantial reorganization that apparently destabilized the folded protein.

To stabilize proteins by introducing a disulfide link, one must identify locations where the disulfide cross link fits in the folded protein without disrupting it. This step requires a structure of the target protein, although a homology model may also be used. Most prediction programs use geometry-only calculations where the distances and angles required for a disulfide crosslink are fit to the current fixed locations of possible amino acid pairs. Three geometry-only programs available as web tools are SSBOND (currently offline),^[36]

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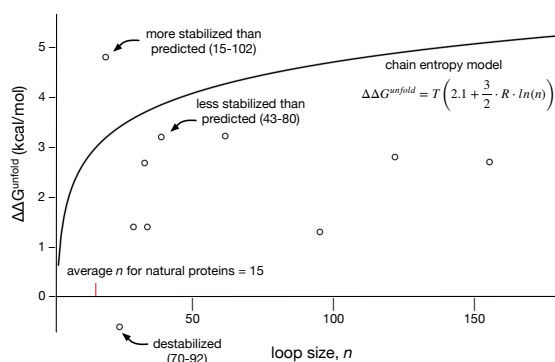


Figure 5.12. The predicted and measured effects of a cross link on protein stability. The open circles are the measured differences in stability between the oxidized protein (the two cysteines form a cross link) and the reduced protein (contains the two cysteines, but no cross link). The line is the predicted difference at 24 °C according to the chain entropy model, eq. Equation 5.11. Loop size, n , is the number of amino acids connected by the disulfide. In all but one case the measured effect is lower than predicted, which may be due to disruption of the folded protein by the cross link. Five data points are for T4 lysozyme,^[32,33] three (marked by arrows) are for *Bacillus* RNase,^[29] and one each for chymotrypsin inhibitor 2^[34] and ribonuclease H.^[35]

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Disulfide by Design,^[37] and YOSSH. ^[38] The web interface requires only a pdb file of the protein structure for the calculation. YOSSH also identifies disulfide cross links that occur in homologs. Since these cross links have been tested by natural selection, they may be prioritized for testing. Figure **Figure 5.13** shows part of the results of an SSBOND prediction.

Typically the programs first identify amino acid pairs with suitable C β -C β distances (2.9-4.6 Å) and second, estimate the energy cost of distorting a disulfide bond to fit. Links with high predicted strain energies should be avoided. For example, conformation 1 in Figure **Figure 5.13** between Ala4 and Asp89 is predicted to be destabilizing. The expected strain energy (total energy, TOTNRG column) is 6.2 kcal/mol, which is larger than the predicted stabilization energy for this link using eq. Equation 5.11 (4.6 kcal/mol). Since disulfide cross links that create larger rings are predicted to be more stabilizing by the chain entropy model, one normal favors larger rings over smaller rings.

Since geometry-only calculations ignore interactions between amino acids, one must consider these before choosing sites for mutagenesis. First, most researchers also avoid cross links near the active site to prevent possible disruption of catalysis. Second, geometry-only calculations do not consider possible loss of favorable interactions due to removal of existing amino acids, nor do they consider potential bumping interactions between the cysteines and adjacent amino acids. Examination of the protein structure can identify these potential problems. Some researchers also include a molecular dynamics simulations to generate alternative protein conformations.^[39] These simulations could identify additional locations where a disulfide cross link can fit or identify locations that should be avoided since the cross link causes large shifts in the backbone that may destabilize the folded protein state.

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C_{β} - C_{β} distance test									
NR	RES1	--	RES2	NAME1	NAME2	CB DIST	CA DIST		
1	4	--	89	ALA	ASP	4.561	6.400		
2	15	--	18	LEU	TYR	3.902	5.848		
3	16	--	201	ASP	ALA	3.945	6.239		

C_{β} - S_{γ} , S_{γ} - S_{γ} distance & angle test												
	SGDIST	X1	X2	X3	X2'	X1'	CHINRG	TAUNRG	DISNRG	TOTNRG		
1	CONFORMATION BETWEEN :											
	1	2.030	173.2	-134.8	-132.4	-62.1	-156.9	5.83	.24	.10	6.18	
2	CONFORMATION BETWEEN :											
	1	2.030	69.2	156.5	-78.3	-88.6	120.1	4.40	.53	.42	5.35	
	2	2.030	-156.7	-98.7	-66.0	153.3	-26.6	5.48	.65	.21	6.35	
3	CONFORMATION BETWEEN :											
	1	2.030	34.5	171.5	109.6	84.9	97.0	4.45	.15	.04	4.64	
	2	2.030	-90.4	117.7	85.9	-149.2	-159.6	4.32	.39	.02	4.73	
	3	2.030	1.2	-122.6	-103.9	-75.2	-165.5	5.25	.11	.02	5.37	
	4	2.030	-99.2	174.3	-114.1	137.2	82.4	5.41	.37	.87	6.65	

Figure 5.13. The program SSBOND predicts potential locations for a disulfide link in a protein. A possible pair of amino acids in a protein: Gly and Asp. The program first identifies amino acid pairs with C_{β} - C_{β} distances in the range of 2.9–4.6 Å. In the case of glycine, which does not have a C_{β} atom, SSBOND predicts its possible location from the location of C, N, and C α . An example SSBOND calculation for a haloalkane dehalogenase protein (pdb ID = 1ede) with default settings found 81 pairs that match the C_{β} - C_{β} distances. The first three are shown here. In the second stage, SSBOND checks for suitable C_{β} - S_{γ} distances, S_{γ} - S_{γ} distances, and proper angles; then calculates the energy cost for deviation from ideal geometry (TOTNRG = total energy). SSBOND ignores any potential interactions of the disulfide link with rest of the protein. This second stage eliminated two pairs leaving 79 possible pairs. The third pair (Asp16-Ala201) has predicted strain energy of at least 4.6 kcal/mol, while equation 3 above estimates 5.3 kcal/mol of stabilization at 25 °C. The prediction is stabilization by 0.7 kcal/mol. An experimental test revealed an increase in the melting transition of 5.2 °C^[39] which corresponds to 1.5 kcal/mol of stabilization according to $\Delta G^{unfold}(\text{kcal/mol}) \sim \Delta T_m(^{\circ}\text{C})/3.6$.^[18]

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Disulfide bonds usually form spontaneously upon air oxidation of the cysteines. If the application requires the protein to remain in the cytoplasm, which is a reducing environment, the disulfide bonds may not form spontaneously making this stabilization approach unsuitable. Mutant strains of *E. coli* where the enzyme thioredoxin reductase is inactivated can form disulfide cross links within the cytoplasm.^[40]

While disulfides are the most common way to cross link the amino acid chain for protein stabilization, there are other possibilities.^[41] For example, connecting the N- and C-termini into a ring also restricts the main chain motion of the unfolded protein and stabilizes proteins to unfolding. Formation of this ring requires special methods as well as a protein fold where the N- and C-termini are near one another.

Introduce proline residues. One can also reduce the backbone flexibility of the unfolded ensemble without creating crosslinks by selective amino acid substitutions. Proline is the least flexible amino acid because its ring limits its backbone conformations. Similarly, glycine is the most flexible amino acid since no side chain limits the possible backbone conformations. Replacing any amino acid with proline is expected to reduce the flexibility of the denatured ensemble and stabilize the folded protein. Replacing glycine with any other amino acid should have a similar effect. To achieve a net stabilization, these substitutions should not otherwise stabilize the denatured ensemble and must not destabilize the folded form with unfavorable interactions. As in other cases, one should avoid changes in the active site of an enzyme to prevent disrupting catalysis.

flexibility: Gly > Ala & 17 other aa > Pro

Stabilization starts by finding a location where introducing a

proline does not disrupt the native structure or interactions. Substitution of an amino acid with proline at such a location should cause no change in the Gibbs energy of the native state, Figure **Figure 5.14**. The Gibbs energy of the denatured state for the proline variant will be higher than for the wild type due to reduced flexibility of the denatured state.

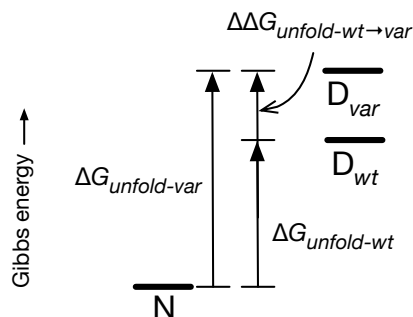


Figure 5.14. Hypothetical free energy changes upon stabilizing a protein with an alanine to proline substitution. Unfolding the native (N) wild-type protein to the denatured state (D) has a Gibbs energy of unfolding of $\Delta G_{unfold-wt}$. The location for the alanine to proline substitution is chosen so that the proline fits without altering the stability of the native state. The proline substitution limits the flexibility of the denatured state, thus the Gibbs energy of the denatured variant state is higher than the wild type and the Gibbs energy of unfolding the variant is higher. The increase in stability of the variant, $\Delta\Delta G_{unfold-wt\rightarrow var}$, is positive indicating that it is less favorable to unfold the variant as compared to wild type.

Introducing proline is a reliable approach to stabilizing proteins. Replacing a typical amino acid with proline reduces the number of conformation in the denatured ensemble by an estimated factor of 7.5 or a ΔS of -4 cal/mol·deg,^[42] eq. Equation 5.12, which corre-

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sponds to a Gibbs energy contribution of 1.2 kcal/mol at 298 °K.

$$\Delta S = R \ln \left[\frac{\text{\# of conformations for typical aa}}{\text{\# of conformations for proline}} \right] = R \ln (7.5) = 4 \text{ cal/mol}\cdot\text{deg} \quad (5.12)$$

This approach requires identifying a suitable location for the proline. The main chain angles for proline fit well in three places: near the start of an α -helix, the $i+1$ position in a type I or II β -turn^[42] or the i position of a type II β -turn.^[43] These choices of proline location consider only main chain angles; the success of any substitution also assumes minimal changes to side chain interactions. The WHAT-IF web tool can identify these locations in a protein structure (Click on ‘mutation prediction’, then ‘Suggest proline mutations’). The first example of protein stabilization by proline substitution was an Ala82Pro substitution in T4 lysozyme, which occurs near the start of an α -helix and stabilized the protein by 0.8 kcal/mol, Table **Table 5.2**.^[44] The restricted backbone angles of proline fit well near the start of an α -helix and the rest of the structure also fit a proline residue. The first four residues of a helix lack a partner to which the main chain N-H can donate a hydrogen bond, so the lack of an N-H in proline is not a disadvantage at the start of a helix. The typical success rate for stabilization upon proline substitution is $\sim 50\%$.

Table 5.2. Stabilization of lysozyme from phage T4 by substitutions that reduce the flexibility of the denatured state.

Protein	Location of substitution	T_m , °C	$\Delta\Delta G^{unfold}$ (kcal/mol)
wild type	none	64.7	0

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Protein	Location of substitution	T_m , °C	$\Delta\Delta G^{unfold}$ (kcal/mol)
Ala82Pro	near start of α -helix	66.8	0.8
Gly77Ala	near start of α -helix	65.6	0.4

Although replacing glycine with alanine also reduces the flexibility of the denatured state, other considerations limit the effectiveness of this substitution. Replacing glycine residues with alanine reduces the number of conformations available to the unfolded ensemble by approximately a factor of three, which corresponds to an Gibbs energy contribution of 0.4 kcal/mol at 298 °K, which is smaller than that of proline substitution, 1.2 kcal/mol. As with the addition of proline, one must ensure that the glycine to alanine substitution does not strain the folded form. Since glycine can adopt conformations that are not accessible to alanine (e.g., a left-handed helix conformation which occurs in some β -turns), replacements at these locations would destabilize proteins. Second, alanine can create stabilizing interactions within the denatured state that counterbalance the reduced flexibility so the net effect may be zero.^[45] The Gly77Ala example in Table **Table 5.2** above stabilized the protein, but the x-ray structure suggests that it stabilized the folded form due to the burial of the hydrophobic methyl group and not due to changes in flexibility of the denatured state.

5.4.3 Stabilize the folded form

Strengthening interactions between amino acids in the folded protein is the most obvious approach to stabilizing proteins. Starting from the three-dimensional structure of the protein, one designs

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various improved interactions. One difficulty with this approach is that proteins already create stabilizing interactions, so this design searches for additional interactions. One must predict both the location for the changes and the replacement amino acids.

Flexible regions identify weak spots in proteins. The effect of flexible regions in proteins can be confusing. Flexibility is a weakly stabilizing feature because it increases entropy.^[46] This flexibility may contribute a factor of two, or 0.4 kcal/mol, to stability. While this flexibility is stabilizing, it also indicates that interactions with other amino acids are weak. Creating strong interactions between amino acids such as hydrogen bonds and hydrophobic interactions (several kcal/mol) in place of the weak stabilization of flexibility can be a net gain for stability, Figure **Figure 5.15**. Thus, it is not removing flexibility that stabilizes the protein, but the addition of new interactions between the amino acids. Loops on the surface and the N- and C-termini of the protein are often the most flexible.

There are two steps to this stabilization approach: first to identify the flexible regions of the protein and second, to identify stabilizing substitutions. One way to identify flexible areas is molecular dynamics simulation, which directly models protein motion.^[39] Another way to identify flexible regions is the high B-factors in the x-ray crystal structure.^[47] B-factors describe the spreading of electron density assigned to that atom. This spread may be due to movement of the atom during the x-ray analysis (temperature-dependent atomic vibrations), or may be due to the atom occupying several fixed positions in the structure (static disorder), which suggests motion in solution. Errors in model-building can also increase B-factors. The B-FITTER program identifies the twenty amino acids in a protein that have the highest B-factors. B-FITTER calculates the average of the B-factors for all atoms (excluding hydrogen) for each amino acid and displays a list of the twenty amino acid residues with the highest B-factors.

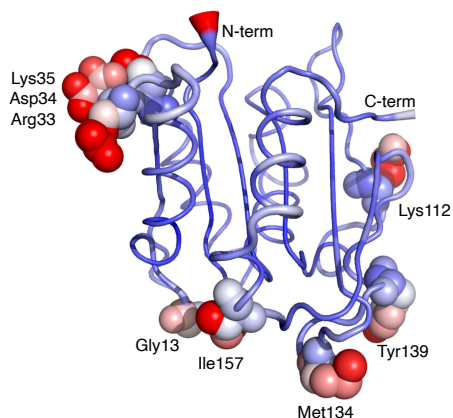


Figure 5.15. X-ray structure of a lipase (pdb code 1isp) colored to show the B-factor from low (blue) to high (red). The most flexible regions are the N- and C-termini (on the left) and several loop regions at the top. Reetz and coworkers²⁹ replaced the residues in eight regions with the highest flexibility (B-factor), excluding the N- and C-termini, using random mutagenesis. Stabilizing substitutions (2-4 °C increase in T_m) occurred in the six regions where the residues shown as spheres.

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The second step is to identify stabilizing substitutions. One can make changes within the flexible region or next to flexible region. Interactions between amino acids require partners so either approach can yield improvements, but one comparison found larger stabilization with the replacements in the neighboring areas.^[48] Some researchers targeted the flexible regions with random mutations,^[47] while others used modeling to predict stabilizing substitutions.^[48]

Copy features from thermotolerant homolog. Proteins from microorganisms that live in extreme environments (such as high temperatures >80 °C; extremes of pH, high salt, high pressure) tolerate their extreme environment better than homologous proteins from mesophiles (organisms that grow best at moderate temperatures). One protein stabilization strategy is to transfer the amino acids responsible for this extra stability in proteins from extremophiles to the corresponding proteins from mesophiles. This difficulty of this strategy is identifying which amino acids to transfer since some sequence differences impart increased stability, but most of the differences reflect random genetic drift. One approach relies on amino acid sequence comparison within the extremophiles. The expectation is that stabilizing amino acids are conserved in the proteins from extremophiles, but missing from the target protein from mesophiles. This approach differs from the consensus sequence approach, identifies residues conserved among all homologs, including both those from extremophiles and mesophiles. One pitfall of this approach is that residues may be conserved within extremophiles because they are closely related, not because they contribute to stability. For example, comparison of a pectate lyase with four homologs from thermophiles identified nine substitutions common to the enzymes from thermophiles, but absent in the target enzyme.^[49] However, only one of these nine substitutions (Arg236Phe) significantly increased the thermostability of the target lyase (12-fold), likely by improving hydrophobic

packing.

A second approach relies on the structures of proteins from extremophiles and the identification of stabilizing interactions within them. The stabilizing interactions are not apparent from the structure, so identifying them requires modeling. For example, researchers identified which ion pairs in adenylate kinase from a thermophile contribute to stability using molecular dynamics simulations.^[50] During the simulated movement of the protein some ion pairs broke apart, while three of them remained tightly paired. Transfer of these tightly-paired ion pairs into a less stable homolog increased its melting temperature by 0.8 to 3.3 °C.

Stabilizing interactions are not apparent from a comparison of structures because no single factor dominates.^[51] Each protein uses a unique mixture of interactions to increase protein stability. For example, many proteins from thermophiles contain more extensive hydrogen bond networks to strengthen electrostatic interactions, but some proteins from thermophiles do not include more extensive hydrogen bond networks but are nevertheless stable at high temperature. Similarly, some, but not all, proteins from thermophiles contain increased atomic packing to strengthen hydrophobic interactions, increased numbers of ion pairs, shortened loops to minimize interactions with the solvent, increased occurrence of Ala in helices and increased oligomerization to enhance interactions between amino acids in the folded state. Other stabilizing features are not apparent in the folded structure because their effect is to destabilize the denatured state. This wide range of features strengthens the argument that there are multiple paths to protein stabilization.

Remove destabilizing electrostatic interactions on the protein surface. Charged residues create favorable electrostatic interactions between residues with opposite charge and unfavorable interactions between residues with the same charge. Optimizing

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these interactions in the folded protein, especially by removing destabilizing interactions on the surface of the protein, stabilizes the protein. For example, five substitutions on the surface of an acylphosphatase increased the Gibbs energy of unfolding by 2.2 kcal/mol without affecting catalytic activity.^[52] Three of these substitutions reversed the charge of the residues and two substitutions introduced new charges.

The two reasons to choose substitutions on the protein surface are that they are likely to be far from the active site and that they are likely to maintain good solvation. Avoiding mutations near the active site increases the probability that the modified protein will maintain catalytic activity (or binding in the case of a binding protein). Choosing residues with good solvation for substitution makes the prediction of stabilizing or destabilizing more accurate. The effect of a charged amino acid residue on protein stability depends on the differences in both electrostatics and solvation in the folded versus unfolded states. Since all residues are assumed to be well-solvated in the unfolded state, choosing residues that are well solvated in the folded state allows the prediction to ignore solvation in the comparison. One reason to avoid substitutions on the protein surface is that they may have a smaller effect on stability than substitutions in the interior of the protein. The environment of residues on the protein surface changes less drastically upon unfolding than the environment of residues in the interior of the protein. Nevertheless, residues on the surface move apart when the protein unfolds, thereby changing the electrostatic interactions.

TKSA-MC is a web-tool to identify residues for replacement due to unfavorable electrostatic interactions.^[53] It calculates the electrostatic contribution of all the charged residues in the protein and suggests replacing them if they make an unfavorable electrostatic contribution and lie on the protein surface. It does not suggest what the replacement amino acid should be, but uncharged

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polar residues or oppositely charged residues are the obvious choices. The electrostatic calculation requires a protein structure and considers the pH of the solution, the distance between the charges, different dielectric constants for the protein and solvent, and the fraction of the residue exposed to solvent. For example, TKSA-MC predicts that replacement of residues Asp36, Asp40, and Glu42 in Staphyloccocal binding protein would stabilize it, Figure **Figure 5.16**.

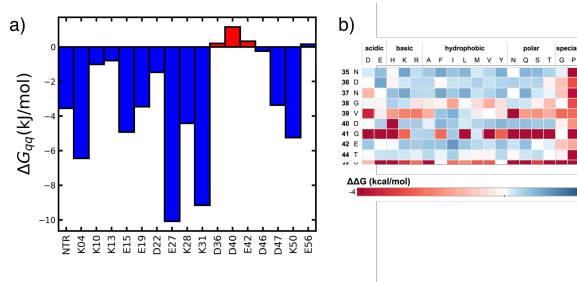


Figure 5.16. Stabilization of Staphyloccocal binding protein by replacement of destabilizing residues on the protein surface. a) Prediction of electrostatic interactions at pH 7 by web tool TKSA-MC based on the x-ray crystal structure (PDB id: 1pga). Positive values indicate destabilizing interactions. Replacement of the destabilizing residues filled in red (Asp36, Asp40, and Glu42), which indicates those on the surface, is predicted to stabilize this protein. b) Blue squares indicate replacements of amino acids in Staphyloccocal binding protein that stabilize the protein. In agreement with predictions, most replacements of Asp36, Asp40, and Glu42 lead to a more stable protein.

Most electrostatic interactions are stabilizing and only a few are destabilizing, so this approach identifies only a few locations for mutagenesis. The ability to predict new stabilizing interac-

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tions would yield additional locations for mutagenesis, but this approach has not been tested with this web tool.

Modeling to identify stabilizing substitutions. Rosetta^[54] and FoldX^[55] are modeling programs widely used to predict protein stability. Both are available online. The programs model the physical interactions of the atoms in the folded form, but Rosetta adds statistical analysis of different properties extracted from protein databases.

FoldX is force field specifically for predicting protein stability. FoldX requires an experimental structure of a protein as the starting point. It does not do geometry optimization or conformational searching; it only calculates unfolding Gibbs energy for the protein and variants of the protein. For protein engineering, it predicts substitutions that stabilize or destabilize a protein. The force field equation contains terms for steric clash and electrostatic interaction like the physics-based force fields, but it also contains terms for solvation and entropy, Equation 5.13.

$$\begin{aligned}
 \Delta G = & a \cdot \Delta G_{vdw} + b \cdot \Delta G_{solvH} + c \cdot \Delta G_{solvP} + d \cdot \Delta G_{wb} \Rightarrow \text{solvation terms} \\
 & + e \cdot \Delta G_{hbond} + f \cdot \Delta G_{el} + g \cdot \Delta G_{kon} \Rightarrow \text{electrostatic terms} \\
 & + h \cdot T\Delta S_{mc} + k \cdot T\Delta S_{sc} \Rightarrow \text{entropy terms} \\
 & + l \cdot \Delta G_{clash} \Rightarrow \text{steric clash term}
 \end{aligned}
 \tag{5.13}$$

These added terms estimate the Gibbs energies of various interactions of the amino acids with each other in the folded form as compared to interaction with solvent, which mimics the unfolded form. For example, the four solvation terms estimate the attractive van der Waals interaction between water and the protein (from the energy of removal of a protein atom from water to vapor phase), the desolvation of hydrophobic and polar groups upon folding

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(from the energy of removal of a protein atom from water to an organic solvent), and specific interactions with water when the water molecules that make more than two hydrogen bonds with the protein. In these cases the water molecules are included as atoms in the calculation. The terms in the FoldX force field were scaled with respect to each other to match the predicted Gibbs energy to experimental measurements of >1000 stabilizing and destabilizing amino acid substitutions. A FoldX calculation is part of the HotSpot Wizard web tool (Sumbalova et al., 2018) at: <http://loschmidt.chemi.muni.cz/hotspotwizard>.

Rosetta's equation for energy is a hybrid of simple terms that model physical interactions combined with knowledge-based terms to improve accuracy. These knowledge-based terms come from statistical analysis of known protein structures. For example, while FoldX estimates electrostatic interactions using Coulomb's law and hydrogen bonding terms, Rosetta also weights the estimate by the probability that the two charged atoms occur nearby in PDB structures. Molecular modeling methods typically identify substitutions to increase hydrophobic interactions and various electrostatic interactions. For example, Rosetta predicted improved hydrophobic interactions in cytosine deaminase with three substitutions (A23L, I140L, V108I), which increased the apparent melting temperature by 10 °C.^[56] The supporting information contains a detailed example of using Rosetta to identify a stabilizing substitution.

The energies from a molecular mechanics calculation like Rosetta are energy relative to a hypothetical unstrained molecule. A geometry optimization gives you a stable, reasonable structure, but the energy value needs a comparison value to make sense. For example, if you calculate the energy of the wt and a mutant, you can conclude which one of those folded structures is more stable. (The folded state is a collection of many conformations, so you are assuming that this single geometry-optimized structure is a good representative all folded structures that contribute to the folded state.)

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If you further assume that the energies of the unfolded states are identical, then the difference between the energies of the wt and mutant is the difference in unfolding energies. The units in Rosetta are 'Rosetta energy units' which are similar to kcal/mol.

The disadvantage of computational predictions of stabilizing substitutions is their low accuracy, often below 10%. Computer predictions are often incorrect either because the calculated energy of the molecular structure is incorrect or because the structure itself is incorrect and the protein favors a different conformation.

The first type of error is due to approximations used to speed up calculations (e.g., fixing the protein backbone, omitting explicit water molecules, ignoring entropic effects) and due to training of computers using small proteins; in short, the force field is inaccurate. To minimize this first type of error, Floor and coworkers^[48] suggested checking for the following in computer-predicted substitutions: 1) steric clashes, 2) internal cavities, 3) solvent-exposed aromatic residues, 4) solvent-exposed methionine, 5) a hydrogen bond donor or acceptor that has fewer hydrogen bonding interactions than in wild type, 6) a large hydrophobic patch on the protein surface, 7) uncompensated removal of a salt bridge, and 8) destabilization of an α -helix by removal of α -helix capping. The supporting information contains PyMOL scripts to check for some of the problems. Even with these checks the success rate for predicting substitutions to stabilize a haloalkane dehalogenase was only 9.2%.

The second type of error, an incorrect structure, is an error of conformational sampling. Computer modeling usually adjusts the residues near the substitution, but in reality a substitution may also affect more distant regions or cause more dramatic rearrangements in the main chain positions. Simulating alternative conformations using molecular dynamics with explicit water molecules adds these interactions and improves the success rate to

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13-19%.^[39,57] Finding these changes is computationally expensive since it requires long molecular dynamics simulations of protein motions. Machine learning approaches may be a faster alternative by using statistical patterns to accelerate conformational searching.^[58]

Even with a perfect structure of the folded protein, errors in predicting the effect of a substitution remain because the computations ignore the effect of the substitution on the unfolded protein state. Protein stability depends on the difference between the folded and unfolded states; ignoring the unfolded state is a major approximation.

Another approach to increasing success is to average the predictions of numerous methods. The assumption is that errors arising from different simplifications will cancel out. For example, the web server PROSS (**P**rotein **R**epair **O**ne-**S**top **S**hop^[59]) combines the homology search used by the consensus sequence approach with computational design. First, PROSS searches for homologs to identify which substitutions are allowed at each position. The rationale for this evolution-based constraint is to favor variants that maintain their original molecular function. While the consensus sequence approach predicts stabilizing substitutions from the frequency of occurrence, PROSS adds an energy calculation to predict which substitutions are best. This calculation requires a three-dimensional structure of target protein. The rationale for adding energy calculations is that individual proteins differ so that an amino acid residue that is suitable in most homologs may not be suitable in the target protein. For example, stabilization of acetylcholine esterase involved a replacement for glycine at position 416. There was no clear choice for a replacement based on the frequency of occurrence because nine amino acids, including glycine, appeared at this position with similar frequency. Computational modeling of the replacements using Rosetta eliminated the most commonly-occurring histidine because it fit poorly and

5.5 Stabilization to stepwise unfolding

suggested third-most-frequently occurring glutamine because it formed a hydrogen bond with a nearby tyrosine.

5.5 Stabilization to stepwise unfolding

So far the discussion assumed that the protein is a single domain protein that unfolds cooperatively, reversibly, and in a single step. Multiple domain proteins likely unfold stepwise with one domain unfolding first followed by other domains. For these proteins, stabilizing substitution must stabilize the domain that unfolds first. Proteins may also associate into oligomers. Oligomerization of folded proteins creates stabilizing interactions between amino acids at the protein-protein interface. (If the interactions were destabilizing, the proteins would not associate.) Creating additional interactions at the protein-protein interface is a new stabilization strategy available for oligomeric proteins.

5.5.1 Multiple domain proteins

Multiple domain proteins may unfold in a single step like single domain proteins, but more often they unfold stepwise with each domain unfolding separately. Stepwise unfolding creates an metastable intermediate with part of the protein folded and part unfolded. The stabilization must focus on the domain that unfolds first.

Bacterial cocaine esterase consists of three domains: an α/β -hydrolase domain, a cap domain, and a jelly roll domain. Molecular dynamics simulations indicated that the cap domain unfolded first. Two substitutions in the cap domain (T172R/G173Q) increased the half-life from ~ 0.2 h to 6 h at 37 °C,^[60] which corresponds to stabilization of 2.1 kcal/mol. Modeling suggested that the T172R

substitution strengthened interactions within the cap domain, while the G173Q substitution strengthened interactions between the cap and α/β -hydrolase domains by creating a new hydrogen bond.

Eijsink and coworkers dramatically stabilized a neutral protease from *Bacillus*, NprT, to self-degradation by stabilizing it to unfolding, Table **Table 5.3**.^[61] Proteolytic degradation proceeds from the unfolded form and does not depend strongly on amino acid sequence for this broad specificity protease, so stabilization to unfolding slows the proteolytic degradation. The substitutions increased the apparent melting temperature of the protein. Upon melting, the protease self-degrades rapidly. The stabilizing substitutions were either copied from a heat tolerant homolog or rationally designed to stabilize the protein. The half-life at 70 °C increased dramatically from <0.5 to 170 min, which corresponds to 4.0 kcal/mol. The researchers named this stabilized enzyme ‘boilysin’ because of its ability to remain active at 100 °C.

Table 5.3. Substitutions that stabilize neutral protease NprT to unfolding and self-proteolysis at >100 °C.

ΔT_m , °C	Substitution	How identified	Molecular basis for stabilization
+12.3	Ala69Pro Thr63Phe	heat tolerant homolog heat tolerant homolog	less flexible unfolded domain hydrophobic interaction

5.5 Stabilization to stepwise unfolding

$\Delta T_m, ^\circ\text{C}$	Substitution	How identified	Molecular basis for stabilization
+7.6	Gly58Ala Thr56Ala Ala4Thr	heat tolerant homolog heat tolerant homolog heat tolerant homolog	less flexible unfolded domain not determined not determined
+6.9	Ser65Pro 8-60Cys link	rational design rational design	less flexible unfolded domain less flexible unfolded domain

The stabilizing mutations all cluster in one region of the protein, Figure **Figure 5.17**. This multiple domain protein unfolds stepwise, and this region of the protein is the part that unfolds first and is then cleaved by other protease molecules. Stabilizing this region to unfolding was the key to preventing proteolytic degradation.

5.5.2 Oligomeric proteins

Dimerization or oligomerization of monomeric native state proteins creates new protein-protein interactions while decreasing protein-water interactions and therefore stabilizes the folded native form, Figure **Figure 5.18**. In contrast, if unfolded or partially unfolded forms dimerize and oligomerize, then these interactions stabilize the unfolded structures, which eventually leads to aggregation.

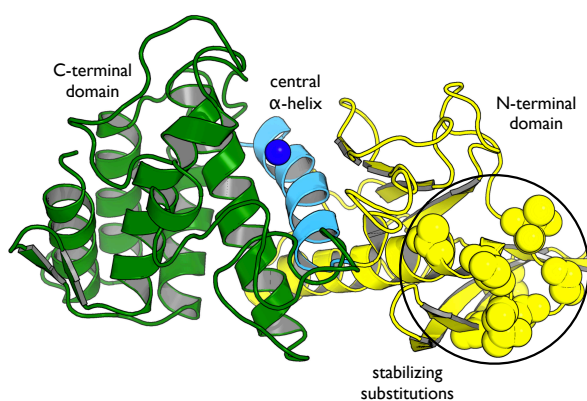


Figure 5.17. Cartoon representation of a model of thermolysin-like protease. The active site zinc ion (blue sphere) lies near the central α -helix (light blue) between the N-terminal domain (yellow) and C-terminal domain (green). The protease's stability depends on partial unfolding, so stabilizing substitutions (spheres within circle) cluster in that region: the N-terminal domain of the protein, in particular in the 55–69 surface loop. This model of protease NprT (P06874) created by SwissModel using the structure of thermolysin as the template, which has 87% identical amino acids.

5.5 Stabilization to stepwise unfolding

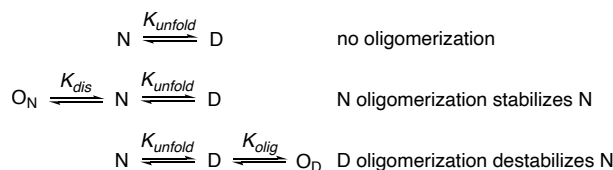


Figure 5.18. Dimerization or oligomerization of the native state (N) stabilizes the native state by shifting the overall distribution of conformations away from the denatured state (D). Strengthening interactions between native monomers stabilizes proteins to denaturation. In contrast, dimerization or oligomerization of the denatured state shifts the distribution of conformations away from the native state and eventually leads to aggregation. K_{dis} is the equilibrium constant for dissociation of native oligomers, O_N , to monomers; K_{olig} is the equilibrium constant for association of denatured form into oligomers, O_D .

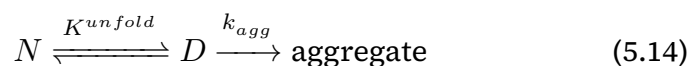
Substitutions at the oligomer interface that strengthen the interaction between monomers will stabilize proteins. In dramatic example, a replacement to remove a negative charge at the dimer interface of malate dehydrogenase (Glu165Gln or Glu165Lys) increased the melting temperature by 24 °C.^[62] Another approach is to covalently link the monomers into dimers by introducing a disulfide link. Comparison of the structures of a peptidase and a thermostable homolog identified a cross-link between monomers in the thermostable homolog. Adding this link to the peptidase increased its denaturation temperature by 30 °C.^[63]

5.6 Weaknesses other than unfolding

While unfolding is the weak point of most proteins, in some cases, other weaknesses dominate protein instability. Cooking irreversibly aggregates egg white proteins so that they do not refold and restore a liquid upon cooling. Other examples are chemical modification of proteins such as oxidation or degradation by proteases. These other weaknesses may be related to unfolding. Proteins unfold at least partially before they aggregate or are degraded by proteases. Chemical modification can promote unfolding. A different type of protein instability is short serum half-life, which depends on biochemical processes that clear proteins from the bloodstream.

5.6.1 Protein aggregation

Protein aggregation is the irreversible association of partially or fully unfolded proteins into insoluble particles. Aggregation-prone regions assemble by intermolecular β -structured interactions to form the core of the aggregate. Aggregation can occur when unfolded proteins encounter one another. For example, cooking an egg first unfolds the egg white proteins to expose hydrophobic regions to the solvent. These regions associate with similar exposed hydrophobic regions of other egg white protein molecules leading to oligomerization and eventually insoluble aggregates, eq. Equation 5.14, Figure **Figure 5.19**. Here k_{agg} is the aggregation rate constant.



5.6 Weaknesses other than unfolding

Aggregation can also occur under mild conditions. For example, overexpressing protein in bacteria creates high concentrations of unfolded protein. If protein folding is slow, the unfolded proteins can aggregate into insoluble particles called inclusion bodies. Biotherapeutic proteins can also aggregate during storage, which creates a danger of an immune response to the aggregates.

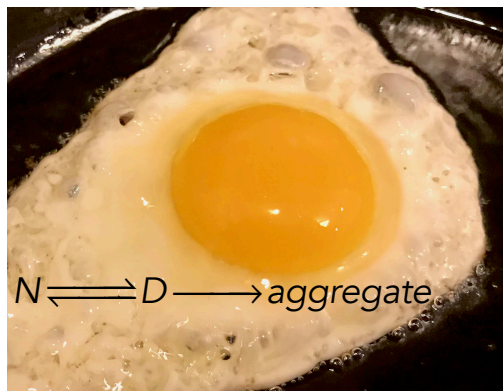


Figure 5.19. Cooking an egg involves heat-induced unfolding of the egg white proteins, followed by their aggregation into an insoluble gel. Egg white contains 10 wt% protein in water; this high concentration of protein promotes aggregation after unfolding.

Measuring a loss of function after heating identifies aggregation as a contribution to protein instability. For example, one measures the enzyme activity at room temperature, incubates the sample at an elevated temperature, then cools it to room temperature and measures enzyme activity again. The decrease in activity reveals the fraction of enzyme irreversibly unfolded due to heating. The values reported are typically the half-life at a specific temperature. For non-catalytic proteins, the change in intrinsic fluorescence can measure the amount of natively folded protein remaining. The ac-

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tivity loss depends on both how much of the protein unfolds upon heating (its inherent stability) and on its ability to refold upon cooling (which may be prevented by aggregation), eq. 3.x above.

To convert the measured loss of activity to half-life, researchers assume the inactivation follows first-order kinetics, eq. Equation 5.15, where A is the activity at time t , A_0 is the initial enzyme activity and k is the inactivation rate constant. The natural logarithm of enzyme activity ($\ln A$) decreases linearly with time. A plot of several measurements of the natural logarithm of the remaining activity at different times yields a straight line, with a slope of $-k$ and a y-intercept of $\ln A_0$.

$$\ln A = -k \cdot t + \ln A_0 \quad (5.15)$$

By measuring the inactivation rate constants for both the wild-type enzyme and the engineered variant, eq. Equation 5.16, yields the change in Gibbs energy of activation, $\Delta\Delta G^\ddagger$, for the rate of enzyme inactivation. If the mutant is more stable, then $\frac{k_{mut}}{k_{wt}}$ is less than one, so the value of $\Delta\Delta G^\ddagger$ is positive, indicating an increase in the activation energy to unfold the protein.

$$\Delta\Delta G^\ddagger = -RT \ln \left[\frac{k_{mut}}{k_{wt}} \right] \quad (5.16)$$

In most cases, the rate determining step in enzyme inactivation is the unimolecular unfolding step, so the assumption of first-order kinetics is justified. However, refolding of the enzyme may also depend on aggregation of the protein into an insoluble precipitate, so the rate determining step may involve several enzyme molecules. In these cases, the inactivation will not fit eq. Equation 5.15, which assumes first order kinetics. Another way to measure irreversible

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denaturation by heat is to measure an apparent melting temperature, $T_{m,app}$. It is not a true melting temperature, only an apparent one, because it is an irreversible process.

Two strategies to reduce aggregation are 1) to reduce unfolding (K_{eq} in eq. Equation 5.14) or 2) slow down aggregation (k_{agg} in eq. Equation 5.14). Minimizing unfolding of the protein will use strategies above in Section 5.4. For example, a combination of three substitutions designed to stabilize the native cytosine deaminase slowed its aggregation 30-fold.^[56]

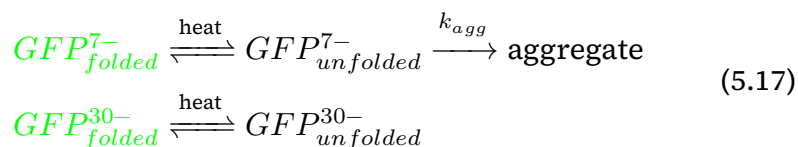
Since aggregation occurs mainly through hydrophobic interactions, one method to slow aggregation is to minimize the hydrophobicity of solvent-exposed regions. To reduce the aggregation of antibodies, Chennamsetty and coworkers^[64] first identified hydrophobic areas on the surface including hydrophobic patches that become exposed as the protein moves. Replacing hydrophobic residues with hydrophilic ones within these patches reduced aggregation. Several web tools predict substitutions needed to reduce protein aggregation: Aggrescan3D^[65] and Solubis.^[66] After predicting the aggregation-prone regions (hydrophobic, β -sheet forming stretches) the web tools predict substitutions to 1) reduce their unfolding and exposure to solvent by strengthening interactions between the aggregation-prone area and the rest of the protein or 2) eliminate the aggregation propensity of the region. These substitutions are charged residues (R, K, D, or E) which reduce hydrophobic interactions needed for aggregation or proline, which hinders the formation of β -sheet structures in aggregates. The tools use FoldX to choose the substitution that is expected to best stabilize the folded form.

Hindering aggregation can even overcome a decrease in stability. Substitution A281E in *Candida antarctica* lipase B decreased its melting temperature from 58 to 51 °C indicating a decline in stability. However, this substitution increased the half-life of this

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enzyme at 70 °C 22-fold demonstrating a reduced propensity to aggregate.^[67] This substitution makes a hydrophobic region of the enzyme less hydrophobic and less likely to aggregate.

A more drastic approach to reducing the association between protein molecules is supercharging. Supercharging is extensive substitutions on the protein surface to increase the net charge as high as +48, so the protein molecules repel one another and cannot aggregate.^[68] Introducing large numbers of charge residues on the protein surface also reduce its hydrophobicity as charged amino acids replace uncharged amino acids. For example, green fluorescent protein (GFP, net charge = -7) loses its fluorescence at 100 °C due to unfolding, Equation 5.17. Cooling to room temperature does not restore fluorescence because the unfolded protein has aggregated and precipitated. In contrast, green fluorescent protein variants engineered to have a +36 or -30 net charge also lost their fluorescence upon heating to 100 °C due to unfolding, but upon cooling, they did not aggregate and regained 62 and 28%, respectively, of their original fluorescence. This supercharging did not make the GFP variants more stable; in contrast, the GFP variants unfolded more readily than wild-type in urea, but supercharging did reduce their propensity to aggregate in the unfolded state.



Two disadvantages of supercharging are the potential to destabilize the protein and to disrupt its function. Residues with the same charge create unfavorable electrostatic interactions, which destabilize proteins. A change in electrostatic environment can also shift the pK_a of residues in the active site, which may affect binding or catalysis. For example, catalysis (k_{cat}) by a

5.6 Weaknesses other than unfolding

supercharged glutathione-S-transferase (net charge -40 for the dimer) was three-fold slower than wild-type (net charge +2 for the dimer).

5.6.2 Chemical modification

The side chains of three amino acids readily undergo spontaneous chemical modification: 1) asparagine deamidate to aspartate, 2) methionine oxidizes to a sulfoxide and 3) cysteine oxidizes to disulfide protein oligomers as well as sulfur oxides such as sulfenic acids (RS-OH). Replacement of problematic residues with a non-reactive residue eliminates the possibility of modification, but replacement of every asparagine, methionine, and cysteine in a protein is rarely needed. Many of the residues may not undergo modification, and even when they do, some modified residues will have little effect on protein properties. Spontaneous deamidation of asparagine to aspartate is a common chemical modification of proteins. The most reactive are Asn-Gly sequences in sterically unhindered regions which have half-lives as short as 6 d in physiological conditions.^[69] The least reactive asparagines are 10^5 -fold more stable. Deamidation converts the neutral Asn residue to a negatively charged Asp residue. This change can impair function, destabilize the protein or have no effect. The web tool at www.deamidation.org predicts asparagine deamidation rates based on the 3-D structures of proteins. Glutamine can also deaminate to glutamate, but the rate is about a thousand-fold slower, except for the case of an N-terminal Gln, which deaminates at a rate similar to Asn. Oxidation of sulfur atoms in methionine and cysteine alters a proteins properties and may destabilize or inactivate it. The first protein engineering of an industrial enzyme was the removal of an oxidation sensitive methionine from the detergent protease subtilisin.^[70] The existing subtilisin could tolerate most of the harsh conditions of laundry (heat, high pH, and detergent),

but it could not tolerate bleach. Bleach oxidized a methionine near the active site to the methionine sulfoxide (R-S(O)-CH₃), which hindered binding of the substrate proteins to inactivate the protease. Replacement of this problematic methionine with alanine created a bleach-tolerant protease. A cysteine-to-serine replacement stabilizes several cytokine drugs. This change does not affect biological activity, but avoids oligomerization by the formation of non-native intermolecular disulfide links between proteins. The specific activity of interferon- β , when expressed in *E. coli*, was about 10-fold lower than the native protein. The researchers hypothesized that oligomerization through intermolecular disulfide bonds caused the lower activity.^[71] Replacement of Cys17 with Ser eliminated oligomerization and restored the specific activity to that of the native protein. The commercial drug, Betaseron[®], contains this substitution. A similar Cys125Ser substitution stabilizes human interleukin 2, marketed as Proleukin[®].

Some applications require proteins to work under conditions where other chemical modifications are possible. For example, glucose, an aldehyde, can react with lysine to form an imine. The enzymatic conversion of corn syrup (glucose) to high fructose corn syrup to increase its sweetness requires the enzyme xylose isomerase to tolerate high concentrations of glucose. Stabilization involved replacing a surface lysine that reacted with the glucose.^[72] At very high or low pH, irreversible chemical reactions can degrade the protein. For example, base-catalyzed β -elimination at pH > 8 alters cystine (disulfide) residues' side chains. Peptide backbone links next to aspartic acid residues can hydrolyze at pH < 4.

5.6.3 Serum half-life

Extending the serum half-life of a therapeutic protein enhances its efficacy because it remains active for a longer time. It also lowers cost because less therapeutic protein is required and improves delivery because injections are less frequent.^[73,74] Three biochemical processes limit the serum half lives of proteins: (i) degradation by proteolytic enzymes, (ii) rapid filtration by the kidneys and (iii) receptor-mediated endocytosis. Small proteins (<50 kDa) typically have short serum half lives (5–50 min).

To stabilize proteins to proteolysis researchers identify the cleavage sites and modify them to slow hydrolysis. For example, human glucagon-like peptide-1 (GLP-1) has a serum half-life of only 2-3 minutes, in part due to rapid proteolytic cleavage by dipeptidyl peptidase 4 between the 8Ala-9Glu linkage, **Figure 5.20a**. To slow down proteolysis researchers replaced the alanine residue with glycine (removal of the methyl group at C α) or with 2-aminoisobutyric acid, Aib (addition of another methyl group at C α).^[75] Researchers replaced the alanine, not the glutamate because proteases are more selective for the amino acid on the carbonyl side of the amide bond rather than the amino acid on the amine side of the amide bond. Other protease that occur in serum besides dipeptidyl peptidase 4 are neutral endopeptidase, plasma kallikrein, and plasmin.

To slow filtration by the kidneys, researchers increase the molecular weight of the therapeutic protein to above ~50 kDa. In the case of GLP-1, researchers increased the molecular weight by adding an octadecanoic diacid group, which binds to serum albumin, which has a molecular weight of 66.5 kDa, **Figure 5.20b** above. The engineering of additional glycosylation sites or the chemical addition of poly(ethylene glycol) chain are other ways to increase the molecular weight of a therapeutic protein.

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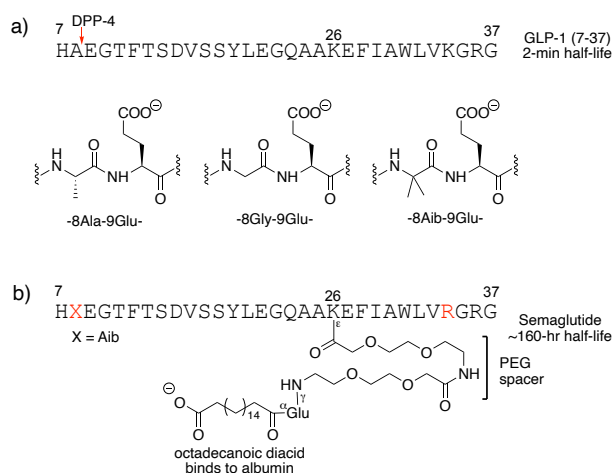


Figure 5.20. Structural modification of GLP-1 to increase its serum half-life. a) To prevent hydrolysis by the serum protease dipeptidyl peptidase 4, which favors hydrolysis at 8Ala-9Pro (marked by red arrow), researchers replaced 8Ala with Gly or the unnatural amino acid Aib. b) To prevent filtration by kidneys and to minimize receptor-mediated endocytosis, researchers attached a hydrophobic C18 group that binds to serum albumin. The red letters indicate changes in the peptide sequence from GLP-1.

5.6 Weaknesses other than unfolding

The choice of endogenous albumin as a binding protein also minimizes the third process that lowers serum half-life, receptor-mediated endocytosis. Endothelial cells lining the bloodstream internalize proteins via receptor-mediated endocytosis, which transfers the proteins from the blood to adjacent cells. However, some proteins, such as serum albumin and immunoglobulin G (IgG) are recycled back to the bloodstream. Upon endocytosis proteins transfer to the endosome, then to the lysosome for degradation. However, the endosome is acidic and contains FcRn receptor proteins, which tightly bind albumin and the Fc region of IgG proteins. Instead of degradation, this complex moves back to the cell surface where the higher pH of the blood weakens the binding between albumin or IgG and FcRn, causing release of the albumin or IgG back to the bloodstream. This recycling mechanism accounts for the long half-life of albumin and IgG in the blood. The modified form of GLP-1 in Figure **Figure 5.20b** above has a half-life of ~160 hours (3,200-fold longer than GLP-1) is an treatment for type 2 diabetes (Ozempic[®]) and obesity (Wegovy[®]).

Etanercept is another example where receptor-mediated recycling extends the serum half-life, but in this case using the Fc fragment of IgG, Figure **Figure 5.21**.^[76] The active protein is the extracellular domain of p75 tumor necrosis factor receptor, which binds the tumor necrosis factor cytokine to reduce inflammation in rheumatoid arthritis. Two copies of the active protein increase its affinity 100-4000-fold over monomeric counterparts, likely by better mimicking the dimeric structure of the receptor. These two active proteins are fused to the Fc fragment of human IgG, which increases serum half-life by enabling recycling of filtered protein back to the bloodstream.

Another approach to increasing the serum half-life of therapeutic proteins is to create insoluble depot of the protein that slowly releases into the bloodstream. One example is the noncovalent entrapment of a GLP-1 homolog in poly(D, L-lactide-co-glycolide) to

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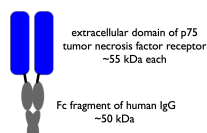


Figure 5.21. Etanercept is a fusion protein of two copies of the extracellular domain of tumor necrosis factor receptor (blue) with a single copy of the Fc fragment of IgG (gray). It is used to reduce inflammation of rheumatoid arthritis.

form 60 μm microspheres. Diffusion and hydrolysis of the polymer release the GLP-1 homolog with a half-life of 5-6 days. Another example is a long-acting insulin called insulin glargine. The protein sequence has been modified to alter the isoelectric point so that it precipitates as microcrystals at physiological pH and then slowly dissolves over twenty-four hours. The modified insulin is formulated in pH 4 solution where it is soluble, but then precipitates upon injection in the muscle.

5.7 Concluding remarks

Most single substitutions increase the stability of a protein by ≤ 1 kcal/mol (3–4 $^{\circ}\text{C}$ increase in melting temperature), so large stabilizations require multiple stabilizing mutations. In many cases, especially when the substitutions are far from one another, their stabilization effects will be approximately additive. For example, a heat-stabilized α -amylase contains at least ten amino acid substitutions, Table **Table 5.4**.^[77] This enzyme catalyzes the hydrolysis of starch to glucose oligomers in the manufacture of corn syrup. High temperatures (90 $^{\circ}\text{C}$) increase the solubility of the starch and speed up hydrolysis, but require a heat-tolerant α -amylase. Stabilizing substitutions include replacement of amino acid residues that can un-

5.7 Concluding remarks

dergo chemical modification, substitutions to stabilize the native form, substitutions to destabilize the unfolded form and substitutions discovered by random mutagenesis where the stabilization mechanism is unknown.

Table 5.4. Heat-stabilizing substitutions in α -amylase from *Bacillus licheniformis*.

Approach	Specifics
prevent chemical modification	remove oxidation (M197) and deamidation sites (N188, Q284)
stabilize N	bury Ca^{2+} ion (A181T), minimize electrostatic destabilization (H156Y)
destabilize D	introduce proline (R124P)
random mutagenesis	M15T, H133I, N199S, A209V

In another example, twelve substitutions in a halohydrin dehalogenase increased its apparent melting temperature by 28 °C and increased its ability to tolerate organic solvents.^[57] The researchers first identified stabilizing single substitutions and then combined the best twelve into a single variant. The increased stability was mainly due to redistributed surface charges and improved interactions between subunits in this homotetrameric enzyme.

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Supporting Information

SI 5.1. Derivation of equation relating spectral changes and equilibrium constant, eq. Equation 5.4.

Deriving the relationship between spectral changes and equilibrium constant is straightforward. The protein is either in the native state or in the denatured state, so the sum of the two fractions is one: $F_N + F_D = 1$. The observed fluorescence at each urea concentration, Y_{obs} , is the sum of the fluorescence contributions from the native state, $Y_N \cdot F_N$, where Y_N is the fluorescence intensity from

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the pure native state, and the denatured state, $Y_D \cdot F_D$, where Y_D is the fluorescence intensity from the pure denatured state.

$$Y_{obs} = Y_N \cdot F_N + Y_D \cdot F_D$$

Substituting $F_N = 1 - F_D$ yields:

$$F_D = (Y_{obs} - Y_N) / (Y_D - Y_N)$$

Similarly substituting $F_D = 1 - F_N$ yields:

$$F_N = (Y_D - Y_{obs}) / (Y_D - Y_N)$$

Dividing these two equations yields the equation below, which is the same as eq. Equation 5.4.

$$K^{unfold} = \frac{F_N}{F_D} = \frac{Y_{obs} - Y_N}{Y_D - Y_{obs}}$$

SI 5.2. Example of linear extrapolation of protein unfolding data in urea Protein A was dissolved in solutions of different concentrations of urea and allowed to reach equilibrium. The fluorescence spectra of these solutions showed an increase in fluorescence at 2-5 M urea, Figure **Figure 5.22**. Using this fluorescence data, calculate the Gibbs energy of unfolding in pure water for protein A.

To solve this problem, first estimate the fluorescence of the native and denatured states. The fluorescence of the native state is the flat part of the curve before unfolding begins. The average of the values at 0 and 1 M urea yields $Y_N = 71$. The fluorescence of the denatured state is the flat part of the curve after unfolding is complete. The average of the values at 6-9 M urea yields $Y_D = 155$.

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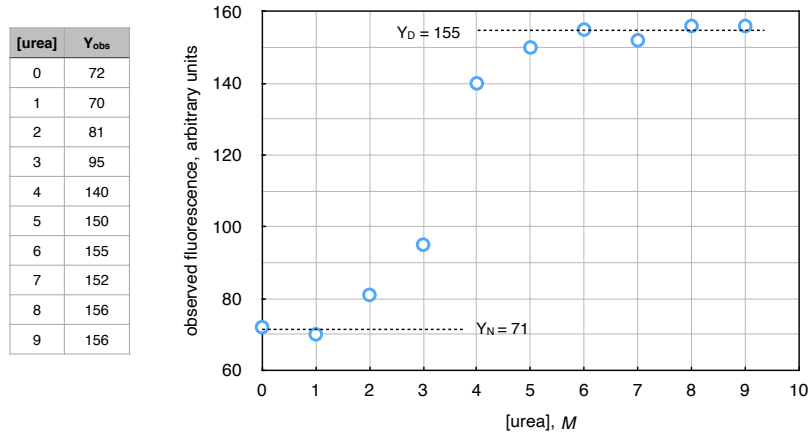


Figure 5.22. Typical experimental data from a urea-unfolding experiment to measure the Gibbs energy of unfolding of a protein. Depending on the protein and the wavelength used to monitor the fluorescence, the intensity of fluorescence can either increase or decrease upon protein unfolding; in this example, it increases.

Next, estimate the equilibrium constant between the native and denatured forms at each urea concentration in the unfolding region. For this example: $K^{unfold} = (Y_{obs} - 71)/(155 - Y_{obs})$, so one can calculate K^{unfold} at different urea concentrations by substituting the experimental values of Y_{obs} . This procedure yields the values in Table **Table 5.5**. Convert the equilibrium constants to Gibbs energy using $R = 1.987 \text{ cal/mol } ^\circ\text{K}$ and $T = 298 \text{ } ^\circ\text{K}$, which yields the ΔG^{unfold} values in the table.

Table 5.5. Calculated values of K^{unfold} and ΔG^{unfold}

[urea] (M)	K^{unfold}	ΔG^{unfold} (kcal/mol)
2.0	0.14	1.2
3.0	0.40	0.54
4.0	4.6	-0.90
5.0	16	-1.6

Finally, plot the ΔG^{unfold} values as a function of urea concentration, fit the data to a straight line and extrapolate to pure water ([urea] = 0), Figure **Figure 5.23**.

The extrapolation yields a y-intercept of +3.3 kcal/mol, which is the Gibbs energy of unfolding of this protein in pure water. This protein is less stable than typical proteins, but even for this one, the equilibrium amount of denatured protein in pure water is only 0.38%, which would be difficult to measure directly.

Problems

5.1. The following are examples of interactions that can stabilize proteins: a) improved core packing, b) higher backbone rigidity, c)

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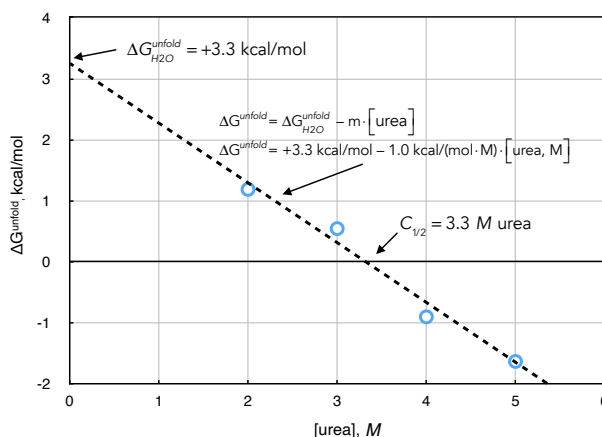


Figure 5.23. Linear extrapolation plot of the data from Table **Table 5.5** above. The best fit line to the data is $\Delta G^{unfold} = 3.3 - 1.0 \cdot [\text{urea}]$, which indicates that $\Delta G_{H_2O}^{unfold}$ is 3.3 kcal/mol.

increased surface polarity, d) increased salt bridges, e) increased hydrogen bonds between amino acids, f) replace residues with proline, g) increase number of charged amino acids. For each interaction, account for the stabilizing effect by describing the effect that it has on the folded protein and on the unfolded protein.

5.2. a) Floor and coworkers* used the ‘Dynamic Disulfide Discovery algorithm’, which they developed previously,[†] to predict disulfide cross link locations. That algorithm yielded seven residues pairs. (You do not need to look up the details of this algorithm.) The Dynamic Disulfide Discovery algorithm excludes residue pairs that

*Floor, R. J., Wijma, H. J., Colpa, D. I., Ramos-Silva, A., Jekel, P. A., Szymanski, W. *et al.* (2014) Computational library design for increasing haloalkane dehalogenase stability. *ChemBioChem* **15**, 1660-72.

[†]Wijma *et al.* (2014) Computationally designed libraries for rapid enzyme stabilization. *Prot. Eng. Des. Sel.* **27**, 49-58; <http://doi.org/10.1093/protein/gzt061>

are within 15 positions of each other. Explain why this is a good idea. If you apply this criterion to the results of SSBOND, how many residues pairs are still suitable for engineering?

- b) Besides excluding residues pairs that are within 15 positions of each other, suggest another criterion that you could use to exclude a suggested pair. Provide a rationale for your criterion.
- c) The apparent melting temperatures for the wild type and two of the variants are listed below. In which variant did the disulfide bond stabilize the protein? Explain how you came to your conclusion. For the variant that did not stabilize the protein, suggest a reason why it did not work based on the data in the table. Explain your reasoning. *Hint*: Consider the structure of amino acids R and D.

Substitutions	T_m (oxidized)	T_m (reduced)
none (wt)	51 °C	49 °C
A5C/A185C	56 °C	49 °C
R20C/D70C	50 °C	45.5 °C

- d) Compare a prediction to a measured value. For the prediction, use equations 5.10 and 5.11 in the text to predict the expected Gibbs energy of stabilization for the two variants in part d above. For the experimental values, estimate the measured Gibbs energy of stabilization from the change in T_m using equation 5.8 (page 104) of the text. Comment on the match or mismatch between predictions and measured values. For the temperature, use the melting temperature for the variant that you are comparing.

5.3. *Predicting stabilizing single substitutions with Rosetta.* Jones and coworkers identified several substitutions that stabilize the esterase SABP2.^[78] Table **Table 5.7** lists four of the best substitutions. The increase in thermostability was measured two ways. The first method measured the time for irreversible inactivation at 60 °C. The sample was incubated at 60 °C for 15 min, then cooled to room temperature and the remaining catalytic activity was measured. The loss in catalytic activity was assumed to be first order in enzyme concentration and the expected half-life was calculated. Wild-type SABP2 had a half-life of 3.5 min. The single substitution variants in Table **Table 5.7** all showed a longer half-life. The second two variants showed a larger increase in the half-life than the first two variants. The second measure of thermostability involved unfolding of the protein in urea and compared the concentration of urea where the protein was half unfolded, [urea]_{1/2}. Wild-type SABP2 was half-unfolded at 2.23 M urea. The variants unfolded at higher concentrations of urea. The increase in urea concentration needed to half-unfold the protein was higher for the first two variants than for the second two.

Table 5.7. Two different measures of which single substitutions increase the thermostability of esterase SABP2.

Substitution	Increase in half-life	
	at 60 °C	Increase in [urea] _{1/2}
L65F	1.2-fold	+1.5 M
E215P	1.5-fold	+1.0 M
Q221M	6.6-fold	+0.4 M
A230V	3.9-fold	+0.4 M

This question tests how well Rosetta predicts these four stabilizing

substitutions. A webtool for Rosetta is available at ROSIE2 (Rosetta Online Server that Includes Everyone). In particular, we will use the tool stabilize-PM (point mutations), which predicts stabilizing single substitutions.^[79] Using the tool is simple; one specifies the pdb code for the protein (1y7i for SABP2) and which positions you would like to mutate (A[65,215,221,230]), which means the four positions listed in chain A). By default, Rosetta tests 19 amino acids (all except cysteine) as replacements. During the calculations, Rosetta allows the two amino acids on either side of the mutated position to move freely, all amino acids within 10 Å of the mutated position to move under weak constraints and all remaining amino acids to move only slightly (strong constraints). The results of the calculations (heat map in Figure **Figure 5.24**) show only one predicted stabilizing substitution, Leu65Phe, which is one of the experimentally identified stabilizing substitutions.

- a) Thermostability was measured two different ways, which gave differing results about which substitution was the most stabilizing. Which of these experimental values should be compared to the Rosetta calculations? Explain your choice.
- b) Rosetta failed to predict the Glu215Pro stabilizing substitution. Explain what was omitted from the Rosetta calculation that could have caused this error.

You may explore the Rosetta web tools for your own protein engineering project. The webtool requires login with a Github account, but is otherwise simple to use.

Answers

5.2. a) The stabilization increases with the size of the ring formed. If the residues are within 15 positions, the expected stabilization is only 3.1 kcal/mol (using eq 3.13 in the text, $n = 16$, $T = 300$ °K) and

5 Engineering More Stable Proteins

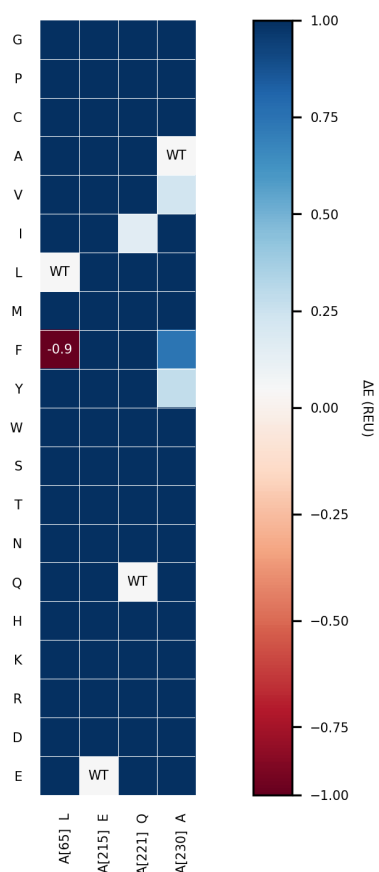


Figure 5.24. Heatmap shows the difference in score between the wild-type residue and each substitution (y-axis) for a given position (x-axis). Numerical values for the stabilizing scores (negative values) are shown, but only varying shades of blue are shown for the destabilizing scores (positive values). Scores greater than +1 are assigned the same navy color. This calculation predicted only one stabilizing substitution: Leu65Phe.

the actual stabilization is often less, so focusing on the larger rings increases the likelihood of success.

- b) Avoid pairs near the active site. Avoid pairs like 2 where you replace an ion pair (Arg, Asp) which stabilizes the folded state. The replacement would destabilize both the folded and unfolded states leading to no net stabilization.
- c) The A5C/A185C cross link stabilizes the protein because the melting temperature of the oxidized form increases by 5 °C as compared to wt. The R20C/D70C cross link destabilizes the protein because the melting temperature of the oxidized form decreases by 1 °C as compared to wt. Replacing the R/D pair with a cystine cross link removes an ion pair from the folded protein. The replacement destabilized the folded form (mp of reduced form is 3.5 °C lower than for wt).
- d) A5C/A185C, $n = 181$, $T = 327.5$ °K, predicted $\Delta\Delta G = 5.8$ kcal/mol; observed $5/3.6 = 1.4$ kcal/mol; also poor agreement. The cysteines seem to fit because the mp of the reduced form is unchanged from wt. The crosslink may not restrict the flexibility as much as expected.

R20C/D70C, $n = 51$, $T = 323.5$ °K, predicted $\Delta\Delta G = 4.5$ kcal/mol; observed $-1/3.6 = -0.28$ kcal/mol. The poor agreement may be due to removal of a ion pair from the folded form. Removal destabilizes folded form, but has no effect on the unfolded form because unfolding breaks the ion pair.

6 Engineering Tighter Binding

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Summary. Tighter binding of proteins like antibodies to their target can increase the therapeutic effectiveness of biopharmaceuticals and lower the detection limits of diagnostic tests. Engineering a tighter-binding protein involves increasing the complementarity between the protein binding site and target. Both the shapes of the interacting surfaces and non-covalent interactions between them should match.

Key learning goals

- The relative stabilities of the bound state, where the protein and ligand interact with each other, and unbound state, where the protein and ligand interact with water, determine the strength of binding. Non-covalent interactions between the binding site and ligand favor the bound state, while solvation of the binding site and ligand favors the unbound state.
- Equilibrium dissociation constants, K_d , which have units of molarity, reveal the binding strength. K_d corresponds to the target concentration where half of the protein is bound to a ligand. Lower dissociation constants indicate tighter binding. Equilibrium dialysis is one technique to measure equilibrium dissociation constants.

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- Increasing the affinity of a protein for a ligand requires increasing the shape and interaction complementarity between the two partners. Close contact between the surfaces creates non-covalent interactions including van der Waals interactions, hydrophobic interactions, electrostatic interactions, and hydrogen bonds. The high cost of desolvating polar groups often makes burying them at the binding-site-ligand interface unfavorable even when they make favorable interactions like hydrogen bonds.
- Binding selectivity is the ability of a protein, often an antibody or receptor, to distinguish between two ligands. Binding selectivity depends on the ratio of the equilibrium association constants for the two binding possibilities.

I am always doing what I can't do yet in order to learn how to do it. — Vincent van Gogh

6.1 Introduction

Tighter binding of a protein to its ligand is a common goal of protein engineering.^[1] A ligand is any species (usually a small organic molecule, an ion, a protein or an oligonucleotide) that binds selectively, stoichiometrically and reversibly to a protein. Increased binding of a diagnostic antibody to its antigen can lower detection limits (e.g., a pregnancy test could detect pregnancy earlier), while increased binding of a therapeutic antibody can decrease drug dosage or increase drug efficacy. Traditional methods to improve the binding of antibodies rely on the natural processes that generate antibodies. Mice are immunized with the antigen followed by screening for antibodies with higher affinity. These *in vivo* methods, called affinity maturation, reach a limit where affinity stops increasing due to biological limits of the innate

6.1 Introduction

immune system. A common protein engineering goal is improving the affinity of an antibody from good binding to great binding. Figure **Figure 6.1** defines the different parts of an antibody.

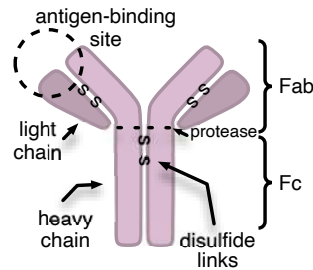


Figure 6.1. Antibodies consist of four protein chains: two identical heavy chains (~ 440 aa each) and two identical light chains (~ 220 aa each). Disulfide bonds connect the two heavy chains to each other and the light chains to their neighboring heavy chain. Antibodies contain two antigen-binding sites, each composed of the tips of the light and heavy chains. One of the two antigen binding sites is circled. Treatment with proteases cleaves both heavy chains as marked into three fragments: two Fab fragments (Fragment antigen-binding) and one Fc portion (Fragment crystallizable). The two protein chains that make up the Fab and Fc fragment remain linked to each other via disulfide cross links.

The strength of binding a ligand to a protein depends on the equilibrium between the bound and unbound states, Figure **Figure 6.2**. In the bound state the protein and ligand interact with each other, while in the unbound state they interact with solvent. Stronger non-covalent interactions with each other as compared to solvent favors binding.

Two features influence the interactions between protein and

6 Engineering Tighter Binding

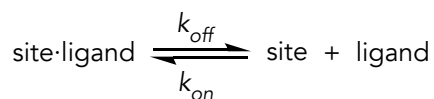


Figure 6.2. The bound state (ligand bound to the binding site) equilibrates with the unbound state (solvated ligand and site).

ligand. First, the size and shape of the binding site in the protein must match the size and shape of the ligand. This match creates strong van der Waals interactions between them. Second, the chemical properties of regions within the binding site match those of ligand. Hydrophobic regions match and electrostatic interactions in polar regions are favorable. X-ray structures of antibody-antigen complexes reveal that 15-20 amino acids from the complementarity determining region of the antibody contact the antigen. Most of the contacts are hydrophobic and both the antibody and antigen adjust their conformations slightly upon complex formation to maximize the strength of the interactions. In the perfect case, no adjustments would be necessary since the most stable conformations of both protein and antibody match each another exactly.

The two comparison states that determine ligand-protein binding strength are the unbound state where water solvates the independently diffusing ligand and protein and the bound state where the ligand and protein interact with each other, Figure **Figure 6.3**. The solvation of polar and charged atoms at the surface stabilize the unbound state. In addition, the ability of the ligand and protein to diffuse independently in solution favors the unbound state. In contrast, the ordering of water at exposed hydrophobic regions of the protein and ligand destabilizes the unbound state as do unfavorable electrostatic interactions in high net charge regions.

In contrast, the bound state is stabilized by van der Waals attrac-

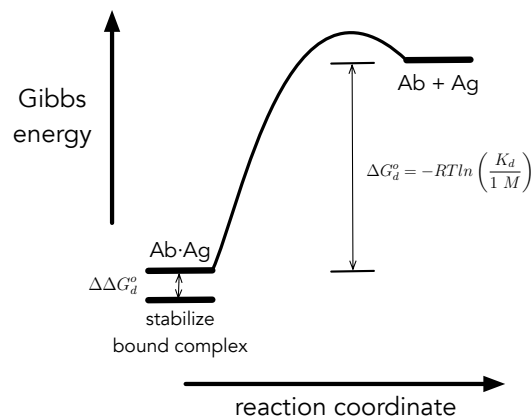


Figure 6.3. Gibbs energy changes associated with binding an antigen (Ag) to an antibody (Ab). The complex, Ab·Ag, is lower in energy than the free species, Ab + Ag, indicating that binding is favorable. Stabilizing the Ab·Ag complex increases the binding strength by an amount $\Delta\Delta G_d^o$.

6 *Engineering Tighter Binding*

tion between touching atoms of the ligand and protein due to shape complementarity, the release of ordered water from hydrophobic regions of ligand and protein, electrostatic interactions between oppositely-charged atoms, and hydrogen bond formation between protein and ligand. The bound state is destabilized by desolvation of buried polar and charged atoms, electrostatic interactions between like-charged atoms, and the loss of independent diffusion for ligand.

Increasing binding requires increasing the energy difference between the bound and unbound states by stabilizing the bound state, destabilizing the unbound state or both. For example, one could destabilize the unbound state by adding exposed hydrophobic surface or high net charge regions within ligand-binding region. Ligand binding can alleviate this destabilization by binding a complementary hydrophobic surface or opposite charges. This alleviation of the destabilization when the ligand binds is critical to the success of this approach. If the destabilization persists in the bound state, then both states will have been destabilized and the binding strength will remain unchanged. Thus, the destabilizing changes must influence the ligand-binding region so that binding can alleviate the destabilization. One can stabilize the bound state relative to the unbound state by increasing the shape and charge complementarity between the protein and ligand stabilizes the bound state, while having little effect on the unbound state.

In all cases one must consider the effect of change to both the bound and unbound states. For example, engineering a hydrogen bond between the ligand and protein comes at the cost of reduced solvation of the polar atoms involved in the hydrogen bond. If the hydrogen bond is solvent accessible, then it remains solvated in the bound state and is often stabilizing. In contrast, buried hydrogen bonds are usually destabilizing because the cost of desolvating the polar atoms exceeds the gain from the hydrogen bond.

6.2 Measuring binding

Defining binding strength. The strength of binding is measured by the equilibrium constant for the reverse reaction, the ligand dissociation equilibrium constant, K_d , Figure **Figure 6.2** and eq. Equation 6.1. For tighter binding, the dissociation equilibrium constant should be more unfavorable, so lower values of K_d indicate tighter binding. K_d has the units of molarity.

$$K_d = \frac{k_{off}}{k_{on}} = \frac{[site] \cdot [ligand]}{[site \cdot ligand]} \quad (6.1)$$

The value of K_d corresponds to the free ligand concentration when half of the sites contain a bound ligand and the other half are empty, eq Equation 6.2. Starting from eq. Equation 6.1, when concentration of empty sites, $[site]$, equals the concentration of filled sites, $[site \cdot ligand]$, these two terms cancel in eq. Equation 6.1 yielding eq. Equation 6.2.

$$K_d = [ligand] \text{ when } [site] = [site \cdot ligand] \quad (6.2)$$

Antibody-antigen complexes are common cases of protein-ligand binding. For example, the mouse monoclonal antibody called 26-10 binds digoxin with high affinity. Digoxin, a hydrophobic steroid with an attached carbohydrate, is used to treat congestive heart failure. Anti-digoxin antibodies have been used to measure serum levels of digoxin and also to neutralize digoxin in cases of an overdose. Since antibody 26-10 contains two independently-acting binding sites for digoxin, the concentration of sites is twice the concentration of antibody. Antibody 26-10 binds digoxin very tightly ($K_d = 10^{-10} M$) and is half-saturated with digoxin when the concentration of free digoxin is $10^{-10} M$ (only 100 picomolar).

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To convert the equilibrium constant in eq. Equation 6.1 into a Gibbs energy, one introduces the notion of a reference state. The purpose of the reference state is to cancel out the molarity unit on the equilibrium constant since one can only apply the logarithm function to dimensionless quantities. The standard reference state is 1 M and the value of K_d is divided by this reference yielding a dimensionless quantity, eq. Equation 6.3.

$$\Delta G_d^o = -RT \ln(K_d/1M) \quad (6.3)$$

The values of K_d for stable protein-ligand binding interactions are always less than 1 M, so the ratio $K_d/1M$ is always less than one. The logarithm of a value less than one is negative, so the Gibbs energy for dissociation, ΔG_d^o , eq. Equation 6.3, is positive indicating that dissociation is unfavorable. The 'o' superscript on the ΔG_d^o indicates that the K_d is compared to a reference state of 1 M. The free energy of dissociation of the digoxin-antibody 26-10 complex, ΔG_d^o , is +14.2 kcal/mol indicating that dissociation is highly unfavorable.

The Gibbs energy diagram shows that dissociation of an antibody-antigen complex is unfavorable, Figure **Figure 6.3** above. To improve binding one seeks to make dissociation even more unfavorable. This approach is analogous to stabilizing a protein by making the unfavorable unfolding reaction even more unfavorable in the previous chapter.

Measuring K_d . One method to measure the strength of binding between a protein like an antibody, and a ligand (or antigen) is equilibrium dialysis. The antigen equilibrates between two compartments separated by a dialysis membrane, Figure **Figure 6.4**. The small pore diameter of the dialysis membrane allows the antigen to pass through, but not the antibody. At equilibrium, both compartments contain equal amounts of free antigen, but the

6.2 Measuring binding

antibody-containing compartment contains additional antigen as an antibody-antigen complex.

Measuring the equilibrium dissociation constant involves a series of equilibration experiments with different antigen concentrations. At lower concentrations of Ag, less Ab·Ag complex will form. The molar amount of antigen is always in excess of antibody. Since the equilibrium dissociation constants are very low, these experiments may require detecting very low antigen concentrations. For example, measuring the affinity of antibody 26-10 for digoxin requires measuring digoxin concentrations in the range of 100 picomolar.

Each equilibration experiment yields a ratio, Y , of sites complexed with ligand compared to the total number of sites. The concentration of sites complexed with ligand is the difference in the ligand concentrations between the antibody-containing and the antibody-free compartments, eq Equation 6.4. The total concentration of sites is known from the amount of antibody added to the experiment. For the example in Figure **Figure 6.4** above, four of the binding sites are occupied by ligand and the total number of sites is eight, so $Y = 4/8 = 0.5$.

$$Y = \frac{\text{sites complexed with ligand}}{\text{total sites}} = \frac{[site \cdot ligand]}{[site] + [site \cdot ligand]} \quad (6.4)$$

Plotting the measured fraction saturation, Y at different concentrations of ligand as a function of free ligand on the x-axis yields a saturation binding curve, Figure **Figure 6.5**. The sites become saturated with ligand as the concentration of free ligand increases.

Equation Equation 6.5 defines the relationship between Y , the free ligand concentration and the equilibrium dissociation constant, K_d . At half-saturation, $Y = 0.5$, the free ligand concentration

6 Engineering Tighter Binding

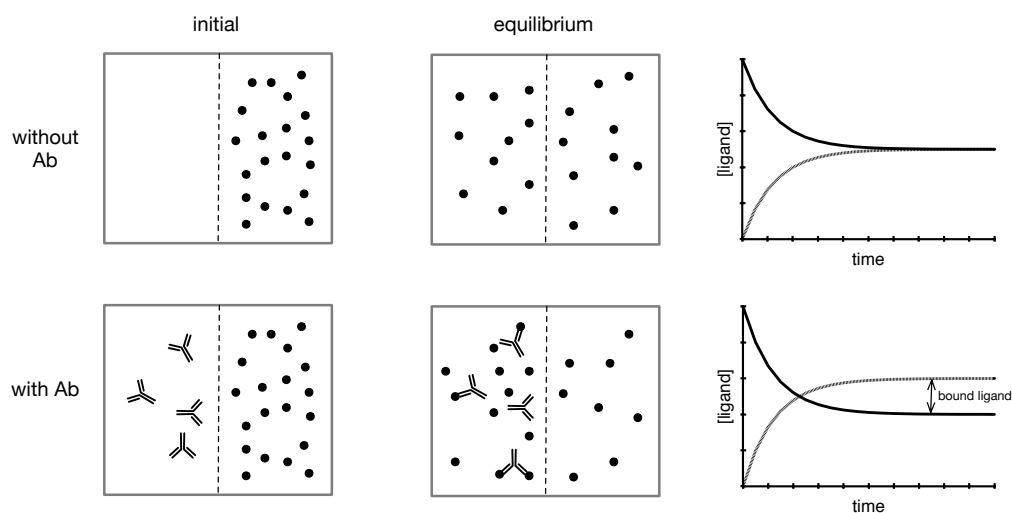


Figure 6.4. Measuring ligand (antigen) binding to an antibody by equilibrium dialysis. A dialysis membrane (dotted line) separates two compartments. Top row shows a control experiment without antibody. The ligand (twenty black circles) that is initially added to the right compartment distributes equally between both compartments (ten in each) because it passes freely through the membrane. Bottom row shows an experiment with antibody (Y-shapes) in left compartment. The dialysis membrane retains the antibody in its compartment. At equilibrium, both compartments contain the same concentration of free ligand (eight in each), but the antibody-containing compartment additionally contains antibody-bound ligand (four bound to the Y-shapes). The difference in the number of ligand molecules in the two compartments is the number of ligand molecules bound to antibody.

6.2 Measuring binding

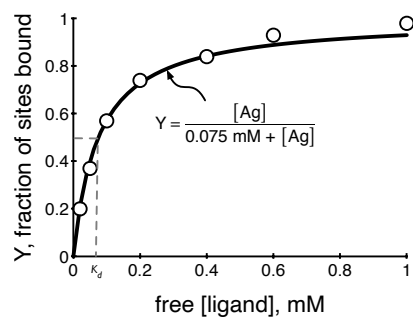


Figure 6.5. Fraction of antibody, Ab, bound to antigen, Ag, increases with increasing concentration of Ag until all of the antibody is saturated. Fitting equation Equation 6.5 to this data yields the equilibrium dissociation constant, $K_d = 0.075 \text{ mM}$. Inspection of the data table reveals that a bound antigen fraction of 0.5 occurs between 0.05 mM and 0.10 mM of free antigen, so the value of K_d lies within this range.

6 Engineering Tighter Binding

equals the equilibrium dissociation constant, 0.075 mM for this example.

$$Y = \frac{[ligand]}{K_d + [ligand]} \quad (6.5)$$

This equation is derived by rearranging eq Equation 6.1 to define the concentration of free sites, eq Equation 6.6, and substituting it into eq Equation 6.4 above.

$$[site] = \frac{K_d \cdot [site \cdot ligand]}{[ligand]} \quad (6.6)$$

Finding K_d from the data in Figure **Figure 6.5** involves adjusting the value of K_d so that the eq Equation 6.5 best fits the experimental data.*

Another way to measure the binding of a ligand to a protein is fluorescence polarization.^[2] This method requires a fluorescent ligand, but does not require the two-compartment apparatus needed for equilibrium dialysis. The solution of ligand plus protein is irradiated with plane-polarized light to excite the fluorescent ligand. The emitted light (fluorescence) from the ligand is measured both parallel and perpendicular to the plane of the irradiated light. If the ligand does not turn during the few nanoseconds between the absorption and emission of light, then most of the light will be emitted parallel to the plane of the irradiated light. However, if the ligand turns during the few nanoseconds between the absorption and emission of light, then less of the light will be emitted parallel to the plane of the irradiating light. A ligand free in solution tumbles rapidly,

*Non-linear least squares fitting of eq. Equation 6.5 to the data starts with an initial guess for the dissociation constant followed by iteration to find the best value. The supporting information includes a Python function for this fit.

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but a ligand bound to a macromolecule tumbles more slowly. Fluorescence polarization detects whether a ligand is bound or not by measuring how much of the emitted light is parallel to the plane of irradiated light.

One can extend this method to also measure the binding of non-fluorescent ligands by adding the fluorescent ligand and a competing non-fluorescent ligand simultaneously. The non-fluorescent ligand competes with the fluorescent ligand for the protein binding site. Measuring the decrease in binding of the fluorescent ligand at varying concentrations of the non-fluorescent ligand reveals the relative strength of the two ligands. Since the binding strength of the fluorescent ligand is known, the binding strength of the non-fluorescent ligand can be calculated.

Practical details on binding measurements are available.^[3] Other ways to measure binding include surface plasmon resonance and isothermal titration calorimetry.

Increasing binding strength requires increasing the difference in energy between the bound and unbound states. Engineering the antibody for tighter binding lowers the Gibbs energy of the Ag·Ab complex, raises the Gibbs energy of free Ag and Ab or both. The value of ΔG_d^o measures improvement of binding, Figure **Figure 6.3**. This value is the difference in the Gibbs energy change for the two dissociation reactions, eq Equation 6.7.

$$\begin{aligned}\Delta\Delta G_d^o &= \Delta G_{d-new}^o - \Delta G_{d-original}^o \\ &= -RT \ln \left(\frac{K_{d-new}}{K_{d-original}} \right) = -RT \ln(\text{improvement factor})\end{aligned}\tag{6.7}$$

Tighter binding corresponds to a larger positive Gibbs energy of dissociation for the improved variant and thus a positive value of

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ΔG_d^o . The value of K_{d-new} is a smaller (dissociation less favorable) than $K_{d-original}$, so $\left(\frac{K_{d-new}}{K_{d-original}}\right)$ is a value less than 1. For example, equilibrium dissociation constant may decrease from $10^{-6}M$ in the original to $10^{-8}M$ in the engineered case. The improvement factor is 10^{-2} and $\ln\left(\frac{K_{d-new}}{K_{d-original}}\right)$ is -4.61 . Then $-RT \ln\left(\frac{K_{d-new}}{K_{d-original}}\right)$ is $+2.73$ kcal/mol at $298^\circ K$.

6.3 Engineering tighter binding

Engineering tighter binding involves modification of the antibody binding site to increase the complementarity between the antibody and antigen, Table **Table 6.1**. The first requirement of a binding site is that its size and shape must match the size and shape of the target ligand. This matching excludes solvent water by placing ligand atoms near binding-site atoms to create the opportunity for favorable interactions. The second criteria is that the binding site be pre-organized for binding, that is, that the lowest energy conformation be the one that binds the ligand. The remaining criteria refer to optimizing different types of non-covalent interactions: hydrophobic interactions, hydrogen bonds and electrostatic interactions.

Table 6.1. Strategies to increase non-covalent interactions between protein and ligand.

Strategy	Reasoning
size and shape matching	creates favorable van der Waals interactions minimizes bumping releases bound water allows direct interactions

6.3 Engineering tighter binding

Strategy	Reasoning
preorganization	binding conformation favored even before binding no rearrangement needed for binding
hydrophobic matching	increases hydrophobic effect
optimize H bonds	compensates for loss of H bonds to solvent avoids buried H bonds due to poor solvation
optimize electrostatics	minimizes interactions with like charges maximizes interactions with unlike charges

Size and shape matching. Matching the size and shape of the protein binding site to the target ligand maximizes their contact surface in order to maximize favorable non-covalent interactions, Figure **Figure 6.6**. Size matching ensures that all of the ligand interacts with the binding site. If the cavity is too large or too small, then part of the ligand remains exposed to solvent and the interaction between ligand and binding site will be less than the maximum possible. Shape matching refers to the surface contours of the ligand and binding site. Surface complementarity means the surfaces touch to create favorable van der Waals interactions. The close contact also makes additional non-covalent interactions between the ligand and binding site possible.

The intricate shapes of molecules prevents most of the atoms in two contacting molecules to directly contact one another. The sphere representation of a molecule shows its van der Waals surface. The radius of each sphere is the van der Waals radius for the atom and the overlapping spheres show the van der Waals

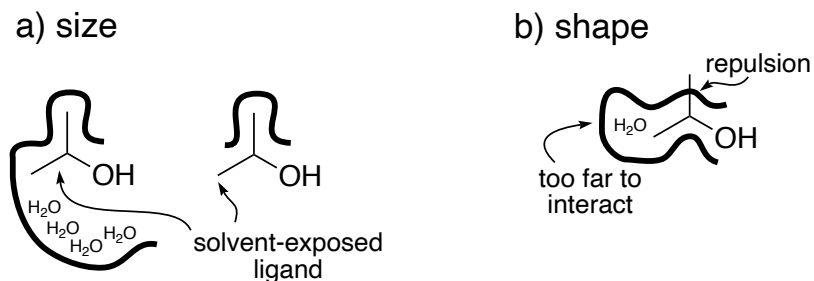


Figure 6.6. To maximize interactions with the target ligand (isopropyl side chain) the binding site must be the same size as the ligand and the contours of the binding site must match the shape of the ligand. a) A binding site that is larger or smaller than the ligand leaves part of the ligand exposed to solvent, which misses opportunities for interactions between ligand and binding site. b) Shape mismatches between the target ligand and contours of the binding site create repulsive interactions in regions where the binding site is too small and an inability to make favorable interactions in regions where the binding site is too large to contact the ligand.

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surface of the molecule, Fig **Figure 6.7**. This surface is intricate and includes narrow crevices too small to fit a water molecule. A water molecule cannot touch the regions within these crevices and a larger molecule will be excluded from even more regions. The *solvent contact surface* is a hypothetical smoother surface that eliminates crevices that are inaccessible to water. The solvent contact surface is the surface mapped by the edge of sphere with a radius of 1.4 Å as it rolls on the van der Waals surface of a molecule. The radius of 1.4 Å corresponds to the van der Waals radius of a water molecule. This surface shows smooth areas where the water molecule rides over the van der Waals surface as well as regions of protruding atoms where the water can directly touch the molecule.[†]

The majority of the surface atoms, even for highly complementary molecules, will not contact one another. The intricacy of molecular structures make complete direct contact geometrically impossible. Good complementary shapes will exclude solvent water from the interface and make many direct contacts.

An example of high surface complementarity is the interaction between antibody fragment 26-10 complexed with digoxin,^[4] **Figure 6.8**. The structure of 26-10 shows a deep pocket that surrounds the hydrophobic steroid portion. The hydrophilic carbohydrate groups of digoxin remain exposed to solvent and do not contribute to binding. The complementarity between the solvent-accessible surface of the aglycone (compound without the glyco-

[†]Another surface, the solvent accessible surface, or Connolly surface, lies 1.4 Å outside the solvent contact surface. This surface represents the center, not the edge, of a water molecule on the surface. The solvent accessible surface is more convenient in some cases because positions on this surface will match the x,y,z coordinates of the rolling water molecule.

[‡]Charles Cullen, a nurse and serial killer convicted in 2003, used digitalis (and other drugs) to kill possibly forty hospital patients. This story and that of Amy Loughren, the ICU nurse who helped convict Cullen, is told in the book *The Good Nurse* (2013), and a 2022 movie by the same title currently on Netflix.

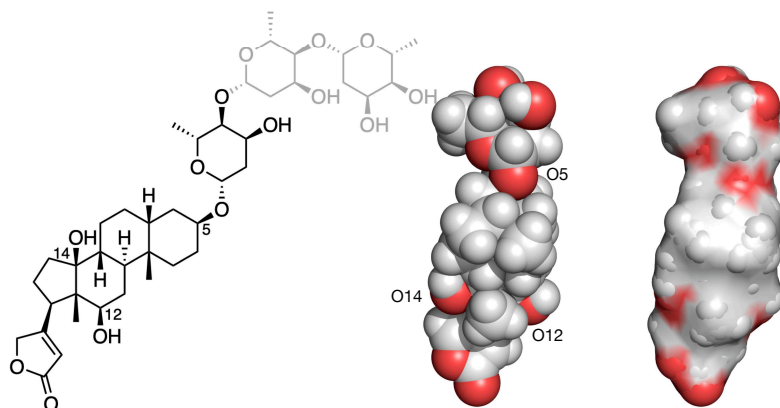


Figure 6.7. Structure of digoxin (digitalis), a cardiac glycoside.[‡] Left: The chemical structure of digoxin consists of a steroid (the aglycone) linked to a trisaccharide. The two sugar moieties in gray are not included in the two structures to the right. Middle: The overlapping van der Waals spheres of each atom create the van der Waals surface of digoxin. It contains numerous crevices that are too small to fit a water molecule. A water molecule would be slightly larger than one of the oxygen atoms shown as red spheres. Right: The solvent contact surface of digoxin is a hypothetical surface that shows the closest approach of a water molecule. Water can directly contact only a small fraction of the van der Waals surface.

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side moiety) of digoxin and 26-10 is imperfect, even in regions that are in contact. The shape complementary is close enough that there are no buried cavities in the complex that are large enough to contain a water molecule. This exclusion of water ensures a strong contribution of the hydrophobic effect to the binding. There are some regions exposed to the bulk water that are large enough to contain water molecules. The conformations of antibody and steroid remain the same in both unbound and bound states indicating that the low energy unbound shapes match each other.

Fragment 26-10 binds digoxin tightly, $K_d = 0.1$ nM. The binding is entirely due to the hydrophobic effect and van der Waals interactions. No hydrogen bonds or salt bridges form between antibody fragment 26-10 and digoxin. One consequence of the lack of interactions with the hydroxyl groups at C12 and C14 is a lack of selectivity for analogs without these hydroxyl groups. For example, 26-10 binds digitoxigenin (which lacks the hydroxyl group at C12) with equal affinity to digoxigenin.[§]

Sculpting a complementary surface will likely require multiple mutations and involve readjustment of the main chain and side chains. The smallest change that one can introduce by amino acid substitutions is the addition or removal of a methyl or hydroxyl group. This is a large change - about the size of a water molecule - and likely too large to precisely sculpt a surface. To make smaller changes one needs to make multiple substitutions including substitutions outside the surface being engineered so that a succession of adjusted positions make more subtle changes to the complementary surface. Precisely predicting the adjustments caused by multiple substitutions is more difficult making such sculpting difficult.

Substitutions often introduce readjustments of the main chain and side chains. A larger side chain causes readjustments when it bumps into nearby atoms. A smaller side chain may also cause

[§]The -genin suffix on these two compounds indicates the aglycone only.

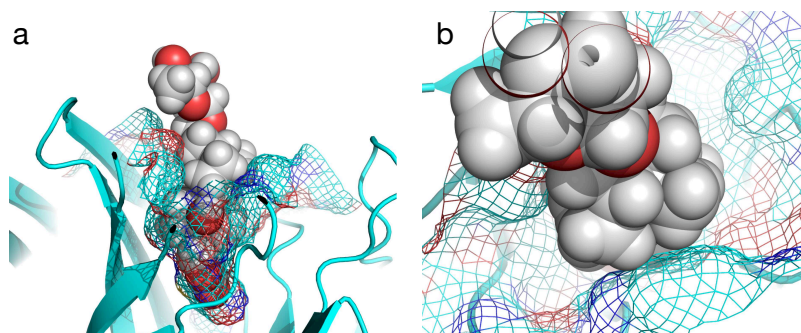


Figure 6.8. The binding pocket of antibody fragment 26-10 (mesh surface, backbone shown as a cartoon) binds the steroid digoxin (spheres) by creating a complementary surface. a) The two terminal saccharides extend into solvent and were not resolved in this structure. b) A top view shows close contact between the 26-10 and the aglycone in several parts, but much of the 26-10 surface does not directly contact the aglycone. The figure was created using PyMOL from the x-ray crystal structure of the complex (pdb id = 1igj^[4]). The surface of the antibody pocket was calculated using the web tool CASTp^[5]

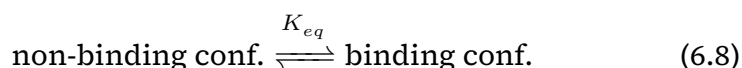
6.3 Engineering tighter binding

readjustments because enlarging a hydrophobic pocket is unfavorable. Replacing a large buried hydrophobic side chain with a smaller one destabilizes the protein because the smaller hydrophobic surface 1) reduces the contribution of the hydrophobic effect to protein folding and 2) reduces the favorable van der Waals contacts between the side chain and the rest of the protein. Structural readjustments often decrease the size of the created space to minimize these changes. The estimated decrease in stability due to removing one buried CH_2 group is 1.1 ± 0.5 kcal/mol.^[6] For example, a Leu to Ala substitution removes three carbon atoms from the protein, so this substitution decreases protein stability by an estimated 3.3 ± 1.5 kcal/mol. If the protein structure readjusts, the destabilization will be at the smaller end of the estimate range, while if the structure does not readjust, it will be at the larger end of the estimated range. For example, a Leu to Ala mutation in T4 lysozyme destabilized the protein by 5.0 kcal/mol at position 99, but by only 2.7 kcal/mol at position 46. The x-ray structures revealed that at position 99 the structure did not readjust upon substitution leading to a larger cavity and a larger loss in van der Waals interactions, while at position 46 the structure relaxed leading to a smaller increase in the cavity size and a smaller loss in the van der Waals interactions.^[7]

In a few cases, readjustments of the nearby residues reverse the intended change in size. For example, an ancestral hydroxynitrile lyase contains phenylalanines at positions 121 and 178 in the active site, while the corresponding modern enzyme from the rubber tree contains smaller leucines at these positions. One would expect that the substrate-binding site of the rubber tree enzyme would be larger, but the x-ray structures showed the opposite. In the rubber tree enzyme the side chain of Trp128 readjusts into the active site creating a smaller substrate binding site,^[8] which is the opposite of what might be expected from comparison of the sizes of phenylalanine and leucine.

Preorganization. Preorganization of a dynamic protein means creating a global energy minimum conformation where the atoms in the apo protein are already oriented to interact with the partner atoms in the ligand. In the ideal case, this conformation has the correct structure with a minimum of flexibility. It must have the correct structure to avoid the energy cost of moving the protein atoms from their most stable positions into the less stable interacting positions. This energy cost lowers the overall favorability of the binding. Flexibility of the apo protein also creates an energy cost. If protein flexibility decreases in the protein-ligand complex, then there is an entropy cost to binding, which also lowers the favorability of binding. Thus, the preorganized apo protein should be no more flexible than the protein-ligand structure.

Binding sites exist as an equilibrium mixture of non-binding and binding conformations, eqs. Equation 6.8 and Equation 6.9. The equilibrium constant (or the Gibbs energy) between the binding and not-binding conformations measures the propensity of the binding site to be preorganized for binding. It is convenient to express this equilibrium as the probability that the binding site is in a binding conformation, eq. Equation 6.10, which is a value between 0 and 1. For example, if the equilibrium constant is 10 in favor of the binding conformations, then the probability that the binding site exists in a binding conformation is $10/11 = 91\%$. This probability is sometimes called the Boltzmann factor or Boltzmann weight.



$$K_{eq} = \frac{[\text{binding conf.}]}{[\text{non-binding conf.}]} \quad (6.9)$$

6.3 Engineering tighter binding

$$\text{probability} = \frac{[\text{binding conf.}]}{[\text{non-binding conf.}] + [\text{binding conf.}]} = \frac{K_{eq}}{1 + K_{eq}} \quad (6.10)$$

Preorganization excludes stable non-binding conformations of the binding site, Figure **Figure 6.9**. These conformations hinder binding because they trap the binding site into a non-binding conformation. The binding site must first escape the trap, then bind the ligand, which lowers the binding interaction by the amount needed to escape the trap. The non-binding, low energy conformation stabilizes the unbound state by $\Delta\Delta G_{preorg}$, thereby reducing the Gibbs energy of dissociation for the complex.

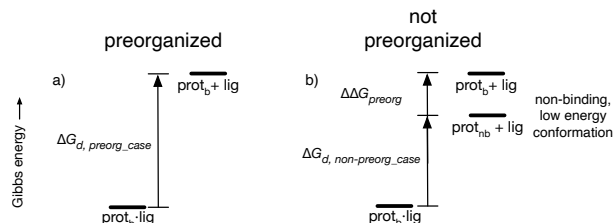


Figure 6.9. Preorganization excludes stable, non-binding conformations of the protein. a) When a protein is preorganized for binding, the binding site conformation in the ligand-free state has the same conformation as in the ligand-bound state. b) When the protein is not preorganized for binding, the protein exists in lower energy conformations that cannot bind the ligand. Binding requires the protein to first shift to a higher energy binding conformation. The cost of shifting to the binding conformation lowers the overall binding energy by $\Delta\Delta G_{preorg}$.

Constraining a flexible ligand into the three-dimensional shape it adopts when bound to a receptor often strengthens the binding

interaction. For example, the phosphotyrosine-containing pseudopeptide **6.1**, Figure **Figure 6.10**, bound tightly to the SH2 domain of a growth receptor binding protein, $K_D = 2.2 \mu M$. Introducing a cyclopropyl ring to create compound **6.2** restricted the flexibility of the ligand so that its structure matches the bound conformation. The binding interaction tightened to $K_D = 0.36 \mu M$.^[9] Similar improvements are expected by restricting the flexibility of the binding site.

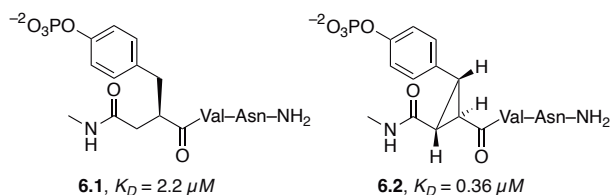


Figure 6.10. Phosphotyrosine containing pseudopeptides bind tightly to the SH2 domain of a growth receptor binding protein. The cyclopropane ring in compound **6.2** restricted its flexibility and increased its affinity as compared to compound **6.1**.

The preorganized protein structure for binding refers to both side chain and backbone orientations. The side chains must point in the correct direction to interact with the ligand, while the backbone must position these side chains at the correct distances. Incorrect backbone positions create binding sites that are too large or too small for the ligand. For example, a molecular dynamics simulation of digoxigenin-binding proteins found that favored conformation of the non-binding proteins was too large to interact with the entire digoxigenin surface due to incorrect backbone positions.^[10]

Hydrophobic matching. Besides matching the shape of the ligand and binding site to maximize van der Waals interactions, match-

6.3 Engineering tighter binding

ing additional non-covalent interactions further strengthens the interaction. For example, matching hydrophobic regions of the ligand and binding site avoids unsatisfied polar interactions. The structure of antibody fragment bound to methamphetamine showed a bound water molecule facing a hydrophobic region of the methamphetamine antigen.^[11] Researchers hypothesized that releasing this water molecule would increase the hydrophobic character of the binding pocket and increase binding. Replacement of SerH93, which formed a hydrogen bond to the water molecule, with the slightly larger residue threonine increased the binding affinity 3.1-fold from $K_d = 0.79$ to 0.25 nM. A crystal structure of the improved variant confirmed that this substitution forced out the water molecule.

Optimize hydrogen bonds. Hydrogen bonds between the binding site and ligand come at the cost losing hydrogen bonds to solvent and desolvation of the hydrogen bonding partners. Buried hydrogen bonds are rarely favorable, but solvent-exposed hydrogen maintain solvation so are more likely to be favorable. Hydrogen bonds often contribute to selectivity of binding.

Electrostatic interactions. Electrostatic interactions can contribute to the tighter binding of the protein and ligand in two ways. One way is to directly stabilize the protein-ligand complex; the other way is to speed up the formation of this complex, Figure **Figure 6.11**.

Direct stabilization of the protein-ligand complex refers to favorable interactions between oppositely charged atoms in the protein-ligand complex after it has formed. For example, improved electrostatic interactions increased the binding of monoclonal antibody hu3F8 (naxitamab) to the surface of cancer cells ~7-fold.^[12] The antibody binds to two negatively charged sialic acid groups on glycolipid ganglioside GD2. An Asp32His substitution replaced a negatively charged amino acid residue with a positively charged

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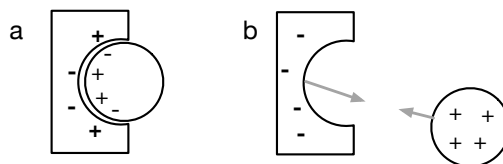


Figure 6.11. Two types of electrostatic interactions strengthen binding between partners. a) Partners with low or zero net charge associate with one another. Electrostatic interactions within the complex strengthen binding. b) Partners have opposite net charges. Favorable, long-range electrostatic interactions steer these oppositely charged molecules toward each other (gray arrows). Uncharged molecules of the same size would likely miss each other because their motions are random.

one just outside the binding site. Computer modeling predicted ~0.5 kcal/mol improvement in electrostatic interaction with the sialic acid groups in good agreement with the measured increase in binding. In another example, computer modeling predicted five substitutions that could strengthen electrostatic interactions between epidermal growth factor receptor and an antibody targeted to it (cetuximab, Erbitux®). Experiments confirmed that three of the five increased binding affinity. Substitutions Ser26Asp and Thr31Glu replaced polar residues with negatively charged residue, while Asn93Ala replaces a polar residue with an uncharged residue. All substitutions were in the antibody light chain. The combination of all three substitutions increased the binding 10-fold by decreasing K_d decreased from 490 pM to 52 pM.^[13] The researchers noted that solvent-exposed electrostatic interactions and hydrogen bonds were more stabilizing than buried ones because they remain solvated. The loss of solvation energy made most buried electrostatic interactions and hydrogen bonds unfavorable.

6.3 Engineering tighter binding

Electrostatic interactions can also strengthen binding by long-range interactions that occur before binding occurs. Electrostatic steering increases binding strength by increasing the collision frequency between oppositely charged proteins and ligands, Figure {Figure 6.11}b above. Binding requires the protein and ligand to bump into one another as they randomly move through the solution. Electrostatic forces alter the collision rates between molecules because they act over long distances, that is, beyond van der Waals contact distances (see Chapter 2). This longer range of electrostatic interactions effectively makes charged molecules larger when they interact with other charged molecules. Nearby oppositely charged molecules can move toward each other to make van der Waals contact, while uncharged molecules must meet within their van der Waals radii. This increase in the number of collisions between oppositely charged molecules increases the rate of association of protein and ligand and thereby strengthens the binding interactions. Increasing the electrostatic interaction between separated partners as compared to the complex by 1 kcal/mol increases the association rate by a factor of 2.8.^[14] Charges that steer partners toward each other may also contribute to stronger binding after the complex forms. In contrast, partners with low net charge will not be steered even though they form favorable electrostatic interactions after binding.

To engineer such an increase, one increases the charge difference between the protein and ligand. For example, Selzer and coworkers^[14] increased the affinity between β -lactamase (negative net charged) and its inhibitor (no net charge) 250-fold by giving the inhibitor a positive net charge. Four substitutions increased the positive charge by six units. A web tool predicts the increase in association rate upon mutagenesis: <https://webhome.weizmann.ac.il/home/bcges/PARE.html>.

The components of binding Gibbs energy, ΔH and $-\Delta S$, do not map simply onto specific molecular events, making mechanistic

interpretation challenging. Enthalpy (ΔH) reflects the balance of non-covalent interactions formed and broken during binding. Favorable contributions ($-\Delta H^\circ$) arise from hydrogen bonds, van der Waals contacts, and electrostatic interactions between protein and ligand. Unfavorable contributions ($+\Delta H^\circ$) result from desolvation penalties—breaking hydrogen bonds between the ligand and water, and between binding site residues and water. The observed ΔH° is the net sum of these competing effects.

Entropy ($-\Delta S^\circ$) reflects changes in the number of accessible microstates of the system. Unfavorable contributions ($-\Delta S^\circ$) include loss of translational and rotational freedom when the ligand binds, and loss of conformational flexibility in both protein (side chain immobilization, induced fit) and ligand (freezing of rotatable bonds). Favorable contributions ($+\Delta S^\circ$) primarily arise from release of ordered water molecules from the binding site and ligand surface into bulk solvent—the hydrophobic effect. Additionally, protein conformational changes may sometimes increase local disorder.

The observed Gibbs energy of binding represents the net balance of all these opposing entropic and enthalpic terms, complicating direct assignment to specific binding mechanisms.

6.4 Computational design of tighter binding

Docking models protein-ligand complexes. Designing improved binding interactions relies on computer modeling of the interactions. An accurate structure of the ligand complexed to the binding protein is the best starting point for computer modeling, but such structures are rarely available. Protein-ligand docking is a molecular modeling technique to create a model of a protein-ligand complex from a structure of the binding protein. SwissDock is a web

6.4 Computational design of tighter binding

tool for protein-ligand docking.^[15] The most common application of docking is drug design, but it is also useful for protein design. Docking predicts where the ligand binds to the protein, the geometry of the bound ligand and the strength of the binding interaction.

As with all protein modeling, the two challenges are sampling and scoring. Sampling finds possible binding locations, orientations and conformations. Scoring evaluates and ranks these possibilities to identify the best ones. If the sampling does not generate the best-binding structure, then scoring cannot identify it.

For sampling, SwissDock uses an approach called Attracting Cavities.^[16] It replaces the complex protein with a simplified set of points that represent attractive interactions with a protein cavity. This simplification eliminates bumping interactions from the protein atoms, which allows more efficient sampling of the conformations to find those that maximize the attractive interactions.

For scoring, SwissDock restores the protein structure and calculates the energies using the CHARMM force field, which is similar to AMBER and intended for modeling biological molecules. Solvation calculations use an implicit solvent model, which lacks individual solvent molecules. The energy contributions to scoring are the ligand-protein interaction energy, the conformational energy of the ligand and protein and the desolvation energy of the protein and ligand. The calculated energies are estimates of the Gibbs energy of binding.

SwissDock accurately reproduced the experimental structures (rmsd ≤ 2 Å) of 285 protein-ligand complexes in 73% of the cases.^[17] It performs best for ligands with limited numbers of possible conformations (≤ 10 rotatable bonds), and for binding orientations where most of the ligand surface ($\geq 85\%$) interacts with protein. The conformations of large solvent-exposed regions of the ligand are difficult to predict accurately.

Computational design of binding proteins. Predicting substitutions that improve the affinity of a protein for a target ligand has shown some success, but remains challenging. Computational design using Rosetta predicted seventeen proteins that bind to the steroid portion of digoxin called digoxigenin.^[18] Two of these proteins showed micromolar affinity for the target, which is relatively weak affinity, but the ability to predict such proteins is an impressive accomplishment. Further optimization did not use computation, but directed evolution methods (see Chapters 6 and 7) and yielded proteins with picomolar affinity for digoxigenin. The designed protein included hydrogen bond interactions with the hydroxyl groups of digoxigenin, so it showed selectivity against analogs lacking these hydroxyl groups. Other computational design were even less successful. Morin et al.^[19] designed twelve possible binding sites in *endo*-1,4- β -xylanase for the antibiotic vancomycin, but none of the designs bound vancomycin. X-ray structure analysis of four designs showed the predicted conformation, but molecular dynamics showed that the designs were highly flexible. The authors suggest that the designed sites spent most of their time in conformations not suitable for binding.

6.5 Selective binding

Define selectivity. Selectivity, S , is the ability to distinguish between alternatives. Binding selectivity is the ability of a protein to bind one ligand more tightly than another one, Figure **Figure 6.12**. The selectivity value, S , is the ratio of the affinities of the protein for the preferred and non-preferred ligands. This value depends on comparison molecules, so they should always be included with the number. For many proteins, their ability of proteins to favor binding one ligand in the presence of competing ligands is what makes them valuable. For example, the cardiovascular antibody

6.5 Selective binding

drug abciximab selectively binds to a receptor involved in platelet aggregation to inhibit formation of blood clots.

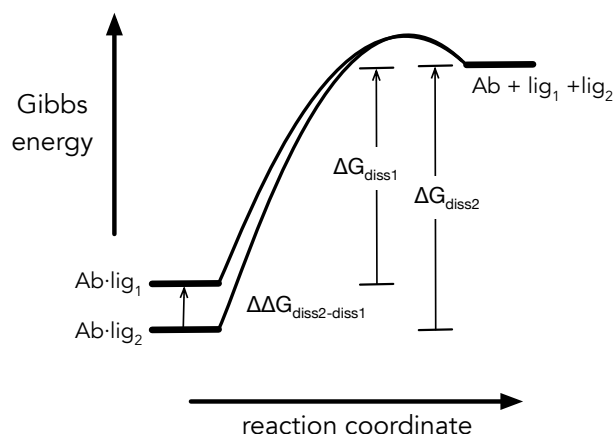
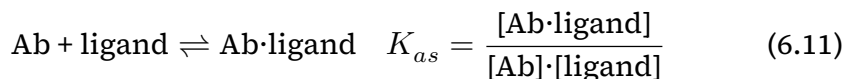


Figure 6.12. Gibbs energy diagram of selective binding of lig- and 2 (lig₂) over ligand 1 (lig₁) to an antibody (Ab)

The affinity of a protein for a ligand is given by the equilibrium association constant, K_{as} , for the binding reaction, eq Equation 6.11. It is the inverse of the dissociation constant defined in Figure **Figure 6.2** above.



Binding selectivity, S_{bind} , is the ratio of the affinities for the two protein-ligand complexes being compared, eq. Equation 6.12. By convention, the equilibrium association constant for the tighter-binding ligand, t, is in the numerator and that for the weaker-binding ligand, w, is in the denominator yielding binding selectivity values that are greater than one. The binding selectivity

6 Engineering Tighter Binding

should also mention which ligands are being compared. A selectivity of ten corresponds to ten-fold tighter binding of one ligand as compared to the other ligand. A selectivity of one corresponds to no selectivity; that is, equal affinity for both.

$$S_{bind} = \frac{K_{as-t}}{K_{as-w}} = \frac{\frac{1}{K_{d-t}}}{\frac{1}{K_{d-w}}} = \frac{K_{d-w}}{K_{d-t}} \quad (6.12)$$

Binding selectivity can also be expressed in terms of the equilibrium constants for dissociation, which are the inverse of the equilibrium constants for association. The dissociation constant for the weaker-binding ligand is in the numerator while that for the tighter-binding ligand is in the denominator. The selectivity of an protein for two competing ligands can be measured by measuring each binding affinity separately, then dividing the two values.

The Gibbs energy change associated with binding selectivity, eq. Equation 6.13, is the difference between the Gibbs energy changes of the two association reactions. The Gibbs energy change is negative since association of the tighter-binding ligand is more favorable.

$$\Delta\Delta G_{S_{bind}} = \Delta G_{as-t} - \Delta G_{as-w} - RT \ln \left(\frac{K_{as-t}}{K_{as-w}} \right) = -RT \ln (S_{bind}) \quad (6.13)$$

To engineer a protein for increased selectivity between a preferred and a non-preferred ligand, one can either decrease the affinity of the protein for the non-preferred ligand (negative selection) or increase the affinity of the protein for the preferred ligand (positive selection). The best approach will depend on the details of the application.

Glossary

Boltzmann factor or weight is the probability that the binding site conformation when unbound is the same as that when it is bound. The Boltzmann factor is a way to measure the preorganization of a binding site for binding.

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Supporting Information

Code Block S4.1. Python script to fit data for an equilibrium dialysis experiment (Fig **Figure 6.5**)

```
#!/usr/bin/env python
# coding: utf-8

# also run the command below in terminal
# sudo chmod +x file-name.py

import numpy as np                #import math functions to use arrays
from scipy import optimize        #import non-linear fit function
import matplotlib.pyplot as plt  #import plotting function

# enter your data here
freeAg = np.array([0.02, 0.05, 0.10, 0.2, 0.4, 0.6, 1])
Y = np.array([0.20, 0.37, 0.57, 0.74, 0.84, 0.93, 0.98])
Kd = 0.1 # initial guess for Kd

# define equation to be fit
def satCurve(freeAg, Kd):
    return freeAg/(freeAg + Kd)

# fit the data to the equation using non-linear least squares with
# default fit settings
popt, pcov = optimize.curve_fit(satCurve, freeAg, Y, Kd)
```

```

print("Kd =", "{0:.3f}".format(popt[0]))

# plot data and best fit curve
plt.scatter(freeAg, Y)
xfit = np.linspace(0,1)
plt.plot(xfit, satCurve(xfit, popt[0]), 'r-')
plt.xlim(0, 1)
plt.xlabel('free antigen concentration')
plt.ylabel('fraction of antibody containing bound antigen')
plt.show()

```

Problems

1. The equilibrium dissociation constant, K_d , of an antibody and its ligand (a peptide) is $5 \mu M$ at pH 5.0 and 25 °C.
 - a) At what concentration of the ligand is half of the protein bound?
 - b) What fraction of the protein is bound at ligand concentration of $1.25 \mu M$?
 - c) When the pH was raised to 6.5, the K_d increased to $20 \mu M$. Is the binding tighter or weaker at this pH compared to pH 5.0? Explain why.
 - d) What functional groups/residues are most likely responsible for this change in the binding affinity with pH?

The problems below use the experimental data from a paper reporting the design of a protein (DIG10.3) that tightly binds

6 Engineering Tighter Binding

digoxigenin^[18] to test the ability of docking calculations, Swiss-Dock[¶] to predict the affinity of a protein for a ligand.

2. *Prepare the wt protein and Tyr34Phe variant for docking.* The x-ray structure of DIG10.3 (pdb id = 4J9A) contains 9 monomers in the unit cell. For simplicity, we will delete eight of these leaving only chain A. The structure also contains bound digoxigenin, which we will remove so that SwissDock can predict the binding position. Finally, we will save the modified file. Download this structure from the Protein Data Bank. Open the file with a text editor and read the notes at the beginning. In particular, note whether any amino acids are missing from the structure of chain A. Next, open the file using PyMOL. Enter the following commands on the input line (ignore comments after #).

```
remove (not chain A) # deletes all chains except chain A
remove (resn DOG) # deletes residue named 'DOG'
save DIG103_A.pdb # saves your modified structure as a pdb file
```

The file is saved in your working directory, usually your home directory; mine is '/Users/romas'. Search to find the file, if needed. Next, create the Tyr34Phe variant using the instructions in Homework 1 (Wizard => Mutagenesis => Protein ...). Use the suggested default rotamer of the Phe replacement. The following commands may be helpful.

```
show sticks, resi 34 # shows Tyr34 so that you can select it
save DIG103_A_Y34F.pdb # saves file once you have made the substitution.
```

[¶]Bugnon *et al.* (2024) SwissDock 2024: Major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. *Nucleic Acids Res.* **52**, W324-W332. <https://doi.org/10.1093/nar/gkae300>

Additional troubleshooting: If SwissDock fails to verify your file of the mutated protein, check that your pdb file contains only the mutated protein. One student's file contained three proteins, not one. Open your file in a text editor. Check to see that a line that reads 'TER' occurs only once in your file and that it is the penultimate line. If you find 'TER' more than once, you must delete both the extra 'TER' and the atoms associated with that extra protein. Your file should contain 905 atoms and the last four lines of your file should read:

```
ATOM 904 O ARG A 123 0.634 -10.220 7.065 1.00 48.12 O
ATOM 905 CB ARG A 123 3.588 -8.274 5.874 1.00 50.26 C
TER
END
```

3. *Docking with default settings.* Submit the calculation to dock digoxigenin to the wild-type protein and to the Tyr34Phe variant at <https://www.swissdock.ch/>. Leave the default 'Docking with Attracting Cavities' selected. In section 1 - Submit a ligand, click on advanced search and enter 'digoxigenin', then click on the digoxigenin structure and click apply. This brings you back to the starting page. Click on 'Prepare ligand'. You should get a green check mark.

In section 2 - Submit a target, click on choose file, then select the DIG103_A.pdb file that you created above to upload it. This brings you back to the starting page. Click on 'Prepare target'. You should get a green check mark.

In sections 3 & 4, no changes are needed. Click on 'Check parameters'. You should get a green check mark and an estimate of how long the calculation will take, about 10 min. It usually takes a bit longer than this estimate; be patient.

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In section 5 - Start docking, enter a docking name like DOG_DIG103_A, then click on 'START DOCKING'.

The result of the docking calculation is a set of ten clusters where each cluster contains a set of similar binding orientations. The tightest binding cluster is cluster 0 and usually contains eight binding orientations. We will compare the best binding orientations (most negative AC score in cluster 0) and assume that it is in kcal/mol.

- a) *Predicted ΔG_d , $\Delta\Delta G_d$, and K_d .* Compare the AC score for the best binding orientation in cluster 0 of digoxigenin to the wt protein and to the Tyr34Phe variant. Assume the score corresponds the Gibbs energy of binding of the ligand to the target in kcal/mol. Calculate the predicted equilibrium dissociation constants, K_d . Calculate the predicted difference in binding affinity of the two proteins. Assume the temperature is 25 °C. Recall that favorable binding corresponds to a positive dissociation energy. Remember to compare *differences* in Gibbs energies, but *ratios* of dissociation constants. SwissDock yields negative numbers, which correspond to association energies. Reverse the sign on these values to convert them to dissociation energies.
- b) *Compare predicted values to experimental values.* The experimental inhibition constants for DIG binding to DIG10.3 is 0.65 ± 0.03 nM and to DIG10.3-Tyr34Phe is 59 ± 6 nM (Figure 4, panels b & d)^[18] The experiments involved displacement of DIG-PEG₃-Alexa488 from the protein with DIG, so they are called inhibition constants, not binding constants. Assume these inhibition constants are accurate measures of K_d . Calculate the same values as in 1a using these experimental values and comment on any differences.
- c) Compare the predicted and experimental difference in the

affinities of the two proteins for DIG to the expected contribution of one hydrogen bond to the binding. Discuss the lack of agreement between the calculated and experimental values. Suggest two reasons that could contribute to the lack of agreement.

4. *Docking with increased sampling of conformations.* Repeat the calculation in question 1, but increase the number of random initial conditions (RIC) from the default of 1 to 3.
 - a) Compare the predicted values here with those in question 1. Rationalize any differences.
 - b) Compare the predicted values here with the experimental values. Rationalize any differences.
 - c) Discuss whether this docking web tool could be used in place of the Rosetta tool used by Tinberg *et al.*^[18] to design a protein to that binds digoxigenin.

Answers

1. a) $5 \mu M$ b) using equation 6.5, $Y = 1.25/6.25 = 20\%$ c) weaker because the K_d has increased. d) A group with a pK_a in the range of 5-6.5; aspartate, glutamate or histidine side chains are possible. Since the pK_a of functional groups changes when embedded into a protein environment, other groups may also be responsible and one could make a better estimate if the structure of the protein were available.
2. a) WT AC scores: -13.96097, -13.91753, -12.76379, -12.47792, -7.96029, -5.14183, 13.89088, 14.30629; AC scores: -20.42112, -16.18831, -14.83161, -14.22944, -14.00921,

6 Engineering Tighter Binding

-13.65794, -13.34280, -12.30052 b) experimental: wt ($K_d = 0.653 * 10^{-9} M$); Tyr34Phe variant $\approx$ 100-fold less tightly ($K_d = 59 * 10^{-9} M$). A factor of 100 corresponds to 2.8 kcal/mol at 298 K. A hydrogen bond is expected to contribute 0-2 kcal/mol to binding (p 25 of text), which corresponds to a factor of 1-30, so a factor of 100 suggests that interactions beyond a hydrogen bond are involved. SwissDock calculations yield *association constants*, that is the inverse a dissociation constant, so use +14 kcal/mol to calculate $K_d = 5 * 10^{-11} M$, which is about 20-fold lower than the experimental value for wt. The value for Tyr34Phe, is $K_d = 10^{-15} M$, which is off by a factor of 50 million. SwissDock using default settings incorrectly predicts that the Tyr34Phe variant binds more 50,000-fold more tightly wt.

- c) The lack of agreement may be a sampling failure due to insufficient search of conformations such as those that may require movement of the protein. The lack of agreement could be due to the inability of the force field used by SwissDock to accurately calculate the binding energy (a scoring failure).
4. a) increase the number of random initial conditions (RIC) from the default of 1 to 3 WT AC scores: -20.83291, -16.41948, -16.21564, -14.90450, -14.35758, -14.31006, -14.29569, -14.01725; Y34F AC scores: -16.303044, -14.243850, -14.050263, -13.886717, -13.784884, -13.460208, -12.810371, -12.024448
- b) Now the ranking correctly predict that wt binds more tightly than Tyr34Phe, but numbers differ significantly from the experimental values. The SwissDock-predicted $K_d^{wt} = 5.5 * 10^{-16}$, which is a million-fold too high; SwissDock $K_d^{Y34F} = 1.1 * 10^{-12}$, which is 50,000-fold too high. The ratio

of K_d 's overestimates the experimental ratio by about a factor of 20. This difference could be due to sampling (SwissDock did not find a representative sample of the true binding orientations), perhaps because even more of the protein must move than the 5 Å permitted here or due to a scoring error, where the Gibbs energy of the interaction is not calculated correctly.

- c) Rosetta predicted an interaction energy of ≈ 15.5 REU for DIG10 and the ligand (Figure 1, panel b). If we assume that REU's (Rosetta energy units) correspond to kcal/mol, then this prediction corresponds to a K_d of $0.004 * 10^{-9}$, which is 3 million-fold stronger binding than the experimental value of $12,000 * 10^{-9}$. Thus, neither Rosetta, nor SwissDock accurately predict the binding constant. SwissDock did rank the wt and Tyr34Phe variant correctly; the Tinberg *et al.* paper did not calculate the relative binding strengths, they just measured them. Rosetta calculated two things that SwissDock did not: average H-bonding Boltzmann weight and the protein stability. Both of these are characteristics of the protein: is it preorganized for binding and is it stable. Both of these are important for design and missing from SwissDock. SwissDock assumes that the given structure is correct, while Rosetta predicts the structure. SwissDock cannot be used for protein design, but it may be helpful to predict small modifications.

7 Engineering Faster Enzymes

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Summary. Engineering faster catalysis enables new substrates to react and reduces the amount of enzyme needed. Most enzyme-catalyzed reactions show saturation kinetics where the reaction rate initially increases with increasing substrate concentration, but later levels out and does not increase further. Two kinetic constants define this curve: V_{max} , the maximum rate of reaction at high substrate concentration, and K_M , the concentration of substrate at half of V_{max} . The definition of these constants differs for different reaction mechanisms. The simplest reaction mechanism is the Michaelis-Menten model, which proposes that the substrate first reversibly forms an enzyme-substrate complex, then reacts to form product. In most cases, the physical steps involved in enzyme catalysis are more complex. Engineering faster enzymes requires speeding up the physical steps that limit catalysis. Engineering faster enzymes may involve improving binding of the substrate to the enzyme, reorienting the substrate, adjusting the pK_a of acids and bases, relieving product inhibition or other steps in the reaction mechanism. Distant residues contribute to catalytic activity by positioning active site residues and altering tunnels that control substrate entrance and product release.

Key learning goals

7 Engineering Faster Enzymes

- Most enzymes follow saturation kinetics defined by two kinetic constants: V_{max} and K_M . The physical steps that contribute to these constants differ when the enzyme reaction mechanisms differ.
- The Michaelis-Menten model for enzyme catalysis involves formation of the enzyme-substrate complex ($E \cdot S$) followed by the chemical step that converts substrate to product. Enzymes speed up this chemical step by stabilizing the transition state for the reaction.
- Two limiting cases of enzyme-catalyzed reactions are low and high substrate concentration. Here low and high are relative to the K_M of the enzyme-substrate complex. At low substrate concentration, the reaction rate can be improved either by improved binding (lower K_M), which increases the fraction of enzyme that contains a bound substrate, or by a faster chemical step (k_{cat}). At high substrate concentration all of the enzyme molecules already contain a bound substrate, so only a faster chemical step (k_{cat}) can increase the reaction rate.
- Improved binding requires increasing shape and interaction complementarity between substrate and enzyme. Some reaction steps may also contribute to K_M .
- Speeding up the conversion of the enzyme-substrate complex to product requires increased stabilization of the transition state. Depending on the reaction mechanism details, this stabilization can involve 1) pre-organizing the geometry of the $E \cdot S$ complex for reaction, 2) ensuring the correct protonation state of any needed acids and bases, 3) stabilizing the charge and shape of the transition state, and 4) breaking up difficult mechanistic steps into several easier ones. When tunnels limit the transport of substrates or products in and out of the active site, then this transport becomes the transition state and engineering must focus there.

7.1 Introduction

Faster enzymes are a common goal in protein engineering since faster enzymes require less protein for the application. Faster may refer to an existing substrate for the enzyme, and it may also refer to expanding the substrate range of an enzyme to new, previously unreactive, substrates.

Chemical reactions are processes that break and make chemical bonds. Catalysts can not change the Gibbs energies of the reactants and products, nor the equilibrium constant between them, **Figure 7.1**. A catalyzed reaction reaches the same equilibrium state as a non-catalyzed reaction, but reaches this equilibrium more rapidly. Catalysts increase the rates of both the forward and reverse reactions. Enzymes are remarkably efficient catalysts and increase the rates 10^5 to 10^{15} -fold over the non-catalyzed reactions.

The highest energy structure along the path from substrate to product is the transition state, **Figure 7.1**. The energy gap between the substrate and transition state limits the rate of the reaction. The size of this gap, $\Delta G^\ddagger$, is the activation energy for the reaction. The transition state is a peak in the energy diagram, so it has a fleeting existence. Only structures that correspond to valleys on the energy diagram can exist for a finite time.

Catalysts act by stabilizing the transition state and lowering the activation energy. For example, bond breaking or making in a transition state often creates partial charges. A catalyst can orient the substrate for reaction and provide compensating charges, thereby stabilizing the transition state, lowering the barrier from substrate to product and speeding up the reaction.

Transition state theory connects the rate constant for the reaction, k , to the Gibbs energy of the transition state, $\Delta G^\ddagger$, eq. Equation 7.1.

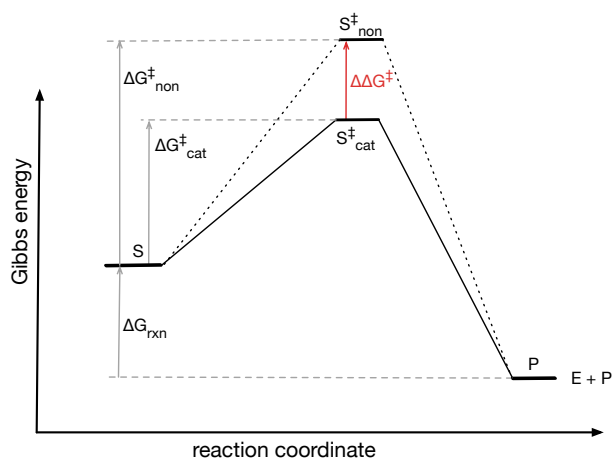


Figure 7.1. Gibbs energy changes associated with a chemical reaction. The Gibbs energy change for the transformation of substrate S to product P, ΔG_{rxn} , is negative indicating that the reaction is favorable. The barrier between the starting material and product limits the rate of the reaction. The top of this barrier is the highest energy structure along the bond-making and bond-breaking path of the reaction and is called the transition state. $\Delta G_{non}^{\ddagger}$ and $\Delta G_{cat}^{\ddagger}$ are both positive because the transition states, $S_{non}^{\ddagger}$ and $S_{cat}^{\ddagger}$, are higher in energy than S. The catalyst stabilizes the $S_{non}^{\ddagger}$ transition state by an amount $\Delta\Delta G^{\ddagger}$, which lowers the barrier to product and speeds up the reaction.

7.1 Introduction

Transition state theory assumes a quasi-equilibrium* between the substrate and transition state. The constants in eq. Equation 7.1 account for the fraction of molecules that react once they reach the top of the barrier. The activation Gibbs energy is a positive value because the value of $\frac{k}{\text{constants}}$ is <1, which makes $\ln\left(\frac{k}{\text{constants}}\right)$ a negative value.

$$\Delta G^\ddagger = -RT \ln \left(\frac{k}{\text{constants}} \right) \quad (7.1)$$

A lower activation energy corresponds to a lower barrier and a faster reaction (larger k). A faster rate, or larger k , decreases the absolute value of $\ln \frac{k}{\text{constants}}$, which leads to a smaller activation energy. For example, the $\ln(0.001)$ is -6.9, while the $\ln(0.1)$ is -2.3. The decrease in activation energy for a catalyzed reaction as compared to a non-catalyzed reaction is proportional to the natural logarithm of the ratio of the rate constants for the catalyzed and non-catalyzed reactions, eq Equation 7.2.

$$\begin{aligned} \Delta G_{cat-non}^\ddagger &= -RT \ln \left(\frac{k_{cat}}{\text{constants}} \right) - \left(-RT \ln \left(\frac{k_{non}}{\text{constants}} \right) \right) \\ &= -RT \ln \left(\frac{k_{cat}}{k_{non}} \right) \end{aligned} \quad (7.2)$$

Transition state stabilization accounts for most of the catalytic effect of enzymes.^[1] Experimental evidence that enzymes stabilize transition states includes the finding that transition state analogs are potent enzyme inhibitors and that antibodies to transition state analogs can catalyze reactions, albeit slowly.

*It is not a true equilibrium because some of the molecules that reach the transition state continue along the reaction coordinate to form product. In a true equilibrium, they would always return to the substrate species.

7.2 Measuring reaction rates

Consider the rate of a unimolecular chemical reaction, eq. Equation 7.3.



The rate of reaction, V , is the disappearance of substrate over time, $-d[S]/dt$, and has units of concentration per unit time. Researchers measured the initial rate of the reaction using different concentrations of substrate, $[S]$, and found that the rate increases linearly as $[S]$ increases, **Figure 7.2**.

The proportionality constant, k , for this rate increase is the rate constant and has units of time^{-1} , eq. Equation 7.4.

$$V = -\frac{d[S]}{dt} = k \cdot [S] \quad (7.4)$$

This rate constant reveals the inherent propensity of the substrate to react. If another reaction has a larger rate constant, then it is faster. It will convert more molecules into product over the same time at the same concentration of substrate.

The rate of an enzyme-catalyzed reaction, eq. Equation 7.5, also varies with the concentration of substrate, but the relationship between V and $[S]$ is not linear, but curved, see **Figure 7.2** above.



The equation that describes this curve, eq. Equation 7.6, contains two constants: V_{max} and K_M . These constants reveal the inherent speed of an enzyme-catalyzed reaction.

7.2 Measuring reaction rates

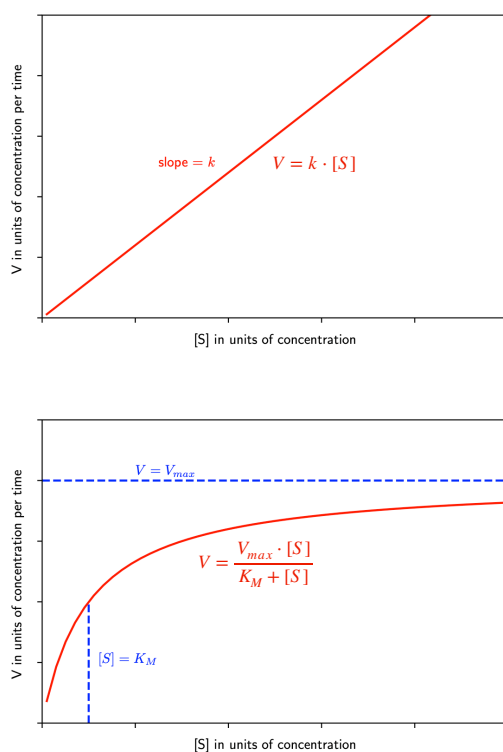


Figure 7.2. The rate of reaction for a unimolecular chemical reaction (top) and a single substrate enzyme-catalyzed reaction (bottom). In both cases the rate of reaction, V , increases with increasing substrate concentration, $[S]$, but this increase flattens for the enzyme-catalyzed reaction. The slope of the line, k , describes the speed of a unimolecular chemical reaction, while two constants are needed to describe the speed of the enzyme-catalyzed reaction. The constant K_M in units of concentration is the concentration at which the reaction rate reaches half of the maximum rate. K_M also indicates the substrate concentration at which the curve starts to flatten. The constant V_{max} in units of concentration/time indicates the maximum rate.

7 Engineering Faster Enzymes

$$V = \frac{V_{max} \cdot [S]}{K_M + [S]} \quad (7.6)$$

V_{max} is the reaction rate at high substrate concentration, while K_M is the substrate concentration at which the reaction rate is half of V_{max} . For a chemical process, the physical meaning of the constant k is the inherent rate of the spontaneous reaction of the substrate. For the enzymatic process, the physical meanings of the constants V_{max} and K_M depend on the mechanism of the enzyme-catalyzed reaction and will be discussed below. At this time, they are simply kinetic constants.

Consider the two limiting cases of low substrate concentration and high substrate concentration. Here low and high are in comparison to K_M for the reaction. At low substrate concentration, $[S]$ is much smaller than K_M , so the denominator of eq. Equation 7.6 simplifies to K_M . Equation Equation 7.6 simplifies to eq. Equation 7.7.

$$V_{low[S]} = \frac{V_{max}}{K_M} \cdot [S] \quad (7.7)$$

This equation resembles the equation for a chemical reaction, eq. Equation 7.4 above, where the ratio V_{max}/K_M corresponds to the rate constant for the reaction. Thus, at low substrate concentration, one compares the rates of enzyme-catalyzed reactions by comparing V_{max}/K_M . Higher values correspond to faster reactions.

At high substrate concentration, K_M is much smaller than $[S]$, so the denominator of eq. Equation 7.6 simplifies to $[S]$. Next, $[S]$ cancels out since it appears as a factor in both the numerator and denominator. Equation Equation 7.6 simplifies to eq. Equation 7.8.

$$V_{high[S]} = V_{max} \quad (7.8)$$

7.2 Measuring reaction rates

At high substrate concentration, the reaction rate is independent of substrate concentration. To compare enzymes at high substrate concentrations, one compares V_{max} values. Higher values correspond to faster reactions.

The comparison of reaction rates described in this section is steady-state kinetics. The steady-state refers to constant conditions over the time of the experiment. The temperature, pH and substrate concentration are all constant over the course of the experiment. To keep the substrate concentration constant, researchers measure the initial rates of the reaction only while the first few percent of the substrate react, that is, the initial rate of the reaction. There is a large excess of substrate as compared to enzyme. Each enzyme molecule completes many catalytic cycles over the course of the measurement. The large excess of substrate also ensures that formation of an enzyme-substrate complex does not significantly alter the concentration of free substrate.

To measure the steady-state kinetic parameters for a reaction, K_M and V_{max} , one measures the initial rate of the reaction at varying substrate concentrations and find the values of K_M and V_{max} for equation Equation 7.6 that best fit the data. Huitema & Horsman (2018) provide scripts for this fitting using the statistical program R. The supporting information of this chapter provides a Python script to find the best values of V_{max} and K_M .[†] For a good fit one needs data points that define both the low [S] and high [S] regions of the curve; that is, measurements at substrate concentrations both above and below K_S .

[†]An older approach used a rearranged form of equation Equation 7.6 to fit the experimental data. This rearrangement allowed fitting a line to the data instead of a curve. This Lineweaver-Burke plot was more convenient when computers to fit data to a curve were not available, but has the disadvantage that it overweights data at low substrate concentration, which can lead to small errors. See supporting information for details.

7.2.1 The Michaelis-Menten model

The simplest model of the physical steps within an enzyme-catalyzed reaction is the Michaelis-Menten model, **Figure 7.3**. The substrate, S, binds reversibly to the enzyme, E, to form an enzyme-substrate complex, $E \cdot S$. The two physical steps are association of enzyme and substrate with rate k_1 and dissociation of the enzyme-substrate complex with rate k_{-1} . Next, the enzyme substrate complex reacts irreversibly to form and release the product with rate k_2 .

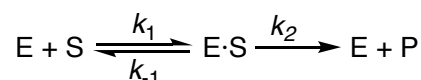


Figure 7.3. The Michaelis-Menten model of an enzyme-catalyzed reaction involves two physical steps: reversible equilibration of free substrate and enzyme with the enzyme-substrate complex, $E \cdot S$, followed by reaction of $E \cdot S$ to form and release the product, P.

Starting from this simple model, one can derive an equation with the same form as eq. Equation 7.6. This derivation will assign physical meanings to the constants V_{max} and K_M .

In the Michaelis-Menten model, the rate of product formation, V , depends on the amount of enzyme-substrate complex, $[E \cdot S]$, and the rate at which it is converted to product, k_2 , eq. Equation 7.9.

$$V = k_2 \cdot [E \cdot S] \quad (7.9)$$

The amount of $[E \cdot S]$ is unknown, so we express $[E \cdot S]$ in terms of known quantities. One known quantity is the amount of added enzyme, $[E]_0$. The total of $[E \cdot S]$ and $[E]$ must be equal to the amount of enzyme added, $[E]_0$.

7.2 Measuring reaction rates

$$[E]_0 = [E] + [E \cdot S] \quad (7.10)$$

Second, we assumed that $[E]$ and $[S]$ are in equilibrium with $[E \cdot S]$, so we may write the following equation.

$$\frac{k_1}{k_{-1}} = \frac{[E \cdot S]}{[E] \cdot [S]} \quad (7.11)$$

Combining these two equations and solving for $[E \cdot S]$ yields eq. Equation 7.12,

$$[E \cdot S] = \frac{[E]_0 \cdot [S]}{\frac{k_{-1}}{k_1} + [S]} \quad (7.12)$$

Combining eq. Equation 7.9 and eq. Equation 7.12 yields the Michaelis-Menten equation, eq. Equation 7.13 where $K_D = \frac{k_{-1}}{k_1}$

$$V = \frac{k_2 \cdot [E]_0 \cdot [S]}{K_D + [S]} \quad (7.13)$$

This equation assigns the following physical meaning to V_{max} and K_M .

$$V_{max} = k_2 \cdot [E]_0 \quad (7.14)$$

$$K_M = K_D = \frac{k_{-1}}{k_1} \quad (7.15)$$

V_{max} corresponds to the product of k_2 , the inherent reactivity of the $[E \cdot S]$ complex, and the amount of added enzyme. K_M corresponds

to the equilibrium dissociation constant of the $[E \cdot S]$ complex, K_D .

Warning: $K_M = K_D$ only for simple Michaelis-Menten mechanisms where $k_2 \ll k_{-1}$. For most enzymes, K_M is a complex kinetic parameter, not a binding constant.

We can now draw a physical picture of what occurs at different substrate concentrations in an enzyme-catalyzed reaction, **Figure 7.4**. The Michaelis-Menten model simplifies enzyme-catalyzed reactions to a binding step described by K_M or K_D that forms the enzyme-substrate complex and a reaction step described by $k_2 \cdot [E]_0$ that converts the substrate to product. Improving enzymes catalysis may require improvements to either or both of the two steps. When the substrate does not bind to the enzyme or binds poorly, then improved binding can increase the enzyme-catalyzed reaction rate. When the substrate binds, but reacts slowly, then the chemical reaction step must be improved by stabilizing the transition state.

7.2.2 Which enzyme is faster?

The metric used to identify the faster enzyme depends on the application. At low substrate concentrations, the ratio k_{cat}/K_M , often called the catalytic efficiency, is a good metric. It measures the ability of the enzyme to both bind the substrate and to catalyze the chemical step. In vivo applications such as therapeutic enzymes typically act at low substrate concentrations, so catalytic efficiency is a good measure. Typical values of k_{cat}/K_M for enzyme-catalyzed reactions are $\sim 10^5 M^{-1}s^{-1}$ and the limit is $\sim 10^9 M^{-1}s^{-1}$. This limit comes from the requirement for two species (E and S) to combine to make the $E \cdot S$ complex. This rate cannot be faster than diffusion.

In many biocatalysis applications, the users control the substrate concentrations and usually choose high concentrations so the reac-

7.2 Measuring reaction rates

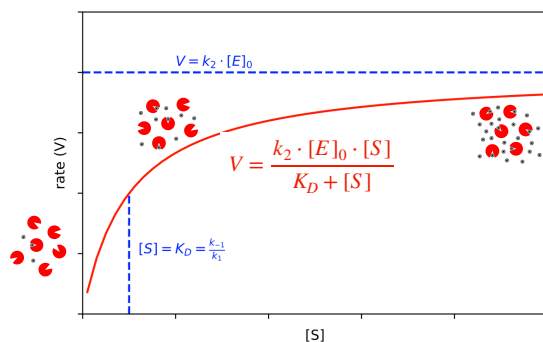


Figure 7.4. The Michaelis-Menten model adds a physical interpretation to the observed variation in rate of an enzyme-catalyzed reaction with substrate concentration. At low substrate concentration, the rate is slow because only a few enzyme molecule contain bound substrate (black dots). The rate increases as the substrate concentration increases because an increasing fraction of enzyme molecules contain substrate. At high substrate concentration, all the enzyme molecules contain a bound substrate and the enzyme-catalyzed reaction rate reaches a maximum and no longer varies with substrate concentration.

7 Engineering Faster Enzymes

tion rate is high and the reaction vessel is small. At high substrate concentrations, k_{cat} determines the reaction rate, so the enzyme with the higher k_{cat} value is the faster one. For example, the isomerization of glucose to fructose is a continuous process in glucose syrup as solvent. The glucose concentration (~5 M) is much higher than the K_M of glucose for the isomerase (0.09-0.9 M).^[2] The value of k_{cat} determines the reaction rate.

In chemical manufacturing using batch processes, the substrate concentration decreases while product concentration increases as the reaction proceeds leading to changes in reaction rate. The value of k_{cat} may control the reaction rate initially, but later k_{cat}/K_M controls the reaction. Substrate and product inhibition, not discussed above, may also influence the reaction rate. Fox and Clay^[3] proposed an average velocity metric that compares how long it takes for a reaction to reach completion. An example where this approach may be useful is the manufacture of a pharmaceutical or pharmaceutical intermediate.

The following sections examine molecular strategies to alter k_{cat} and K_M to speed up enzyme-catalyzed reactions. Identifying whether k_{cat} or K_M or both need improvement is a good first step, but it is not enough. Both binding and the reaction steps may include multiple physical steps for different catalytic mechanisms. Engineering requires identifying which physical steps limit catalysis and then speed up these steps. For example, a substrate may bind poorly to an enzyme, which could indicate a mismatch in the shapes of the substrate and active site. When a substrate reacts slowly, there are many possible causes. The substrate may orient in a non-reactive orientation, the required acids or bases may be in the incorrect protonation state, the charges generated during bond-breaking and bond making may not be stabilized, a required conformational change of a protein loop may be slow as well as other causes.

7.2.3 Gibbs energy diagrams for enzyme-catalyzed reactions

Michaelis-Menten model. Enzymes catalyze reactions by first binding the substrate and then transforming the substrate to product, so a Gibbs energy diagram includes these steps to show how K_M and k_{cat} affect an enzyme-catalyzed reaction,^[4] **Figure 7.5**. First, the Gibbs energy of the transition state for the non-catalyzed reaction, $S^\ddagger$, is higher than that for the enzyme-catalyzed reaction, $E \cdot S^\ddagger$ indicating that the enzyme-catalyzed reaction is faster. The formation of the $E \cdot S$ complex is favorable by an amount ΔG_{K_M} ; then transformation of $E \cdot S$ to the transition state $E \cdot S^\ddagger$, is uphill by $\Delta G_{k_{cat}}$.

More complex reaction mechanisms. The Gibbs energy diagrams include additional states for more complex reaction mechanisms. For example, a reaction that includes an enzyme intermediate includes a valley that represents that intermediate, **Figure 7.6**.

Even if the Gibbs energy profile is complex, the largest barrier (from valley to peak) is the step that determines k_{cat} and the highest point measured from the starting $E + S$ determines k_{cat}/K_M .

Simplification for limiting cases. Consider two limiting cases of enzyme-catalyzed reactions: high and low substrate concentration, **Figure 7.7**. At high substrate concentration most of the enzyme exists as the $E \cdot S$ complex, while at low substrate concentration most of the enzyme exists as free enzyme.

At high substrate concentration, most of the enzyme exists as the $E \cdot S$ complex, so the starting state is this complex. The barrier from the starting state to product is determined by k_{cat} and the only way to increase the rate of reaction is to increase k_{cat} . Raising the energy of the $E \cdot S$ complex (weakening binding) will increase k_{cat} and speed up the reaction up to a point. Once the binding becomes

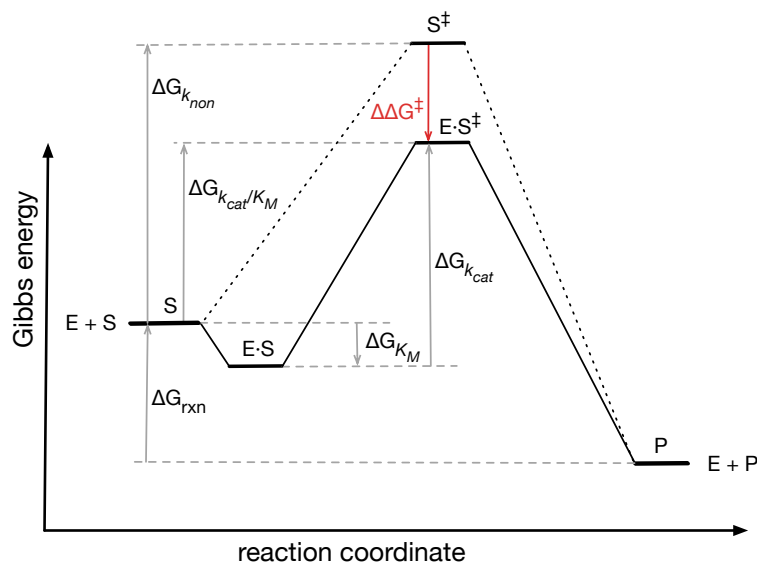


Figure 7.5. Gibbs energy diagram for a non-catalyzed and an enzyme-catalyzed reaction at standard state. The non-catalyzed reaction proceeds via transition state $S^\ddagger$, which lies $\Delta G_{k_{non}}$ above the energy of S . The enzyme-catalyzed reaction is faster because the energy distance from $E + S$ to the transition state, $E \cdot S^\ddagger$, $\Delta G_{k_{cat}/K_M}$, is lower by an amount $\Delta \Delta G^\ddagger$. The path from $E + S$ to $E \cdot S^\ddagger$ involves binding to form $E \cdot S$ (decrease Gibbs energy by ΔG_{K_M}), then an increase to $E \cdot S^\ddagger$ by an amount $\Delta G_{k_{cat}}$.

7.2 Measuring reaction rates

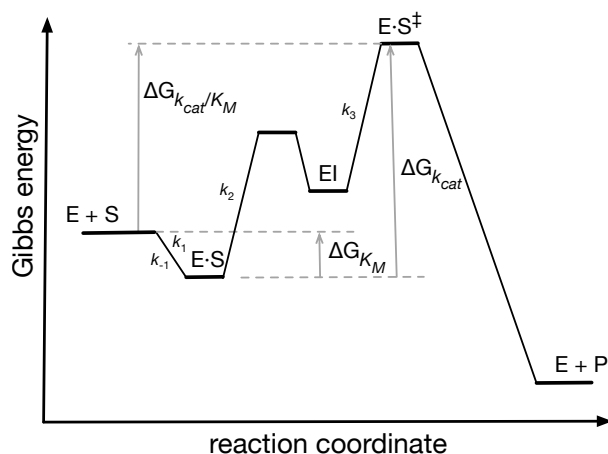


Figure 7.6. Gibbs energy diagram for an enzyme-catalyzed reaction that includes an enzyme intermediate as shown in **Figure 7.8** above. k_{cat} depends on both k_2 and k_3 .

so weak that the reaction is no longer under ‘high substrate concentration’ conditions and most of the enzyme no longer exists as the $E \cdot S$ complex, it will slow down and the limiting case of high substrate concentration no longer applies. Increasing binding (lowering K_M) slows down the reaction when the substrate concentration is high because it increases the distance between the $E \cdot S$ complex and the transition state. Most biocatalysis applications use high substrate concentrations so the rates are limited by k_{cat} .

i Box 7.1: What Determines k_{cat} vs. k_{cat}/K_M in a Gibbs Energy Diagram

The barrier that sets k_{cat} starts from the $E \cdot S$ complex or similar.

7 Engineering Faster Enzymes

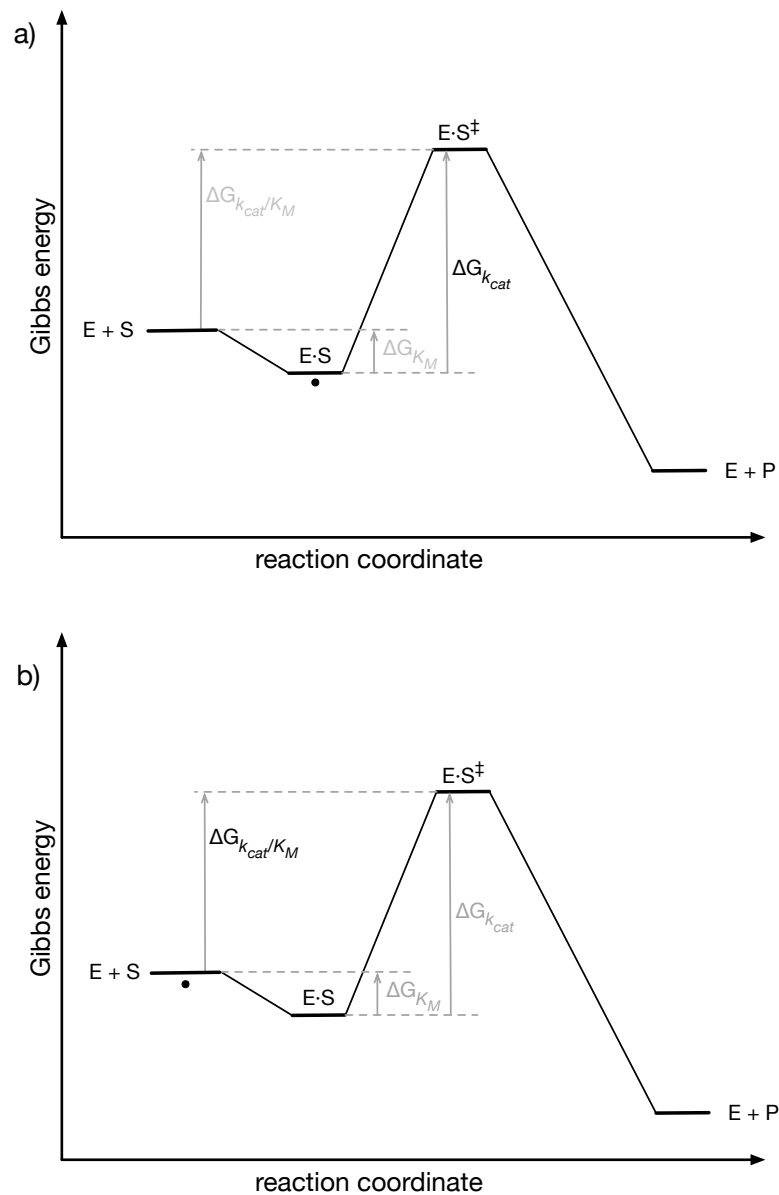


Figure 7.7. Limiting cases for enzyme-catalyzed reactions. a) At high substrate concentration, most of the enzyme contains bound substrate as indicated by the black dot under that state. The reaction rate under these conditions depends only on k_{cat} . b) At low substrate concentration, most of the substrate and enzyme are in the free state as indicated by the black dot under that state. The reaction rate depends on substrate concentration and k_{cat}/K_M at low substrate concentration. The substrate concentration and K_M determine the small fraction of enzyme that forms $E \cdot S$ while k_{cat} determines how fast it is converted to product.

- k_{cat} limits the rate at high substrate concentration, so the ground state (stable state populated by the enzyme at steady state) is the E·S complex or another low energy state after the E·S complex. The highest transition state energy relative to this ground state sets k_{cat} . The vertical distance from this ground state to the highest transition state is the Gibbs energy that sets k_{cat} .
- This highest transition state may be a chemical transformation (bond-making and bond breaking), a product release step, or a conformational change of the protein.
- Example: Peptide hydrolysis catalyzed by serine proteases involves two steps – acyl-enzyme formation followed by acyl-enzyme release. If acyl-enzyme *formation* is rate-limiting, then the enzyme ground state is the E·S complex and the barrier from E·S to the first tetrahedral intermediate determines k_{cat} . If the acyl-enzyme *release* is rate-limiting, then the enzyme ground state is the acyl enzyme and barrier from the acyl enzyme to the second tetrahedral intermediate determines k_{cat} .
- If two barriers have similar heights, then both contribute to k_{cat} .

The barrier that sets k_{cat}/K_M starts at the free E + S state.

- The highest transition state energy relative to free E + S sets k_{cat}/K_M .
- $[S] \cdot k_{cat}/K_M$ sets the rate when at low substrate concentrations. Low substrate conditions may occur where one cannot control the substrate concentration, such as applications where the substrate is a metabolite in living organisms.

At low substrate concentration, most of the enzyme exists as $E + S$.

The distance to the barrier to product is determined by k_{cat}/K_M . Selectively stabilizing $E \cdot S^\ddagger$ will lower the energy barrier and speed up the reaction as it did in the high substrate concentration case. However, stabilizing $E \cdot S$ (decreasing K_M) in the low substrate concentration case also increases the reaction rate by increasing the proportion of enzyme molecules bound to substrate. This approach is limited since once $E \cdot S$ is so stable that all the enzyme is in the $E \cdot S$ state, then the high substrate concentration case applies and only stabilizing $E \cdot S^\ddagger$ can increase the reaction rate. For example, the anti-cancer activity of asparaginase^[5] (mentioned in Chapter 1) depends on its ability to reduce the concentration of asparagine in the blood so that it is not available for cancer cells. The blood contains 2-25 μM asparagine, so lowering the K_M of an asparaginase to this level or slightly below would make the reaction faster, but further decreases in K_M would not.

7.3 More complex models of enzyme-catalyzed reactions

The simple Michaelis-Menten model makes several simplifying assumptions, **Table 7.1**, but these can be removed either by additional experiments or redefining the meanings of k_{cat} and K_M .

Table 7.1. Assumptions within the Michaelis-Menten model of enzyme catalysis and how they can be removed.

Assumption	Removing assumption
single substrate	additional experiments
equilibrium between E , S and $E \cdot S$	redefine K_M
single step reaction	redefine k_{cat} and K_M

7.3 More complex models of enzyme-catalyzed reactions

Assumption	Removing assumption
product release is fast	additional experiments

The Michaelis-Menten model includes only one substrate, but it can be extended to two or more substrates by repeating the measurements under different limiting conditions. To extend the model to two-substrate enzyme-catalyzed reactions one first adds a large excess of one substrate. The other substrate now limits the reaction rate according to eq. Equation 7.6, so varying its concentration reveals its steady-state kinetic parameters. Next, a second set of experiments with saturating amounts of the other substrate allows measurement of the kinetic parameters of the first substrate.

The Michaelis-Menten model assumed that the free enzyme and substrate are in equilibrium with the enzyme substrate complex in eq. Equation 7.11 above. This assumption was needed in order to define the concentration of $E \cdot S$. The Briggs-Haldane approach replaces this assumption with a more relaxed assumption, that the concentration of $E \cdot S$ is constant. The assumption, known as the steady-state approximation, is less restrictive and leads to a different physical meaning for K_M .

The steady state assumption means that the rate of formation of $E \cdot S$ (k_1 term in eq. Equation 7.16) must be equal to the rate of destruction of $E \cdot S$ ($k_{-1} + k_2$ term in eq. Equation 7.16).

$$k_1 \cdot [E] \cdot [S] = (k_{-1} + k_2)[E \cdot S] \quad (7.16)$$

This equation is used in place of eq. Equation 7.10 above to express $[E \cdot S]$ in terms of known quantities. Combining eq. Equation 7.16 with eq. Equation 7.10 and solving for $[E \cdot S]$ yields the equation below.

7 Engineering Faster Enzymes

$$[E \cdot S] = \frac{[E]_0 \cdot [S]}{\frac{k_{-1} + k_2}{k_1} + [S]} \quad (7.17)$$

Combining this equation with eq. Equation 7.9 above yields eq Equation 7.18 below.

$$V = \frac{k_2 \cdot [E]_0 \cdot [S]}{K_M + [S]} \quad (7.18)$$

where $K_M = \frac{k_{-1} + k_2}{k_1}$

The form of the equation and the definition of V_{max} is the same as for the Michaelis-Menten model, but the definition of the constant K_M includes an additional physical step. It is no longer purely a dissociation constant since it contains the rate constant for the chemical step, k_2 , in the numerator. Nevertheless, it still represents an amount of $[E \cdot S]$ since the rate constants in the numerator are for processes that destroy $[E \cdot S]$, while the rate constant in the denominator is for formation of $[E \cdot S]$. If k_2 is slow, the K_M is a true dissociation constant.

Many enzyme-catalyzed reactions involve multiple steps, not just the single step in the Michaelis-Menten model. For example, many enzyme-catalyzed reaction form an intermediate, EI, **Figure 7.8**. Subtilisin-catalyzed hydrolysis of esters involves an acyl-enzyme intermediate. Later in this chapter, **Figure 7.26** shows a detailed mechanism.

The steady-state enzyme kinetics for subtilisin still follow the Michaelis-Menten equation, but both K_M and k_{cat} are combinations of multiple physical steps:

7.3 More complex models of enzyme-catalyzed reactions

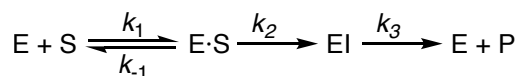


Figure 7.8. Many enzyme-catalyzed reactions involve an enzyme-bound intermediate. These reactions have two steps that involve bond-making and bond-breaking, k_2 and k_3 , instead of just one in the Michaelis-Menten model. These reactions still follow the curve in eq. Equation 7.6, but multiple physical steps contribute to V_{max} and K_M .

$$K_M = \frac{k_3}{k_2 + k_3} \cdot \frac{k_{-1} + k_2}{k_1}$$

$$k_{cat} = \frac{k_2 \cdot k_3}{k_2 + k_3}$$

Measuring steady state kinetics still yields values for K_M and k_{cat} , but their meaning cannot be assigned to a simple step in the catalytic mechanism. In the special case where the deacylation step (k_3) is fast as compared to the acylation step (k_2), then the equations simplify to those for a single step reaction. In this case, k_{cat} represents the slow step of catalysis, k_2 , the formation of the acyl-enzyme intermediate.

$$K_M = \frac{k_2 + k_{-1}}{k_1}$$

$$k_{cat} = k_2$$

Finally, the Michaelis-Menten model assumes that product release is fast, but in practice product inhibition may slow the rate

of enzyme-catalyzed reactions. Additional experiments in the presence of varying amounts of product can identify product inhibition.

In this text, we will use the Michaelis-Menten model to discuss most enzyme-catalyzed reactions. When the mechanisms are more complex, additional physical steps contribute to K_M and k_{cat} , and the Gibbs energy diagrams become more complex.

7.4 Challenges in engineering faster enzymes

Speeding up catalysis requires speeding up the rate-limiting step, which poses two challenges: identifying that step, and finding a way to accelerate it. Both of these problems are challenging.

7.4.1 Physical, non-chemical steps can limit k_{cat}

In the simplest enzymes, k_{cat} reflects the chemical step in which bonds are broken and formed; if this step is slow, it limits the rate, and the methods described in the previous section for stabilizing the transition state can speed it up.

Most enzyme-catalyzed reactions involve multiple steps, both chemical and physical. Serine esterase-catalyzed ester hydrolysis, for example, involves two chemical steps — formation of the acyl-enzyme intermediate and its subsequent hydrolysis — plus several physical steps that involve no bond making or breaking: correct orientation of the substrate before acylation, release of the alcohol product and binding of water after acylation, and release of the acid product after deacylation. Any of these steps, chemical or physical, can limit k_{cat} . This is easy to overlook because one tends to picture the simplest case, where k_{cat} is purely chemical;

7.4 Challenges in engineering faster enzymes

in reality, the measured k_{cat} usually reflects a mixture of chemical and physical rates.

Resolving k_{cat} into its elementary steps requires additional experiments. Chemical steps can be diagnosed with a primary deuterium or ^{13}C kinetic isotope effect (KIE): a KIE substantially greater than one indicates that making or breaking the isotope-sensitive bond is at least partially rate-limiting, and a mutation that reduces the KIE toward one indicates that the chemical step has become faster relative to the others.^[6] Mutations that sharpen the bell-shaped pH-rate profile, or shift an apparent $\text{p}K_a$ toward its optimum, indicate an optimized proton-transfer step.^[7] Linear free-energy relationships—varying substrate electronics systematically—reveal how much the chemical step contributes to the overall rate; for ester hydrolysis, a shallower Brønsted slope indicates that the chemical step no longer dominates.^[8] Pre-steady-state burst kinetics resolve acylation from deacylation directly, uncoupled from product release, so an increased burst rate confirms an improved acylation step.

Physical rate limitations can be intermolecular or intramolecular. Intermolecular steps—substrate association and product release—couple to bulk solvent diffusion, so k_{cat} and k_{cat}/K_M vary with solvent viscosity; the slope of rate versus relative viscosity quantifies the diffusion-controlled contribution to each parameter.^{[9], [10]} Isotope partitioning identifies non-productive binding by measuring the fraction of bound substrate that commits to product rather than dissociating and reassociating.^[11] Intramolecular steps—tunnel transit, active-site preorganization, and conformational gating—require structural or computational diagnostics: crystal structures and MD ensembles report tunnel geometry and bottleneck radius,^[12] and HDX-MS reports changes in conformational dynamics at peptide resolution.^[13] Improvement in a specific step confirms successful engineering even before the overall k_{cat} increases.

The sections below identify ways to speed the various steps of enzyme-catalyzed reactions.

7.5 Increasing reaction rates by engineering tighter binding

The optimum binding of the substrate and enzyme will be just strong enough (K_M value just low enough) so that most of the enzyme contains bound substrate under the reaction conditions. Filling most enzyme active sites with substrate creates the opportunity for each enzyme molecule to contribute to catalysis. Because over-stabilizing the $E \cdot S$ complex eventually slows the reaction (see **Figure 7.5** above), further stabilization beyond this point is undesirable.

In many cases, the target substrate is not the natural substrate of the enzyme and the reaction is slow because little $E \cdot S$ forms. Adjusting the complementarity of binding site to match the target substrate improves binding and increases the reaction rate. The engineering approaches are the same as those described in the previous chapter on binding: preorganization, matching the size and shape and the interactions between the substrate and the binding site. A number of research groups have engineered enzymes for higher catalytic activity toward new substrates by improving binding.^[14]

Matching substrate size. Substrates that are larger than the natural substrate bind poorly when they cannot fit in the active site. For example, alcoholic fermentation by *Lactococcus lactis*, involves the reduction of acetaldehyde to ethanol by NADH catalyzed by alcohol dehydrogenase, **Figure 7.9**, which shows good catalytic efficiency: $K_M = 0.4 \text{ mM}$, $k_{cat} = 35 \text{ s}^{-1}$, $k_{cat}/K_M \sim 90,000 \text{ M}^{-1} \text{ s}^{-1}$.

7.5 Increasing reaction rates by engineering tighter binding

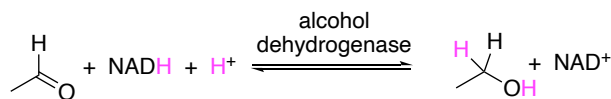


Figure 7.9. Wild-type alcohol dehydrogenase efficiently catalyzes reduction of the natural substrate, acetaldehyde.

As part of a project on biofuels, researchers sought to use this enzyme to reduce isobutyraldehyde to isobutanol, **Figure 7.10**, where the larger isobutyraldehyde replaces acetaldehyde.

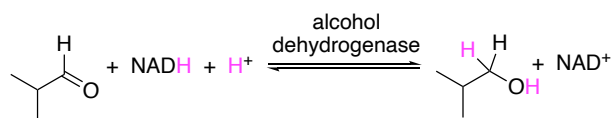


Figure 7.10. Wild-type alcohol dehydrogenase also catalyzes reduction of the substrate analog, isobutyraldehyde, but ~30-fold less efficiently.

The catalytic efficiency of this *Lactococcus* alcohol dehydrogenase is ~30-fold lower toward isobutyraldehyde, mainly due to looser binding of the substrate: $K_M = 12 \text{ mM}$, $k_{cat} = 30 \text{ s}^{-1}$, $k_{cat}/K_M \sim 3,000 \text{ M}^{-1} \text{ s}^{-1}$. The researchers identified several improved variants by random mutagenesis and screening.^[15] The structure of one variant showed an expanded binding site caused by substituting Leu264 with valine to remove a methylene group and substituting Tyr50 with phenylalanine to remove an oxygen atom.^[16] The modified enzyme bound the larger substrate more tightly ($K_M = 1.6 \text{ mM}$) and showed a ten-fold higher catalytic efficiency ($k_{cat}/K_M \sim 32,000 \text{ M}^{-1} \text{ s}^{-1}$). These substitutions made the substrate-binding site larger by removing two non-hydrogen atoms (a carbon and an oxygen) to match the new substrate, which is larger by two non-hydrogen atoms (two carbons).

In an industrial example, Savile and coworkers engineered a

transaminase to accept a precursor for a diabetes drug by expanding the substrate-binding site with multiple substitutions to fit the larger substrate.^[17]

Although smaller substrates fit in the active site, they may bind poorly because only part of the smaller substrate contacts the surface of the binding site. Decreasing the size of the binding site will increase the contact surface and strengthen the binding interaction.

Matching interaction complementarity. Engineering a suitable binding site often requires adjusting the non-covalent interactions as well as the size. For example, accommodating the new substrate, a glutamine residue, in a site that normally accepts lysine or arginine required adjusting the size, the charge and the hydrogen bonding partners of the binding site. Gluten, a protein found in wheat, rye, and barley, contains a high proportion of proline and glutamine. Digestive proteases do not cleave peptides after a Pro-Gln sequence so oligopeptides enriched in the dipeptide sequence proline-glutamine accumulate upon digestion of gluten. In celiac disease patients, these oligopeptides are believed to trigger an autoimmune response, which causes diarrhea and other gastrointestinal problems. One potential solution is adding a protease that can degrade Pro-Gln containing peptides to the meals of celiac patients. Engineering such a protease started with kumamolisin, which catalyzes hydrolysis after a Pro-Arg/Lys sequence in a peptide.^[18] The goal was to increase the reaction rate for hydrolysis after a Pro-Gln sequence, **Figure 7.11**.

The changes involved replacing two negatively-charged Glu residues in the binding site with neutral residues to favor binding of the neutral Gln instead of the positively charged Arg or Lys. In addition, a glycine in the binding site was replaced by serine to decrease the size of the binding site because the side chain of Gln is smaller than the side chain of Arg or Lys. The side chain of the

7.5 Increasing reaction rates by engineering tighter binding

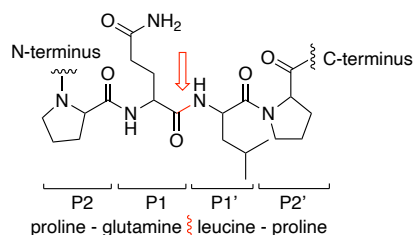


Figure 7.11. Target tetrapeptide used for computational design of a protease that cleaves immunogenic gluten peptides. Cleavage occurs between the P1 (Gln) and P1' (Leu) amino acid residues as marked by the red arrow and the wavy line. The starting protease, kumamolisin, favors hydrolysis after a Pro-Arg/Lys sequence. The engineering goal was to increase catalysis of hydrolysis after a Pro-Gln sequence such as the one shown.

newly-added serine can also make a hydrogen bond with the side chain of Gln. Several other substitutions were also introduced outside the binding site. The rate of hydrolysis (k_{cat}/K_M) of a test peptide (R-ProGlnProGln~LeuPro-R') where the '~' marks the cleavage site, increased 120-fold from 4.9 to 570 M⁻¹ s⁻¹. In this example, creating a complementary binding site required changing the size of the binding site, the electrostatic character of the binding site, and the hydrogen bonding partners in the binding site. A further-engineered variant of this protease is currently undergoing clinical trials.^[19]

Slow binding due to tunnels or flexible loops. In some cases, the rate of association of substrate and enzyme is slow; this slow rate can limit the rate of reaction. In this case improving binding will speed up the reaction.

The most likely cause of slow binding is a deeply-buried active site. Reaching the active site may require the substrate to diffuse

through a tunnel or require a flexible loop to move out of the way to allow the substrate into the active site. This slower binding hinders formation of the $E \cdot S$ complex and thus raises its energy. Increasing the rate of formation of the $E \cdot S$ complex can increase the amount of complex that forms. For example, to increase the ability of a transaminase to accept larger substrates up to 120-fold, Pavlidis and coworkers^[20] both expanded the active site with two substitutions of Tyr with Phe and increased the flexibility at the opening of the binding site with a Pro to His substitution to allow larger substrates to enter.

Tunnels and flexible loops can also contribute to the selectivity of enzymes.^[21] Tunnels can prevent access to large substrates or to substrates that interact unfavorably with the tunnel (hydrophilic substrate may not pass through a hydrophobic tunnel) or substrates that interact too favorably (hydrophobic substrate may bind strongly to a hydrophobic tunnel leading to non-productive binding). Flexible loops can also exclude solvent from the active site to prevent undesirable side reactions.

7.6 Stabilizing the chemical transition state to speed up reactions

After binding, the enzyme must stabilize the transition state for the reaction. This stabilization promotes the bond-making and bond-breaking steps. Enzymes use different strategies for this stabilization depending on what is required by the reaction, **Table 7.2**.

7.6 Stabilizing the chemical transition state to speed up reactions

Table 7.2. Contributions to the stabilization of transition states by enzymes.

catalytic principle	rationale	implementation
preorganize $E \cdot S$ geometry for reaction	$E \cdot S$ complex adopts catalytically productive conformation	-model near attack complex -remove non-productive binding
optimize acids & bases	acids & bases usually needed for catalysis	-adjust pK_a of acids & bases
stabilize charge & shape of transition state	bond rearrangements alter charge & shape of substrate	-add complementary charges -adjust shape of active site
enable new mechanistic steps	enzymes can create new pathways for reaction	-chemical reasoning -introduce new functional groups

The first requirement is a reactive orientation of the substrate, which is called preorganization. This preorganization is a binding orientation where the substrate adopts a ready-to-react geometry close to the catalytic residues of the enzyme. In contrast, holding the substrate in a non-reactive orientation slows down the reaction. Preorganization may require the substrate to adopt a conformation different from its low energy conformation in solution.

Once the substrate is oriented for reaction, additional interactions can stabilize the bond-breaking and bond-making process. Moving the substrate from the polar environment of water ($\epsilon = 80$), to

the non-polar environment of protein with a 10-fold lower dielectric constant ($\epsilon = 4\text{-}20$) strengthens all electrostatic interactions 10-fold because they vary inversely with dielectric constant.^[22] If the reaction involves proton transfers, the required acids must be protonated and the bases deprotonated for catalysis. The enzyme can further stabilize the new molecular shapes and charges as electrons reorganize in the transition state. In some cases, enzymes create new mechanisms as compared to the non-catalyzed reaction. These new mechanisms break up a high energy path into several low energy steps.

Individual enzymes vary in the contributions that they use for catalysis because reactions differ in the reasons that they are slow. For example, chorismate mutase stabilizes the transition state mainly by folding the substrate into a reactive conformation (preorganization), while a designed eliminase catalyzes the Kemp elimination reaction by providing a base and stabilizing the charges formed in the transition state. Enzymes must provide whichever catalytic contributions are needed to stabilize the transition state for each reaction. A cooking analogy is that adding salt sometimes improves the taste of a dish, but other times adding something sour or spicy is needed. Each dish differs on what improves it.

To qualitatively predict the transition state for an enzyme-catalyzed reaction, start with an arrow-pushing mechanism to identify the bond-making and bond-breaking steps. Identify the distances and angles that define a catalytically productive conformation. Draw a shape with partial bonds that lies in between the shapes of the substrate and product. Label any partial charges that form in the transition state. Identify any acids and bases required for the mechanism. Quantitative prediction of a transition state requires quantum-chemical modeling.

To identify how an enzyme accelerates the rate of reaction, com-

7.6 Stabilizing the chemical transition state to speed up reactions

pare the enzyme-catalyzed mechanism and transition state to a similar analysis of the non-catalyzed reaction. Does the enzyme preorganize the reacting atoms for the reaction? Does the enzyme supply acids and bases as needed for any proton transfer steps? Does the enzyme stabilize the unique shape of the transition state and any partial charges that form in the transition state? If the enzyme-catalyzed reaction includes covalent intermediates, then the enzyme also changes the mechanism of the reaction as compared to the non-catalyzed reaction.

7.6.1 Preorganization of $E \cdot S$ for reaction (binding correctly)

Preorganization is the bringing of atoms into the correct orientation to react. Preorganization lowers the Gibbs energy of the transition state, $E \cdot S^\ddagger$. The preorganized state (sometimes called a near-attack complex) involves both the conformation of the substrate and the orientation of substrate relative to the enzyme. This preorganized state is like a turnstile through which the substrate must pass to reach the transition state. Preorganization involves not just binding, but catalytically productive binding. In an efficient enzyme, formation of the $E \cdot S$ complex also organizes the complex for reaction such that binding and preorganization are the same processes. K_M can be improved by any type of binding, but k_{cat} can only be improved by catalytically productive binding. Preorganization may occur within the $E \cdot S$ state or along the path to the transition state, $E \cdot S^\ddagger$.

Holding reactive groups close to each other in the correct orientation for the reaction enormously speeds up reactions as compared to freely diffusing molecules containing the same groups. Even when the reacting groups are in the same molecule, reducing flexibility to favor a reactive orientation speeds up the reaction rate tremendously. For example, the intramolecular ring closure

reaction in **Figure 7.12** varies 50,000-fold by varying the number of rotatable bonds between the reacting groups.^[23] The chemical steps are the same in all cases, but the faster ones orient the substrate better for reaction than the slower ones.

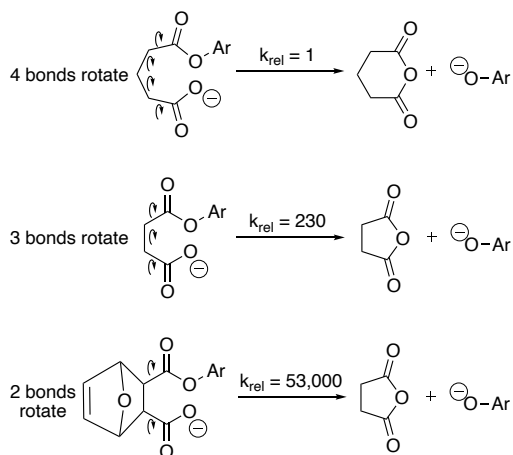


Figure 7.12. The rates of a ring closure reaction increase dramatically as the connecting fragments restrict possible rotations to favor the reactive conformation. The ring closure reaction requires attack of the carboxylate oxygen at the ester carbonyl carbon (Ar = 4-bromophenyl). Similarly, an active site that positions substrate and enzyme functional groups in a reactive orientation is a better catalyst than one that positions the groups in a non-reacting orientation. k_{rel} = relative rate constant.

In solution, substrates are usually flexible and adopt many different conformations by rotations along single bonds. Only some conformations orient the reactive groups such that reaction can occur. This reactive conformation might be rare in solution. The changes in structure force the carboxylate and aryl ester groups close to

7.6 Stabilizing the chemical transition state to speed up reactions

each other in **Figure 7.12** above. Binding to the enzyme can similarly shift the balance toward a reactive conformation. The balance shifts because the enzyme restricts the available space for the substrate to move and interacts with it using hydrogen bonds, the hydrophobic effect, and electrostatic interactions.

For example, the enzyme chorismate mutase catalyzes the rearrangement of chorismate by binding it in a conformation that rarely exists in solution. Chorismate mutase catalyzes a Claisen rearrangement, which rearranges bonds within a cyclic transition state, **Figure 7.13**. The transition state requires the two carbons that form a bond in the product to be near each other.

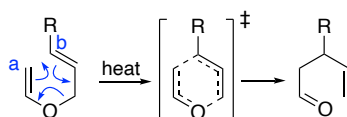


Figure 7.13. The Claisen rearrangement requires the carbon atoms marked a and b to be nearby each other in the transition state (marked ‡) so that a new bond can form between them.

The favored conformation of chorismate in solution places the two carboxylate groups far from each other, **Figure 7.14**. This separation of the carboxylates minimizes electrostatic repulsion and allows water to solvate each carboxylate. The Claisen rearrangement forms a bond between carbon 1 and 9 of chorismate, so these carbon atoms must be nearby in the transition state. Chorismate mutase holds chorismate in a folded, reactive conformation using two positively charged arginine side chains to position the carboxylates and two hydrophobic side chains to restrict their motion.^[24] Molecular dynamics simulations indicate that chorismate adopts the correct preorganized conformation only 0.00007% of the time in solution, but 32% of the time while bound to the enzyme. This preorganization is the primary way by which chorismate mutase stabilizes

the transition state and catalyzes this rearrangement.

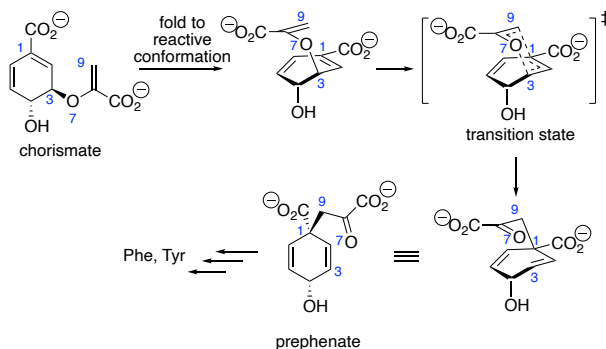


Figure 7.14. Chorismate mutase catalyzes a Claisen rearrangement that converts chorismate to prephenate in the pathway to make aromatic amino acids. In the transition state, the carbon atoms marked 1 and 9 must be nearby, but in solution, these carbon atoms lie far from each other to minimize electrostatic repulsion between the two carboxylate groups and to maximize their solvation. Chorismate mutase stabilizes the transition state for reaction by binding the substrate in a folded conformation that places carbon atoms 1 and 9 nearby for reaction.

Many enzyme-catalyzed reactions involve proton transfers. For these reactions, preorganization involves positioning the donor and acceptor atoms to make a hydrogen bond. Once the hydrogen bond is formed, the hydrogen is ready to transfer from the donor to acceptor atom.

An example of a reaction involving proton transfers is the Kemp elimination, **Figure 7.15**. Researchers designed many enzymes to catalyze this reaction, but testing revealed that most of them were inactive.^[25] To model preorganization of the substrate-enzyme complex, researchers calculate the ensemble of conformations

7.6 Stabilizing the chemical transition state to speed up reactions

adopted by this complex. If the substrate-enzyme complex is preorganized for reaction, then many of the conformations will have the catalytic atoms in the enzyme and the reacting atoms of the substrate site positioned (distances and angles) for reaction. Molecular dynamics simulations are the traditional way to calculate these ensembles. They revealed that most of the active enzymes were preorganized for reaction, while most of the inactive enzymes were not,^[26] **Figure 7.16**. Preorganization was defined by the presence of a hydrogen bond between the substrate and the catalytic histidine. Researchers suggested that future enzyme designs could include molecular dynamics simulations to more accurately predict preorganization.

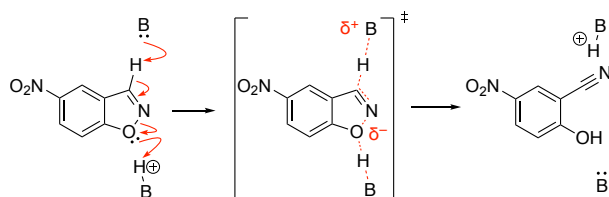


Figure 7.15. The Kemp elimination reaction is a one-step reaction involving proton transfers, breaking of the N–O bond and formation of the C–N triple bond. This reaction proceeds via a single transition state involving a base and an acid. The transition state differs from the substrates in increased partial charges at atoms marked by δ and by altered bond lengths marked by red dashed lines.

A newer, faster way to calculate the ensemble of conformations in the enzyme-substrate complex is neural-network-based calculations such as PLACER.^[27] PLACER is a graph neural network trained on structures of small organic molecules from the Cambridge Structural Database and the Protein Data Bank. PLACER generates conformational ensembles of small organic molecules bound to

7 Engineering Faster Enzymes

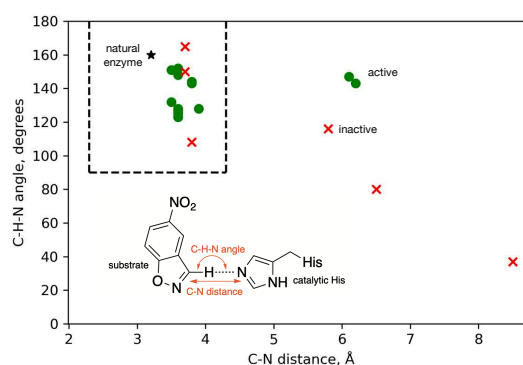


Figure 7.16. Preorganization of the substrate and catalytic histidine for proton transfer correlates with catalytic activity for the Kemp elimination. The reaction involves the catalytic histidine deprotonating the carbon atom indicated. Preorganization involves positioning the donor C-H of the substrate and acceptor N of the catalytic histidine at a suitable distance (x-axis) and angle (y-axis) for a hydrogen bond (boxed region). The angles and distances come from molecular dynamics simulations of the enzyme-substrate complex. The star marks the orientation of donor and acceptor atoms between catalytic groups in the natural enzyme cathepsin.

7.6 Stabilizing the chemical transition state to speed up reactions

proteins. Researchers tested PLACER by predicting the improved reactivity of a retro-aldolase. The designed retro-aldolase (RA95.0) was slow, but an engineered variant was $\sim 10^5$ -fold faster (RA95.5-8F). Previous molecular dynamics simulations showed that the substrate adopted many different conformations in the active site of the original designed enzyme, but in the faster, engineered enzyme the range of conformations narrowed to those where the substrate interacted with the catalytic residues, that is, the enzyme-substrate complex was preorganized for reaction.^[28] The PLACER calculations gave similar results, see Figure 4C of the Anishchenko paper.

When preorganization does not limit the reaction rate, then other features, such as faster product release, need to improve to make the enzyme faster. In such a case, examining an ensemble of conformations of the enzyme-substrate complex from a molecular dynamics or PLACER calculation will not reveal why the rate improved because the calculations look for preorganization and that is not the feature that improved. For example, researchers evolved a variant, KE07 R7-2 (pdb id 6CT3), that was 100-fold faster (k_{cat}) than the designed Kemp eliminase mentioned above, KE07 R1 (pdb id 5D2W). A PLACER calculation shows no improvement in preorganization. A Google colab page is available to test this calculation at <https://colab.research.google.com/drive/1BjbisGQSlqC5n3YnGox2k9valxdElGN>. Other experiments revealed that the reason for the faster rate was better substrate binding and product release associated with changes outside the active site.^[29]

The opposite of preorganization is non-productive binding where the enzyme binds the substrate in an unreactive mode at the active site, **Figure 7.17**. Non-productive binding increases the distance from the $E \cdot S$ complex to the transition state and therefore slows down the reaction. The $E \cdot S$ complex must first reorient into reactive orientation before it can react.

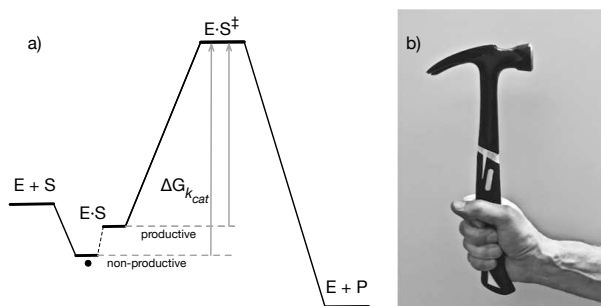


Figure 7.17. Non-productive binding slows down catalysis. This hypothetical Gibbs energy diagram for an enzyme-catalyzed reaction shows the $E \cdot S$ complex as an equilibrating mixture of two $E \cdot S$ species. The more stable one is non-productive because it lies earlier along the reaction coordinate. If most of the $E \cdot S$ complex exists in the non-productive form (indicated by the black dot), then the reaction rate is slower (ΔG_{cat} increases) than if it existed in the productive form. This case is similar to over stabilization of the $E \cdot S$ complex mentioned previously. b) A hammer is ineffective in the wrong orientation and will not drive nails. An active site that positions the substrate in a nonreactive orientation is also ineffective.

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Non-productive binding slows the thermolysin-catalyzed coupling of *N*-benzyloxycarbonyl-L-aspartate (Cbz-L-Asp) with L-phenylalanine methyl ester (L-Phe-OMe), **Figure 7.18**. This reaction is a condensation to form an amide link, not the usual hydrolysis of a peptide. Precipitation of the product dipeptide drives this reaction in the otherwise unfavorable direction.^[30] This condensation is a critical step in the manufacture of aspartame (L-aspartyl-L-phenylalanine methyl ester), a low-calorie sweetener.

X-ray structures of substrates bound to thermolysin revealed the reason for this slow reaction, **Figure 7.18**.^[31] The active site of thermolysin contains a catalytic zinc ion as well as separate regions to bind the carboxyl and amine donors. The carboxyl donor (Cbz-L-Asp) binds to the wrong site, the amine donor site, instead of the carboxyl donor site. The rotation of the Cbz-L-Asp substrate into the correct site is slow, but required before the amine donor substrate (L-Phe-OMe) can bind to make the catalytically productive complex. Replacement of a histidine residue in the amine binding region with aspartate improved the binding of the amine donor substrate approximately 2.5-fold, likely due to favorable electrostatic interactions, and increased the yield of aspartame derivative approximately 2-fold.^[32]

Substrates smaller than the natural one often react poorly due to non-productive binding. Although the substrate fits in the binding site, it adopts a non-productive orientation as it adjusts within the larger active site to find favorable non-covalent interactions. The difficulty in protein engineering is that one rarely knows the nature of this non-productive orientation and how to prevent it. For example, expanding the active site with a Ile303Ala substitution expanded the substrate range of a ligase from the native substrate 4-chlorobenzoate to a larger substrate 3,4-chlorobenzoate, **Figure 7.19**. However, a smaller regioisomer, 3-chlorobenzoate, reacted about 110-fold slower. The researchers hypothesized that

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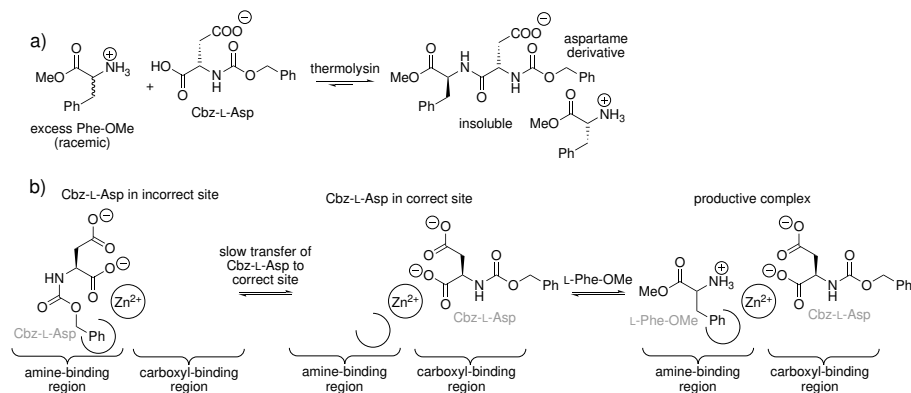


Figure 7.18. Thermolysin-catalyzed coupling yields an aspartame derivative. a) Thermolysin catalyzes the coupling of L-phenylalanine methyl ester (L-Phe-OMe) with Cbz-L-Asp to yield an aspartame derivative. This derivative precipitates from the aqueous reaction mixture by forming a salt with D-phenylalanine methyl ester. The starting Phe-OMe is racemic. Thermolysin selects the L-enantiomer for reaction, while the D-enantiomer is the counter ion for salt formation. b) This coupling is slow because the carboxyl donor, Cbz-L-Asp, initially binds in the incorrect location. It binds in the amine donor site, thereby blocking access of L-Phe-OMe. Once the Cbz-L-Asp rotates and moves out of the way, L-Phe-OMe can bind and the reaction proceeds. Cbz = benzyloxycarbonyl.

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the smaller 3-chlorobenzoate must fit in the active site, but bind in a non-productive orientation. Additional substitutions to close the space for the 4-chloro substituent (Phe184Trp) and orient 3-chlorobenzoate with a hydrogen bond to the 3-chloro substituent (Val209Thr) improved k_{cat}/K_M about 5-fold.

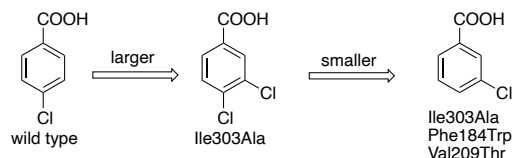


Figure 7.19. Expanding the substrate range of 4-chlorobenzoate:coenzyme A ligase to a larger substrate (3,4-chlorobenzoate) and to a regioisomer (3-chlorobenzoate). Enlarging the active site improved reaction of the larger substrate, but not of the regioisomer, likely due to non-productive binding. Additional adjustments to match the shape of the regioisomer improved its reactivity 5-fold. The reaction rates with the unnatural substrates were 100-1000-fold lower than the natural substrates even with the engineered variants.

Molecular modeling of preorganization and non-productive binding uses docking and molecular dynamics to model the different orientations of the substrate within the active site. Plots of key distances and angles during the simulation identify the favored geometry and classifying the conformations as productive or non-productive, Box 7.1. Protonation states, discussed in the next section, must also be correct for catalysis to occur. Substitutions that favor formation of the near-attack complexes are expected to increase k_{cat} .

i Box 7.2: Preorganization – Geometry, Conformation, and Protonation

Check substrate geometry and conformation by modeling substrate–enzyme complex using molecular dynamics.

Geometric criteria

- **Nucleophilic attack:** Attacking atom 2.5–3.5 Å from electrophile, approach angle 90–120° to leaving group
- **Proton transfer** Donor–acceptor distance 2.4–3.0 Å, D–H···A angle >140°

Conformation criteria

- Substrate adopts reactive conformation >10% of the time.
- No low-energy, non-productive binding modes competing with productive mode.

Check pK_a of catalytic residues with PROPKA.

Protonation criteria

- Catalytic residues maintain correct protonation >90% of the time.

Improving preorganization

- Distance or angle wrong → Resize or reshape binding pocket (may require substitutions outside active site)
- Wrong conformation or non-productive binding → Block incorrect orientation with bulky residues
- Wrong protonation state → Adjust pK_a of catalytic residues.

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7.6.2 Optimize acids & bases

Many reactions involve the transfer of protons, so acids (proton donors) and bases (proton acceptors) must be in the correct protonation state for the reaction to occur. For example, serine esterases require the histidine of the catalytic triad to act as a base in the first chemical step, **Figure 7.20**. Only the neutral form of the histidine side chain can act as a base. The protonated histidine has already accepted a proton (acted as a base) and cannot accept a second proton. At low pH, this histidine is mostly protonated so only the tiny fraction that is neutral can contribute to catalysis, **Figure 7.20**. As the pH increases, the fraction of neutral histidine increases, and the observed rate increases. The inflection point of the increase in activity corresponds to the pK_a of the catalytic histidine. Once the pH increases beyond the pK_a of the histidine, the fraction of neutral histidine approaches 100% and the reaction proceeds at the maximum rate. Thus, the reaction rate varies with pH because the fraction of enzyme in the catalytically active form varies.^[33]

β -lactam antibiotics inactivate penicillin-binding proteins by forming a covalent acyl-enzyme intermediate with an active site serine. For effective inhibition, this adduct must be long-lived, which requires slow hydrolysis. Hydrolytic cleavage depends on a general base activating a water molecule to attack the acyl-carbonyl, but Tyr150's pK_a (~10) is too high relative to physiological pH (~7), leaving it predominantly protonated and unable to efficiently deprotonate water. This inefficient base catalysis results in slow hydrolysis, prolonging enzyme inactivation—precisely what makes β -lactams effective antibiotics.^[34] In contrast, β -lactamase resistance enzymes have evolved efficient catalytic machinery, typically using a glutamate residue (pK_a ~4.2) that is predominantly deprotonated and competent as a general base at physiological pH, enabling rapid hydrolysis of the acyl-intermediate and destroying the antibiotic before it can inhibit its target.

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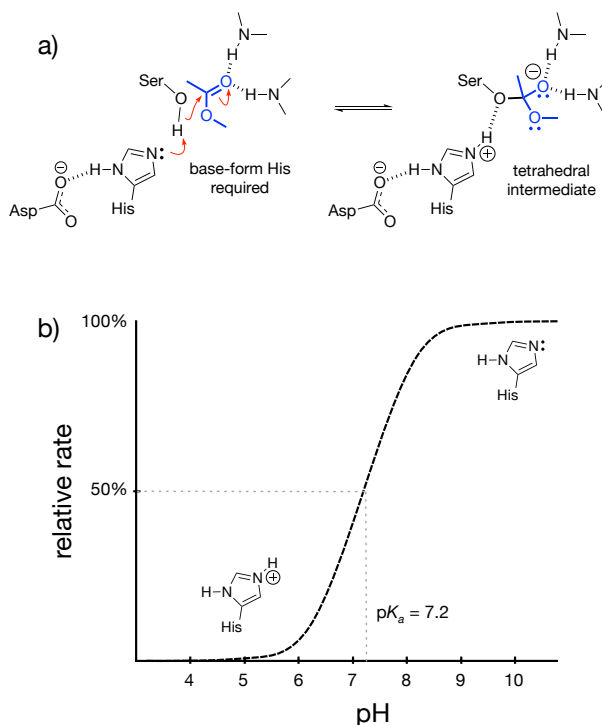


Figure 7.20. The start of the catalytic mechanism of serine esterases requires the deprotonated form of the catalytic histidine. a) The first chemical step of a serine esterase-catalyzed hydrolysis of an ester requires the catalytic histidine to act as a base and to deprotonate the catalytic serine. b) The activity of the esterase varies with pH. The esterase reaches full activity when the histidine is fully deprotonated. The inflection point corresponds to the pK_a of the catalytic histidine. Here the pK_a of the catalytic histidine is 7.2, slightly higher than that for the free histidine due to the environment of the active site.

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A general equation for pH-dependent behavior is

$$k_{obs} = \frac{k_{A^-} \cdot K_a + k_{HA} \cdot [H^+]}{K_a + [H^+]} \quad (7.19)$$

where k_{A^-} is the rate constant of the deprotonated form, k_{HA} is the rate constant for the protonated form and K_a is the ionization constant of the acid. If the reaction requires the basic form, then one sets k_{HA} to zero and equation Equation 7.19 simplifies to equation Equation 7.20 below. The observed rate constant, k_{obs} , is the rate constant for the deprotonated form, k_{A^-} , scaled by the fraction of the molecule in the deprotonated form.[‡]

$$k_{obs} = k_{A^-} \cdot \frac{K_a}{K_a + [H^+]} \quad (7.20)$$

[‡]The fraction deprotonated, $\frac{[A^-]}{[A^-] + [HA]}$ at a given pH depends on the pH and the acid dissociation constant, K_a . The acid dissociation constant is defined as:

$$K_a = \frac{[H^+] \cdot [A^-]}{[HA]}$$

Solving for $[HA]$ yields:

$$[HA] = \frac{[H^+] \cdot [A^-]}{K_a}$$

Substitution into the definition for fraction deprotonated yields:

$$\text{fraction deprotonated} = \frac{[A^-]}{[A^-] + \frac{[H^+] \cdot [A^-]}{K_a}}$$

which simplifies to

$$\text{fraction deprotonated} = \frac{K_a}{K_a + [H^+]}$$

The pK_a of a functional group within a folded protein may differ from that in free solution due to changes in solvation, hydrogen bonding and electrostatic interactions with the protein. For example, the phenolic hydroxyl of tyrosine has a pK_a of ~ 10 in solution, but Tyr80 in xylanase has a predicted pK_a of ~ 19 . This increase in pK_a corresponds to a billion-fold shift in the equilibrium between phenol and phenoxide toward the phenol, ensuring Tyr80 remains in the neutral, protonated form throughout the physiological pH range. This is functionally essential because the neutral phenol acts as a structural scaffold that positions a catalytic glutamate for optimal activity. If Tyr80 were deprotonated to the negatively charged phenoxide, it would electrostatically repel the glutamate carboxylate and disrupt the precise active site geometry required for catalysis. The folded protein environment achieves this dramatic pK_a elevation by surrounding Tyr80 with hydrophobic residues that severely destabilize the charged phenoxide form through poor solvation, guaranteeing that Tyr80 functions as a neutral hydrogen bond donor/acceptor rather than as an ionizable catalytic residue.^[35]

Besides solvation, hydrogen bonding and electrostatic effects also shift the pK_a of ionizable groups in proteins. When hydrogen bond donors or acceptors interact with ionizable groups and the interaction is unequal for the protonated and deprotonated forms, then their pK_a will shift, **Figure 7.21**. The pK_a 's also shift when nearby charged groups stabilize or destabilize the charges associated with protonation changes. The key to these changes is that the interactions are unequal – either the protonated or deprotonated state interacts more strongly.

PROPKA predicts shifts in pK_a . PROPKA estimates the pK_a of ionizable groups in a protein ($pK_a^{protein}$) by adjusting the pK_a of the groups in solution ($pK_a^{solution}$) for their different environment in the folded protein.^[36] The estimate requires a protein structure and assumes that the structure is

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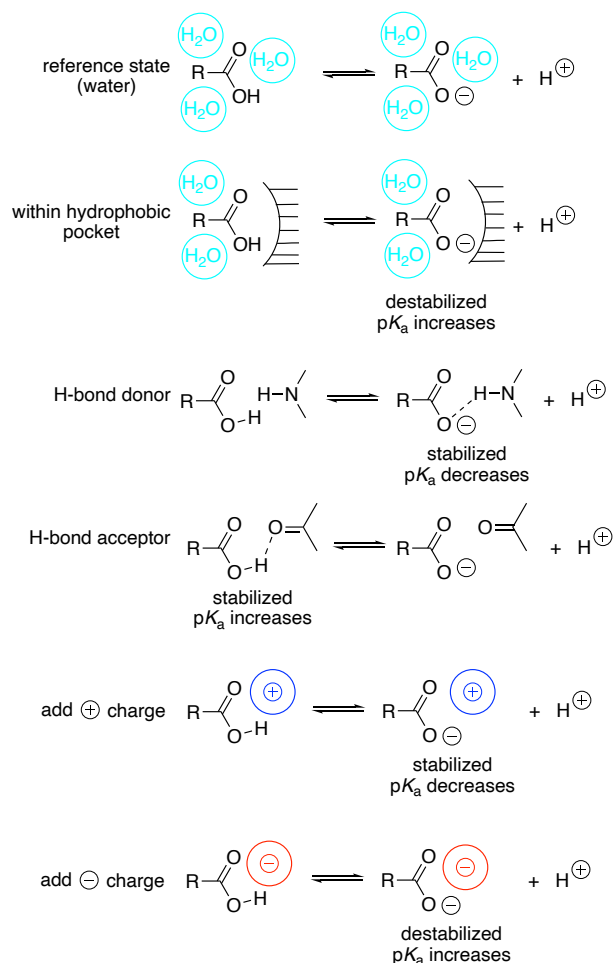


Figure 7.21. The pK_a of a carboxylic acid changes in a folded protein as compared to solution due to differences in the interactions between the protein and the two protonation states. Placing the carboxylic acid in a hydrophobic pocket destabilizes the charged form more than the neutral form, thereby shifting the equilibrium toward the neutral form. Hydrogen bond donors and acceptors stabilize different protonation forms and shift the pK_a in opposite directions. These hydrogen bond donor and acceptors can also interact with the carboxylic acid/carboxylate in other ways; the interactions shown are those that cause differences in the interactions. Electrostatic effects also shift the pK_a of the carboxylic acid.

static. The program uses empirical equations and parameters to account for desolvation ($\Delta pK_a^{desolvation}$, disfavors charged form), formation of hydrogen bonds ($\Delta pK_a^{hydrogen\ bonds}$, H-bond donors favor deprotonated form, H-bond acceptors favor protonated form) and electrostatic effects of nearby charged groups ($\Delta pK_a^{electrostatic}$, oppositely charged groups favor charged form), eq. Equation 7.21. A web interface of PROPKA is available at <https://server.poissonboltzmann.org/pdb2pqr>.^s

$$pK_a^{protein} = pK_a^{solution} + \Delta pK_a^{desolvation} + \Delta pK_a^{hydrogen\ bonds} + \Delta pK_a^{electrostatic} \quad (7.21)$$

A more detailed look at the predicted altered pK_a of Tyr80 in the active site of xylanase, **Figure 7.22**, shows that, besides desolvation mentioned above, hydrogen bond differences and Coulombic interactions also contribute. The phenoxide is disfavored because desolvation destabilized the phenoxide as mentioned above, because Glu172 makes a favorable hydrogen bond with the phenol form, and because two nearby negatively-charged glutamates create unfavorable Coulombic interactions with the phenoxide, eq. Equation 7.22. PROPKA predictions typically fit experimentally determined values with an rmsd <1 pK_a unit.

$$pK_a^{Tyr80} = 10 + 3.55 + 0.79 + 4.63 = 18.98 \quad (7.22)$$

Shifting an enzyme's pH-rate profile. When the pH optimum of the enzyme differs from the pH of the application, then engineering a shift in the pH optimum toward the pH of the application will increase the reaction rate. For example, some subtilisins (proteases for laundry applications) reach their maximum activity above pH

^sRun the PDB2PQR configuration calculation using default settings, then look in the log file to find the PROPKA calculations.

7.6 Stabilizing the chemical transition state to speed up reactions

RESIDUE	pKa	BURIED	DESOLVATION REGULAR	EFFECTS RE	SIDECHAIN HYDROGEN BOND	BACKBONE HYDROGEN BOND	COULOMBIC INTERACTION
TYR80 A	18.98	100 %	3.55 632	0.00 0	0.79 GLU 172 A	0.00 XXX 0 X	-0.27 ARG 112 A
TYR80 A					0.00 XXX 0 X	0.00 XXX 0 X	0.68 TYR 65 A
TYR80 A					0.00 XXX 0 X	0.00 XXX 0 X	0.16 TYR 166 A
TYR80 A					0.00 XXX 0 X	0.00 XXX 0 X	2.03 GLU 78 A
TYR80 A					0.00 XXX 0 X	0.00 XXX 0 X	2.03 GLU 172 A

Figure 7.22. Part of a PROPKA prediction of the pK_a 's of ionizable groups in a xylanase from *Bacillus circulans* (pdb id: 1XNB). PROPKA predicts that the pK_a of the phenolic hydroxyl of Tyr80 is 18.98, much higher than the solution value of 10. The side chain is buried disfavoring the charged anion, which raises the pK_a by 3.55 units. This estimate comes from the 632 (under desolvation effects 'regular') non-hydrogen protein atoms within 15.5 Å of the phenolic oxygen, which indicates how deeply the group is buried. The desolvation effects 're' is a small correction (zero in this case) to account for the non-hydrogen-bonding electrostatic effects of replacing water with protein atoms. The Tyr80 phenol hydrogen bonds to the anionic Glu172, but the phenoxide loses this hydrogen bond, which raises the pK_a by 0.79 units. Coulombic interactions with five nearby groups raise the pK_a by a total of 4.63 units. Negative charges on two nearby glutamates strongly disfavor formation of the negatively-charged phenoxide. Smaller contributions come from a more distant positively-charged arginine and partial negative charges on two nearby tyrosines.

10, but some wash applications use pH 8 where the enzyme shows low activity. Phytase, a feed additive enzyme, must be active in digestive tract, which is pH 2-5 for pigs and poultry. Other reasons to shift the pH optimum of an enzyme are to get enzymes with different pH optima to work together in a multi-step pathway, to allow a reaction pH that alters the solubility of substrates or products or partitioning into an organic phase, or to reduce susceptibility of microbial contamination by running the process outside of the neutral range.^[37]

Since solvation, hydrogen bond networks and electrostatic effects all influence the pK_a of the ionizable groups, researchers can, in principle, use any of these to engineer a shift in the pH optimum. However, since the first two involve changes near the ionizable group, there is a danger of disrupting catalytic activity and researchers avoid these approaches. Most engineering to shift the pH optimum uses remote electrostatic effects. Electrostatic effects decrease proportional to $1/r$, so charges even 10 Å away still affect the pK_a of an ionizable group. Being further away from the group makes it less likely that the changes disrupt orientation of the catalytic residues or the shape of the active site. Shifting the pH optimum of an enzyme usually involves changing the type and number of charges outside the active site.

The strategy to alter the pH optimum by changing electrostatics is as follows, **Figure 7.23**.

- Add $\oplus$ charge or remove $\ominus$ charge = lower pH optimum
- Add $\ominus$ charge or remove $\oplus$ charge = raise pH optimum

The example below considers the effect of added charges on the pK_a of histidine. The reader should confirm that the pK_a of a negatively-charged side chain like glutamate also shifts in the same direction.

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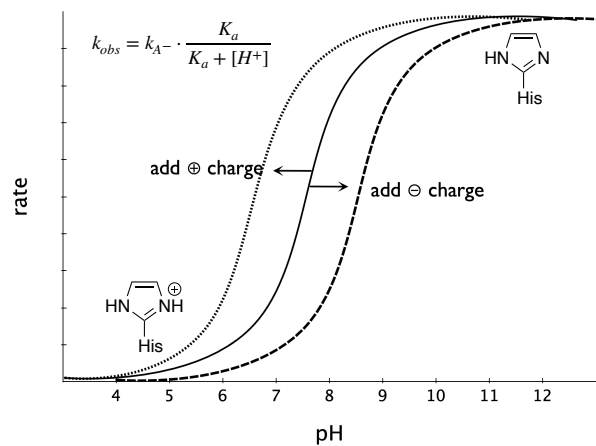


Figure 7.23. Shifting the pK_a of histidine using electrostatic effects. The curve of rate versus pH assumes that catalysis requires the base form of histidine. Adding a nearby positive charge destabilizes the protonated form of histidine so the curve shifts to lower pH. Similarly, adding a nearby negative charge shifts the curve to higher pH.

For example, to increase the activity of a serine hydrolase (subtilisin) at lower pH, the engineering must shift the pK_a of the catalytic histidine to lower pH. Adding positive charges or removing negative charges will destabilize the protonated histidine, so that the inflection point for deprotonation occurs at lower pH. Replacing a distant (~ 12 Å away) glutamate or a distant aspartate with lysine shifted this inflection point to lower pH by ~ 0.5 pH units.^[38] The effects were approximately additive, so replacing both residues with lysine decreased the pH of the inflection point by ~ 1.0 pH units. The lysine destabilized the positively charged (inactive) form of catalytic histidine so the shift to the active form occurred at lower pH. Increasing the ionic strength lowers the effect of surface charges due to their association with counterions. These measurements were made at a low ionic strength. Immobilization of enzymes on charged surfaces can similarly shift the pH optimum.

Catalysis often requires several ionizable groups in the correct protonation state. For, example, glycosidases require two acids, one in the acidic form, the other in the basic form, **Figure 7.24**.

Eq. Equation 7.23 below predicts the fraction of the enzyme that exists in the catalytically active form at different pH. The k_{obs} is the rate constant at the pH in question, while k_{best} is the rate of the catalytically active form and K_{a1} and K_{a2} are the acid ionization constants of the two ionizable groups. The active form of the enzyme requires the deprotonated form of group 1 and the protonated form of group 2.

$$k_{obs} = k_{best} \cdot \frac{[H^+]K_{a1}}{[H^+]K_{a1} + [H^+]^2 + K_{a1}K_{a2}} \quad (7.23)$$

If the pK_a 's of the two groups are widely separated, e.g., 6.0 and 10.0, then the maximum activity can reach to nearly k_{best} , **Figure 7.25**. The optimum at pH 8.0 is broad, halfway between the two pK_a 's. If

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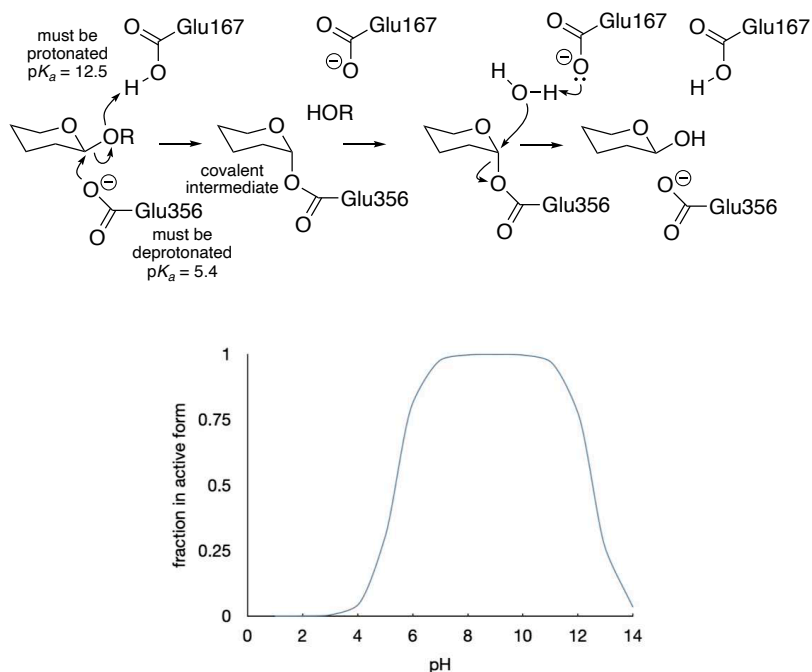


Figure 7.24. A glycosidase requires one glutamate in its acid form and another in its basic form. Their catalytic activity peaks in the range where both are in the correct protonation state. The first step of the catalytic mechanism requires Glu167 to act as an acid and Glu356 to act as a nucleophile, which requires the basic form. If the $pK_a^{Glu356} = 5.4$ and $pK_a^{Glu167} = 12.5$, then the glycosidase is the fully catalytic form between pH 7 and 11. At lower pH Glu356 is not deprotonated, while at higher pH Glu167 is not protonated. The curve is drawn from equation Equation 7.23.

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the pK_a 's of the two groups closer to each other, e.g., 6.0 and 7.3, then the maximum activity can never reach k_{best} . At best 69% of the enzyme molecules are in the correct protonation state. The pH optimum is narrow.

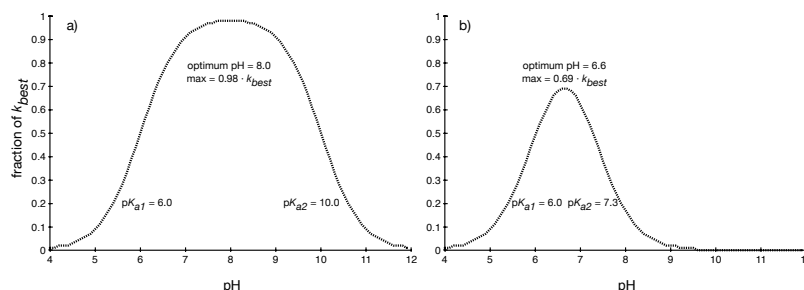


Figure 7.25. Narrowing the pH range for optimum activity reduces the catalytic activity. a) If the pK_a 's of the two groups are well separated (6.0 and 10.0), then the pH optimum is broad and the activity reaches the maximum possible. b) If the pK_a 's of the two groups are closer to each other (6.0 and 7.3), then the pH optimum is narrow and the activity only reaches part way to the maximum possible because only a fraction of the enzyme can contain the correctly ionized groups.

If protein engineering to shift the pH optimum also narrows the gap between the two pK_a 's, then the overall activity may decrease. For example, adding an arginine decreased the pH optimum of the glycosidase xylanase from 6.5 to 5, but unfortunately also decreased the catalytic activity 100-fold.^[39] The added arginine was closer to one of the carboxylic acids and strongly shifted its pK_a . The narrowing of the difference between the two pK_a 's made the correct protonation state for catalysis rarer thereby slowing the reaction.

A common confusion is the notion of regions of high pH or low pH. There are no such regions; the pH is constant throughout the solu-

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tion since proton transfer is fast. The protonation state of all groups is at equilibrium and reflects its pK_a . However, the same groups in different environments may differ in their pK_a and therefore have different protonation states. A carboxyl group in a non-polar environment may be uncharged while one in a polar environment may be charged. The pK_a of those two carboxyl groups is different, but the pH is identical in both places.

Attempts to shift pH optima by introducing charged residues can fail if the new residue doesn't have the expected charge at the target pH. Consider shifting a serine hydrolase pH optimum from 6.5 to 7.5 by adding a negative charge near the catalytic histidine. The strategy is to stabilize protonated histidine (His-H⁺), raising its pK_a . Introducing a glutamate or aspartate should work since their side chain pK_a (~5) means they'll be negatively charged at pH 6.5-7.5. However, if the carboxylate is placed in a hydrophobic, poorly solvated environment, its pK_a can increase from 5 to 9. This creates a problem: at pH 6-7.5 (the target range), the histidine is transitioning from charged to neutral, but the carboxylate is still mostly protonated (neutral COOH) because its elevated pK_a means it doesn't become negatively charged until pH >9. By the time the carboxylate finally becomes charged at pH ~9, the histidine is already fully neutral. Since electrostatic stabilization requires both groups to be charged simultaneously, and they're never both charged at the same pH, the carboxylate cannot stabilize the protonated histidine. The histidine pK_a remains unchanged, and the pH optimum doesn't shift. Introduced charges only shift active site pK_a values if they're actually charged at the relevant pH. The local protein environment can dramatically alter ionization behavior, undermining the design strategy.

In summary, the effect of proteins on a ionizable groups is predictable, but complex, because all charges interact. Engineering to shift the pK_a of catalytic groups can shift the pH optimum of

proteins while maintaining high catalytic activity. These shifts are typically ≤ 1 pH unit.

7.6.3 Stabilize shape and charge of transition state

To selectively stabilize the transition state over the enzyme-substrate complex, the enzyme must stabilize features that are unique to the transition state and not present in the enzyme-substrate complex. Those features are likely to be the charge and shape of the transition state. For example, consider the transition state for hydrolysis of an ester catalyzed by a serine hydrolase, **Figure 7.26**. Hydrolysis of an ester involves an acyl-enzyme intermediate. After binding the substrate and formation of the first tetrahedral intermediate, the acyl-enzyme intermediate forms, then a second tetrahedral intermediate forms, and finally the product forms and dissociates. The transition state (highest energy species) will be either the acylation or deacylation step. The structure of the transition state is not shown, but it is assumed to be similar to the first or second tetrahedral intermediate, **Figure 7.27**.

The charge distribution on the transition state differs from the enzyme-substrate complex. First, positive charge on the imidazole ring increases in the transition state as it is protonated. Second, the negative charge on the substrate carbonyl oxygen increases in the transition state as it forms an oxyanion. Serine hydrolases stabilize the positive charge on the imidazole ring with the carboxylate side chain of the catalytic aspartate. Serine hydrolases stabilize the negative charge on the oxyanion with two hydrogen bonds from main chain N-H's.

The shape of the transition state also differs from the enzyme-substrate complex. The transition state is tetrahedral at the carbonyl carbon, while the enzyme-substrate complex is planar

7.6 Stabilizing the chemical transition state to speed up reactions

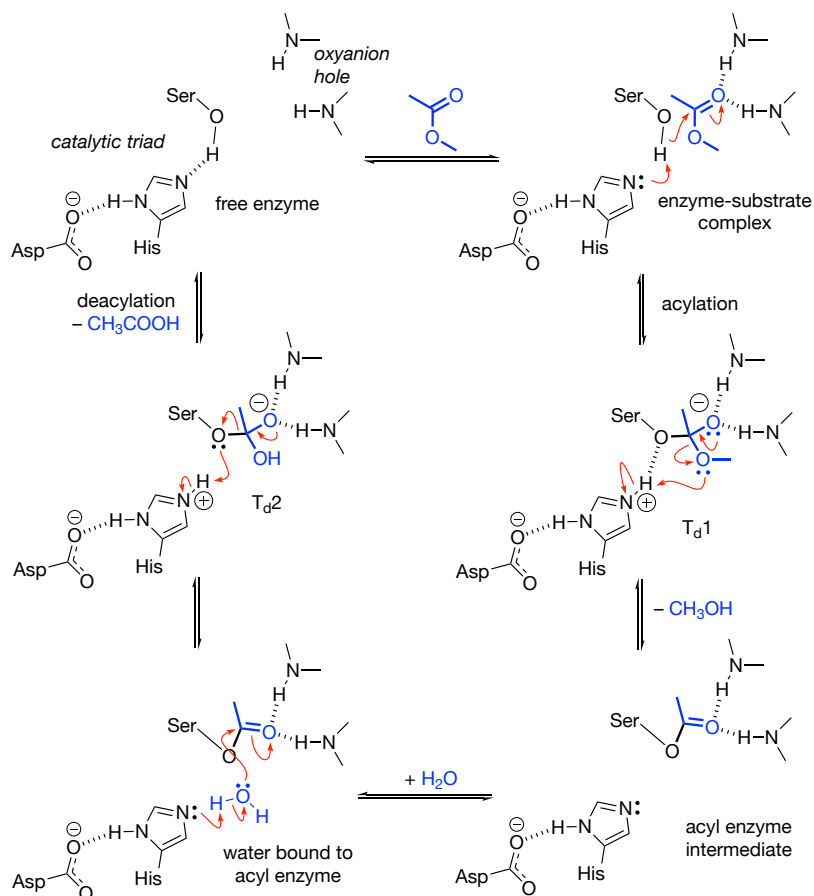


Figure 7.26. Serine esterases use a multi-step mechanism for hydrolysis of esters. Hydrolysis proceeds via an acyl-enzyme intermediate (bottom right structure). After formation of the enzyme-substrate complex (top equilibrium), the catalytic serine γ -oxygen adds to the carbonyl to form a tetrahedral intermediate. Collapse of this intermediate forms the acyl-enzyme. A similar set of steps hydrolyze this acyl enzyme to complete the cycle.

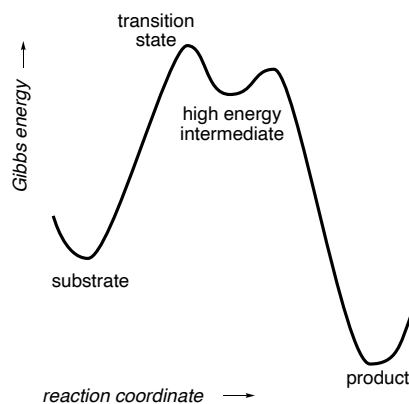


Figure 7.27. Both transition states and high energy intermediates lie in similar regions of the reaction coordinate diagram and therefore resemble each other. Transition states correspond to peaks in the diagram indicating that they have a fleeting existence, while intermediates correspond to valleys indicating that they have a finite lifetime.

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at the carbonyl carbon. If the shape of the active site matches the tetrahedral arrangement better than the planar arrangement, then this shape selectively stabilizes the transition state.

7.6.4 Create new mechanistic steps

The rate-acceleration mechanisms above assume that the transition states for the enzyme-catalyzed and the non-catalyzed reactions are the same; that is, that the mechanisms for the two reactions are the same. However, enzymes often change the mechanism of the non-catalyzed reaction. These changes split the high energy step of the non-catalyzed reaction into multiple lower-energy steps, **Figure 7.28**. These mechanistic changes may involve covalent enzyme intermediates, the participation of cofactors, such as pyridoxal-5'-phosphate or metal ions, and acid-base catalysis not available to the non-catalyzed reaction.

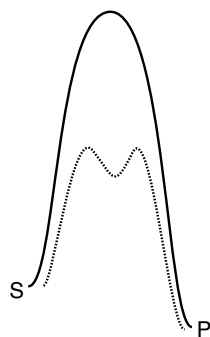


Figure 7.28. One way to lower the energy of a transition state is to break an high energy step into two lower energy steps. This change corresponds to changing the mechanistic steps involved in the reaction.

One type of new mechanism is the formation of a covalent intermediate with the enzyme. Both glycoside hydrolases, **Figure 7.24**, and

serine esterases **Figure 7.26** form covalent intermediates: a glycosyl enzyme link to the catalytic glutamate residue and an acyl enzyme link to the catalytic serine residue, respectively. Instead of water directly attacking the substrate, a catalytic residue within the enzyme attacks the substrate and releases part of it while forming a covalent intermediate. In the next step, water attacks this covalent intermediate to release it from the enzyme. In contrast, the acid catalyzed hydrolysis of glycosides and esters involves direct attack of water on the substrate and formation of a high energy intermediate, **Figure 7.29**. This mechanism is higher in energy for two reasons. First, simultaneously positioning the water and substrate for reaction is entropically costly. The enzyme uses a stepwise approach: position the substrate in the first step and the water in the second step. Second, the charged intermediate is much less stable than the neutral covalent intermediates formed in the enzyme-catalyzed reaction.

A comparison of binding strengths to enzyme rate acceleration shows the importance of these mechanistic changes to enzyme catalysis. If binding the transition state were the only way to stabilize the transition state, then transition state stabilization energies would be similar to binding constants. The maximum strength of binding between receptors and drugs, antibodies and antigens, or enzyme and drugs corresponds to $K_d \sim 10^{-13}$ M, so the maximum rate acceleration by enzymes would be similar. In contrast to this expectation, the maximum rate acceleration by enzymes corresponds to a binding constant a billion times lower: 10^{-22} M.^[40] This additional rate acceleration comes from the ability of enzymes to change the catalytic mechanism.

7.6 Stabilizing the chemical transition state to speed up reactions

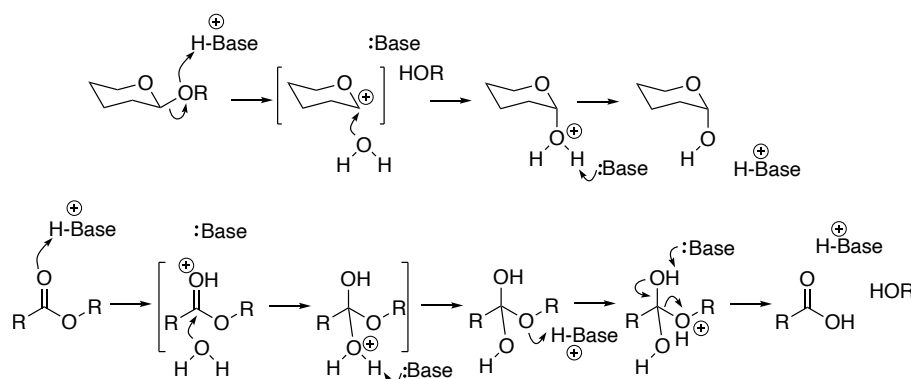


Figure 7.29. Acid-catalyzed hydrolysis of a glycoside (top) or an ester (bottom) involves direct attack of water on the substrate and formation of a high energy charged intermediates, shown in brackets. Enzymes avoid formation of these high energy intermediates by breaking the reaction into two steps where a catalytic residue first attacks the substrate to form a covalent intermediate and then water attacks the covalent intermediate. The neutral covalent intermediate is lower in energy than the charged intermediates shown here. See **Figure 7.24** and **Figure 7.26** for the enzyme-catalyzed mechanisms.

7.6.5 Computational Modeling

Most enzyme-catalyzed reactions involve multiple steps. Steady-state kinetics hides this complexity into the kinetic constants, each of which depend on multiple elementary steps. In contrast, computational modeling focuses on one elementary step at a time, e.g., substrate binding, formation of near attack complex, reaction step, or product release. The accuracy needed to design the multiple steps and interactions that stabilize transition states is at the edge of the capabilities of current computational methods.^[41] One approach is using experimental methods to identify the rate limiting step (there may be several), then focus computational methods on fixing those steps.

Glossary

Acyl-enzyme intermediate covalent intermediate formed between the catalytic serine and the substrate's acyl group during serine hydrolase-catalyzed hydrolysis; hydrolyzed in a second step to release product and regenerate free enzyme.

Activation energy ($\Delta G^\ddagger$) Gibbs energy difference between a ground state and the transition state; sets the rate constant via transition state theory ($\Delta G^\ddagger = -RT \ln(k/\text{constants})$). For an uncatalyzed reaction the ground state is S. For an enzyme-catalyzed reaction the relevant ground state depends on the reaction conditions: E·S (or a later low-energy state) sets the barrier for k_{cat} , which limits the rate at high substrate concentrations, while free E + S sets the (larger) barrier for k_{cat}/K_M , which limits the rate at low substrate concentrations.

Briggs-Haldane (steady-state) approximation kinetic treatment that replaces the Michaelis-Menten assumption of rapid equi-

librium between E, S, and E·S with the weaker assumption that $[E \cdot S]$ is constant; redefines K_M to include the rate constant of the chemical step (k_2), so K_M is no longer purely a dissociation constant.

Desolvation effect shift in the pK_a of an ionizable side chain caused by burying it away from water; typically destabilizes the charged form and raises the pK_a of acids.

Effective molarity the rate enhancement from holding reactive groups in proximity and correct orientation (as in an intramolecular reaction or an enzyme active site) relative to the same groups reacting intermolecularly; illustrated by the 50,000-fold rate variation with rotatable-bond count in ring-closure reactions.

Kemp elimination a base-catalyzed proton transfer/ring-opening model reaction (breaks an N-O bond, forms a $C\equiv N$) with no natural enzyme catalyst; commonly used to benchmark de novo-designed catalysts.

k_{cat} rate constant that, multiplied by $[E]_0$, gives V_{max} ; in the simple Michaelis-Menten model equals k_2 , the rate of the chemical step. Sets the reaction rate at saturating (high) substrate concentration, where the ground state is $E \cdot S$.

k_{cat}/K_M (**specificity constant**) apparent second-order rate constant that sets the reaction rate at low (subsaturating) substrate concentration ($V = k_{cat}/K_M \cdot [S] \cdot [E]$); determined by the highest transition-state energy relative to free E + S.

K_M (**Michaelis constant**) substrate concentration at half V_{max} . In the Michaelis-Menten model it equals the dissociation constant K_D of E·S; in more complex mechanisms it is a composite of several rate constants and no longer purely reflects binding affinity.

Lineweaver-Burk plot double-reciprocal linearization of the Michaelis-Menten equation ($1/V$ vs. $1/[S]$); mathematically convenient but overweights low- $[S]$ data, giving less accurate V_{max} and K_M estimates than nonlinear regression.

Michaelis-Menten model simplest kinetic mechanism for enzyme catalysis: reversible formation of the E·S complex (rates k_1 , k_{-1}) followed by irreversible, rate-limiting conversion of E·S to product (rate k_2).

Near-attack complex a preorganized conformation of E·S with the reacting atoms positioned at the distance and angle required for reaction; a required orientation to get from E·S to the transition state. A near-attack complex need not be an energy minimum and may be a fleeting conformation along the reaction path.

Non-productive binding binding of substrate to the enzyme, at or near the active site, in an orientation that does not lead to reaction; competes with productive binding and lowers k_{cat} .

pH optimum (pH-rate profile) the pH at the peak in a plot of enzyme activity variation with pH, reflecting the fraction of enzyme molecules with catalytic acid/base residues in the correct protonation state; can be shifted by engineering catalytic-residue pK_a 's.

pK_a shift change in a side chain's ionization constant caused by the folded-protein environment (desolvation, hydrogen bonding, electrostatics) relative to its value for the free amino acid in solution; can span many pK_a units (e.g., Tyr80 in xylanase, pK_a ~19 vs. ~10 in solution).

Preorganization structural arrangement of an enzyme's active site that positions catalytic residues and substrate to favor conversion to product.

Product inhibition slowing of the reaction rate by accumulated product; violates the Michaelis-Menten assumption that product release is fast and requires additional experiments (varying product concentration) to characterize.

PROPKA empirical program that predicts a protein's pK_a values by adjusting solution pK_a 's for desolvation, hydrogen-bonding, and Coulombic contributions from the surrounding structure.

- Rate constant (k)** proportionality constant relating reaction rate to substrate concentration for an uncatalyzed unimolecular reaction ($V = k[S]$); units of time^{-1} .
- Saturation kinetics** the characteristic leveling-off of reaction rate with increasing substrate concentration shown by most enzyme-catalyzed reactions, in contrast to the linear rate increase of an uncatalyzed reaction.
- Specific activity** an enzyme's reaction rate under a defined set of assay conditions, in U/mg ($\mu\text{mol product/min/mg enzyme}$); a quick single-assay measure that gives much less information than full steady-state kinetic constants (k_{cat} , K_M), which require 5-10 assays.
- Transition state** the highest-energy structure along the reaction coordinate from substrate to product; a peak (not a valley) on the Gibbs energy diagram, and therefore has only a fleeting existence.
- Transition-state stabilization** geometric and electronic matching between the enzyme active site and the reaction transition state; achieved through preorganization, acid-base optimization, stabilizing interactions with transition-state charge/shape, and enabling new reaction steps.
- V_{max} the reaction rate at saturating (high) substrate concentration; equals $k_{cat} \cdot [E]_0$.

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Problems

1. The reaction rate of an enzyme-catalyzed reaction was faster with variant A than with variant B at a low substrate concentration, but was slower at a high substrate concentration. How is this possible?
2. Predict the pK_a of the catalytic histidine in serine esterase with pdb id 1va4. The catalytic triad consists of Ser94, His251,

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and Asp222. This structure contains 6 monomers in the unit cell, which exceeds the maximum of 1000 atoms for the PROPKA calculation. Download this structure from the Protein Data Bank and open the file using PyMOL. Delete five of the chains and the bound glycerol and water molecules by entering the following commands on the input line (ignore comments after #).

```
# optional, changes your default directory to the Desktop; files that you do
cd ~/Desktop
# downloads the 'biological assembly' pdb file (only chain A) for structure 1
fetch 1va4, type=pdb1
# deletes residues named 'GOL' (glycerol) and water molecules
remove (resn GOL or resn HOH)
# saves your modified structure as a pdb file
save 1VA4_chainA.pdb
```

Submit this file to the site <https://server.poissonboltzmann.org/pdb2pqr>. Registration is optional; you can use the software without registering. The calculation takes about 10 min. Open the log file, scroll up to find the PROPKA calculation for His 251.

- What is the predicted pK_a for His251? How does this value compare to the value for the histidine side chain in an unstructured peptide?
- How does burying the histidine in a hydrophobic active site affect the pK_a ?
- How does the third member of the catalytic triad, Asp222, affect the pK_a ?

If you run into difficulty with the website, you may use the data below to answer the questions.

RESIDUE	pKa	BURIED	DESOLVATION		EFFECTS		SIDECHAIN		BACKBONE		COULOMBIC
			REGULAR	RE			HYDROGEN BOND		HYDROGEN BOND		
HIS 251 A	5.96	100 %	-3.59	638	0.00	0	1.60 ASP 222 A		0.00 XXX	0 X	-0.58 HIS 218 A
HIS 251 A							0.00 XXX	0 X	0.00 XXX	0 X	2.03 ASP 222 A

Answers

1. At high substrate concentration, the rate of reaction depends only on k_{cat} , so variant B has a higher k_{cat} . At low substrate concentration, the rate of reaction depends on $\frac{k_{cat}}{K_M} \cdot [S]$. For variant A to react faster, it must have a lower value of K_M so it reaches its maximum rate at a low substrate concentration, while variant B is still limited by incomplete saturation of the active sites with substrate.
2. The pK_a is 5.96, which is lower than the normal pKa of 6.51. Desolvation lowers the pK_a of His251 by 3.6 units due to destabilization of the protonated (charged) form. Asp 222 counteracts this shift by 3.57 units by stabilizing the protonated (charged) form by accepting a hydrogen bond (1.6 contribution) and providing a nearby negative charge (2.03 contribution). Finally a nearby positively charged histidine slightly destabilizes the protonated (charged) form of His251 by 0.58 units leading to a net decrease in pK_a of 0.52.

Supporting Information

Fitting steady-state kinetic data using Lineweaver-Burke plots (linear fit) yields less accurate estimates of V_{max} and K_M .

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The text recommends finding V_{max} and K_M by fitting the Michaelis-Menten equation below, eq. Equation 7.24, directly to the experimental data (measured rates, V , at different substrate concentrations, $[S]$). This equation describes a curve since it has the form $y = \frac{ax}{b+x}$ so fitting this equation to the data requires a non-linear fitting program. Convenient examples include Microsoft Excel,^[42] R scripts,^[43] or the Python script given below. Non-linear least squares fitting of eq. Equation 7.24 to the data starts with initial guesses for V_{max} and K_M followed by iteration to find the best values.

$$V = \frac{V_{max} \cdot [S]}{K_M + [S]} \quad (7.24)$$

Before the widespread use of computers, fitting a curve to experimental data was difficult, so an alternative was to rearrange the Michaelis-Menten equation to a form that yields a straight line, eq. Equation 7.25.

$$\frac{1}{v} = \frac{1}{S} \cdot \frac{K_M}{V_{max}} + \frac{1}{V_{max}} \quad (7.25)$$

Plotting $1/v$ on the y-axis and $1/[S]$ on the x-axis yields a straight line with a slope of K_M/V_{max} and y-intercept of $1/V_{max}$. This plot is known as the Lineweaver-Burke plot. Fitting a line to experimental data is mathematically simpler, so it was often used in the past. The disadvantage of fitting the Lineweaver-Burke line to the data is that the estimates of V_{max} and K_M are less accurate when the data are imperfect.

Both methods would yield the same values of K_M and V_{max} if the data were perfect. Typically, errors in measuring reaction rates are approximately constant; for example, one might measure V

= 5 ± 1 at low [S] and 50 ± 1 at high [S]. That is, rates measured at low [S] have a higher percentage error (20% for this example) than rates measured at high [S] (2%). When fitting the data to the Michaelis-Menten curve, the deviation of each data point from the curve counts equally. This is the best approach when the data points have equal absolute errors as suggested above.

In contrast, fitting the data to eq. Equation 7.25 minimizes the deviation of each $1/V$ data point from the line. In our example, the error for the value at low [S], $1/V = 0.20 \pm 0.04$, is 10-fold higher than the error for the value at high [S], 0.0200 ± 0.0004 . Since deviations from either data point count equally, but the uncertainty is higher for data point at low [S], this approach yields estimates of V_{max} and K_M that weight the data points at low [S] too heavily. These data have high errors, but are treated as having the same errors as values at high [S].

For example, fitting the Michaelis-Menten curve and the Lineweaver-Burke line to the imperfect data below yields different estimates for V_{max} and K_M . These simulated data start with a perfect fit to the Michaelis-Menten equation and add Gaussian random error to simulate experimental error.^[42]

```
S = (0, 2.5, 5, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100)
    # units = mM
V = (-2, 153, 231, 342, 396, 438, 467, 505, 523, 523, 539,
     548, 555, 554) # units = mM/s
```

Fitting the Michaelis-Menten curve, eq. Equation 7.24, to the data yields $V_{max} = 598 \pm 3 \text{ mM/s}$, $K_M = 7.6 \pm 0.2 \text{ mM}$, while fitting the Lineweaver-Burke line, eq. Equation 7.25, to the same data yields $V_{max} = 591 \pm 7 \text{ mM/s}$, $K_M = 7.3 \pm 0.2 \text{ mM}$, **Figure 7.30**. The V_{max} differs by 1.2%, while K_M differs by 4.5%. The first data point ($S = 0$, $V = -2$) was omitted from the linear fit because $1/0$ is undefined. The

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differences are within the estimated errors (standard deviations) for the values.

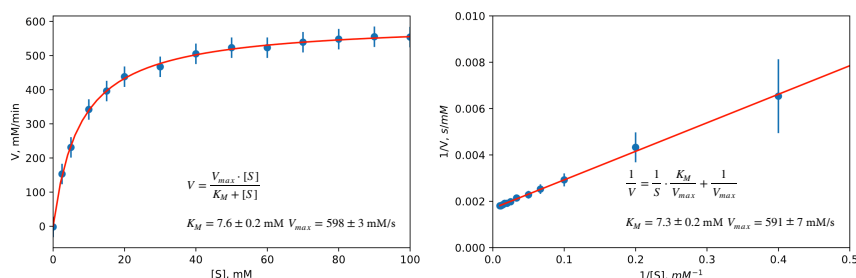


Figure 7.30. Fitting the Michaelis-Menten equation of a curve (left) to experimental data and fitting the Lineweaver-Burke line to the same data (right). The best fit yields slightly different values of K_M and V_{max} . Error bars correspond to a fixed error of ± 30 ($\sim 5\%$ of V_{max}) for the values of V .

The code blocks below generate the plots shown in **Figure 7.30** so that you can experiment with your own data. Another requirement for accurate estimates of K_M and V_{max} is that the data include data points at substrate concentrations both below and above K_M . Otherwise, the shape of the curve is poorly defined and the result will be inaccurate regardless of which equation is fit the data.

Code Block S5.1 Python script to fit experimental data to the Michaelis-Menten curve. This approach is the preferred method for fitting steady-state kinetic data to extract K_M and V_{max} .

```
"""
This script fits the nonlinear Michaelis-Menten curve to the
experimental data by adjusting the initial guesses for Km and
Vmax until the calculated values of V best match the experimen-
tal ones. This script is the preferred one to calculate Vmax and
```


Supporting Information

```
Km from experimental data.
"""
# fit to Michaelis-Menten curve
import numpy as np          #import math functions to use arrays
from scipy import optimize  #import non-linear fit function
import matplotlib.pyplot as plt #import plotting function

# list of data, assumes units of S are millimolar, units of V are millimolar/min
S = np.array([0, 2.5, 5, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90,
100])
V = np.array([-2, 153, 231, 342, 396, 438, 467, 505, 523, 523,
539, 548, 555, 554])
# set a fixed error for all V measurements
Verr = 30
Km = 5.0 # initial guesses for Km and Vmax
Vmax = 550

# define equation for MM curve to be fit
# This function returns calculated values of V for current
# values of Vmax and Km. These calculated values should match
# the experimental data for V best for when Vmax and Km are
# correct.
def MM(S, Km, Vmax):
    return S*Vmax/(Km + S)

# non-linear fit adjusts Km and Vmax so the calculated values
# of V match the experimental values of V
popt, pcov = optimize.curve_fit(MM, S, V, [Km, Vmax])
perr = np.sqrt(np.diag(pcov))
print("Km =", "{0:.3f}".format(popt[0]), "±", "{0:.3f}".format
(perr[0]), "Vmax =", "{0:.3f}".format(popt[1]), "±", "{0:.3f}".format
(perr[1]))
```

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```
# plot data and best fit curve
plt.scatter(S, V)
xfit = np.linspace(0,100)
plt.errorbar(S, V, yerr=Verr, linestyle = 'None')
plt.plot(xfit, MM(xfit, popt[0], popt[1]), 'r-')
plt.xlim(0, 100)
plt.xlabel('[S], mM')
plt.ylabel('V, mM/min')
plt.show()

# comment line above and uncomment lines below to save plot as
# eps file to Desktop; adjust file path as needed
# changing file extension (e.g., jpg, pdf) changes format of file

# import os
# os.chdir('/Users/romas/Desktop')
# plt.savefig('mm_plot.eps', dpi=600)
```

Code block S5.2 Python commands to fit experimental data to the Lineweaver-Burke linear form of the Michaelis-Menten equation. This approach is common in older texts, but is no longer preferred.

```
"""
This script fits the Lineweaver-Burke line (a linearized form of
the Michaelis-Menten curve) to the experimental data by adjusting
the initial guesses for Km and Vmax. This approach is less accurate
because it overweights data a low [S].
"""

# fit to LB line
import numpy as np                #import math functions to use arrays
from scipy import optimize        #import non-linear fit function
import matplotlib.pyplot as plt  #import plotting function
```

Supporting Information

```
# list of data
S = np.array([2.5, 5, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100])
# delete 0 data point to avoid error of 1/0
V = [153, 231, 342, 396, 438, 467, 505, 523, 523, 539, 548, 555, 554]
Verr = 30
# define as list
Vinv = np.array([1/x for x in V]) # y-axis is 1/V for this fit
# calculate error in Vinv
Vinv_err = np.array([(1/(x-Verr))-(1/x) for x in V])

Km = 5.0 # initial guesses for Km and Vmax
Vmax = 550

# define equations for LB line to be fit
def LB(S, Km, Vmax):
    return (1/S) * (Km/Vmax) + 1/Vmax
    # result should equal data for Vinv when Vmax and Km are correct

# fit the data to the LB line
popt, pcov = optimize.curve_fit(LB, S, Vinv, [Km, Vmax])
perr = np.sqrt(np.diag(pcov))
print("Km =", "{0:.3f}".format(popt[0]), "±", "{0:.3f}".format(
    perr[0]), "Vmax =", "{0:.3f}".format(popt[1]), "±", "{0:.3f}".format(
    perr[1]))
# calculate error in Vinv
Vinv_err = np.array([(1/(x-Verr))-(1/x) for x in V])

# plot data and best fit curve
plt.scatter(1/S, Vinv)
xfit = np.linspace(1,100)
plt.errorbar(1/S, Vinv, yerr=Vinv_err, linestyle='None')
plt.plot(1/xfit, LB(xfit, popt[0], popt[1]), 'r-')
plt.xlim(0,0.5)
```

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```
plt.ylim(0, 0.01)
plt.xlabel('1/[S], $mM^{-1}$')
plt.ylabel('1/V, $s/mM$')
plt.show()

# comment line above and uncomment lines below to save plot as
# eps file adjust file path to your computer, adjust file exten-
# sion as needed
# import os
# os.chdir('/Users/romas/Desktop')
# plt.savefig('lb_plot.eps', dpi=600)
```

8 Engineering enzyme selectivity

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Summary. Enzymes are often selective for one stereoisomer over another (stereoselectivity). This selectivity can refer to substrate selectivity, which involves the separation of existing compounds, or to product selectivity, which involves the creation of new compounds. The ability of enzymes to distinguish stereoisomers relies on their ability to position one isomer for reaction better than another by exploiting differences in their shapes.

Two shape rules — Prelog’s rule for dehydrogenases and a related rule for esterases and lipases acting on secondary alcohols — predict which enantiomer of a substrate reacts faster, or which enantiomer of a product forms, from the relative sizes of the substituents at the stereocenter. The structural basis of these rules points to where a protein can be engineered to change or improve selectivity: within the binding pockets themselves, in the shell of residues nearby that add hydrogen bonds or electrostatic interactions without contacting the substrate, or, for a buried active site, along the tunnel that sets the substrate’s orientation before it arrives. A second, independent strategy, proofreading, raises selectivity further by linking two selective steps in series so that their selectivities reinforce one another; this strategy underlies the fidelity of protein and DNA synthesis and can be adapted to biocatalysis.

Learning goals

8 Engineering enzyme selectivity

- Selectivity is the ability of proteins to distinguish between different molecules. The selectivity value, S , is the ratio of the preferred to non-preferred molecule. This value depends on comparison molecules. Selective reactivity may involve discrimination between two substrates or between two products.
- Stereoselectivity is a type of selectivity where the molecules to be distinguished are stereoisomers; enantioselectivity is a type of stereoselectivity where the stereoisomer to be distinguished are enantiomers. Many pharmaceuticals require pure enantiomers and enzyme-catalyzed reactions are a good way to make these pure enantiomers.
- Discrimination between two substrates occurs when one substrate reacts faster than the other. This substrate selectivity depends on the ratio of k_{cat}/K_M for the two substrates. Tighter binding allows a substrate to occupy a larger fraction of active sites, while faster reaction allows the substrate to react faster.
- When there are equal amounts of the two substrates, the initial ratio of substrates that react or products that form corresponds to the selectivity. At higher conversions, the relative amounts of substrate that reacts or product that forms differ from the selectivity because the amounts of the two substrates is no longer equal.
- Discrimination between two products occurs when a single substrate forms two products. This product selectivity depends only on the relative values of k_{cat} for the two products. The relative amount of the two products is constant throughout the reaction and corresponds to the selectivity.
- Shape rules predict which enantiomer of a substrate reacts faster, or which enantiomer of a product forms, from the relative sizes of two substituents at the stereocenter. Prelog's rule

applies to ketone reduction by dehydrogenases; a similar rule applies to hydrolysis or acylation of secondary alcohols by esterases and lipases. Both rules describe an active site built from a large and a medium (or small) pocket, and both predict the sign of the selectivity but not its magnitude.

- Rational protein engineering can change or improve selectivity by targeting residues at three levels: within the binding pockets that a shape rule identifies, in the shell of residues adjacent to those pockets that add a hydrogen bond or shift electrostatics without contacting the substrate directly, or, for a buried active site, along the tunnel that orients the substrate before it arrives.
- Proofreading increases selectivity beyond what a single binding site can achieve by linking two selective steps in series, so their selectivities multiply. Because the second step also destroys some correctly made product along with the errors, proofreading always trades some yield for higher purity. This strategy underlies the fidelity of protein and DNA synthesis and can be engineered into biocatalytic processes using a second enzyme or the same enzyme acting twice.

8.1 Introduction

Selectivity, S , is the ability to distinguish between alternatives. This ability is the key to the value of many proteins. For example, the cardiovascular antibody drug abciximab selectively binds to a receptor involved in platelet aggregation to inhibit formation of blood clots.^[1] In another example, a transaminase with high selectivity for the (*S*)-amine yields the required stereoisomer of a cardiovascular drug precursor with high purity.^[2]

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Binding selectivity is the ability to bind one ligand more tightly than other one and was covered in Chapter 6. This chapter focuses on reaction selectivity.

Reaction selectivity is the ability to react with one substrate more quickly than another (substrate selectivity) or form one molecule more quickly than another (product selectivity). The classification as substrate or product selectivity depends on whether there are multiple possible substrates or multiple possible products.

Reaction selectivity stems from differences in reaction rates, which originate in differences in transition state energies for the competing reactions. The ratio of rates is the selectivity for that pair of molecules, Equation 8.1.

$$S_{reaction} = \frac{rate_{fast}}{rate_{slow}} \quad (8.1)$$

The words non-selective (or promiscuous), selective and highly selective describe degrees of selectivity. Avoid using the word specific to mean highly selective because specific has a different meaning in chemistry. The reaction mechanism of a selective reaction favors the formation of a particular isomer, but both isomers can form. For example, epoxidation of norbornene (bicyclo[2.2.1]hept-2-ene) favors formation of the *exo* diastereomer (oxygen close to the one-carbon bridge) over the *endo* diastereomer (oxygen close to the two-carbon bridge), (**Figure 8.1 a**). The selectivity originates the different steric hindrance between the two sides and varies with the oxidant used. A highly selective reaction may form only one detectable product, but the reaction mechanism could also form the other one. In contrast, specific reactions are those that require formation of a particular isomer for mechanistic reasons. For example, epoxidation of *cis*-2-butene can only form the *cis*-epoxide, (**Figure 8.1 b**). Likewise, epoxidation

of *trans*-2-butene can only form the *trans*-epoxide. The reaction is a concerted addition of oxygen to the double bond without an intermediate that permits rotation of the carbon-carbon bond. The reactions in this chapter all refer to selectivity.

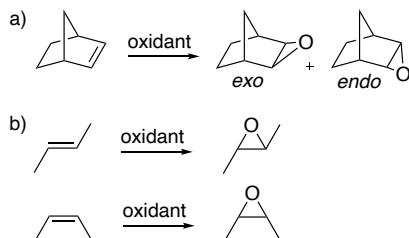


Figure 8.1. Selective reactions favor one possible product, but specific reactions form only one product because the reaction mechanism makes formation of the other one impossible. a) Epoxidation of norbornene is a selective reaction. The oxygen atom can add to either face of the double bond. b) Epoxidation of 2-butene is a specific reaction: *cis*-2-butene yields only the *cis*-epoxide, while *trans*-2-butene yields only the *trans*-epoxide.

8.2 Stereoisomers

Isomers are non-superposable molecules with the same molecular formula. Isomers can differ in their atom connectivity (constitutional isomers) or only in shape (stereoisomers), (**Figure 8.2**). Classifying a person as a father, son, brother or uncle depends on who the person is being compared to. In the same way, isomerism classifies relationships between molecules. A given molecule may be a diastereomer, an enantiomer or a constitutional isomer depending on the comparison molecule. For example, one may recognize that

a molecule is chiral, but until one has a comparison molecule, one cannot classify the relationship between the molecules.

The concept of a stereogenic unit is also useful. A stereogenic unit is the part of the molecule that makes stereoisomers possible. Chiral molecules must contain at least one stereogenic unit. For example, the stereogenic unit in L-alanine is the α -carbon atom containing four different substituents: HOOC- , $\text{H}_2\text{N-}$, $\text{H}_3\text{C-}$, and H . Occasionally one may encounter the term regioisomer, which refers to a special type of constitutional isomer. Regioisomers can form when a chemical reaction has different orientations or sites to choose from. For example, electrophilic aromatic substitution of bromobenzene can yield *ortho*-, *meta*-, or *para*-substituted products. These isomeric products are regioisomers.

Researchers often use the high stereoselectivity of enzymes, especially their high enantioselectivity to prepare pure stereoisomers. Most of the examples in this chapter involve enzymes distinguishing between stereoisomers.

8.2.1 Stereoisomeric purity

There are two commonly used measures for stereoisomeric purity: the ratio of stereoisomers and the stereoisomeric excess. The enantiomer ratio, *er*, or diastereomer ratio, *dr*, is simply the relative amounts of the two stereoisomers.

$$er \text{ or } dr = \frac{\text{major stereoisomer}}{\text{minor stereoisomer}}$$

For example, a 9:1 mixture of *cis*- and *trans*-2-butene has a diastereomeric ratio of 9. Analytic methods like HPLC or GC with columns containing a chiral stationary phase reveal the amounts of each enantiomer.

8.2 Stereoisomers

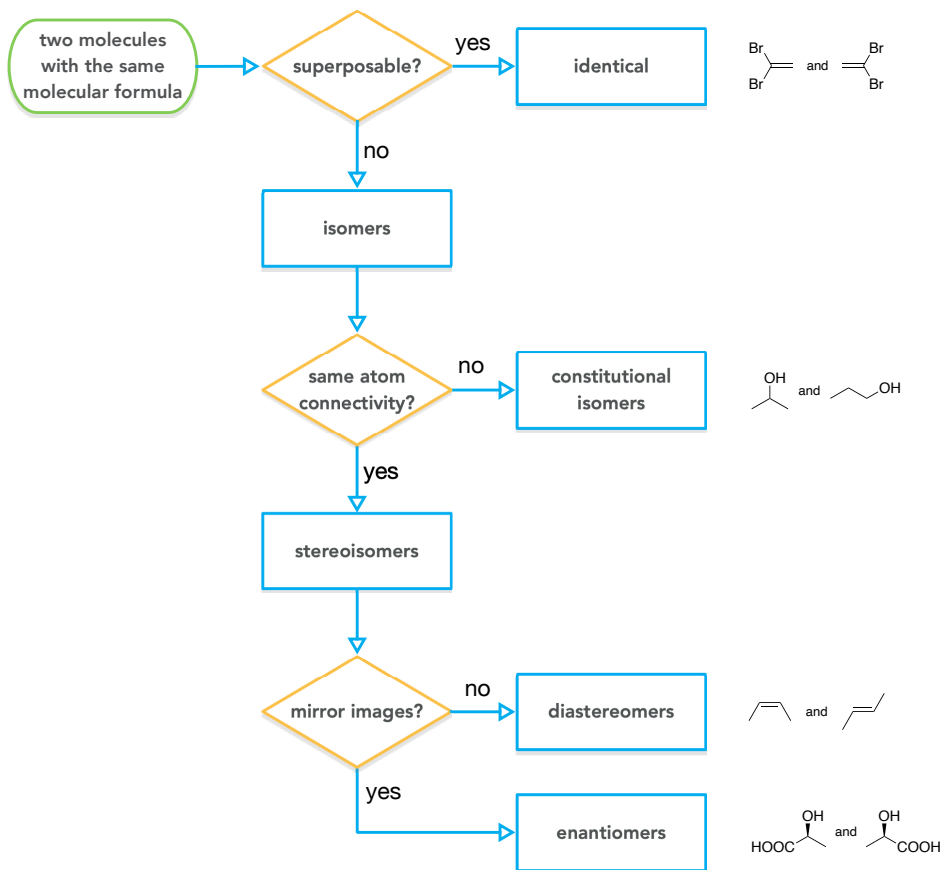


Figure 8.2. Flowchart to identify common relationships between pairs of molecules. Constitutional isomers differ in their atom connectivity, while stereoisomers differ only in their shape. Stereoisomers can be either diastereomers or enantiomers.

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Another measure of purity is enantiomeric excess, ee , usually expressed as a percent instead of a number between 0 and 1. Enantiomeric excess is the excess of the major enantiomer over the minor enantiomer in the sample, Equation 8.2. For example, a 9:1 mixture of two enantiomers has an 80% ee . Diastereomeric excess, de , is defined similarly.

$$ee \text{ or } de = \frac{major - minor}{major + minor} \quad (8.2)$$

This unusual way to express purity stems from the historic use of optical rotation to measure enantiomeric purity. Enantiomers rotate plane-polarized light in opposite directions. A racemic sample shows no rotation because the opposite rotations of the two enantiomers cancel out. A racemic sample has an ee of 0%. If one enantiomer is present in excess, then the excess amount of that enantiomer causes a rotation. The enantiomeric excess in percent is the size of that rotation as a fraction of the rotation for the pure enantiomer.

To convert enantiomeric excess to amounts of the individual enantiomers, one can imagine the sample to consist of an excess amount of the major enantiomer, ee , plus an amount of racemate $1 - ee$. The total amount of major enantiomer is the excess amount plus half of the racemate, Equation 8.3.

$$major_{total} = major_{excess} + \frac{racemate}{2} = ee + \frac{1 - ee}{2} \quad (8.3)$$

The total amount of the minor enantiomer is half of the amount of the racemate, Equation 8.4.

$$minor_{total} = \frac{racemate}{2} = \frac{1 - ee}{2} \quad (8.4)$$

For example, a sample with an enantiomeric purity of 80% ee consists of 80% major enantiomer in excess plus 20% racemate. Thus, the sample contains 90% of the major enantiomer and 10% of the minor enantiomer.

8.3 Product selectivity

Product selectivity occurs when an enzyme converts a substrate into several possible products and favors one or more of the possibilities. Product selectivity involves the *creation* of two possible products from a single substrate. For example, an enantioselective transaminase catalyzes the key step in the synthesis of sitagliptin, a type 2 diabetes drug,^[3] **Figure 8.3**. The reaction is highly selective for one of the two possible enantiomers. Enantioselectivity is a particularly useful type of selectivity since pure enantiomers are difficult to make with traditional chemical methods. Enantiomers can differ in their biological effects, so most chiral drugs are marketed as single enantiomers.

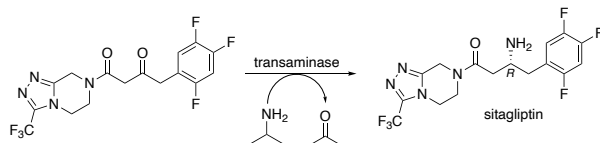


Figure 8.3. An enantioselective transaminase converts the achiral starting ketone into the (*R*)-amine. This reaction is an example of product enantioselectivity since the other possible product, the enantiomeric (*S*)-amine (not shown), forms only in trace amounts.

Product selectivity depends on the ratio of k_{cat} values for the two products, Equation 8.5. The subscript *f* refers to the faster-reacting

8 Engineering enzyme selectivity

substrate and subscript s refers to the slower reacting substrate. There is no contribution from binding, K_M , because there is only one substrate, so no competition is possible, **Figure 8.4**. The Gibbs energy difference between the two transition states is proportional to the natural logarithm of the selectivity, Equation 8.6.

$$S_{product} = \frac{(k_{cat})_f}{(k_{cat})_s} \quad (8.5)$$

$$\Delta\Delta G_{f-s} = -RT\ln(S_{product}) \quad (8.6)$$

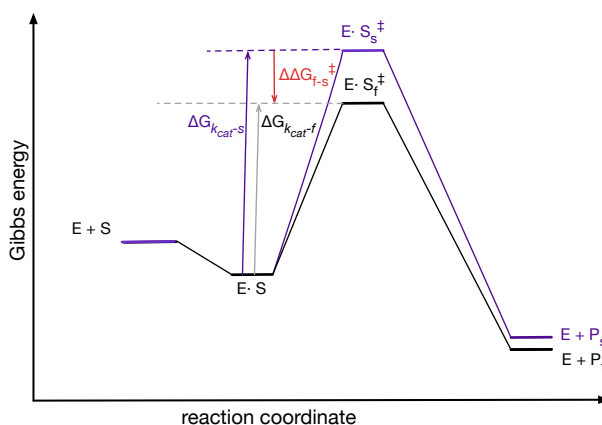


Figure 8.4. Gibbs energy diagram for product selectivity. The enzyme can convert a single substrate into one of two products. One product, P_f , forms faster because the transition state to form that product is lower in energy than the transition state to form the other product, P_s . The difference in Gibbs energy for the two transition states is the origin of the selectivity.

To measure product selectivity one measures the relative amounts of the products formed. The ratio of the products corresponds

to the selectivity. This ratio remains constant as the reaction proceeds because the starting material never changes; it just decreases in amount. The two equations in Equation 8.7 are convenient way to calculate the enantiomeric ratio from a measured enantiomeric excess or to predict an enantiomeric excess from a known enantiomeric ratio.

$$E = \frac{1 + ee}{1 - ee} \quad \text{and} \quad ee = \frac{E - 1}{E + 1} \quad (8.7)$$

8.4 Substrate selectivity

Substrate selectivity occurs when two substrates compete in the same solution. An enzyme encounters similar substrates, but reacts with some of them more rapidly than others. Proteases are a good example of substrate selectivity. Substrate selectivity involves a *separation*; the two substrates already exist and the selective reaction will transform one of them. A peptide contains multiple peptide links each of which is a different potential substrate. Proteases often favor hydrolysis of some of these links over others. For example, trypsin favors hydrolysis of peptide links after lysine or arginine residues, **Figure 8.5 a**. The example peptide contains fifteen amide links (fifteen possible substrates), but trypsin cleaves only two of these due to its substrate selectivity.

Protease nomenclature describes how the peptide orients in the protease when cleavage occurs,^[4] **Figure 8.5 b**. The numbering of the binding sites, S, indicates their distance from the catalytic site where cleavage occurs:S3-S2-S1-S1'-S2'-S3'... The sites numbered without primes bind the N-terminal part of the peptide, while the sites numbered with primes bind the C-terminal part of the peptide. The sites bind both the side chains and main chain of the peptide. The catalytic site lies between the S1 and S1' sites

8 Engineering enzyme selectivity

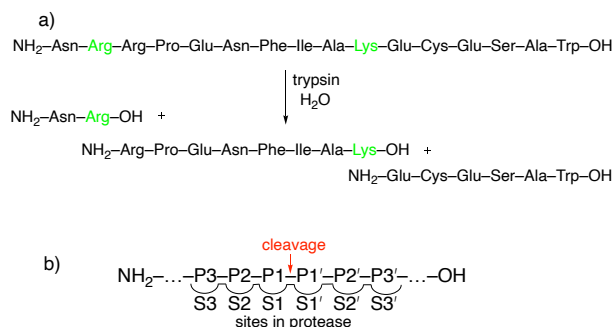


Figure 8.5. Proteases are often selective in which peptide links they cleave. a) Trypsin cleaves only two of the fifteen amide links in peptide shown. Trypsin is selective for amide links on the C-terminal side of lysine and arginine residue, except when followed by proline. b) The binding sites in the protease are numbered to indicate their distance from the catalytic site. The sites S1, S2, S3, etc. bind the N-terminal part of the peptide, while the sites S1', S2', S3', etc. bind the C-terminal part of the peptide. The peptide residue naming matches the names of the binding sites: P1, P2, P3, etc. in the direction of amino terminus of the peptide and P1', P2', P3', etc. in the direction of the carboxy terminus. The peptide residue naming changes as different amide bonds are positioned at the cleavage site.

8.4 Substrate selectivity

and cleaves the peptide bond connecting the two residues bound in those sites. The numbering of the enzyme sites is fixed by the location of the catalytic residues. The numbering of the peptide substrate varies when different amide links are positioned at the cleavage site. The peptide numbering matches the binding sites at which it binds: $\text{NH}_2\text{....P3-P2-P1-P1'-P2'-P3'...COOH}$. Trypsin is selective for peptide links with lysine or arginine at the P1 position, except when the P1' amino acid is proline. The proline at P1' presumably disrupts catalytically productive binding.

Substrate selectivity originates from both the ability of the enzyme to bind the good substrate and also to catalyze a reaction on the good substrate. The good substrate must succeed at both, while the poor substrate may fail at either step. Poor substrates either do not bind to enzyme or bind in a way that does not lead to reaction. The selectivity between the two depends on both k_{cat} and K_M , Equation 8.8, where subscript f refers to the faster-reacting substrate and subscript s refers to the slower reacting substrate.

$$S_{\text{substrate}} = \frac{(k_{cat}/K_M)_f}{(k_{cat}/K_M)_s} \quad (8.8)$$

The tighter binding substrate will occupy a larger fraction of the enzyme active sites, while the faster catalytic step will convert more of that substrate to product. The Gibbs energy diagram, **Figure 8.6**, shows these two contributions to substrate selectivity graphically. In this diagram, both k_{cat} and K_M favor the faster-reacting substrate, but it is also possible that they favor different substrates, in which case the competing effects lead to a lower net selectivity.

For the trypsin example, the different K_M 's refer to the binding of different amide bonds in the catalytic site. To estimate K_M for the different amide bonds one could measure K_M using substrate analogs that contain only one type of amide bond.

8 Engineering enzyme selectivity

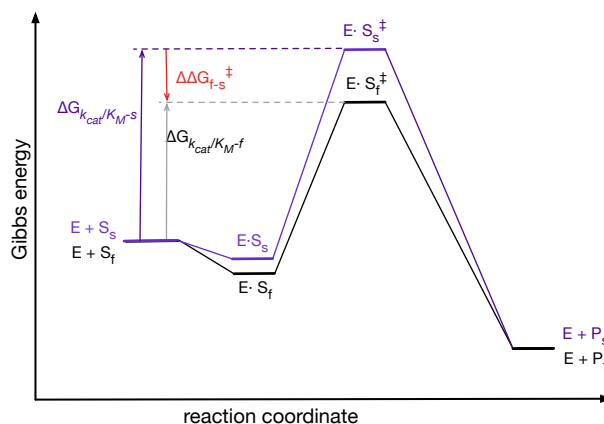


Figure 8.6. Gibbs energy diagram for two competing substrates in an enzyme-catalyzed reaction. Both tighter binding and faster reaction contribute to the selectivity. S_f = faster-reacting substrate; S_s = slower-reacting substrate.

A common special case of substrate selectivity occurs when the two competing substrates are enantiomers. In this case the reaction is a kinetic resolution. The word resolution refers to a separation of enantiomers, while kinetic specifies that the separation relies on differences in the rates of reaction. For example, the manufacture of diltiazem, a calcium-channel blocker uses a kinetic resolution to make an enantiomerically-pure precursor,^[5] **Figure 8.7**. Starting from a racemic mixture, the lipase catalyzes the hydrolysis of the unwanted enantiomer to the carboxylic acid, which spontaneously decarboxylates.

In some cases, reversing the reaction reverses its classification as substrate selective or product selective. Reversing the reaction in **Figure 8.3** by starting with a racemic mixture of amines and converting them to the ketone would create a case of substrate selectivity. The transaminase would convert the favored (*R*)-amine

8.4 Substrate selectivity

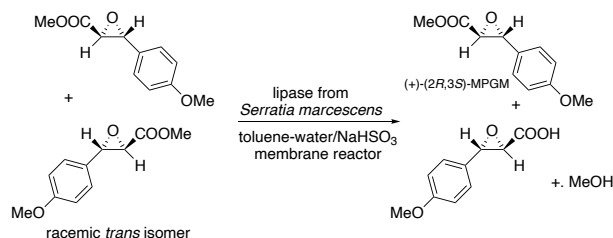


Figure 8.7. Lipase-catalyzed kinetic resolution to manufacture of an enantiopure precursor to diltiazem, a calcium-channel blocker drug. The lipase catalyzes the hydrolysis of the unwanted enantiomer to the carboxylic acid, which spontaneously decarboxylates to the aldehyde. The membrane reactor separates the charged carboxylate from the uncharged aldehyde.

to the ketone leaving the (*S*)-amine unchanged and an example of a kinetic resolution. The disadvantage of a kinetic resolution is that the maximum yield of each enantiomer is 50%. If the purpose of the reaction is the manufacture of a precursor to make an enantiomerically-pure drug, then only one of the enantiomers will be useful. In contrast, the maximum yield of pure enantiomer in the product selectivity case is 100%.

8.4.1 Measuring selectivity in kinetic resolutions

One way to measure reaction selectivity in a substrate selectivity case is to measure the amounts of substrates consumed or products formed at very low conversion (<5%), eqs. Equation 8.9 or Equation 8.10. At low conversion

8 Engineering enzyme selectivity

$$S_{substrate} = \frac{(\text{amount of substrate reacted})_f}{(\text{amount of substrate reacted})_s} \text{ at low conversion} \quad (8.9)$$

$$S_{substrate} = \frac{(\text{amount of product formed})_f}{(\text{amount of product formed})_s} \text{ at low conversion} \quad (8.10)$$

The low conversion is required so that the ratio of the two substrates remains constant over the time of the measurement. The equations Equation 8.9 and Equation 8.10 above also assume equal starting concentrations of the two substrates.

It is often inconvenient to measure the amounts of substrate consumed or product formed at low conversion. In these cases one must use more complex equations. These equations account for the fact that the composition of the substrate changes as the reaction proceeds. The relative amount of the slow-reacting substrate increases as the reaction depletes the fast-reacting substrate, **Figure 8.8**.

The amount of enrichment varies depending on the enantioselectivity of the reaction, **Figure 8.9**

The equation below^[6] predicts the enantioselectivity, E , from the conversion (c , value between zero and one) and the enantiomeric excess of the product, ee_p , Equation 8.11.

$$E = \frac{\ln[1 - c(1 + ee_p)]}{\ln[1 - c(1 - ee_p)]} \quad (8.11)$$

To measure enantioselectivity, one measures two of the three variables: c , ee_p , ee_s . Then, use Equation 8.11 to calculate E from c and

8.4 Substrate selectivity

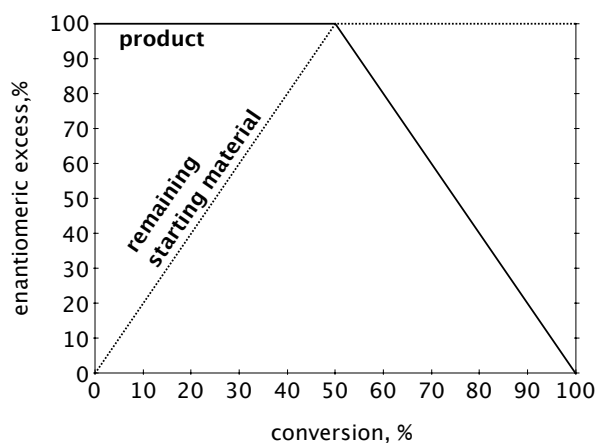


Figure 8.8. Variation of enantiomeric excess for a perfect kinetic resolution as a function of conversion. As the reaction proceeds from 0% to 50% conversion, the remaining starting material is enriched in the slow reacting enantiomer. At 50% conversion, the product consists of the fast-reacting enantiomer, while the substrate consists of the slow reacting enantiomers. The reaction should be stopped at this point. If the reaction is continued, then the enantiomeric excess of the product will decrease as it becomes contaminated with the slow-reacting enantiomer.

8 Engineering enzyme selectivity

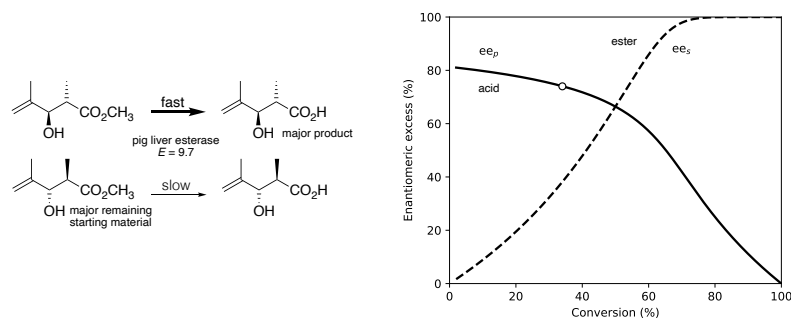


Figure 8.9. Predicted enantiomeric excess of product (ee_p , solid) and remaining substrate (ee_s , dashed) as a function of conversion for an enzyme with a moderate enantioselectivity of $E = 9.7$. Equation 8.11 and experimental measurement of $ee_p = 74\%$ and $c = 34\%$ (circle) reveals the enantioselectivity of this reaction. The theoretical curves shown are calculated from this enantioselectivity. Product ee is highest at low conversion ($\sim 81\%$) and falls toward 0% as conversion approaches 100%, as the faster-reacting enantiomer is depleted. Substrate ee starts near 0% and rises toward 100% as conversion approaches completion, reflecting progressive enrichment of the remaining substrate pool in the slower-reacting enantiomer.

8.4 Substrate selectivity

ee_p , Equation 8.12 to calculate E from c and ee_s , or Equation 8.13 to calculate E from ee_p and ee_s . Note that you can calculate c from ee_p and ee_s using $c = \frac{ee_s}{ee_s + ee_p}$. For example, the data point in **Figure 8.9** shows a measurement of $c = 34.0\%$ and $ee_p = 74.0\%$, which corresponds to $E = 9.7$ and could be used to calculate the theoretical lines.

$$E = \frac{\ln[(1 - c)(1 + ee_s)]}{\ln[(1 - c)(1 - ee_s)]} \quad (8.12)$$

$$E = \frac{\ln \left[\frac{1 - ee_s}{1 - (ee_s/ee_p)} \right]}{\ln \left[\frac{1 - ee_s}{1 + (ee_s/ee_p)} \right]} \quad (8.13)$$

A web tool that encodes these equations simplifies this calculation: <http://biocatalysis.uni-graz.at/biocatalysis-tools/enantio>. Entering any two of the following values ee_p , ee_s , or c , returns the enantioselectivity for the reaction and a graph similar that in **Figure 8.9**.

However, if you already know the enantioselectivity and would like to calculate the conversion needed to get a certain enantiomeric excess of the product, then you must use an iterative method to solve the equation because you cannot rearrange Equation 8.14 to solve for c . For example, if your reaction has an enantioselectivity of 35 and you would like the enantiomeric excess of the product to be at least 92% ee, then

$$35 = \frac{\ln[1 - c(1.92)]}{\ln[1 - c(0.08)]} \quad (8.14)$$

Solving this equation iteratively using the Python script in the supporting information (Code Block S8.1) yields $c = 0.295$. (Less

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elegantly, one can also try different values of c with $ee_p = 92\%$ in the web tool cited above until the calculated enantioselectivity matches 35.) If you try to solve Equation 8.14 for c when ee_p is a higher value of 95% ee, you get a negative value for c indicating that is it impossible. An enantioselectivity of 35 is too low to ever yield product with 95% ee.

Equation 8.11 and the discussion above apply to effectively irreversible reactions like the ester hydrolysis examples given. If the resolution reaction is reversible, it will be less efficient and follow another equation^[7] that takes reversibility and the equilibrium constant into account. An example of such a reversible reaction is the hydrolase-catalyzed ester formation from an acid and alcohol in organic solvents.

8.4.2 Dynamic kinetic resolution

A dynamic kinetic resolution combines a kinetic resolution with an *in situ* racemization of the substrate to overcome the limitation of a maximum of 50% yield, **Figure 8.10**. The enantioselective enzyme converts the fast-reacting enantiomer to product, while the racemization replenishes the fast-reacting enantiomer. At the end of the reaction all of the substrate has been converted to product in up to 100% yield.

The requirements for a dynamic kinetic resolution are: (1) the substrate must racemize faster than the subsequent enzymatic reaction, (2) the product must not racemize, and (3) the enzymatic reaction must be highly enantioselective. The equations relating product enantiomeric purity and enzyme enantioselectivity are the same as those for an asymmetric synthesis, Equation 8.7 above.

8.4 Substrate selectivity

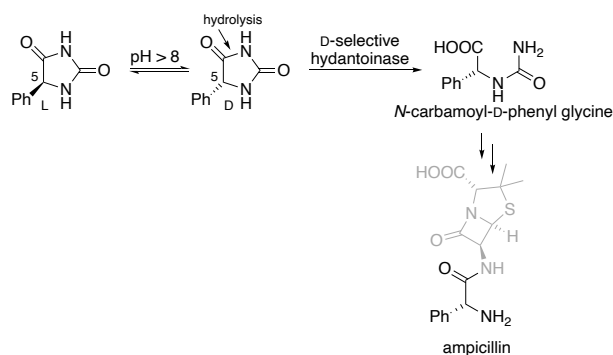


Figure 8.10. Dynamic kinetic resolution adds an in situ racemization to a kinetic resolution, which increases the maximum yield to 100%. The resolution of 5-phenyl hydantoin is a dynamic kinetic resolution because 5-phenyl hydantoin spontaneously racemizes at pH > 8. A D-selective hydantoinase catalyzes the hydrolysis of D-enantiomer, which is later used in the manufacture of ampicillin.

A dynamic kinetic resolution shares elements of both a kinetic resolution and an asymmetric synthesis. Like a kinetic resolution, it relies on an existing stereocenter and the substrate selectivity of the enzyme. Like an asymmetric synthesis it yields up to 100% of one enantiomer. A dynamic kinetic resolution is classified as an asymmetric transformation of the second kind because it is neither a kinetic resolution (only one enantiomer results) nor an asymmetric synthesis (the stereocenter already exists in the substrate).

8.5 Molecular basis of stereoselectivity

8.5.1 Ketoreductases/alcohol dehydrogenases

Since stereoisomers differ only in shape, the only way to distinguish them is by their shapes. In two cases - reduction of ketones and hydrolysis of secondary alcohol esters - researchers have proposed simple models that predict the favored enantiomer based on the relative sizes of the substituents. These models imply that enzymes also choose the favored enantiomer based on the relative sizes of the substituents and that all such enzymes have similarly-shaped active sites.

Ketoreductases/alcohol dehydrogenases are NAD(P)(H)-dependent enzymes that catalyze the interconversion of alcohols and the corresponding carbonyl compounds. The two enzyme names are synonyms. Ketoreductase refers to the reduction of carbonyl compounds with NAD(P)H, while alcohol dehydrogenase refers to the oxidation of carbonyl compounds with NAD(P)⁺. The favored direction depends on the substrate and reaction conditions, not the enzyme. Most of the examples below are reductions that form a stereocenter.

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In the 1960's Prelog carried out many yeast-catalyzed reductions of ketones and noticed that the product alcohols usually had a similar shape. He proposed a simple model or rule based on the relative sizes of the ketone substituents.^[8] The rule states that yeast-catalyzed reductions of ketones yield the alcohol enantiomer shown below where L represents a larger substituent and S represents a smaller substituent, **Figure 8.11 a**. If the priorities of the substituents are $O > L > S$, then the alcohol has the (S)-configuration. The example reactions in **Figure 8.11 b** proceed according to Prelog's rule yielding the alcohols shown with high enantiomeric purity. Note that the stereochemical descriptors for the product alcohols differ because the priority of the small substituent (methyl or chloromethyl) changes. Prelog's rule defines the shape of the molecule, not the (R,S) naming of the alcohol.

To test if a reductase reaction fits Prelog's rule one first orients the product alcohol so that 1) C-O bond points upward to the top of the page and 2) C-O bond points out of the plane of the page toward the reader. In this orientation, if the large substituent lies to the left, then the reaction follows Prelog's rule. Prelog's rule is based on the relative sizes of the substituents, not on Cahn-Ingold-Prelog (CIP) ranking of the substituents, which is based on atomic numbers. When the larger substituent has higher CIP ranking than the smaller substituent, then Prelog's rule predicts formation of the (S)-alcohol, but if the smaller substituent has the higher CIP ranking, then Prelog's rule predicts formation of the (R)-alcohol. The reliable way to use Prelog's rule is to predict the shape first, then assign the descriptor. A second caution is that the effective size of the substituents is conformationally averaged. A flexible *n*-alkyl chain folds and behaves as smaller than a rigid substituent of similar molecular weight, so *n*-hexyl is "small" relative to phenyl.

The transition state for an alcohol-dehydrogenase-catalyzed reduction of a ketone orients the carbonyl carbon of the ketone near the hydride of the reduced nicotinamide, **Figure 8.12**. There are

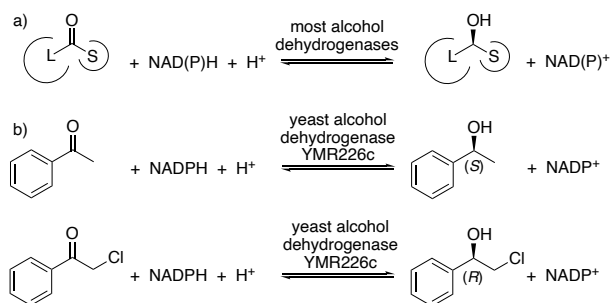


Figure 8.11. Prelog's rule predicts which alcohol enantiomer forms faster during the alcohol-dehydrogenase-catalyzed reduction of ketones. The rule is based on the relative sizes of the ketone substituents. a) Dehydrogenase-catalyzed reduction of ketones favors formation of the enantiomer shown where L is a large substituent and S is a small substituent. b) Yeast alcohol dehydrogenase YMR226c catalyzes the reduction of acetophenone and α -chloroacetophenone according to Prelog's rule. The shape of both product alcohols matches the shape predicted by Prelog's rule despite their opposite (*R,S*) designation the chlorine substituent.

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two ways, differing by a 180° rotation along the carbonyl C–O bond, to orient the ketone. These two orientations yield opposite enantiomers and place the large and small substituents in different regions of the enzyme. Prelog's rule implies that most alcohol dehydrogenases have similarly shaped regions that bind the ketone and that the shape more closely matches the top transition state in **Figure 8.12**

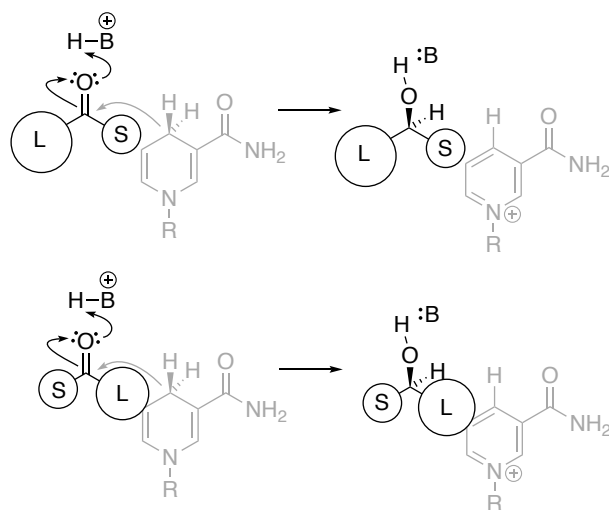


Figure 8.12. Two transition states for the alcohol-dehydrogenase-catalyzed reduction of ketones. In both cases the ketone lies above the plane of the reduced nicotinamide ring and the hydride transfer is to the bottom face of the ketone carbonyl group. The two transition states differ in which face of the ketone carbonyl faces the reduced nicotinamide; that is, they differ by a 180° rotation along the carbonyl C–O bond. The upper transition state is the one that corresponds to Prelog's rule.

The x-ray crystal structures of alcohol dehydrogenases show that

most of them have ketone-binding regions that match Prelog's rule, **Figure 8.13**. These pockets orient the ketone for reaction. In this example, the substrate is not a ketone, but an aldehyde so that both orientations yield an achiral alcohol. Nevertheless, the structure shows the large and small pockets.

Three approaches to increase the enantioselectivity of reductases are substrate modification, screening for a more enantioselective enzyme, and engineering a more enantioselective enzyme. Reductases usually show higher enantioselectivity toward ketones with larger size differences between the two substituents. For example many reductases show high enantioselectivity toward acetophenone, which has phenyl and methyl as the ketone substituents. In contrast, most reductases show low enantioselectivity toward 3-hexanone, which has ethyl and *n*-propyl substituents. Thus, choosing a substrate with larger differences in size in the two substituents will often increase the enantioselectivity.

In some cases, the synthetic goal, say a pharmaceutical precursor, requires reducing a ketone with similarly-sized substituents. In these cases, the choices are screening for more enantioselective reductases or engineering a more enantioselective reductase by modifying the binding pockets. For example, enantioselective reduction of 3-hexanone requires the enzyme to distinguish between an ethyl and an *n*-propyl substituent. Most alcohol dehydrogenases show low enantioselectivity, presumably because they have difficulty distinguishing between the similarly-sized substituents. Koesoema and coworkers found a yeast enzyme that showed high enantioselectivity ($E > 200$ favoring the (*S*)-enantiomer).^[9] An x-ray crystal structure of the enzyme suggested that Trp288 limits the size of the small pocket. Docking suggests that the ethyl substituent fit in the small pocket, but fitting the *n*-propyl group in the small pocket places the ketone group too far from the catalytic zinc. The non-productive orientation prevents formation of the (*R*)-enantiomer.

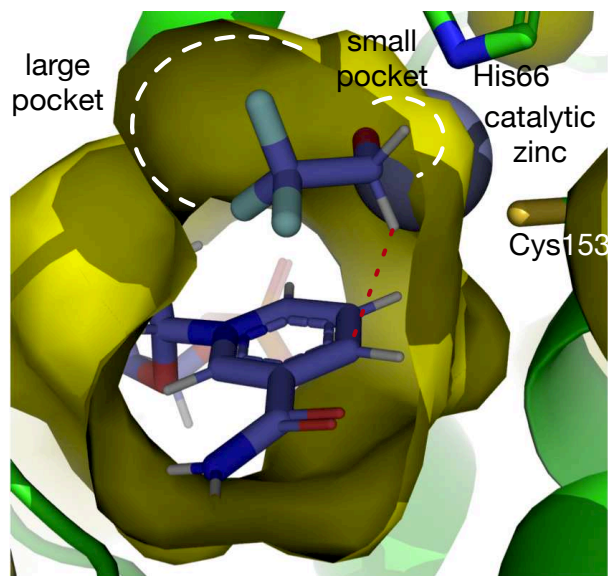


Figure 8.13. Active site of yeast alcohol dehydrogenase containing bound NAD^+ and 1,1,1-trifluoroethanol (pdb id: 5ehv). The nicotinamide ring of NAD^+ is at the bottom, while the 1,1,1-trifluoroethanol is coordinated to the catalytic zinc at the top. The side chains of His66, Cys153 and Cys43 (not visible) coordinate the catalytic zinc. Oxidation of 1,1,1-trifluoroethanol involves transfer of a hydride from the alcohol to the nicotinamide C4 along the red dotted line. The distance from nicotinamide C4 carbon and the C2 of 1,1,1-trifluoroethanol is 3.5 Å suggesting a catalytically productive orientation. The yellow surface shows the limits of the active site pocket. The large pocket containing the trifluoromethyl substituent and the small pocket containing a hydrogen substituent are marked by white dashed curves.

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In an engineering example, the manufacture of an antibiotic required the reduction of tetrahydrothiophene-3-one, where the difference in size stems from a CH_2 versus a sulfur, **Figure 8.14**.^[10] The initial enzyme showed poor enantioselectivity ($E = 4.4$), but evolution to modify the substrate binding pockets dramatically increased the enantioselectivity to $E = 285$.

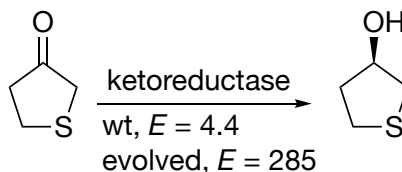


Figure 8.14. The synthesis of an antibiotic required the enantioselective reduction of tetrahydrothiophene-3-one, where the ketone substituents differ subtly in size. The wild-type ketoreductase showed an enantioselectivity of only 4.4, but the evolved enzyme, which contained ten substitutions, showed a 65-fold higher enantioselectivity of 285.

8.5.2 Esterases and lipases

Prelog's rule in the previous section and the rule introduced here are both shape rules: each reduces a three-dimensional recognition problem to a comparison of the sizes of two substituents, and each predicts a favored enantiomer. However, they apply to different enzymes and they answer different questions, **Table 8.1**.

Prelog's rule describes an enantioselective *synthesis*. The substrate is a prochiral ketone without a stereocenter; the dehydrogenase delivers hydride to one face, and the rule predicts which face, and therefore which enantiomer of the alcohol is *created*. The rule below describes a kinetic *resolution*. The stereocenter already exists,

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the substrate is normally racemic, and the rule predicts which of the two enantiomers reacts faster - which one is consumed, not which one is made.

Table 8.1. Two shape rules that answer different questions.

	Prelog's rule	Rule for secondary alcohols
Enzymes	Dehydrogenases, ketoreductases	Esterases, lipases
Substrate	Prochiral ketone; no stereocenter	Chiral alcohol or ester, usually racemic
Question answered	Which face is attacked, hence which enantiomer forms	Which of two existing enantiomers reacts faster
Type of process	Enantioselective synthesis	Kinetic resolution
Maximum yield of one enantiomer	100%	50%
Product <i>ee</i> during the reaction	Constant	Decreases with conversion
Site of reaction	At the stereocenter	Adjacent to the stereocenter

The last row of **Table 8.1** is the one that matters mechanistically, and we return to it in Section 8.5.2.1: because the hydrolase reaction occurs *next to* the stereocenter rather than at it, the disfavored enantiomer has ways to bind non-productively that are unavailable in a dehydrogenase reaction.

Rule for secondary alcohols and their esters

Draw the substrate with the oxygen—the hydroxyl group in an acylation or the ester oxygen in a hydrolysis—pointing out of the page toward the reader. Extend the C–O bond as a line that divides the molecule in two. The enantiomer with the **larger substituent on the right** is the one that reacts faster, **Figure 8.15**.^[11,12] A similar rule applies to primary amines of the type $RR'CHNH_2$, which are isosteric with secondary alcohols.

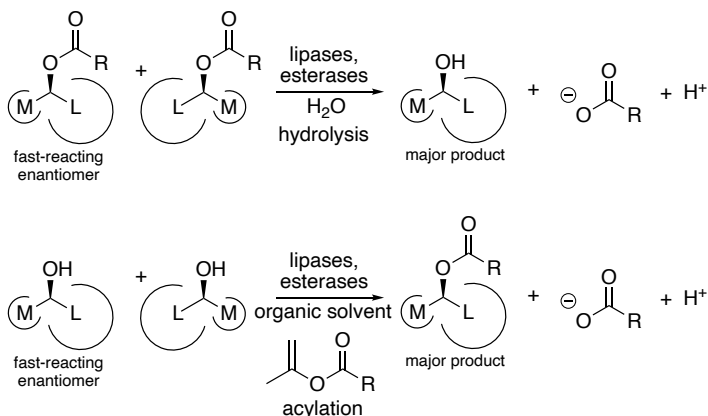


Figure 8.15. The rule for secondary alcohols. M is a medium-sized substituent such as methyl; L is a large substituent such as phenyl. The top reaction is a hydrolysis of a racemic ester in water. The bottom reaction is an acylation of an alcohol in an organic solvent. In both cases the ester or alcohol with the large substituent on the right reacts faster.

Note that the drawing convention is not the one used for Prelog's rule, and the two should not be applied to the same picture. Prelog's rule is set up in the plane of the carbonyl group, looking at a trigonal carbon; this rule is set up along the C–O bond of a tetrahedral

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stereocenter. Sketching the substrate afresh for whichever rule is in play avoids most errors. Like Prelog's rule, the secondary alcohol rule is based on the relative sizes of the substituents, not on Cahn–Ingold–Prelog ranking.

The rule applies to all lipases and esterases, including cholesterol esterase, *Burkholderia cepacia* lipase, lipases from several *Pseudomonas* species, lipase from *Rhizomucor miehei*, lipase B from *Candida antarctica* (CAL-B), and porcine pancreatic lipase. *Candida rugosa* lipase (CRL) follows it for cyclic secondary alcohols such as menthol, but not reliably for acyclic ones. Because lipases are poor amidases, the extension to primary amines is useful only in the acylation direction. As with Prelog's rule, there are exceptions to the rule.

8.5.2.1 The structural basis of lipase enantioselectivity toward secondary alcohols

X-ray structures of phosphonate transition-state analogues bound to CRL show why the rule works.^[13] The alcohol-binding site contains one large hydrophobic pocket and one medium pocket that match the sizes of the two substituents, **Figure 8.16**. The isopropyl-bearing half of the cyclohexyl ring occupies the large pocket; the methyl-bearing half occupies the medium pocket. Thus, the alcohol-binding site of CRL contains pockets to bind medium and large substituents and resembles the shape in the secondary alcohol rule.

The phosphonate inhibitor corresponding to the fast-reacting enantiomer of menthol not only fits in the active site, but also fits in a catalytically productive orientation, **Figure 8.17**. The alcohol oxygen points toward His449 and accepts a hydrogen bond. This hydrogen bond could speed protonation of this oxygen as the alcohol group leaves during CRL-catalyzed hydrolysis. In contrast,

8 Engineering enzyme selectivity

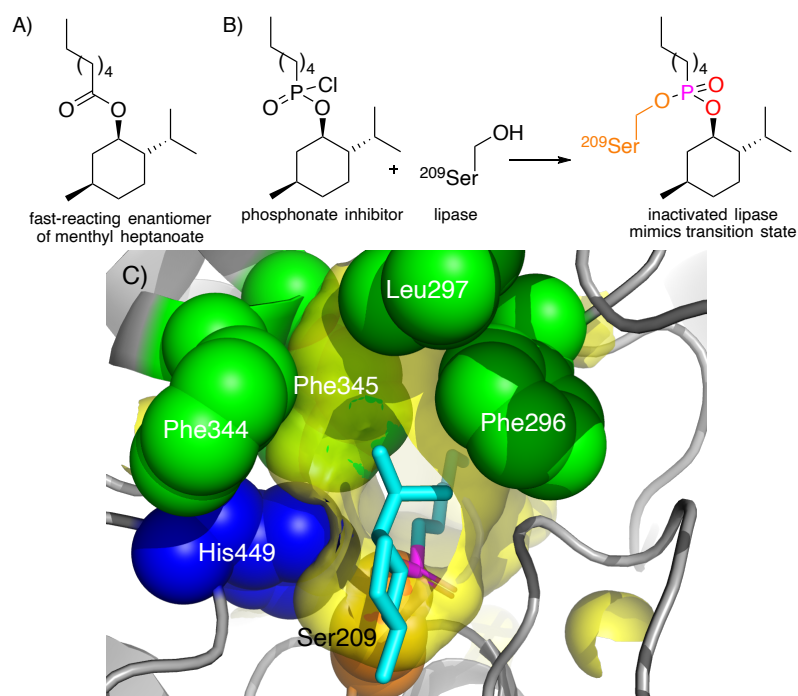


Figure 8.16. X-ray structure of lipase from *Candida rugosa* showing large- and medium-sized pockets that bind substituents of a secondary alcohol. A) CRL catalyzes the hydrolysis of menthyl esters with high enantioselectivity, favoring (1*R*)-menthyl heptanoate. B) Reaction of CRL with the phosphonyl chloride shown covalently binds a transition-state analog of (1*R*)-menthyl heptanoate hydrolysis in the active site. C) An X-ray structure of the CRL-phosphonate reveals a large hydrophobic pocket created by Phe296, Leu297, Phe344, and Phe345 (green spheres) that binds the large isopropyl substituent. A medium-sized pocket binds the methyl-bearing half of the cyclohexyl ring. The phosphorus (magenta sticks) binds covalently to the catalytic Ser209 (orange spheres) and near the catalytic His449 (blue spheres).

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the phosphonate inhibitor corresponding to the slow-reacting enantiomer of menthol fits into the active site, but not in a catalytically productive orientation. The large substituent binds in the large pocket and the medium substituent in the medium pocket, but the catalytic hydrogen bond to His449 is missing. The mirror-image shape of the slow-reacting enantiomer points the alcohol oxygen away from His449, instead of toward His449 as in the fast-reacting enantiomer. In addition, the mirror-image shape of the slow-reacting enantiomer points the isopropyl substituent at His449, causing it to turn and further making formation of the catalytic hydrogen bond difficult. Thus, the slow-reacting enantiomer fits, but its enantiomeric shape creates a non-reactive orientation.

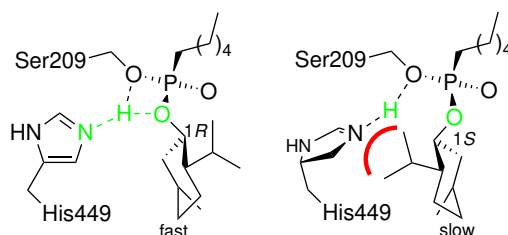


Figure 8.17. The phosphonate mimicking the fast-reacting enantiomer (left) contains a catalytically essential hydrogen bond (green atoms). This hydrogen bond allows the catalytic histidine to protonate the leaving alcohol oxygen as the alcohol group leaves. This hydrogen bond is missing in the phosphonate for the slow-reacting enantiomer (right) because of its different shape. The C–O bond points toward the histidine in the fast-reacting enantiomer, but away from the histidine in the slow-reacting enantiomer.

The fast enantiomer satisfies three features simultaneously: L in the large pocket, M in the medium pocket, and the oxygen oriented

for the hydrogen bond to histidine. The slow enantiomer cannot satisfy all three. It can compromise in four ways, **Figure 8.18**. One is seen in the X-ray structure: the oxygen is not oriented to make the catalytically important hydrogen bond to His449. Another compromise is to place the L substituent in the M pocket and the M substituent in the L pocket. Ensuring the M and L differ significantly in effective size minimizes this possibility. The two rightmost compromises are unlikely because they place the M or L substituent where the hydrogen bond forms in the fast-reacting enantiomer—this region is simply too small to fit an M or L substituent. Consistent with this picture, enantioselectivity in hydrolase reactions usually appears in k_{cat} rather than K_{M} : both enantiomers bind with similar affinity, and selectivity arises from how well the bound complex is positioned to react.

Consequences for engineering

The rule is a design tool as much as a predictive one, in three ways.

It identifies which pocket to change. Because the two enantiomers differ by which substituent sits in which pocket, altering the *relative* sizes of the pockets is the direct route to altering selectivity. Enlarging the medium pocket of CAL-B with a single Trp104Ala substitution inverts the preference toward 1-phenylethanol, from strongly (*R*)-selective wild type to a modestly (*S*)-selective variant.^[14] The asymmetry of that result is instructive: removing a wall is easy, whereas building a new one on the opposite side is not, so reversals typically stop at moderate selectivity unless a second, compensating substitution is added.

It identifies which substrates are worth attempting. If M and L are similar in size, no amount of pocket engineering will separate them cleanly. Choosing or modifying the substrate so that the two substituents differ substantially—or choosing a different acyl group,

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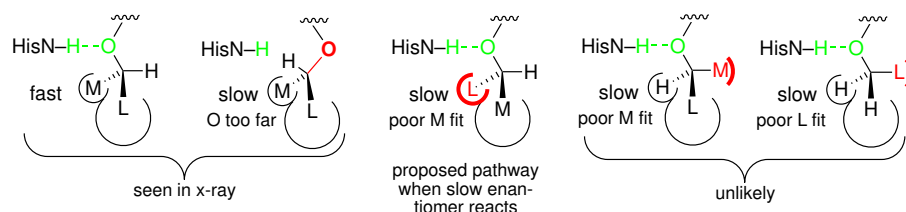


Figure 8.18. Productive orientation of the fast-reacting enantiomer of a secondary alcohol in the lipase active site (far left) and four possible ways that the slow-reacting enantiomer could bind. The second diagram from the left represents the orientation seen in the X-ray where the M and L substituents fit into their respective pockets, but the catalytic hydrogen bond is missing because the oxygen points away from His449. The center structure is one way that the slow enantiomer might react by exchanging the locations of the L and M substituents. The two rightmost orientations are unlikely because they place the M or L substituent in the limited space occupied by the hydrogen bond of the fast-reacting enantiomer.

which changes how deeply the alcohol seats in the pocket—is often cheaper than engineering the enzyme.

It predicts sign, not magnitude. The secondary alcohol rule and Prelog’s rule state which enantiomer is favored and suggest that larger differences in substituent size lead to higher enantioselectivity. Predicting the magnitude of enantioselectivity would require comparing the energies of the transition states for the fast- and slow-reacting enantiomers, including entropy, solvation, and the conformational ensemble. A partial step in that direction is to compare the key catalytic distance between HisN ϵ and O_{alc}.^[15] Slow enantiomers of well-resolved substrates ($E > 100$) showed long distances, suggesting that the slow enantiomer reacted poorly, while slow enantiomers of poorly resolved substrates ($E < 20$) showed distances near the 2.5 Å seen for fast enantiomers, suggesting that the slow enantiomer could react.

8.5.3 Engineering active sites for selectivity

A systematic survey of published enzyme-improvement studies found that mutations close to the substrate-binding site improve enantioselectivity more often, and by a larger amount, than mutations placed elsewhere in the protein.^[16] For example, focusing random mutagenesis on four pocket residues of an esterase from *Pseudomonas fluorescens* (PFE) — Trp28, Val121, Phe198, and Val225 — found beneficial variants far more often than the same amount of screening applied to the whole gene, and reached higher enantioselectivities as well.^[17] A close mutation can influence the transition state directly; a distant one must propagate its effect through the protein before it reaches the substrate.

“Close” could mean a residue that sits next to a pocket, without lining it, that can add an interaction that a model based on size alone

8.5 Molecular basis of stereoselectivity

would miss. *Candida rugosa* lipase (CRL) ordinarily follows the secondary alcohol rule (Section 8.5.2.1), but it reverses to *S* selectivity for *N*-Boc-protected γ -amino alcohols. The reversal traces to Ser450, a residue adjacent to, but not part of, the medium pocket: it forms a hydrogen bond to the carbamate nitrogen of the substrate, and this bond forms only when the substrate is held in the geometry of the *S* enantiomer.^[18] Modeling the tetrahedral intermediate for each enantiomer, rather than only measuring the two pocket volumes, was what revealed this residue.

A related result comes from CAL-B. A region beside its main alcohol pocket, called the “stereoselectivity pocket,” contributes an electrostatic rather than a steric interaction. A single substitution there doubled the enantioselectivity toward 1-halo-2-octanols, and a different single substitution at the same position eliminated it.^[19] Because the effect is electrostatic, it would not show up on a model that only checks whether a substituent fits.

This kind of near-site engineering is not limited to hydrolases. In a secondary alcohol dehydrogenase from *Thermoanaerobacter brockii*, a flexible loop runs alongside the two Prelog pockets. Two substitutions in this loop increased its mobility, which enlarged the effective substrate-binding pocket, allowing the enzyme to accept bulky ketones that the wild-type enzyme reduced only poorly.^[20] A similar strategy reversed or improved the enantioselectivity of a pyridoxal-phosphate-dependent ω -transaminase from *Chromobacterium violaceum* through substitutions near, rather than within, its amine-binding pockets.^[21]

Some selectivity-determining residues line the tunnel leading to a buried active site. In the Rieske oxygenases SxtT and GxtA, which hydroxylate different carbons of the same substrate, swapping the two active-site residues that differ between them only biases site-selectivity. Three residues near the start of a ~33 Å tunnel connecting each protein’s surface to its buried iron center differ

between the enzymes. Changing just these three residues in SxtT shifts hydroxylation toward the GxtA-preferred carbon; combined with the active-site and one loop substitution, selectivity fully inverts.^[22] Once inside, the substrate fills the active site too tightly to reorient, so its final orientation is set as it passes through the tunnel.

These examples suggest a search order for engineering: after adjusting the residues that define the pockets, look at the next shell out — side chains that point toward the pocket or toward the transition state without forming part of the pocket surface. These residues can add a specific hydrogen bond, change local flexibility, or shift electrostatics, often with a single substitution. When the active site is buried and reached through a tunnel, that search can extend further still: residues lining the access route, far from the substrate itself, are also worth screening, since they can gate which substrates, and in what orientation, reach the active site.

8.6 Proofreading

When two substrates are similar, a single binding site discriminates poorly. For example, isoleucine and valine differ by one CH₂. Excluding the larger isoleucine from a valine site is straightforward, since steric hindrance is a strong interaction, but excluding the smaller valine from an isoleucine site is much harder. Valine fits, only less tightly. Protein and DNA synthesis require higher selectivity than one site can provide, so they link two reactions in series. The first step adds the amino acid or nucleotide; a second step, called proofreading or editing, removes the mistakes. Because the two steps select independently, their selectivities reinforce one another. The gain comes at a cost in yield, since the proofreading step destroys some correctly made product along

with the errors. The reinforcement requires the two steps to proceed at comparable rates. If one step is much faster, then the slow step determines the overall selectivity and the selectivity gain of two steps is lost.

8.6.1 Protein synthesis

Accurate translation of mRNA into protein requires correctly loaded aminoacyl-tRNAs.^[23] Aminoacyl-tRNA synthetases load each amino acid onto its correct tRNA using a double sieve mechanism. In isoleucyl-tRNA synthetase, the first step — loading the amino acid onto the tRNA — uses a binding site that excludes amino acids larger than isoleucine, **Figure 8.19**. The proofreading step occurs at a second, smaller site.^[24] Aminoacyl-tRNAs mistakenly loaded with the smaller valine fit into this site and are hydrolyzed, preventing their incorporation into proteins.

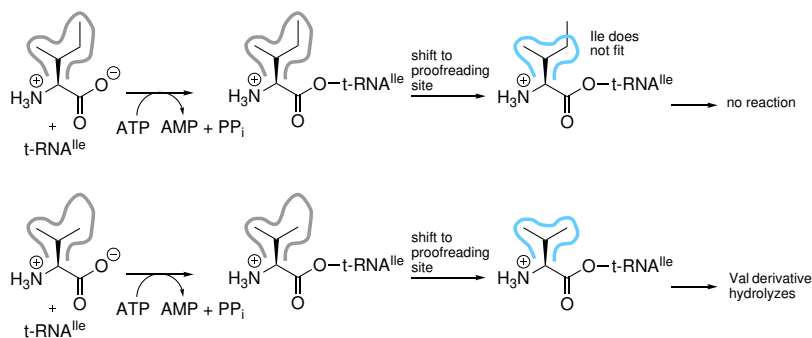


Figure 8.19. Proofreading by the tRNA^{Ile} synthetase increases its selectivity for Ile over valine. The overall selectivity for Ile over Val in proteins is ~3000.^[25]

8.6.2 DNA synthesis

DNA polymerases use the same two-step strategy.^[26] Correct base pairing at the polymerase active site provides the first selection, favoring the correct dNTP by roughly 10^4 - to 10^5 -fold. The second selection is a $3' \rightarrow 5'$ exonuclease in a separate active site, about 30 Å away in *E. coli* DNA polymerase I. The polymerase extends efficiently only from a correctly paired $3'$ end, so a misincorporated nucleotide stalls synthesis; the stall gives the frayed primer terminus time to melt out of the duplex and slip into the exonuclease site, where it is excised. After every nucleotide added, the enzyme partitions between extending and excising, and fidelity is set by that balance. Proofreading contributes roughly another 100-fold to selectivity, bringing the error rate to about one per 10^7 bases.

The classic evidence comes from bacteriophage T4.^[27] Genetic screens identified polymerase variants that replicate less accurately than wild type (“mutators”) and, more surprisingly, variants that replicate more accurately (“antimutators”). The variant differed in the ratio of exonuclease to polymerase activity: mutators had relatively less exonuclease, antimutators relatively more. Shifting the partition toward excision raises fidelity. The antimutators also show the cost, since they hydrolyze more dNTPs per base pair synthesized than wild type — they excise correctly paired termini along with the errors — and antimutator phage grow poorly when nucleotides are scarce.

A third mechanism, mismatch repair, increases the DNA error rate to about one per 10^9 to 10^{10} bases. It differs from proofreading in acting after the polymerase has released the DNA, and by a separate set of proteins. The mismatch itself no longer reveals which of the two bases is the error, since both are ordinary bases; repair must be directed to the newly synthesized strand by a separate, transient marker. In *E. coli* this marker is hemimethylated

GATC sites: the parental strand carries the methyl group, the newly synthesized strand does not, and repair is directed to the unmethylated strand.

8.6.3 Biocatalysis

Proofreading need not be a separate site on a single large protein. The second step can be carried out by a separate enzyme, or even by the same enzyme acting twice on a symmetrical substrate. This is what makes proofreading attractive to an engineer: a proofreading step can be added without engineering the key enzyme itself.

8.6.3.1 Adding a separate enzyme for proofreading

Modular polyketide synthases and non-ribosomal peptide synthetase assembly lines carry substrates as thioesters on a 4'-phosphopantetheine arm attached to their carrier domains. The phosphopantetheinyl transferases that attach substrates to this arm are promiscuous, accepting short acyl-CoAs, so carriers become misprimed with acetyl or other short acyl groups. A misprimed carrier cannot be elongated, and the assembly line stalls.

The fix is a separately encoded enzyme, the type II thioesterase (TEII), an α/β -hydrolase with a Ser-Asp-His triad encoded within the biosynthetic gene cluster but not part of the megasynthase itself. TEII hydrolyzes the aberrant acyl group, regenerating a free thiol ready for the correct extender unit. TEII_{srf} (surfactin) and TEII_{bac} (bacitracin) strongly prefer misprimed acetyl-PCP over aminoacyl- or peptidyl-PCP,^[28] and the TEII of the phoslactomycin PKS, prefers alkyl-ACPs over (alkyl)malonyl-ACPs by up to three orders of magnitude.^[29] This is the same inverted sieve seen in the

synthetases: the loading step cannot fully reject acyl-CoAs, and a separate protein with complementary specificity removes what got through. The cost in yield appears here as well. TEII co-expression raises product yields, but strong overexpression lowers titers because promiscuous TEIIs also strip correct intermediates – an editing activity tuned too aggressively destroys product.

Another example is the use of MutS binding to identify errors in synthetic DNA. The error rate in de novo, template independent DNA synthesis is much higher than in DNA replication in vivo and requires a proofreading step. Denaturing and re-annealing the error-containing synthetic DNA creates heteroduplexes where the errors appear as mismatches. The protein MutS binds these mismatches but does not cut them. The error-containing fraction, with MutS bound, is removed by gel shift or on an immobilized column. This approach reduced errors more than 15-fold, to roughly one per 10,000 bp,^[30] and raised the fraction of correct eGFP clones from 0.93% to 83%.^[31]

8.6.3.2 Using the same enzyme twice on symmetrical substrates

The second selective step need not use a different catalyst at all. A molecule with similar reactive groups can react twice with a single enzyme, so the material passes through two filters instead of one. Two arrangements are possible, and they differ in what becomes of the mistakes.

The first starts from an achiral meso diester.^[32] Hydrolysis of one of the two enantiotopic ester groups desymmetrizes the molecule, giving enantiomeric monoesters that form at rates differing by $E_1 = 15.6$. From this step alone the monoester would have 88% ee, **Figure 8.20**. The enzyme does not stop at the monoester, but further

catalyzes its conversion to the diol. This second hydrolysis also favors the same enantiotopic ester in the monoester ($E_2 = 24.8$), so corrects the errors of the first hydrolysis. Removing the minor enantiomer preferentially raises the enantiomeric purity of the surviving monoester as the reaction proceeds.

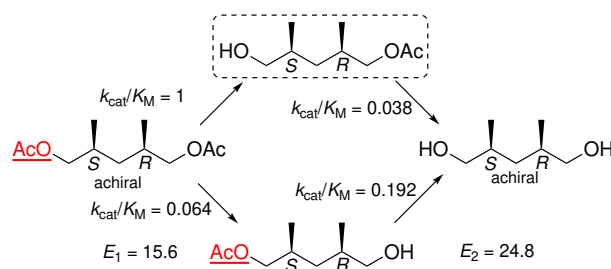


Figure 8.20. Enhancement of stereoselectivity during the desymmetrization of (2R,4S)-2,4-dimethylpentane-1,5-diyl diacetate catalyzed by porcine pancreatic lipase to form (2R,4S)-5-hydroxy-2,4-dimethylpentyl acetate (in box). Both steps favor hydrolysis of the acetate underlined and in red text with similar enantioselectivity: $E_1 = 15.6$, $E_2 = 24.8$. If only step 1 occurred (diacetate to monoacetate), then the overall enantioselectivity would be 15.6 and the product would have 88% ee. At low conversion (<50% yield of monoacetate), the product has ~88% ee. As the reaction continues the second step (monoacetate to diol) selectively converts the minor enantiomer to diol and enhances the enantiomeric purity to 97% ee at 70% yield of monoacetate.

The second example starts from a racemic diester of a C_2 -symmetric diol, where both hydrolyses are enantioselective and the two enantioselectivities reinforce one another, giving an overall selectivity of approximately $(1 + E_1 E_2)/2$.^[33] This is the

clearest demonstration of the requirement for comparable rates. In the hydrolysis of *trans*-1,2-diacetoxycyclohexane by pig liver esterase, the first step ($E_1 = 41$) runs 47 times faster than the second ($E_2 = 2.6$), so the monoacetate accumulates and the slow, poorly selective second step dominates; the diol was obtained with only 58% ee at 44 mol%, **Figure 8.21**. A single-step resolution with an enantioselectivity of 41 would give 90% ee at the same conversion, so the second step was making the resolution worse, not better. Adding a hexane phase, which selectively extracts the more hydrophobic diacetate and thereby slows the first step, raised the enantiomeric purity to 94% ee without any change to the enzyme itself. Screening for an enzyme with a better product of the two selectivities – lipase from *Burkholderia cepacia*, $E_1 > 230$, $E_2 = 17$ – and again equalizing the rates gave the diol in >99% ee.

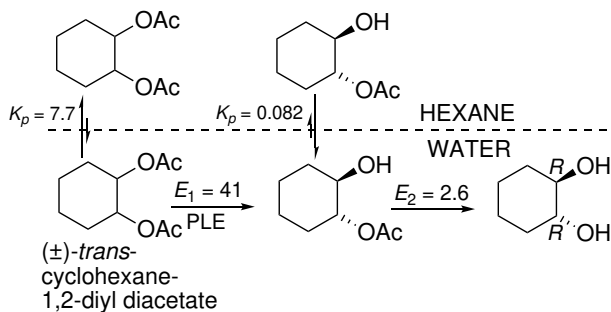


Figure 8.21. Sequential kinetic resolution of a racemic diacetate with pig liver esterase. In water, the first step proceeds 47 times faster than the second step, which prevents the two steps, both of which favor the (*R,R*)-enantiomer, from reinforcing each other. Adding a hexane phase slowed the first step to only 6-fold faster by lowering the concentration of the diacetate in the aqueous phase. The enantiomeric purity of the product diol increased to 94% ee at 34 mol%. K_p is the hexane-water partition coefficient.

The two cases differ in where the mistakes end up. Starting from a meso diester, the desired product is the intermediate monoester and the errors are hydrolyzed away, as in a true proofreading step. Starting from a racemic diester, the desired product is the fully hydrolyzed diol and the errors are left behind in the monoester fraction. In both cases, one enzyme acting twice achieves a selectivity that neither step could reach alone.

8.7 Conclusions

The strategies in this chapter — shape rules, pocket and near-site engineering, tunnel engineering, and proofreading — all rely on knowing, or being able to model, an enzyme's structure and mechanism well enough to predict which residues to change. That knowledge is not always available: many enzymes lack a solved structure, a reliable homology model, or a mechanistic hypothesis linking a specific residue to selectivity, and even a good structure can fail to predict which substitution will work, since selectivity sometimes depends on cooperative effects between residues that no static picture reveals. Directed evolution, covered in the next chapter, requires no such knowledge in advance: it generates a diverse library of gene variants and lets a screen or selection identify which mutations improve selectivity, at the cost of needing a large library and a way to measure the property of interest in each variant. In practice, the two approaches complement each other: the structural insight in this chapter narrows a library to the residues most likely to matter, while directed evolution can find compensating or synergistic mutations that structural reasoning alone would not anticipate.

Glossary

Absolute configuration (R/S) the three-dimensional arrangement of substituents around a stereocenter, assigned using the Cahn-Ingold-Prelog rules.

Asymmetric synthesis a reaction that forms one stereoisomer preferentially by using a chiral catalyst, reagent, or enzyme to bias the reaction pathway.

Chirality a property of a molecule that makes it non-superimposable on its mirror image.

Diastereomer a stereoisomer that is not a mirror image of another stereoisomer.

Diastereoselectivity a preference of a reaction to convert one diastereomer of a substrate or to form one diastereomer of a product. The diastereoselectivity of a reaction is a unitless number greater than one that corresponds to the relative reactivity of the diastereomers.

Enantiomer one of a pair of non-superimposable mirror-image molecules.

Enantiomeric excess (ee) quantitative measure of enantiomeric purity of a substance, defined as $|\%R - \%S|$.

Enantioselectivity a preference of a reaction to convert one enantiomer of a substrate or to form one enantiomer of a product. The enantioselectivity of a reaction is a unitless number greater than one that corresponds to the relative reactivity of the enantiomers.

Kinetic resolution an enzymatic or chemical process that selectively converts one enantiomer of a racemic substrate into product faster than the other.

Meso compound a molecule that contains multiple stereocenters, but is overall achiral because it possesses an internal plane of symmetry. For example, (1S, 2S)-dihydroxycyclopentane is a chiral molecule, but (1R, 2S)-dihydroxycyclopentane is a

meso compound and not chiral.

Optical rotation the ability of an enantiomerically enriched sample to rotate plane-polarized light. The measured optical rotation of a sample reveals its enantiomeric purity if the rotation of the pure enantiomer is known.

Proofreading (editing) is a second reaction that destroys the incorrect products of a synthetic step, typically by hydrolysis at a site whose selectivity is opposite of the first, so that the two selectivities reinforce one another. One of the costs of proofreading is destruction of some of the correct product.

Prochiral center atom that becomes chiral upon replacement of one substituent.

Stereocenter (chiral center): an atom at which exchange of two substituents yields a stereoisomer.

Stereospecificity Reaction behavior in which different stereoisomeric substrates lead to distinct stereoisomeric products via mechanistic constraint. Stereospecificity is rarely encountered in enzyme-catalyzed reactions.

Stereoselectivity unequal formation of stereoisomeric products when multiple are possible, due to differential stabilization of transition states. Enzyme-catalyzed reactions are often stereoselective. Enantioselectivity and diastereoselectivity are sub-classes of stereoselectivity.

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Supporting Information

Code Block S8.1. Python script to calculate the conversion, c , needed to reach a target enantiomeric excess of product, ee_p , for a known enantioselectivity, E (Equation 8.14). This equation cannot be rearranged to solve for c directly, so the script finds it by bisection. To run it, save the code below as `c_from_E_eep.py` and run `python3 c_from_E_eep.py` in a terminal — as written, this executes the “example” line at the bottom, `c_from_E_eep(35, 92)`, which calls the function with $E = 35$ and a target ee_p of 92% and prints the result shown below. To try different numbers, edit that line and re-run the script.

```
#!/usr/bin/env python
# coding: utf-8
"""
This script calculates the conversion,  $c$ , needed to reach a target
enantiomeric excess of product,  $ee_p$ , for a kinetic resolution with a
known enantioselectivity,  $E$ . The equation


$$E = \ln(1 - c(1+ee_p)) / \ln(1 - c(1-ee_p))$$


cannot be rearranged to solve for  $c$  directly, so this script finds  $c$ 
by bisection: repeatedly halving a search interval known to contain
the answer.
"""
import math # load math functions needed to calculate logarithms

def ratio(c, eep):
    # E as a function of c, for a fixed target eep
    return math.log(1-c*(1+eep))/math.log(1-c*(1-eep))

def c_from_E_eep(E, percenteep, tol=1e-6):
```

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```
eep = percenteep/100.0 # convert percent to a fraction between 0 & 1

# As c -> 0, the ratio approaches its smallest possible value,
# (1+eep)/(1-eep). E below this means no conversion ever reaches
# eep -- there is nothing to bisect for.
min_E = (1+eep)/(1-eep)
if E < min_E:
    print('An enantioselectivity of',E,'is too low to ever reach',
          percenteep,'\b% ee (the highest reachable ee needs E of at least',
          round(min_E,1),'\b).')
    return None

# ratio(c, eep) rises from min_E (near c = 0) to infinity (as c
# approaches the conversion where 1 - c(1+eep) reaches zero), so
# bisecting this interval for ratio(c, eep) = E is safe.
lo, hi = tol, 1/(1+eep)-tol
while hi-lo > tol:
    mid = (lo+hi)/2
    if ratio(mid,eep) < E:
        lo = mid
    else:
        hi = mid
c = (lo+hi)/2

print('To reach',percenteep,'\b% ee with an enantioselectivity of',E,
      '\b, stop the reaction at c =',round(c,3),
      '('+format(100*c,'.1f')+'\b% conversion).')
return c

# example: change these two numbers to try a different case
c_from_E_eep(35, 92)
```

Running the script as shown prints: “To reach 92% ee with an enan-

tioselectivity of 35, stop the reaction at $c = 0.295$ (29.5% conversion).” Changing the example line to `c_from_E_ee(35, 95)` and re-running instead prints: “An enantioselectivity of 35 is too low to ever reach 95% ee (the highest reachable ee needs E of at least 39.0).”

Problems

1. Draw an achiral molecule and identify its plane of symmetry. Draw two regioisomers that can result from the electrophilic aromatic substitution of toluene with chlorine. Draw a pair of achiral diastereomers and a pair of chiral diastereomers.
2. An enantioselective transaminase converts an achiral ketone to a chiral amine using isopropyl amine as the amine donor.
 - (a) Draw a balanced equation for this reaction. If the enantioselectivity was infinite, what would be the enantiomeric purity of the product at 50% conversion and the maximum yield product? The enantioselectivity for this reaction was not infinite, it was only 10. What is enantiomeric purity of the product at 50% conversion?
 - (b) Draw hypothetical Gibbs energy diagrams for formation of each enantiomer. Label the Gibbs energy difference that is responsible for enantioselectivity. Is this Gibbs energy difference a ΔG , a $\Delta\Delta G$, or a $\Delta\Delta\Delta G$ value? Can this difference come from k_{cat} , from K_M , or from both? Which of these possibilities does your diagram show?
 - (c) Protein engineering increased the enantioselectivity to 100. To compare the original enantioselectivity, 10, to the new enantioselectivity, 100, do you subtract them or divide them? How large was the change in Gibbs energy for this change? Is this Gibbs energy change a ΔG , a $\Delta\Delta G$, or a $\Delta\Delta\Delta G$ value?

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To compare the Gibbs energy values for the original and the improved enzyme, do you subtract or divide the values? Draw a new Gibbs energy diagram for the reaction catalyzed by the engineered enzyme.

3. A researcher reduced 3-octanone using an alcohol dehydrogenase and isolated the product alcohol with 60% ee. The configuration of the product showed that it followed Prelog's rule.

- (a) What was the enantioselectivity of the reaction?
- (b) Draw the structure of the favored enantiomer.
- (c) To engineer a dehydrogenase with higher enantioselectivity, how could the researcher change the shape of the active site? Explain your reasoning.
- (d) Sketch a transition state for the for the dehydrogenase catalyzed reduction of a ketone. How does the enzyme stabilize the transition state of the favored enantiomer as compared to the transition state for the non-favored enantiomer?

4. A researcher hydrolyzed 3-octyl acetate using a lipase and isolated the product alcohol with 60% ee at 30% conversion. The configuration of the product showed that the favored product followed the secondary alcohol rule.

- (a) Calculate the enantioselectivity of the reaction using the web tool: <http://biocatalysis.uni-graz.at/biocatalysis-tools/enantio>.
- (b) Draw the structure of the favored enantiomer.
- (c) Even with this low enantioselectivity it is possible to isolate unreacted starting material with > 95% ee. Explain how this is possible. Estimate the yield of unreacted starting material

with $> 95\%$ ee using the graph generated in part a of this question.

Answers

1. One example is bromochloromethane. The plane of symmetry passes through the carbon and both hydrogens. The two regioisomers are *o*-chlorotoluene and *p*-chlorotoluene. One example of achiral diastereomers is *cis*- and *trans*-2-butene. One example of chiral diastereomers is D-glucose and D-galactose; they differ in their configuration at C4, but have the same configuration at C2, C3, and C5.
2. a) The reaction is an asymmetric synthesis where an achiral starting material is converted to a chiral product. The ee of the product would be constant throughout the reaction. If the E were infinite, it would be 100% ee. The maximum yield is 100%. If the E was 10, then the ee of the product at any percent conversion would be 81.8% ee.
- b) The enantioselectivity stems from differences in k_{cat} for the two reactions. The K_M cannot contribute because there is only one substrate and therefore only one K_M . The energy difference between the $E \cdot S$ complex and the transition state is $\Delta G_{k_{cat}}$. The Gibbs energy difference between the two ΔG values is $\Delta\Delta G$.
- c) To compare enantioselectivities, you take the ratio, so an enantioselectivity of 100 is 10 times better than an enantioselectivity of 10. The increase in enantioselectivity corresponds to a difference between two $\Delta\Delta G$ values, so it is a $\Delta\Delta\Delta G$. To compare the Gibbs energy values, you subtract the two values. The $\Delta\Delta G$ corresponding to an enantioselectivity of 10 is

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1.37 kcal/mol at 300 K and the $\Delta\Delta G$ corresponding to an enantioselectivity of 100 is 2.74 kcal/mol, so the difference is 1.37 kcal/mol.

3. a) The product of 60%ee corresponds to a 80/20 mixture of enantiomers. The E is therefore 4.
b) The product has the (*S*)-configuration.
c) The minor enantiomer forms when the ketone flips over so that the L and S substituents bind in the S and L pockets, respectively. To prevent this binding mode, one can make amino acid substitutions that decrease the size of the S pocket.
d) The transition state shows hydride transfer from the nicotinamide to the carbonyl carbon and simultaneous protonation of the carbonyl oxygen. Stabilization comes from preorganization and from stabilization of the negative charge on the carbonyl oxygen. The favored enantiomer orients closer and with a better angle because the substituents fit better into the pockets.
4. a) The enantioselectivity is 5.
b) The favored enantiomer has the (*R*)-configuration.
c) By carrying out the reaction to >50% conversion, the enrichment of the remaining starting material in the slow-reacting enantiomer increases. 79% conversion would yield remaining starting material with 99%ee.

9 Introduction to Directed Evolution

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Summary. Directed evolution improves a protein by making many variants and then identifying the better ones, usually over several rounds. When it is not known which residue to change, error-prone PCR spreads substitutions across the whole protein by copying the gene with a deliberately error-prone polymerase. Its reach is limited: because each codon usually changes by only one nucleotide, error-prone PCR samples an average of only 5.7 of the 19 possible replacement amino acids at a position. The genetic code adds further bias: Trp and Met are encoded by a single codon while Leu, Ser, and Arg have six. The polymerase itself favors changes at A·T sites and favors transitions over transversions. Most substitutions also land far from the active site simply because most of the protein is far from the active site. Too few mutations per gene wastes effort on unchanged clones; too many disable the protein, so an intermediate mutation rate works best. Despite being incomplete, these libraries routinely contain useful changes, which makes error-prone PCR well suited to finding stabilizing substitutions that can occur anywhere. When the useful region is known, site-saturation mutagenesis is preferred: mixed-base (ambiguity) codons, which are codons synthesized as mixtures of bases, let a single position encode some or all twenty amino acids, and the resulting variants are screened. It is the better tool for reshaping the active site to expand substrate range or change selectivity. No method can build

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every possible variant because the number is astronomically large, so both are strategies for sampling sequence space efficiently.

Key learning goals

- Explain how DNA polymerase bias and the single-nucleotide-change limit make error-prone PCR libraries incomplete, and predict which amino acid substitutions are accessible from a given wild-type codon.
- Design a site-saturation mutagenesis experiment at a chosen codon, and explain why testing the nineteen alternative amino acids at a substrate-binding residue is an efficient route to altered substrate range or selectivity.
- Interpret IUPAC ambiguity codon notation (e.g., NNK, NDT) and select an appropriate codon scheme for a target set of amino acids.
- Calculate the oversampling required to screen a saturation library to a stated coverage, and explain why codon redundancy makes the number of clones exceed the number of variants.
- Compare error-prone PCR with saturation mutagenesis, and justify which method suits a given engineering goal.
- Calculate the number of possible variants for a given number of positions, replacement amino acids, and protein length, and use the result to argue why exhaustive sampling is impossible.

9.1 Introduction

Directed evolution is the recursive generation of random variations and identification of improved variants, usually by screening or selection. It mimics natural protein evolution, but differs in that it can improve a protein for abilities that do not occur in nature, such

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as reactivity toward pharmaceutical precursors or tolerance to organic cosolvents. The idea is simple enough to write as a short program. Over forty years ago, Eigen & Gardiner^[1] suggested this set of computer instructions:

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10 PRODUCE A MUTANT SPECTRUM OF SELF-REPRODUCING TEMPLATES
20 SEPARATE AND CLONE INDIVIDUAL MUTANTS
30 AMPLIFY CLONES
40 EXPRESS CLONES
50 TEST FOR OPTIMAL PHENOTYPES
60 IDENTIFY OPTIMAL GENOTYPES
70 RETURN TO 10 WITH A SAMPLE OF OPTIMAL GENOTYPES
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The instructions are straightforward, but line 70 carries the weight. Without it, the procedure is a single round of mutagenesis and screening; with it, the enriched genotypes become the templates for the next cycle, and small improvements accumulate over successive rounds. Recursion is what makes the process evolution rather than screening.

Eigen & Gardiner wrote these instructions before the methods to carry them out existed. One difficulty in practice is not the logic but line 10: a mutant spectrum is easy to produce and hard to produce well, since the number of possible variants vastly exceeds the number that can be tested. Most of this chapter concerns how to bias that spectrum toward variants worth screening. The next chapter will cover accumulating or combining mutations.

An early example shows the whole scheme in operation. Chen and Arnold set out to make the protease subtilisin E catalytically useful in high concentrations of dimethylformamide (DMF), an environment no natural subtilisin encounters.^[2] They began with a variant carrying four substitutions (D60N, D97G, Q103R, N218S) that had

been identified separately in earlier screens and combined by site-directed mutagenesis. This variant then served as the parent for three sequential cycles of random mutagenesis by error-prone PCR. Screening was done on plates: secreted subtilisin hydrolyzes casein to give a visible halo, so colonies were lifted onto nitrocellulose, transferred to DMF-containing casein plates, and the clone with the largest halo became the template for the next cycle — line 70 of Eigen and Gardiner's program. After three cycles and roughly 4000 colonies screened, PC3 carried six new substitutions (ten in total). These six directed-evolution substitutions improved catalytic efficiency about 3-fold over the 4M parent in 40% DMF.

9.2 Error-prone PCR

9.2.1 Error rates of DNA polymerases

DNA polymerases occasionally insert an incorrect nucleotide. The polymerase from *Thermus aquaticus* (Taq) makes about 4×10^{-5} errors per nucleotide added — one error per 25,000 nucleotides — and this value ranges from $(1-20) \times 10^{-5}$ depending on reaction conditions.^[3,4] Taq lacks a $3' \rightarrow 5'$ exonuclease, so it cannot correct these mistakes. The polymerases from the archaea *Pyrococcus furiosus* and *Thermococcus kodakarensis* do proofread, lowering their error rates by one to two orders of magnitude. High-fidelity polymerases are the right choice for routine cloning, but the wrong choice for mutagenesis.

The polymerase chain reaction (PCR) copies DNA, and the copying introduces errors. Error-prone PCR (epPCR) increases the error rate enough to generate a library of variants from a long stretch of DNA — an entire gene, or several genes at once. The procedure

is simple, requires no structural information, and distributes substitutions throughout the target sequence. Screening the resulting library identifies the variants with improved properties.

To increase the error rate, error-prone PCR uses an unbalanced pool of dNTPs, elevated free magnesium, and added manganese to raise the Taq error rate about tenfold.^[5] These conditions also slow the polymerase and shorten its useful lifetime, so the amplification is inefficient and a fresh aliquot of Taq is often added after the fifteenth cycle. Pushing the conditions further degrades the amplification, making a tenfold increase in error rate a practical ceiling.

9.2.2 Error rates in the PCR product

The error frequency in the PCR product is higher than the per pass error rate of the polymerase because the errors accumulate in a PCR amplification. Cadwell and Joyce found 6.6×10^{-3} substitutions per position in the product of such a reaction — 6.6 changes per thousand base pairs, or roughly an order of magnitude above Taq's per-pass error rate under error-prone conditions.^[5] Each cycle of PCR uses the products of the previous cycle as templates. Once an error is acquired, it is copied faithfully into all descendants of that molecule. The error frequency in the final product therefore depends on how much the DNA was amplified as well as on the polymerase and the buffer.

Amplification is most conveniently expressed as the number of doublings: a reaction that yields a 10^5 -fold increase over the input template has gone through $\log_2(10^5) \approx 17$ doublings. A perfect PCR would give one doubling per cycle, but real amplifications — especially under error-prone conditions — are less efficient, so the number of doublings is smaller than the number of cycles. Code Block S9.1 in the supporting information calculates the number

of doublings for a given number of cycles when the efficiency is known.

Each doubling gives every position another chance to acquire an error, so substitutions accumulate roughly in proportion to the number of doublings. The mutation frequency per position in the final product is approximately $f \times d$, where f is the per-pass error rate and d the number of doublings. Taking $f \approx 4 \times 10^{-4}$ errors per position for Taq under mutagenic conditions and $d \approx 17$ gives 7×10^{-3} substitutions per position, close to the measured value above. For a gene of 1000 bp, this amounts to about seven nucleotide changes per molecule — an average, since the changes are distributed randomly among molecules and individual clones will carry more or fewer. Note that short sections of DNA, e.g., 10 nucleotides, are unlikely to contain substitutions. Error-prone PCR is suitable for gene-length sections of DNA or longer, but not for short sections because the error rate is not high enough.

9.2.3 Biases that limit the randomness

Four independent factors restrict which amino acid substitutions a random-mutagenesis library can contain (**Table 9.1**). Two arise from the genetic code (codon accessibility and unequal codon degeneracy), one from the chemistry of the polymerase (preferences for some nucleotide replacements over others), and one from the structure of the protein itself. Codon accessibility makes some amino acid replacements highly unlikely with error prone PCR while the additional biases make some substitutions less common than a random distribution would predict, so finding those substitutions requires more screening.

Table 9.1. Four independent factors bias the amino acid substitutions produced by error-prone PCR.

Source of bias	Consequence
Codon accessibility	One nucleotide change reaches only 5.7 of 19 alternative amino acids; the other 13.3 need two changes in one codon, which are ~100-fold rarer.
Unequal codon degeneracy	Trp and Met have one codon each, Leu, Ser and Arg have six, so amino acids are not sampled with equal frequency
Polymerase chemistry	Taq favors substitutions at A·T over G·C sites and transitions over transversion.
Protein architecture	Only a handful of residues line the substrate-binding site, so random substitutions rarely land there.

Codon accessibility. Although there are 64 codons, error-prone PCR can only create a subset of them because it changes only one nucleotide in a codon. At a mutation frequency of 0.7% per position, a substitution somewhere in a given codon has probability $3 \times 0.7\% = 2.1\%$, while two substitutions in the same codon have probability $3 \times (0.7\%)^2 = 0.015\%$, more than a hundredfold lower. Three substitutions are rarer still. If only one nucleotide changes in a codon, then only nine new codons are accessible, (**Figure 9.1**). In the glycine codon GGA, single nucleotide substitutions encode only four new amino acids. Averaged over all sense codons, 24% of

single-nucleotide substitutions are synonymous, 4% create a stop codon, and 71% are missense. Seven nucleotide changes therefore yield about five amino acid substitutions ($7 \times 0.71 = 5$) and at any one position, only 5.7 of the nineteen alternative amino acids are reachable at all.

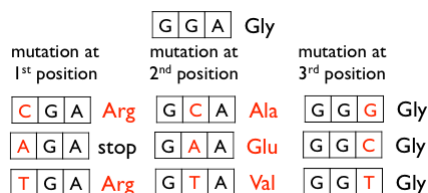


Figure 9.1. Replacement of one nucleotide in the glycine codon GGA yields nine new codons. The three new codons created by replacement of the third nucleotide, A, also code for glycine, so 33% of the single nucleotide substitutions for this codon are synonymous. Five of the nine new codons (56%) are missense codons that introduce a new amino acid. Of these five, arginine occurs twice, so only four new amino acids of the nineteen possible (21%) are encoded. On average, single substitutions in a codon yield missense codons encoding only 5.7 new amino acids.

Unequal codon degeneracy. The genetic code does not assign codons evenly. Trp and Met have one codon each, while Leu, Ser, and Arg have six. Even perfectly random nucleotide substitution would therefore not sample the twenty amino acids with equal frequency: amino acids specified by one or two codons are harder to reach than those specified by four or six.

Polymerase chemistry. Nucleotide misincorporation is not random. If misincorporation were random, then mutations would occur at AT sites at the same frequency as at GC sites. Sequencing of DNA from an error-prone PCR using Taq DNA polymerase shows

that the sites of substitution are non-random,^[6] **Table 9.2.** Most of the errors (>95%) were substitutions rather than frame shifts. Substitutions at AT sites ($40.9 + 7.3 + 23.7 = 71.9\%$) are 3.9 times as common as substitutions at GC sites ($1.4 + 4.5 + 13.6 = 19.5\%$). Random substitutions would correspond to the GC content of the DNA, equal preference occurs when the GC content is 50%; for this example, the GC content was ~49%.

Table 9.2. Fraction of each substitution type expected for random substitutions and observed in error-prone PCR using Taq DNA polymerase or Mutazyme II for DNA sequence encoding *B. lentus* subtilisin.

substitution	fraction expected for random substitutions, %	fraction observed, <i>Taq</i> polymerase, % ^a	fraction observed, Mutazyme II, % ^[7]
A→T and T→A	16.7	40.9	28.5
A→C and T→G	16.7	7.3	4.7
A→G and T→C	16.7	27.6	17.5
G→C and C→G	16.7	1.4	4.1
G→T and C→A	16.7	4.5	14.1
G→A and C→T	16.7	13.6	25.5

^a Data from DNA sequencing of 9800 nucleotides, which identified 286 errors ($\varepsilon_{nt} = 2.9\%$).^[6] The sum of the substitution errors is 95.3%. The remaining errors were frame shifts: insertions 0.3%, deletions 4.2%; the missing 0.2% is rounding. This mutation frequency is roughly fourfold higher than the 0.66% discussed earlier in this chapter; these authors used more aggressive conditions to accumulate enough errors for a reliable count of substitution types.

Another measure of randomness is whether the replacement nucleotides are random. Again, the measured values differ from

those expected for random replacements. There are two measures for this randomness of replacement. First, transitions, which exchange bases of similar size (purine for purine, pyrimidine for pyrimidine) are more common than transversions, which exchange bases of different size (**Figure 9.2**). Four transitions and eight transversions are possible, so random substitution gives a transition/transversion ratio of 0.5. The measured ratio for Taq is higher, 0.76 $((27.6 + 13.6)/(40.9 + 7.3 + 1.4 + 4.5))$.

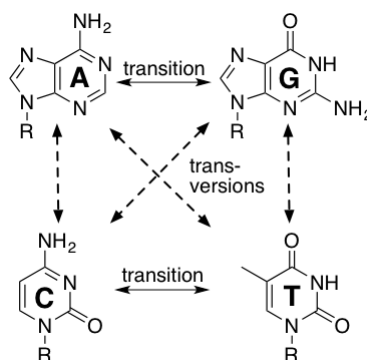


Figure 9.2. Transitions substitute a purine with a purine (A with G and vice versa) or a pyrimidine (C with T and vice versa), while transversions substitute a purine with a pyrimidine (A or G with C or T) or a pyrimidine with a purine (C or T with A or G). There are four possible transitions and eight possible transversions.

The second criterion for randomness of the replacement nucleotides is that AT→GC replacements should equal GC→AT replacements, leaving the GC content of the DNA unchanged. The measured AT→GC replacements 34.9 (7.3 + 27.6) 1.9-fold more common than GC→AT replacements 18.1 (4.5 + 13.6), showing that the GC content of the sequence increased during mutagenesis. Other researchers using different reaction conditions reported

also reported non-random distributions.^{[8], [9]}

Mutazyme II DNA polymerase, a blend of two error-prone polymerases with complementary preferences, reduces some of this bias.^[7] The site preference improves substantially, from 3.9 to 1.2 (50.7/43.7). The GC drift reverses rather than disappears, from 1.9 to 0.56 (22.2/39.6) – the library now loses GC content instead of gaining it. The transition bias increases slightly and remains the largest deviation: 0.84 (43.0/51.4) against a random value of 0.5. No available polymerase makes the substitutions random; the choice is which bias to accept.

Protein architecture. The first three restrictions are properties of the method, this last one comes from the target. Random mutagenesis methods target all amino acid positions equally. Counterintuitively, this random process favors mutations far from the active site.^[10] Only a few residues (<10%) form the substrate-binding site and most lie far away, so most mutations land far from the active site. This inherent preference is known as location bias.

Imagine a spherical enzyme with the active site at the center and the residues arranged in concentric shells around it. Each shell farther out is larger and therefore contains greater numbers of amino acids. The number of amino acids n in a 1-Å-thick shell of outer radius r is the volume of the shell divided by the average volume occupied by a residue, 141 Å³ (**Figure 9.3**). The prediction fits a typical 30 kDa protein well out to about 15 Å; beyond that, the observed number falls below the prediction because the shells start to extend past the edge of the protein.

More than half the residues in this protein lie 13–22 Å from the active site and fewer than 10% lie within 10 Å of it. Larger proteins push the edge farther out, so the imbalance grows with size. Because substitutions anywhere in the protein can contribute to stability, the bias toward distant substitutions is no disadvantage when stability is the target. Substitutions near the active site are

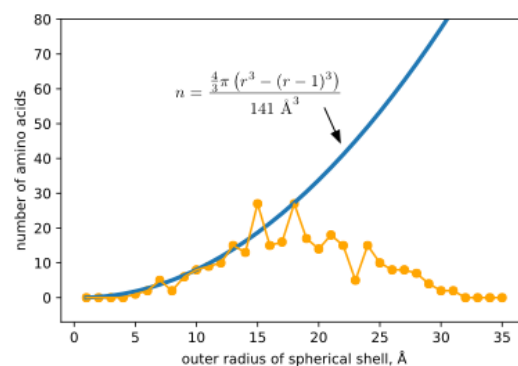


Figure 9.3. The number of amino acids within concentric, spherical shells increases as one moves outward from the center of a protein. The shells are 1 Å thick with outer radius r . The blue line plots the values predicted by the equation shown; the orange circles are observed values for a typical 30 kDa protein (esterase, PDB 1VA4). Agreement is good out to ~15 Å, beyond which the shells reach the outside edge of the protein and contain fewer amino acids than predicted.

far more effective at changing substrate specificity and selectivity, however, and for those goals the location bias is a real limitation.

The subtilisin work described earlier may look like a counterexample, since the six effective substitutions in PC3 clustered near the active site. It is not. The library contained substitutions throughout the gene; screening recovered the small minority that happened to fall in useful positions. What a library contains and what screening returns are different things, and the distinction matters whenever one reasons backward from a set of hits to the diversity that produced them.

These limitations and biases mean that no error-prone PCR library approaches completeness. That is not fatal. As the subtilisin work showed, useful substitutions were common enough that a few thousand colonies sufficed. The practical question is not whether the library is complete — it never is — but whether the improvements sought are frequent enough to be discovered by sampling.

9.2.4 Optimal mutation rate

At a low mutation rate, most of the epPCR library is unmutated wild-type sequence. Screening these wastes effort because they cannot be improved variants. The mutated fraction is dominated by single-substitution variants, and because mutations generated early in the amplification are copied into all their descendants, the same single substitution often turns up in more than one colony. These duplicates likewise waste screening effort: testing a duplicate contributes no new information. For example, a mutation rate, m , of 0.3% in a 1000-bp PCR product is estimated to contain ~25% wild-type or single substitutions. Recall that short sections of DNA are not suitable targets for error-prone PCR because the mutation rate is too low and most of the PCR product contains no substitutions.

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At higher mutation rates, the accumulation of strongly detrimental substitutions limits the ability to find improved variants. Detrimental or inactivating substitutions are those that so strongly hinder protein performance that it cannot be detected. Destabilizing substitutions that hinder protein folding are an example of detrimental substitutions. An improved variant must contain at least one beneficial mutation and, at the same time, no detrimental ones. Assume that mutations are either beneficial, detrimental, or neutral, with fixed fractions F_{ben} , F_{det} , and F_{neu} summing to one, and that they act independently. For a single substitution, the chance of avoiding a detrimental mutation is $(1 - F_{det})$; over m substitutions it is $(1 - F_{det})^m$. By the same reasoning, the fraction containing at least one beneficial substitution is $1 - (1 - F_{ben})^m$. The fraction of improved variants is therefore

$$F_{better} = (1 - F_{det})^m [1 - (1 - F_{ben})^m] \quad (9.1)$$

When $F_{ben} \ll 1$, the second term is approximately $m \cdot F_{ben}$, so

$$F_{better} = (1 - F_{det})^m \cdot m \cdot F_{ben} \quad (9.2)$$

The first term decreases exponentially with m while the second increases linearly, so F_{better} passes through a maximum. Differentiating Equation 9.2 locates it at

$$m_{opt} = \frac{-1}{\ln(1 - F_{det})} \quad (9.3)$$

Note what is absent: F_{ben} does not appear. The optimal number of mutations depends only on how well the starting protein tolerates substitution. The frequency of beneficial mutations sets the *height* of the peak, not its position. This is why a rule of thumb of a

few substitutions per gene transfers across systems whose hit rates differ by orders of magnitude. For a protein in which 30% of substitutions are strongly detrimental ($F_{det} = 0.30$), the optimum is at $m = 2.8$, and if 1% of substitutions are beneficial the peak value of F_{better} is 1% (**Figure 9.4**).

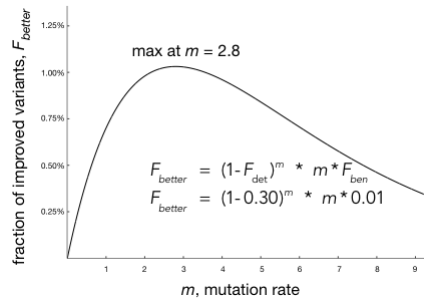


Figure 9.4. Both beneficial and strongly detrimental mutations accumulate with increasing numbers of mutations. An improved variant must be both stable (contain no detrimental mutation) and contain at least one beneficial mutation. If 1% of single substitutions are beneficial and 30% are detrimental, the maximum fraction of improved variants, F_{better} , is 1%, occurring at 2.8 mutations per gene.

Chodorge and coworkers^[11] tested this idea using green fluorescent protein. Their model differs slightly from the one above but yields similar curves. Converting GFP to a blue fluorescent variant requires one specific nucleotide substitution out of the 2151 possible in the 717-bp gene (3×717), so $F_{ben} = 0.00046$. They estimated $F_{det} = 0.20$ from the loss of fluorescence in wild type at different mutation rates, which by Equation 9.3 predicts an optimum at $m = 4.5$. Testing a range of error-prone PCR conditions, they found the highest fraction of blue fluorescent variants at about 5 nucleotide substitutions per gene — close agreement on

the position of the optimum. The observed peak height (1.3%) exceeded the predicted 0.08% by more than an order of magnitude, a reminder that the model captures the shape of the trade-off better than its magnitude. (The predicted peak height of 0.08% comes from evaluating Equation 9.2 at $m = 4.48$: $(0.8)^{4.48} \times 4.48 \times 0.00046$. Worth checking against Chodorge's own model, since theirs differs slightly and may predict a different height.)

Quantitative use of Equation 9.2 is not expected for real problems, since the probability of a beneficial and detrimental mutations is rarely known in advance. The useful result is that an optimum exists, and that its location is set by the mutational tolerance of the starting protein. Indeed several researchers demonstrated that starting with a more stable protein allows discovery of a greater number of beneficial substitutions.^{[12] [13]}

9.2.5 Mutator strains

Another approach to random mutagenesis uses mutator strains of *E. coli*, which carry defects in DNA repair and proofreading that raise the spontaneous mutation rate. The XL1-Red strain, for example, carries mutations in three genes required for accurate replication (nucleotide sanitization, replication fidelity, and post-replicative repair), raising the mutation rate about 5000-fold over wild type, but ~10-fold lower than error-prone PCR.^[9] Random point mutations accumulate as the bacteria replicate. The advantage is that variants are generated *in vivo*, without designing primers or optimizing error-prone PCR conditions. The disadvantage is that mutations are not confined to the gene of interest but occur genome-wide, causing growth defects, changes in expression or copy number, and genetic instability.

To narrow the target, researchers have engineered microorganisms and mammalian cells that replicate plasmid DNA and genomic DNA

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with separate polymerases.^[14–16] An error-prone polymerase copies the plasmid while an accurate proofreading polymerase copies the genome. These strains are more stable than the mutator strains above.

These *in vivo* mutagenesis methods generate many variants in a background of many wild type due to the low mutation rate. Screening would be inefficient, so researchers use selection: the experiment is designed so that improved variants survive or grow faster, and the population itself does the sorting. To evolve resistance to an antibiotic, for instance, one grows the organisms in progressively higher concentrations so that only improved variants persist. Sequencing the survivors then reveals the changes responsible.

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In some cases, researchers know where they want to target mutations, so site-saturation mutagenesis is a better option. Site-saturation mutagenesis is similar to site-directed mutagenesis in that a specific site is targeted, but differs in the use of ambiguity codons. Ambiguity codons are mixtures of codons that code for multiple amino acids, so mutagenesis using ambiguity codons yields a mixture of protein variants. A common application of site-saturation mutagenesis is targeting the substrate-binding site to expand the substrate range or increase the selectivity, **Table 9.3**. Another application is targeting flexible surface residues to create interactions between nearby amino acids to increase protein stability. Increasing reactivity may require reshaping the substrate binding region, but it can also require changes in more distant loops or tunnels. Several examples of targeted site-saturation mutagenesis are below.

Table 9.3. Selected examples of site-saturation mutagenesis to create variants.

protein	region targeted	improvement
D-hydantoinase	substrate-binding site	11-fold increase in activity toward new substrate
nitrilase	all positions ^a	increase enantioselectivity from ~15 to >100
lysine tRNA synthase	substrate-binding site	27.6
phenylacetone monooxygenase	hinge region between domains	expand substrate range
lipase	flexible surface loops	increase stability temperature by 45 °C

^a The researchers did not have a structure of the nitrilase, so they tested all 6,270 single substitutions (19 replacements at 330 positions). Later a model of the nitrilase revealed that the beneficial substitutions were all in the substrate-binding site.

Cheon et al.^[17] expanded the substrate range of D-hydantoinase to include hydroxyphenylhydantoin, a precursor for antibiotic synthesis. Site saturation mutagenesis Phe159 in a loop forming the substrate binding pocket followed by screening of variants identified the Phe159Ala variant with ~6-fold enhanced catalytic activity (k_{cat}) toward this substrate. Subsequent site saturation mutagenesis at Leu65, located on a loop on the other side of the substrate binding pocket, followed by screening identified the Leu65Phe/Phe159Ala variant with a further ~2-fold increase in catalytic activity for a total of an 11-fold increase over the wild-type

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enzyme.

To introduce the non-canonical amino acid L-homoglutamine into proteins, Anderson and coworkers^[18] altered the selectivity of a lysine tRNA synthase to favor L-homoglutamine instead of lysine or any of the other nineteen canonical amino acids. They identified two amino acid residues that interact with the ϵ -amino group of lysine in the active site of the lysine tRNA synthase and varied these residues using site-saturation mutagenesis (NNK codons). Screening identified the Glu41Ile-Tyr268Ser variant as one with the desired activity and selectivity. Subsequent experiments showed that L-homoglutamine was incorporated into the target protein.

To expand the substrate range of phenylacetone monooxygenase (PAMO, from *Thermobifida fusca*), Reetz and coworkers^[19] used a different approach - they targeted substitutions at two sites distant from the substrate binding site. Previous research identified open and closed conformations of PAMO and a hinge between the FAD- and NADP-binding domains that allows the domains to move. The researchers hypothesized that substitutions near the hinge could reshape the substrate-binding site through altered domain motion. Structure analysis identified Gln93/Pro94 in a helical segment at the hinge region and those positions were randomized using NDT codons, which encoded twelve amino acids at each site. Screening identified the Gln93Asn/Pro94Asp variant which reacted with a wide range of previously unreactive substrates and with high enantioselectivity for most of them.

DeSantis and coworkers^[20] used site-saturation mutagenesis to increase the enantioselectivity of a nitrilase for the synthesis of a cholesterol-lowering drug, atorvastatin (Lipitor®), **Figure 9.5**. The starting nitrilase showed an enantioselectivity of only ~15. Their experiments identified the Ala190His variant, which showed an enantioselectivity of >100. This early application of site-saturation mutagenesis differs from modern ones in that the researchers

did not have a structure of the enzyme to guide their choice of locations. Instead they exhaustively applied site-saturation mutagenesis to all 330 positions of the nitrilase ($330 \times 19 = 6270$ possible single-amino-acid substitution variants) to find the Ala190His variant. Indeed, the AlphaFold 3 structure reveals that residue 190 lies in the substrate-binding region, ~ 10 Å from the catalytic Cys162 Sy. If this experiment were repeated today, one would likely use the predicted AlphaFold 3 structure to focus site-saturation mutagenesis into the substrate binding region.

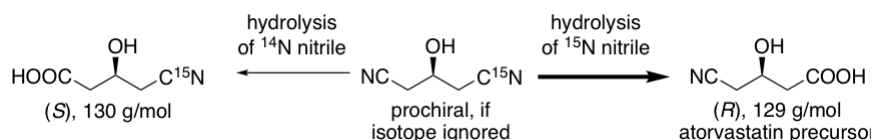


Figure 9.5. A nitrilase catalyzes the hydrolysis of either nitrile in the dinitrile shown in the center. Hydrolysis of the nitrile on the left yields the (*S*)-enantiomer, while hydrolysis of the nitrile on the right yields the (*R*)-enantiomer. To find variants with higher enantioselectivity for the (*R*)-enantiomer, DeSantis and coworkers^[20] used the isotopically labelled dinitrile shown. The two hydrolysis products differed in mass, so mass spectrometry revealed the ratio of the (*R*)- and (*S*)-products.

One advantage of site-saturation mutagenesis is completeness. For example, libraries of single nucleotide substitution variants created by error-prone PCR would likely not include the Ala190His variant. Converting a codon for alanine into a codon for histidine requires two nucleotide substitutions, which makes this substitution unlikely using error-prone PCR.

Targeting site-saturation mutagenesis to several flexible surface loops of a lipase identified stabilizing substitutions increased

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stability. Combining five stabilizing substitutions raised the temperature that halves the catalytic activity by up to 45 °C. The stabilization was presumed to arise through new interactions with neighboring amino acids (e.g., salt bridges, hydrogen bonds, and hydrophobic contacts).^[21]

9.3.1 Ambiguity codons

Site-directed mutagenesis is the replacement of one amino acid with another at a particular location. This replacement involves synthesis of new DNA strands, typically with a DNA polymerase to extend mutagenic DNA primers. Details of one site-directed mutagenesis method were in Chapter 1. These mutagenic primers are made by chemical synthesis, usually by a DNA synthesis company. These primers encode the replacement amino acid instead of the original amino acid, **Figure 9.6**. Site-saturation mutagenesis makes all nineteen possible amino-acid substitutions at a particular location. One approach is nineteen separate site-directed mutagenesis experiments at each amino acid position. This solution is tedious and rarely used.

More commonly, researchers use mixtures of primers that encode multiple amino acids. These primers contain ambiguity codons, which are mixtures of codons prepared by DNA synthesis, **Table 9.4**. For example, the NNK codon represents a mixture of 32 different codons. N represents any of the four nucleotides (A, T, G, C), while K represents the two nucleotides containing a ketone group (G or T), **Figure 9.7**. This approach saves time because only one site-directed mutagenesis procedure is needed to create plasmids encoding all twenty amino acids.

The disadvantage of site-directed mutagenesis using ambiguity codons is that it creates a mixture of protein variants. Transfer of the mixture of DNA into bacteria yields a mixture of bacteria,

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wild type sequence
 5' . . . GATTGCAGCTGCTGTTTTCCACAATTCAGTATTGCCAGACACCGAGC . . .

primer for site-directed mutagenesis His103Val
 5' GATTGCAGCTGCTGTTTTCGTGAATTCAGTATTGCCAGAC 3'

primer mixture for site-saturation mutagenesis His103Xxx
 5' GCTGTTTTCCNNKAATTCAGTATTGCCAGAC 3'

Figure 9.6. Example of primers for site-directed and site-saturation mutagenesis at the underlined codon. The wild type codon is CAC, which encodes histidine. In the primer for site-directed mutagenesis, this codon is replaced by GTC, which encodes valine. The mixture of primers for site-saturation mutagenesis contains thirty-two different primers as represented here by the ambiguity codon NNK.

			A	A
N	N	K	=	T T T
				G G G
				C C

Figure 9.7. The NNK ambiguity codon is a mixture containing 32 individual codons. N (any of the four nucleotides) occupies the first two positions of the codon, while K (G or T) occupies the third position, giving $4 \times 4 \times 2 = 32$ individual codons.

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each one producing one variation at the targeted location. This approach saves time by requiring only one site-directed mutagenesis experiment, but increases the amount of screening since it yields a mixture of variants.

Table 9.4. One-letter ambiguity codes indicate mixtures of several nucleotides.

one-letter code	nucleotides
N (any)	A, T, G, C
B (not A)	T, G, C
D (not C)	A, T, G
H (not G)	A, T, C
V (not T)	A, G, C
K (keto)	G, T
M (amino)	A, C
R (purine)	A, G
S (strong)	G, C
W (weak)	A, T
Y (pyrimidine)	C, T

An example nucleotide sequence containing a ambiguity codon is 5'-NNKATG, which contains the ambiguity codon NNK. Chemists make this nucleotide mixture by adding mixtures of nucleotide precursors at the appropriate step. Chemical DNA synthesis starts at the 3'-end. For this example, the chemist adds the phosphoramidite precursor for G, then the precursor for T, then the precursor for A. At the next step, the chemist adds an equimolar mixture of G and T precursors and in the last two steps the chemist adds an equimolar mixture of A, T, G and C. The result is a mixture

of 32 DNA sequences which differ at the NNK locations. Besides the twenty codons encoding the twenty amino acids, it also contains eleven redundant codons and one stop codon, **Table 9.7** in the supporting information.^[22] Twelve amino acids are encoded once: Asn, Asp, Cys, Gln, Glu, His, Ile, Lys, Met, Phe, Trp, Tyr; five amino acids are encoded twice: Ala, Gly, Pro, Thr, Val and three amino acids are encoded three times: Arg, Leu, Ser.

An alternative to the NNK ambiguity codon is the NNS ambiguity codon, which also encodes all the amino acids, but changes sixteen of the thirty-two codons. K represents G or T, while S represents G or C. The sixteen codons that contain G in the last position are identical in both choices of ambiguity codon. The remaining sixteen codons are NNT for the NNK ambiguity codon, but are NNC for the NNS ambiguity codon. The NNT codons encode the same amino acids as the NNC codons, but using NNS in place of NNK increases the GC content. Higher GC content increases the melting temperature of the primers and, depending on the rest of sequence, may alter the secondary structure of the primers.

The twelve extra codons in the NNK or NNS ambiguity codons beyond the twenty codons needed to encode the twenty amino acids increase the amount of screening needed, see Section 9.3.2. An alternative to this extra screening is to replace the NNK or NNS mutagenic primer with a mixture of four primers: two with ambiguity codons and two with specific codons, **Table 9.5**. The NDT ambiguity primer encodes twelve amino acids: N, S, I, H, R, L, Y, C, F, D, G, V (N = A, T, C, or G and D = A, T, or G). The VMA ambiguity primer encodes six amino acids: E, A, Q, P, K, T (V = A, C, or G and M = A or C). Finally, the ATG codon encodes M and the TGG codon encodes W. Adding the primers in a 12: 6: 1: 1 ratio encodes all twenty amino acids, with no stop codons, no rare codons, and no synonymous codons.^[23]

9.3 Site-saturation mutagenesis

Table 9.5. A 20-codon mixture of two ambiguity codons (NDT encodes twelve amino acids; VMA encodes six amino acids) and two specific codons (ATG and TGG) efficiently randomizes the encoded amino acid. A 12:6:1:1 mixture of primers containing these four codons encodes all 20 amino acids, no stop codons and includes no redundant codons and no rare codon for yeast or *E. coli*.

codon	nucleotide mixture	encoded amino acid
NDT	AAT	Asn
	AGT	Ser
	ATT	Ile
	TAT	Tyr
	TGT	Cys
	TTT	Phe
	GAT	Asp
	GGT	Gly
	GTT	Val
	CAT	His
	CGT	Arg
	CTT	Leu
VMA	AAA	Lys
	ACA	Thr
	GAA	Glu
	GCA	Ala
	CAA	Gln
	CCA	Pro
ATG	ATG	Met
TGG	TGG	Trp

codon	nucleotide mixture	encoded amino acid
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The disadvantage of this approach is the requirement of four mutagenic primers for each location randomized instead of one mutagenic primer for the NNK or NNS approach. For randomization of a single site, the NNK/NNS approach is simpler, but for randomization of multiple sites the time savings of less screening tips the balance toward the four mutagenic primer approach.

9.3.2 Oversampling

Site-saturation mutagenesis using ambiguity codons yields a mixture of bacteria, each producing a different protein. Spreading the mixture on an agar plate and allowing them to grow yields individual colonies where each colony is derived from a single bacterium. The bacterial colonies are picked and each tested separately. Randomly picking colonies results in picking some duplicate variants, so one needs to test more colonies than the number of possibilities to have a high probability of testing each possible variant. This section explain how much oversampling is required. The conclusion will be that for saturation mutagenesis of a single position using an NNK codon, sampling 94 colonies ensures that the rarest codon in the mixture has a 95% probability of being included in the sample. Thus, one can expect that an amino acid encoded only once (e.g., Met or Trp) is represented, but not that all 20 amino acids (32-codons) are included. To ensure that at least one of each codon was tested requires sampling 200 colonies for a 95% probability.

Including the rarest variant, three-fold oversampling. To derive a general equation, consider the probability to pick a blue ball from an equal and infinite mixture of red, white & blue balls. The probability is 1/3, which is the frequency of the blue balls in the mixture.

9.3 Site-saturation mutagenesis

Lets call this frequency F_{blue} . Similarly, the probability to NOT pick a blue ball from this mixture is $1 - F_{blue}$ or $1 - 1/3$ or $2/3$.

If we pick a second ball, the probability to pick a second blue ball is $1/3$, the same as for the first ball since it is an infinite mixture. The probability for both balls to be blue is $(1/3)^2$ or $1/9$. Similarly, the probability for neither ball to be blue after two picks is $(2/3)^2$ or $(1 - 1/3)^2 = 0.436$. In the general case, the probability NOT to pick a blue ball after T picks would be $(1 - F_{blue})^T$. Therefore, the probability of having a least one blue ball is $1 - (1 - F_{blue})^T$. The probability of picking a least one blue ball after two picks from an equal mixture of red, white & blue is $1 - (1 - 1/3)^2 = 0.564$. In the general case, the probability of picking object i from an infinite mixture, P_i , is $1 - (1 - F_i)^T$ where F_i is the frequency of object i in the mixture and T is the number of times an object is selected from the mixture.

In the context of protein engineering, the probability P_i , that a particular variant, i , is among T tested transformants (colonies) is given by:

$$P_i = 1 - (1 - F_i)^{T_{rarest}} \quad (9.4)$$

where F_i is the frequency at which sequence i is present in the library. This frequency can be replaced by $1/N$ where N is the number of replacement codons. These equations assume the variants form independently according to a Poisson distribution.

$$P_i = 1 - \left[1 - \left(\frac{1}{N} \right) \right]^{T_{rarest}} \quad (9.5)$$

A common situation is a site-saturation mutagenesis created using an NNK ambiguity codon. The number of different codons, N , is 32, so the frequency F_i is 0.03125. While some amino acids are encoded

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by two or even three codons in this mixture and their frequency is higher, one is normally concerned about the rarest members, those amino acids encoded by a single codon.

Rearranging Equation 9.5 yields the number of transformants that needed to be tested.

$$T_{rarest} = \frac{\ln(1 - P_i)}{\ln\left(1 - \frac{1}{N}\right)} \quad (9.6)$$

For a 95% probability of including a particular variant in a saturation mutagenesis experiment using an NNK codon, one needs to test 94 colonies, Equation 9.7.

$$T_{rarest} = \frac{\ln(1 - 0.95)}{\ln\left(1 - \frac{1}{32}\right)} = 94 \quad (9.7)$$

Researchers typically screen $3 \cdot N$ transformants with the expectation that they have included the rarest member of the library. This guideline comes from Equation 9.5 above with a probability of 0.95.

$$\begin{aligned} 0.95 &= 1 - \left[1 - \left(\frac{1}{N}\right)\right]^{T_{rarest}} \\ 0.05 &= \left[1 - \left(\frac{1}{N}\right)\right]^{T_{rarest}} \\ \ln(0.05) &= T_{rarest} \ln\left[1 - \left(\frac{1}{N}\right)\right] \end{aligned}$$

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Using the approximation that $\ln(1 - x) = -x$ when x is small, yields:

$$\ln(0.05) = T_{rarest} \left(-\frac{1}{N} \right)$$

$$T_{rarest} = -N \cdot \ln(0.05) \Rightarrow T_{rarest} = 3 \cdot N \quad (9.8)$$

Thus, three-fold oversampling, or testing three times the number of expected variants, gives a 95% probability that any individual variant has been tested. This three-fold oversampling does not mean that the library has a 95% chance of being 100% complete. To ensure with a 95% probability that ALL variants have been tested requires even more oversampling.

Including all the variants. What is the probability of picking at least one of each object? If you pick 9 balls, the probability to have at least one blue ball is $1 - (2/3)^9$ or $1 - (1 - 1/3)^9 = 0.976$. Similarly the probability to have at least one red ball among the nine is 0.976 and the same for the white ball. The probability of having at least one of each color is $(0.976)^3$ or 0.93. In the general case, the

probability of having all possibilities, $P_i = \left[1 - \left(1 - \frac{1}{N} \right)^{T_{all}} \right]^N$

where N is the number of objects in the mixture. Rearranging this equation yields Equation 9.9.

$$T_{all} = \frac{\ln(1 - (P_i)^{1/N})}{\ln\left(1 - \frac{1}{N}\right)} \quad (9.9)$$

While Equation 9.6 could be simplified to the three-fold oversampling rule in Equation 9.8, no similar simplification is possible for Equation 9.9. For typical cases, including all variants requires

six- to ten-fold oversampling. Code Block S9.2 in the supporting information calculates T_{rarest} and T_{all} using Equation 9.6 and Equation 9.9 for different numbers of variants and probabilities.

9.3.3 Library quality

For this discussion, a library is a collection of DNA or protein sequences. Two features define Q , the quality of this library: completeness, C , and uniqueness, U .^[24] Completeness ensures that one does not miss a possible beneficial variant, while uniqueness avoids testing the same variant multiple times. Library quality, defined by Equation 9.10, has a maximum value of one for a library that contains all possible variants without any duplicates.

$$Q = C \cdot U \quad (9.10)$$

The goal of saturation mutagenesis is to test all possible substitutions at the selected locations. An NNK library is complete ($C = 1$) since it encodes all twenty amino acids, but only 20 of the 32 codons (62.5%) supply this information and the twelve extra codons are duplicates or an unwanted stop codon. The uniqueness is 0.625, so $Q = 0.625$, reflecting the disadvantage that extra screening effort will be needed to find all the amino acids. The quality of an NDT library is similar, $Q = 0.6$. It encodes only twelve amino acids, so $C = 12/20 = 0.60$, but $U = 1$ since all twelve encoded amino acid are unique. The disadvantage is that one will miss eight amino acids. The highest quality library is the 20-codon library from **Table 9.5**, where $Q = 1$ since the library is both unique and complete.

If two locations are targeted for saturation mutagenesis, then the quality of both the NNK and NDT libraries decreases, but the quality of 20-codon library stays a $Q = 1$. The maximum possible number

9.4 Number of possible substitution variants

of proteins is $20 \times 20 = 400$ and the 20-codon library encodes all of them with no extras. Saturation mutagenesis using an NNK codon is complete ($C = 1$) since this library encodes all 400 of the possible 400 proteins, but only 39% unique ($U(20 \times 20)/(32 \times 32) = 0.39$), so $Q = 0.39$. Saturation mutagenesis with an NDT codon is only 36% complete ($C = 0.36$) since it encodes only $12 \times 12 = 144$ of the possible 400 proteins, but 100% unique ($U = 1$), so $Q = 0.36$. As the number of targeted locations increases, the quality of the libraries decreases further. The quality of the NNK library decreases because it contains increasing numbers of duplicates, while the quality of the NDT library decreases because it is increasingly incomplete. The 20-codon library maintains its high quality even when multiple locations are targeted.

9.4 Number of possible substitution variants

The most common change in a protein is substitution of one amino acid with another. An important question is how many substitutions are possible. The answer depends on 1) the number of substitutions (k), the number of replacement amino acids (n), and number of amino acids in the chain (L). It is useful to break the problem into two parts: how many replacements are possible and how many locations are possible. The number of possible protein variants with k substitutions, V_k , depends on the product of the number of location possibilities and the number of replacement possibilities, Equation 9.11.

$$V_k = \text{location possibilities} \times \text{replacement possibilities} \quad (9.11)$$

Review: permutations and combinations

Permutations relate to the order of the elements, while combinations refer to selecting elements from a group. Situations where order matters are permutations, while situations where the order does not matter are combinations. A lock sequence such as 5-27-37 is a permutation because the numbers must be entered in that order. (Confusingly, these locks are called combination locks, while permutation locks would be more accurate.) The other thing to consider in the calculating the number of possibilities is whether items can repeat or not. In the lock sequence, the numbers can repeat.^[25]

Permutations with repetition. The lock sequence example is a permutation where repetition is possible. To determine the number of possible lock sequences, there are say, 39 numbers to choose for the first number, then 39 again for the second number and again 39 for the third. The total number of choices is $39 \times 39 \times 39 = 59,319$. In general, for a permutation with repetition, there are n things to choose from and k choices each time so there are n^k possibilities. In protein engineering, the number of possible amino acid replacements at specified locations is a permutation with repetition. For two locations, x and y , an alanine at position x is different from an alanine at position y . Since order is important (substitution at x differs from substitution at y), it is a permutation. It is further a permutation with repetition since both x and y may contain an alanine.

$$\text{number of possibilities} = n^k$$

For example, number of possible double-substitution variants of a dipeptide is $19^2 = 361$ because there are nineteen possible replacement amino acids and two positions that require a choice.

Permutations without repetition. When choices are used up upon being chosen, then repetition is not possible. For example, choos-

9.4 Number of possible substitution variants

ing an order for sixteen billiard balls is an example of a permutation without repetition. The there are 16 choices at first, then 15 remaining balls and so on. There are $16!$ different orders that these balls could be chosen ($16! \approx 2.1 \times 10^{12}$). In general, there are n things to choose from the first time, $n - 1$ the second time, etc. so there are $n!$ choices in total. In some cases, one may choose fewer than the maximum possible. Then the choices are:

$$\text{number of possibilities} = \frac{\text{total number}}{\text{number not used}} = \frac{n!}{(n - k)!}$$

where k is the number choices made.

For example, one could choose only three billiard balls from the sixteen possible. In this case, $k = 3$ and $n = 16$, so the number of possibilities is $16!/13! = 16 \times 15 \times 14 = 3,360$.

Combinations without repetition. In protein engineering, the number of locations in a protein is a combination without repetition. It does not matter what order they are chosen, but once chosen, they cannot be chosen again. The choice x and y is the same as the choice y and x , but the choice x and x is impossible. Here the order does not matter, but choices cannot be repeated. For example, one could choose billiard balls again, but without regard to the order in which they were chosen. We already have the equation above for the case where order does matter, we just need to reduce it by the number of ways the objects could be in ordered. k objects can be ordered in $k!$ (k for the first choice, $k - 1$ for the second choice, etc.), So the number of combinations without repetition when choosing k objects from a total of n is:

$$\text{number of possibilities} = \frac{\text{total number}}{\text{order does not matter} \times \text{number not used}} = \frac{n!}{k!(n - k)!} \quad (9.12)$$

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We will not consider the fourth possibility, combinations with repetition since it does not occur in the context of protein engineering.

Replacement possibilities = n^k

The number of replacement possibilities at k locations with n replacements is n^k . In the general case, n is the number of objects to choose from (amino acids in this case) and k is the number of choices to be made (positions mutated in this case). For example, replacing the amino acids at positions x and y with new amino acids creates $19^2 = 361$ variants. If you include the original amino acids, then number of possibilities is $20^2 = 400$. This value includes the 361 two-substitution variants, 38 single-substitution variants and one wild-type enzyme. In two more examples, the number of nucleotide sequences of length 6 ($k = 6$) are possible from a set of 4 nucleotides ($n = 4$) is 4^6 or 4,096 and the number of possible amino acid sequences of length 300 ($k = 300$; $n = 20$) is 20^{300} or $\approx 2 \times 10^{390}$. These values include single substitution variants, double substitution variants, triple, etc. This simple calculation firmly establishes that you can never make all the variants of a protein. There are only $\approx 10^{70}$ particles in the universe, so 2×10^{390} is an impossibly large number. One will never be able to test all possibilities.

Location possibilities = $L!/(k!(L - k)!)$

The number of location possibilities is given by Equation 9.13, which is the same as Equation 9.12, except n has been replaced by L to indicate sequence length.

$$\text{location possibilities} = \frac{L!}{k!(L - k)!} = \frac{\text{total number of permutations}}{\text{order not important} \times \text{number of permutations}} \quad (9.13)$$

9.5 Conclusions

For example, to choose two locations in a 300 amino acid protein, there are 300 places to put the first substitution, but only 299 places to put the second substitution. The total number of permutations is 300×299 or 89,700. Since order is not important, we divide by 2! or 2.

$$\frac{300!}{2!(300-2)!} = \frac{300!}{2! \times 298!} = \frac{300 \times 299}{2} = 44,850$$

Hint: Large factorials are too large for many calculators. For example, $300! \approx 3 \times 10^{614}$. Simplify the ratio $L!/(L-k)!$ to $L \times (L-1) \times (L-2) \times \dots (L-k+1)$ as shown above to avoid calculating large factorials.

Number of possible variants

Combining the expression for the number of replacements possible with the number of locations possible yields an expression for the number of possible protein variants with k substitutions, V_k :

$$V_k = \frac{L!}{k!(L-k)!} 19^k \quad (9.14)$$

Code Block S9.3 in the supporting information calculates values of V_k . For example, for one substitution anywhere in a 300-aa protein: $V_1 = 300 \times 19 = 5,700$. For two substitutions anywhere in a 300-aa protein: $V_2 = 44,850 \times 361 = 16,190,850$

9.5 Conclusions

No mutagenesis method comes close to making all variants, and none needs to. The practical question is never whether a library

is complete but whether the improvements sought are frequent enough to appear in the sample actually screened — as the subtilisin work showed, a few thousand colonies can suffice. The two methods in this chapter answer that question under two different assumptions. If many beneficial substitutions exist and are scattered through the structure — the usual situation for stability — error-prone PCR’s broad, biased, incomplete sampling of the whole gene is enough to catch some of them, **Table 9.6**. If the useful substitutions are rare and concentrated near the active site, which is the usual situation for substrate range and selectivity, then site-saturation mutagenesis at chosen positions is the efficient choice. Two ideas underlie both. First, the optimal mutation rate is set by how well the starting protein tolerates substitution, not by how common good substitutions are, which is why “a few substitutions per gene” transfers across systems with very different hit rates. Second, what a library *contains* and what screening *returns* are different things: hits recovered near the active site do not mean the diversity was concentrated there.

Table 9.6. Recommended approaches to make single-substitution variants for different protein engineering goals.

goal	approach	rationale
increase stability	epPCR	many solutions likely distributed throughout the protein
expand substrate range	saturation mutagenesis of residues in substrate-binding site	solutions may be rare and most likely within or close to the substrate binding site

goal	approach	rationale
increase selectivity	saturation mutagenesis of residues in substrate-binding site	solutions may be rare and most likely within or close to the substrate binding site
increase reactivity	mixed	after optimizing binding, distant residues may also contribute to catalysis

This chapter treated the single substitution as the unit of improvement. Real directed evolution rarely stops there. Line 70 of Eigen and Gardiner's program, which feeds the best variants back as templates, is what turns screening into evolution, and it raises questions a single round cannot: how substitutions accumulate over successive rounds, whether their individual effects add up when combined, and how to recombine beneficial changes found separately. Those questions, and the interactions between substitutions that complicate them, are the subject of the next chapter.

Glossary

Ambiguity codon is a mixture of codons that encode different amino acids. A ambiguity codon is indicated by one-letter ambiguity codes that denote mixtures of several nucleotides. For example, the codon NAT in a DNA sequence means that the DNA se-

quence contains a mixture of AAT, TAT, GAT, and CAT at that position.

Codon accessibility is the restriction that, because epPCR usually changes only one nucleotide per codon, only ~5.7 of the 19 alternative amino acids are reachable at a given position; the other ~13.3 require two or three changes in one codon and are much rarer.

Directed evolution is the recursive generation of random sequence variation followed by identification of improved variants by screening or selection, iterated over several rounds. The recursion (feeding enriched genotypes back as templates) distinguishes it from a single round of screening.

Error-prone PCR is PCR performed with a non-proofreading polymerase under conditions (unbalanced dNTPs, added Mn^{2+} , elevated Mg^{2+}) that raise the misincorporation rate, distributing substitutions throughout a target sequence.

Library quality, $Q = C \times U$ is the product of completeness (C , fraction of intended variants encoded) and uniqueness (U , fraction of codons that are non-redundant) and has a maximal value of one for a library encoding all target variants with no duplicates or stop codons.

Location bias is the tendency of random mutagenesis to place most substitutions far from the active site, because only a small fraction of residues (<10%) line the substrate-binding site while the bulk of the protein lies in larger, more distant shells.

Mutator strain is a strain (e.g. *E. coli* XL1-Red) carrying defects in DNA repair and proofreading that raise the spontaneous mutation rate, generating variants in vivo but genome-wide, causing growth defects and instability.

Oversampling is screening more colonies than there are variants to raise the probability that the rarest variant is present. Oversampling is needed because random picking produces duplicates. Three-fold oversampling ($3 \cdot N$) gives ~95% probability

that any individual variant is sampled; ensuring all variants requires six- to ten-fold.

Site-saturation mutagenesis is the targeted replacement of a chosen residue with some or all twenty amino acids using ambiguity (mixed-base) codons, yielding a mixture of variants; used to reshape the active site for altered substrate range or selectivity.

Transition / transversion are the two categories of single-base substitution. A transition exchanges bases of the same size (purine $\leftrightarrow$ purine or pyrimidine $\leftrightarrow$ pyrimidine); a transversion exchanges bases of different size. Four transitions and eight transversions are possible, so random substitution gives a transition/transversion ratio of 0.5, but polymerases favor transitions above this.

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Supporting Information

Code Block S9.1. Python script to calculate the number of doublings, or 100% efficient cycles, for a known number of inefficient cycles when the efficiency is known.

Supporting Information

```
# Calculates D, the number of doublings
#  $2^D = (1+EF)^{\text{cycles}}$  or  $D = \text{cycles} * \log_2(1+EF)$ 
# EF = efficiency of PCR (ranges from 0 to 1)
# cycles = number of inefficient PCR cycles

from math import log
EF = 0.7
cycles = 30.0

D = cycles * log(1.0+EF,2)

print(format(cycles, '.0f'), ' PCR cycles with an efficiency of ',
      format(100*EF, '.0f'), '% yields ',
      format(D, '.1f'), ' doublings.', sep = '')
```

With EF = 0.7 and cycles = 30.0, this prints: “30 PCR cycles with an efficiency of 70% yields 23.0 doublings.”

Code Block S9.2. Python function to calculate the number of transformants, T, needed to have tested the rarest one in a collection of N variations with a probability of P. $T = \ln(1-P)/\ln(1-(1/N))$. In the example, N = 32.0 corresponds to site-saturation mutagenesis of a single site using an NNK or NNS ambiguity codon and P = 0.95 corresponds to a 95% probability. Adjust values for N and P as needed.

```
from math import log          # log = natural logarithm
def numTransformants(N, P):  # N = number of codon variations
                              # P = probability
    T_rarest = log(1-P)/log(1-(1/N))
    print('Testing', round(T_rarest), 'colonies gives a',
          '{:.0f}'.format(100*P)+'% probability that any individual variant',
          'in a collection of', '{:.0f}'.format(N), 'variants has been tested.')
    T_all = log(1-(P**(1/N)))/log(1-(1/N))
```

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```
print('Testing',round(T_all),'colonies gives a',
      '{:.0f}'.format(100*P)+'% probability that every variant',
      'in a collection of',{:.0f}'.format(N),'variants has been tested.'

# example
numTransformants(32*32, 0.95)
```

Calling `numTransformants(32*32, 0.95)` prints: “Testing 3066 colonies gives a 95% probability that any individual variant in a collection of 1024 variants has been tested.” and “Testing 10134 colonies gives a 95% probability that every variant in a collection of 1024 variants has been tested.”

Code Block S9.3. Python script to calculate V_k , the number of variants with k substitutions. $V_k = \text{number of location possibilities} * \text{number of substitution possibilities}$. $nlp = L!/[k! * (L - k)!]$ and $nsp = n^k$ where L = length of sequence, k = number of substitutions and n = the number of replacements, usually 19 for proteins and 3 for nucleotides. Adjust these values as needed. The loop runs from $i = 0$ to $i = k-1$ and calculates the ratio $= L!/(L - k)! = L * (L - 1) * (L - 2) * \dots * (L - k + 1)$.

```
from math import factorial
def Vk(L, n, k):
    # L = length of sequence
    # n = 19 for proteins, 3 for nucleotides
    # k = number of substitutions
    # Loop calculates L!/(L-k)! = L*(L-1)*(L-2)*...*(L-k+1)
    ratio = 1
    for i in range(0,k):
        ratio = ratio*(L-i) # runs from i = 0 to i = k-1
    # number of location possibilities = L!/[k!*(L-k)!]
    nlp = ratio/factorial(k)
    # number of substitution possibilities = n^k
```

```

nsp = n**k
Vk = (nlp*nsp)
print ('The number of possible variants for',k,
      'substitutions using',n,
      'replacements anywhere in a sequence of length',L,
      'is','{:.3G}'.format(Vk))

# example
Vk(4000, 3, 3)

```

With `Vk(4000, 3, 3)`, this prints: “The number of possible variants for 3 substitutions using 3 replacements anywhere in a sequence of length 4000 is 2.88E+11”

Table 9.7. The mixtures of thirty-two codons represented by NNK or NNS both contain the sixteen NNG codons, but differ in the remaining sixteen. NNK contains the sixteen NNT codons, while NNS contains the sixteen NNC codons. In each case, the thirty-two codons encode the same amino acids, which includes all twenty amino acids, one stop codon and eleven redundant codons.

sequence NNK	sequence NNS	encoded amino acid	comment
	AAG	Lys	only codon for this amino acid
AAT	AAC	Asn	only codon for this amino acid
	ACG	Thr	rare codon for yeast
ACT	ACC	Thr	
	AGG	Arg	rare codon for <i>E. coli</i> & yeast
AGT	AGC	Ser	
	ATG	Met	only codon for this amino acid

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sequence NNK	sequence NNS	encoded amino acid	comment
ATT	ATC	Ile	
	CAG	Gln	only codon for this amino a
CAT	CAC	His	only codon for this amino a
	CCG	Pro	rare codon for yeast
CCT	CCC	Pro	CCC is a rare codon for <i>E. co</i>
	CGG	Arg	rare codon for <i>E. coli</i> & yeas
CGT	CGC	Arg	
	CTG	Leu	only codon for this amino a
CTT	CTC	Leu	
	GAG	Glu	only codon for this amino a
GAT	GAC	Asp	only codon for this amino a
	GCG	Ala	rare codon for yeast
GCT	GCC	Ala	
	GGG	Gly	
GGT	GGC	Gly	
	GTG	Val	
GTT	GTC	Val	
	TAG	stop	
TAT	TAC	Tyr	only codon for this amino a
	TCG	Ser	
TCT	TCC	Ser	
	TGG	Trp	only codon for this amino a
TGT	TGC	Cys	only codon for this amino a
	TTG	Leu	
TTT	TTC	Phe	only codon for this amino a

Rarely-used codons can create protein expression problems, but NNK largely avoids this problem, and NNS avoids it in *E. coli* but not in yeast. There are eight rare codons in *E. coli*: CGA, CGG, AGA, AGG for Arg; CTA for Leu; ATA for Ile; GGA for Gly; CCC for Pro.^[22] The tRNAs that recognize rare codons are also present in low amounts. Normally, this low abundance has no effect, but over expressed proteins may deplete the existing pool of charged tRNAs and cause premature termination or misincorporation of amino acids. Two rare codons occur in the NNK library and both code for Arg. However, the NNK library includes a third codon for Arg (CGT), which is not a rare codon. There are eight rare codons in yeast: AGG, CGA, CGG, CGC for Arg; CCG for Pro, CTC for Leu, GCG for Ala; ACG for Thr.^[22] The five rare codons appear in the NNK library, but, like the *E. coli* case, the NNK library also includes synonymous non-rare codons for each of the amino acids. The same analysis gives a different result for NNS. Because NNS replaces the T of NNK with C in the third position, it loses access to the NNT codons (CGT, CCT, GCT, ACT) that supply the common synonyms above. For *E. coli* this causes no problem — every amino acid still retains a non-rare codon under NNS. In yeast, however, the only arginine codons available under NNS are CGC, CGG, and AGG, all three of which are rare; NNS therefore provides no common arginine codon in yeast. For this reason NNK is generally preferable to NNS, particularly for expression in yeast.

Problems

9.1. PCR cycles, doublings, and mutation frequency.

- a) How many PCR cycles are required for 17 doublings if the amplification efficiency E is 1.0 (perfect), 0.8 (typical for a well-optimized PCR), or 0.47 (typical for an error-prone PCR)? Am-

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plification efficiency is defined by $(1 + E)^n = A$ for n cycles and A -fold amplification. Efficiency also falls as the reaction approaches plateau, so this is an average over the run rather than a per-cycle constant.

- b) Using $d \approx 17$ doublings and a per-pass error rate $f \approx 4 \times 10^{-4}$, estimate the mutation frequency per position in the product. Does running more cycles at the lower error-prone efficiency raise this frequency? Explain.

9.2. Oversampling an NNK library. Show that screening 203 colonies gives a 95% probability that *every* codon of a single-site saturation library made with an NNK ambiguity codon has been sampled. Section 9.3.2 shows that only 94 colonies are needed for the same 95% probability of sampling any *one* chosen variant. In one sentence, state what each number guarantees and why they differ.

9.3. Number of possible dipeptides.

- a) How many possible dipeptides are there?
- b) Starting from the dipeptide Asp-Phe, how many single-substitution variants are possible?
- c) How many double-substitution variants are possible?
- d) Show that the sum of wild-type, single-, and double-substitution variants equals the total.

9.4. Transitions versus transversions. Write out all possible base substitutions to show that random mutagenesis should yield twice as many transversions as transitions. Taq's measured transition/transversion ratio is 0.76 (Table 9.2). Compared with the random expectation, does Taq favor or disfavor transitions, and by how much?

9.5. The impossibility of exhaustive sampling. Consider a small protein of 50 amino acids. How many variants are possible if all 20

amino acids are allowed at every position? If the average molecular weight is 5,500 g/mol, what is the mass, in grams, of a collection containing one molecule of each variant? Compare this with the mass of the earth, $\approx 6 \times 10^{27}$ g.

9.6. Codon accessibility from a wild-type codon. A position carries wild-type Phe (codon TTT).

- List every amino acid reachable by a single nucleotide change.
- Of the other hydrophobic residues (Leu, Ile, Val, Met, Trp), which are reachable by one change and which are not?
- Explain why an error-prone PCR library screening for Phe→Trp or Phe→Met is likely to fail, and what method you would use instead.

9.7. Interpreting the NDT ambiguity codon.

- How many distinct codons does NDT represent, and which amino acids does it encode?
- Explain why NDT produces no stop codon and no synonymous duplicates, unlike NNK.
- Using $Q = C \cdot U$, compare NNK and NDT for a two-site library. Which requires less screening, and what is the trade-off?

9.8. Choosing a mutagenesis strategy. For each goal, state whether error-prone PCR or site-saturation mutagenesis is the better first approach and give a one-sentence rationale:

- raise the melting temperature by 8 °C
- accept a bulkier substrate at the active site
- invert enantioselectivity toward the (*R*)-product
- improve activity when no structure and no homology model is available

Answers

9.1. a) With $(1 + E)^n = 2^{17}$, $n = 17 / \log_2(1 + E)$: at $E = 1.0$, $n = 17$ cycles; at $E = 0.8$, $n \approx 20$ cycles; at $E = 0.47$, $n \approx 31$ cycles. Lower efficiency needs more cycles to reach the same amplification. b) Mutation frequency $\approx f \cdot d = 4 \times 10^{-4} \times 17 = 6.8 \times 10^{-3}$, about 0.7% per position (~7 changes per 1000 bp). It depends on the number of *doublings*, not the number of cycles: because errors accumulate per doubling, a low-efficiency reaction that reaches the same fold-amplification gives the same mutation frequency – it just takes more cycles to get there.

9.2. Using $P_i = [1 - (1 - 1/N)^T]^N$ with $N = 32$:

$$T_{all} = \frac{\ln(1 - (0.95)^{1/32})}{\ln(1 - \frac{1}{32})} = \frac{-6.437}{-0.03175} = 203.$$

The 94-colony figure guarantees a 95% chance that *one particular* codon (e.g., the single codon for Met) appears; the 203-colony figure guarantees a 95% chance that *all* 32 codons appear. Covering every codon is stricter, so it requires more than twice the sampling.

9.3. a) $20^2 = 400$. b) $2 \times 19 = 38$. c) $19^2 = 361$. d) $1 + 38 + 361 = 400 = 20^2$.

9.4. Transitions (4): A→G, G→A, C→T, T→C. Transversions (8): A→C, A→T, G→C, G→T, C→A, C→G, T→A, T→G. Random substitution therefore gives 8 transversions to 4 transitions, a ratio of 0.5. Taq's measured ratio of 0.76 is $0.76/0.5 = 1.5$ times the random value, so Taq favors transitions about 1.5-fold over a random polymerase.

9.5. Variants $= 20^{50} \approx 1.1 \times 10^{65}$. Mass $= (1.1 \times 10^{65} \times 5,500) / (6.02 \times 10^{23}) \approx 1 \times 10^{45}$ g, which is $\approx 1 \times 10^{45} / (6 \times 10^{27}) \approx 2 \times 10^{17}$ earths. Exhaustive sampling of even a small protein is physically impossible.

9.6. a) Single changes to TTT reach nine new codons encoding six new amino acids: Ile (ATT), Val (GTT), Leu (CTT, TTA, TTG), Tyr (TAT), Cys (TGT), Ser (TCT); TTC is a synonymous Phe. No stop codon is one change away. b) Reachable in one change: Leu, Ile, Val. **Not reachable: Met (ATG) and Trp (TGG)** — each needs two nucleotide changes from TTT. c) Phe→Trp and Phe→Met each require two simultaneous changes in one codon, which error-prone PCR produces ~100-fold less often than single changes, so these substitutions are rarely present in an epPCR library. Use site-saturation mutagenesis, which installs any target amino acid regardless of codon distance.

9.7. a) $NDT = (A/T/G/C)(A/T/G)(T) = 4 \times 3 \times 1 = 12$ codons, encoding twelve amino acids, one each: Asn, Ser, Ile, Tyr, Cys, Phe, Asp, Gly, Val, His, Arg, Leu. b) Fixing the third position as T excludes the stop codons TAA, TAG, and TGA (all end in A or G), and the twelve codons were chosen so that each specifies a different amino acid — hence no synonymous duplicates. c) For two sites: NNK is complete ($C = 1$) but only $U = 400/1024 = 0.39$ unique, so $Q = 0.39$; NDT is fully unique ($U = 1$) but only $C = 144/400 = 0.36$ complete, so $Q = 0.36$. NDT requires far less screening (12 codons per site versus 32, and no duplicates or stop codons), at the cost of sampling only 12 of the 20 amino acids at each position.

9.8. a) **Error-prone PCR** — stabilizing substitutions are common and distributed throughout the structure, so whole-gene sampling is efficient. b) **Site-saturation mutagenesis** of substrate-binding residues — a bulkier substrate needs reshaping of the pocket, where useful substitutions are rare and localized. c) **Site-saturation mutagenesis** of substrate-binding residues — selectivity changes cluster near the active site (cf. the nitrilase Ala190His and PAMO examples). d) **Error-prone PCR** — with no structure or homology model there is no way to localize the search, so whole-protein sampling is the practical starting point.

10 In revision

This chapter is in revision and will be posted here once it is ready.

